# LOCAL EXCLUSION OF RESIDUAL TYPE I NAVIER–STOKES SEREGIN LIMITS

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Abstract. This paper proves the residual-class hypothesis left open by the companion setup paper for local pointwise Type-I Navier–Stokes blow-up. Starting from the setup Seregin extraction, every remaining bounded centered ancient profile with retained local velocity $L^3$-concentration is assigned to an ordered residual decomposition. Axisymmetric bounded-circulation profiles are excluded by a centered angular-circulation absorption argument. Rotational and stationary-hull profiles are reduced to a terminal concentration theorem. The generic terminal branch is closed by a local concentration-set decomposition, diffuse-defect and critical-tail compactifications, a minimal mesoscopic-scale rigidity theorem, exclusion of separated retained profiles, and an endpoint sequence-$L^3$ Liouville argument for bounded mild ancient solutions. No global critical-norm estimate or global Liouville theorem for bounded centered profiles is used.

## Contents



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## 1. Introduction and main theorem

The incompressible Navier–Stokes equations on $\mathbb{R}^3 \times (0, T)$ are written as

$$\partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = 0, \qquad \nabla \cdot u = 0. \tag{1.1}$$

The residual analysis in this paper begins after the standard Type I blow-up construction of Seregin has been performed [8]. Thus the basic solution is a smooth bounded ancient solution of the centered equation

$$\partial_\tau V - \Delta V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V + (V \cdot \nabla)V + \nabla P = 0, \qquad \nabla \cdot V = 0, \tag{1.2}$$

on $\mathbb{R}^3_y \times \mathbb{R}_\tau$. The equation is obtained from the physical ancient representative $(U, Q)$ by

$$y = \frac{x}{\sqrt{-t}}, \qquad \tau = -\log(-t), \qquad V(y,\tau) = \sqrt{-t}\, U(x,t), \qquad P(y,\tau) = (-t)Q(x,t), \tag{1.3}$$

where $t < 0$ and pressures are always understood modulo addition of a function of time. Conversely,

$$U(x,t) = (-t)^{-1/2} V\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \qquad t < 0. \tag{1.4}$$

The purpose of this paper is to present the refined residual decomposition as a single standard PDE argument, using only the reductions already established in the companion setup paper. In this two-paper presentation, the current paper depends only on the imported setup results stated below and on cited Navier–Stokes regularity and Liouville literature [4, 8]. The rotational-flux estimates, terminal concentration-set decompositions, exclusions of local compactness failure and diffuse exterior concentration, and finite separated-profile exclusions are proved in this paper from local estimates rather than assumed as global estimates.

This paper does not reconstruct the Seregin extraction or the endpoint mildness theory. Those are imported from the setup paper. The sole new task is to prove that the residual alternative left open there is empty, namely

$$\mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing.$$

1.1. **Proof architecture.** The retained normalized residual class $\mathcal{R}^{\#}(\mathcal{S}; I, J)$ of Definition 1.7 is the complement, inside the normalized Seregin collection $\mathcal{S}$, of the alternatives already removed in the setup paper. After the previously treated small, stationary $L^3$, uniformly $L^3$-tight, and closed structured/decay alternatives have been removed (Definition 1.6), $\mathcal{R}^{\#}(\mathcal{S}; I, J)$ is divided in Definition 1.8 into four analytic classes – axisymmetric bounded-circulation profiles $\mathcal{C}_{\rm ax}$, rotational relative equilibria $\mathcal{C}_{\rm rot}$, degenerate stationary-hull profiles with retained local velocity concentration $\mathcal{C}^{\rm ret}_{\rm stat}$, and the generic terminal residual $\mathcal{C}_{\rm gen}$ – and the terminal generic case is itself refined into the affine/parasitic quotient stratum $\mathcal{C}_{\rm aff}$, the log-diffuse and Young critical-tail alternatives $\mathcal{C}_{\log\,\rm diff}, \mathcal{C}_{\rm Young}$, the coherent homogeneous, log-periodic, and aperiodic critical-tail alternatives $\mathcal{C}_{\rm homcrit}, \mathcal{C}_{\rm logper}, \mathcal{C}_{\rm ap}$, and the residual remainder $\mathcal{C}_{\rm gen}$.

The closure of these classes is organized as follows.

1. $\mathcal{C}_{\rm ax}$ is excluded directly by a centered angular-circulation equation and an axis-absorption Liouville theorem (Theorem 4.4).
2. $\mathcal{C}_{\rm rot}$ is reduced by a local Coriolis-flux identity to either lower classes already excluded by the setup paper or the terminal stratification assembly (Proposition 5.8); its closure is obtained only after the terminal assembly has excluded retained velocity concentration profiles.
3. $\mathcal{C}^{\rm ret}_{\rm stat}$ is reduced, by scale reselection for time-translation hulls, to a stationary profile in a degenerate stationary family which occurs as a Seregin limit and still carries local velocity concentration, and is then excluded by the terminal stratification assembly theorem (Proposition 7.6).
4. $\mathcal{C}_{\rm gen}$ and the critical-tail subclasses are closed by covariant tail normalization, concentration-set extraction, diffuse-defect compactness, critical-tail compactification, recurrence for chains of concentration profiles, exclusion of finite separated concentration profiles, indecomposable-profile sequence-$L^3$ completeness, parasitic-free mildness inheritance, and the Albritton–Barker endpoint Liouville theorem.

The paper distinguishes statements inherited from the setup paper, local compactness estimates and reduction results proved here, and the minimal mesoscopic-scale reduction theorem (Theorem 9.81) which excludes the critical-tail boundary defects. The affine stratum and the coherent critical-tail alternatives are closed here; the log-diffuse and Young alternatives are both reduced to the same minimal mesoscopic-scale theorem, which is then proved by pressure compactness and a bounded ancient variance Liouville argument. No global Coriolis–Leray Liouville theorem for bounded profiles or global critical-norm estimate is used in the residual closure. A short reader's guide is given in Section 1.3 below.

**Definition 1.1** (Type I convention used in the chain)**.** In this paper and in the final two-paper chain, a terminal singular point $z_* = (x_*, T)$ is called *local pointwise Type I* if there are $\rho > 0$ and $M < \infty$ such that

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\, \|u(t)\|_{L^\infty(B_\rho(x_*))} \le M.$$

This is the local pointwise Type I hypothesis used in the setup paper and in Section 1.4. For suitable weak solutions this envelope lies in the endpoint weak Serrin class $L^{2,\infty}_t L^\infty_x$. The Albritton–Barker weak-Serrin local Type I estimate, recorded below as Theorem 1.24, upgrades it on smaller cylinders to the scale-invariant local energy bounds needed for terminal velocity no-escape. Thus the terminal no-escape condition used by the setup paper is automatic for the local pointwise Type I singularities considered here. The paper does not use the phrase "Type I" to mean an unrelated global $L^\infty$, local CKN, weak-$L^3$, or Serrin scale condition unless that condition is explicitly stated.

*Remark* 1.2 (Translation table for standard Type I hypotheses)*.* The global pointwise condition

$$\sup_{t<T} \sqrt{T-t}\,\|u(t)\|_{L^\infty(\mathbb{R}^3)} < \infty$$

implies Definition 1.1 at every terminal point. Any stronger local criterion that yields the displayed $L^\infty$ envelope on a neighborhood of $x_*$ enters the same argument, provided the solution is suitable in the corresponding cylinder. The endpoint weak Serrin estimate then gives the local scale-invariant energy bounds and terminal no-escape. Purely CKN-type, weak-$L^3$, or other critical criteria are not silently identified with this hypothesis; they require their own implication into the local pointwise envelope, or directly into the local Type I estimates, before the present exclusion applies.

### 1.2. **Imported setup results and remaining hypothesis.**

**Theorem 1.3** (Imported setup results)**.** *The companion setup paper proves the following facts, in the exact theorem numbers listed below.*

(1) *Positive terminal local velocity concentration at every local pointwise Type I singular point.*
*Theorem 5.7 of the setup paper*
*Proposition 5.9 of the setup paper*

(2) *Local pointwise Type I control gives an admissible Seregin extraction sequence with pressure gauges, no-escape, and retained unit-scale velocity concentration.*
*Theorem 8.3 of the setup paper*
*Corollary 8.4 of the setup paper*
*Lemma 8.6 of the setup paper*

(3) *Every normalized Seregin limit produced by this extraction is a bounded ancient suitable weak solution, admits the pressure atlas of the setup paper, and has the mild ancient representative required by the endpoint Liouville theorem.*
*Theorem 9.1 of the setup paper*
*Proposition 11.12 of the setup paper*
*Proposition 12.2 of the setup paper*
*Lemma 11.2 of the setup paper*
*Lemma 11.3 of the setup paper*
*Lemma 11.4 of the setup paper*
*Lemma 15.3 of the setup paper*
*Lemma 20.4 of the setup paper*
*Lemma 25.4 of the setup paper*

(4) *The previously classified small, stationary $L^3$, uniformly tight, and known structured/decay alternatives are empty.*
*Theorem 15.8 of the setup paper*
*Theorem 15.10 of the setup paper*
*Theorem 26.1 of the setup paper*
*Theorem 25.8 of the setup paper*

(5) *The endpoint and mildness results include the Albritton–Barker input and the setup paper proves local pointwise Type I exclusion conditionally on the residual-class hypothesis.*
*Theorem 25.5 of the setup paper*
*Hypothesis 16.1 of the setup paper*
*Theorem 33.1 of the setup paper*

*The purpose of the present paper is to prove the residual closure $\mathcal{R}^\#(\mathcal{S}; I, J) = \varnothing$. This verifies Hypothesis 16.1 of the setup paper.*

*Remark* 1.4 (Use of imported labels)*.* Below, the phrase "by the imported setup results" means that the terminal profile is bounded, locally suitable, pressure-normalized by the setup pressure atlas, parasitic-free modulo the affine class, and mild after finite terminal shift, with the exact imported labels listed in Theorem 1.3. Paper IV proves only the residual closure $\mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing$.

(1) Local concentration and terminal no-escape entry: positive local concentration depends on CKN theory and the setup paper; terminal no-escape follows from the endpoint weak Serrin local Type I bounds in Theorems 1.24 and 1.26.
(2) Seregin extraction and parasitic-free normalization: follows from the setup paper and the mildness and compactness arguments in this paper.
(3) Small and stationary $L^3$ strata: handled in the setup paper.
(4) Uniformly tight stratum: handled in the setup paper.
(5) Closed structured/decay strata: handled in the setup paper.
(6) Axisymmetric and rotational residual strata: proved in this paper.
(7) Terminal concentration-set decomposition, recurrence for concentration chains, and diffuse-defect compactness: proved in this paper.
(8) Critical-tail compactification and realized critical-tail decomposition: proved in this paper.
(9) Residual indecomposable-profile closure: proved in this paper, with the final sequence-$L^3$ endpoint supplied by Albritton–Barker.

**Theorem 1.5** (Admissible Type I Seregin extraction theorem)**.** *An admissible finite-energy Type I singular point produces a normalized Seregin ancient limit $(V, P)$ satisfying (1.2), with*

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le M < \infty,$$

*local suitability and local pressure-gauge compactness, and a compact parabolic cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ for which*

$$\iint_K |V|^3 \, dy \, d\tau \ge \eta_0 > 0. \tag{1.5}$$

*The collection of all normalized Seregin limits generated by the original solution is denoted by $\mathcal{S}$.*

*Proof.* This is the base admissible Seregin extraction theorem included in Theorem 1.3: velocity nonvanishing, local suitability, and local pressure compactness are imported setup conclusions. The present paper uses this inherited reduction and does not reprove the Type I blow-up compactness theorem. □

**Definition 1.6** (Earlier alternatives inherited from the setup paper)**.** Within a fixed normalized Seregin collection $\mathcal{S}$, the small-amplitude, stationary $L^3$, uniformly $L^3$-tight, and closed structured/decay classes are the non-residual alternatives already established in the setup paper. The small-amplitude and stationary $L^3$ alternatives are closed there; the uniformly tight class is excluded there; and the closed structured/decay class is precisely the part proved empty there. Any structured profile not belonging to that closed class is residual by definition and is handled by the present paper. The present paper does not reprove those earlier alternatives and does not add any hypothesis to their statements.

**Definition 1.7** (Coarse residual class)**.** Let $I, J$ be the index sets used in the base Type I residual criterion. The pre-quotient coarse residual class

$$\mathcal{R}_{\mathrm{pre}}(\mathcal{S}; I, J)$$

is the complement in $\mathcal{S}$ of the already removed alternatives in Definition 1.6. The retained normalized residual class

$$\mathcal{R}^{\#}(\mathcal{S}; I, J)$$

is obtained from $\mathcal{R}_{\mathrm{pre}}(\mathcal{S}; I, J)$ by passing to the canonical affine/parasitic representative and by removing any representative assigned to the affine/parasitic lower stratum. Unless the word

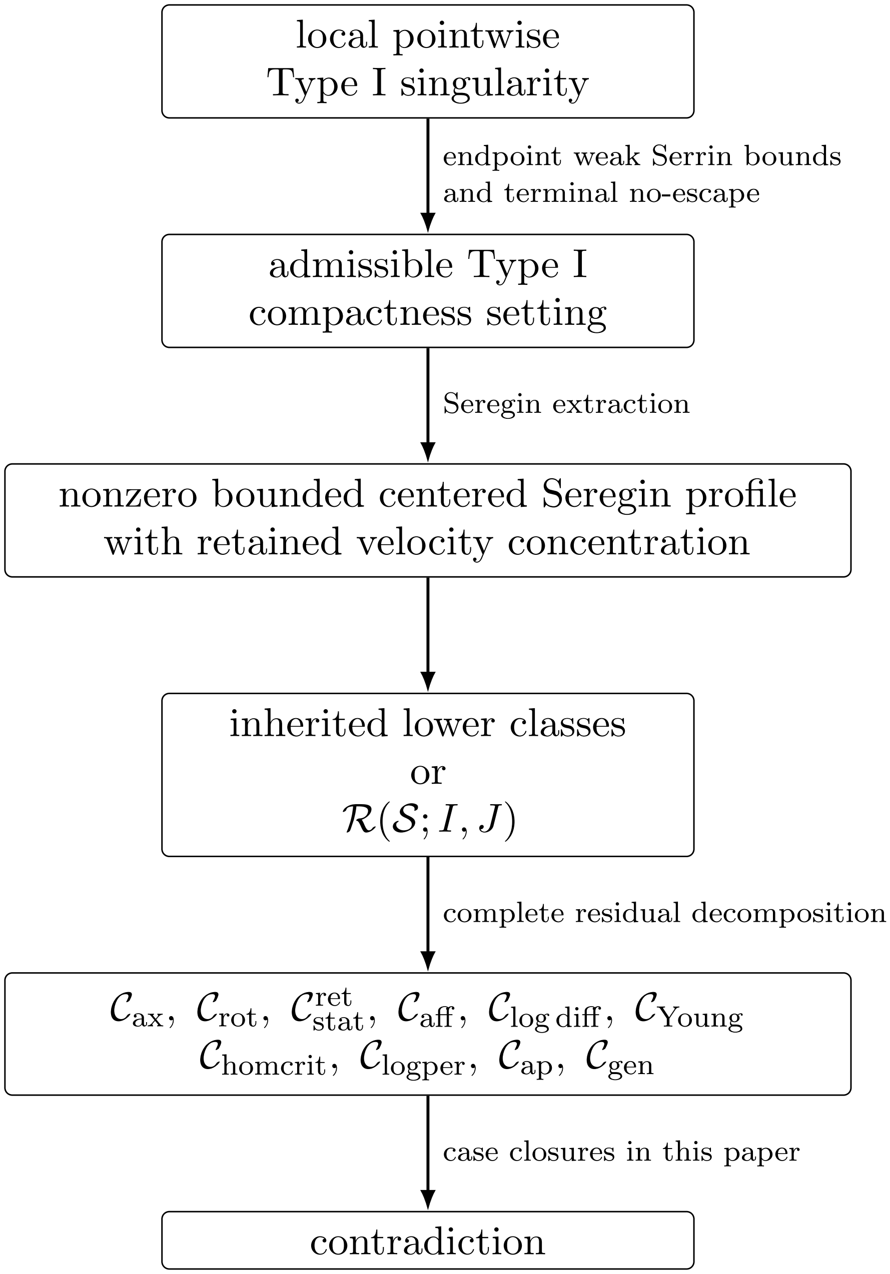


FIGURE 1. Decision diagram for the Type I profile decomposition. The first arrow uses the local pointwise Type I to terminal no-escape implication supplied by the endpoint weak Serrin local Type I estimate. The inherited lower classes in the split box are already closed in the setup paper. The residual alternatives in the boxed list are excluded by the case-exclusion theorems cited in the proof of Theorem 1.14.

unreduced is used explicitly, $\mathcal{R}(\mathcal{S}; I, J)$ denotes this retained normalized residual class. Thus an "empty affine/parasitic stratum" below always means that no retained normalized residual representative remains in that quotient stratum.

**Definition 1.8** (Refined residual decomposition)**.** The refined residual decomposition is an ordered definition, not a separate compactness theorem. Starting with the pre-quotient coarse residual class $\mathcal{R}_{\mathrm{pre}}(\mathcal{S}; I, J)$, one removes the following alternatives in order; the affine/parasitic alternative is then recorded as a quotient stratum with no retained normalized representative in $\mathcal{R}^{\#}(\mathcal{S}; I, J)$. After the first nine alternatives have been removed, whatever remains is by definition the generic terminal residual $\mathcal{C}_{\mathrm{gen}}$:

$$\mathcal{C}_{\mathrm{ax}}, \qquad \mathcal{C}_{\mathrm{rot}}, \qquad \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}, \qquad \mathcal{C}_{\mathrm{aff}},$$
$$\mathcal{C}_{\mathrm{log\,diff}}, \qquad \mathcal{C}_{\mathrm{Young}}, \qquad \mathcal{C}_{\mathrm{homcrit}}, \qquad \mathcal{C}_{\mathrm{logper}}, \qquad \mathcal{C}_{\mathrm{ap}}, \qquad \mathcal{C}_{\mathrm{gen}}.$$

Here $\mathcal{C}_{\mathrm{ax}}$ is the axisymmetric bounded-circulation class, $\mathcal{C}_{\mathrm{rot}}$ is the rotational relative-equilibrium class, $\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$ is the degenerate stationary-hull class with retained local velocity concentration, $\mathcal{C}_{\mathrm{aff}}$ is the affine/parasitic lower stratum selected by affine-normalized oscillatory entry, $\mathcal{C}_{\mathrm{log\,diff}}$ and $\mathcal{C}_{\mathrm{Young}}$ are respectively the log-diffuse and Young critical-tail alternatives, $\mathcal{C}_{\mathrm{homcrit}}, \mathcal{C}_{\mathrm{logper}}, \mathcal{C}_{\mathrm{ap}}$ are the coherent homogeneous, log-periodic, and aperiodic critical-tail alternatives, and $\mathcal{C}_{\mathrm{gen}}$ is the generic terminal residual class. The inclusion below is part of this ordered definition, written for the retained normalized residual class; the appearance of $\mathcal{C}_{\mathrm{aff}}$ records the quotient alternative before retained normalization, and $\mathcal{C}_{\mathrm{aff}} \cap \mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing$ is proved in Theorem 9.89. The substantive content is the closure of each non-quotient alternative, especially $\mathcal{C}_{\mathrm{gen}}$. Equivalently,

$$\mathcal{R}(\mathcal{S}; I, J) \subset \mathcal{C}_{\mathrm{ax}} \cup \mathcal{C}_{\mathrm{rot}} \cup \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{log\,diff}} \cup \mathcal{C}_{\mathrm{Young}} \cup \mathcal{C}_{\mathrm{homcrit}} \cup \mathcal{C}_{\mathrm{logper}} \cup \mathcal{C}_{\mathrm{ap}} \cup \mathcal{C}_{\mathrm{gen}}. \tag{1.6}$$

**Definition 1.9** (Profiles realized from the original blow-up sequence)**.** Fix a normalized Seregin collection $\mathcal{S}$ and a retained residual candidate $(V, P) \in \mathcal{R}(\mathcal{S}; I, J)$. An auxiliary profile or limiting measure used in the residual proof is *realized from the original blow-up sequence* if it is obtained from $(V, P)$, or from the terminal sequence which generated $(V, P)$, by a finite or diagonal sequence of the following operations:

(i) exact centered symmetries $T_a$, $\Theta_s$, and the covariant observer maps $\mathcal{R}_z$;
(ii) unreduced parabolic recenterings before passing to centered variables, with the local pressure gauges supplied by Seregin compactness;
(iii) retained tail limits or concentrating successors with $\mathcal{V}_1 \geq \eta_*$ after the fixed admissible pressure gauges;
(iv) normalized diffuse or critical-tail defect limits obtained by restricting the associated defect measure to a positive-measure window and renormalizing it;
(v) minimal mesoscopic blow-downs selected from such retained or normalized defect windows.

When the shorter phrase "realized from the original sequence" is used below, it refers to this definition and to the original singularity-generated candidate $(V, P)$, not merely to the immediately preceding normalized profile.

**Definition 1.10** (Retained descendants and normalized support profiles)**.** A *retained descendant* is a descendant with a genuine inherited positive unit velocity concentration bound

$$\liminf_{n} \mathcal{V}_1(\cdot; 0) \geq \eta_* > 0.$$

A *normalized support profile*, also called a *support profile descendant*, is a normalized profile in the support of a defect measure or normalized window measure associated with the original blow-up sequence. It may arise from unnormalized mass tending to zero and therefore need not carry the original velocity threshold. Log-window and Young/variance normalizations are called retained only when the displayed unit velocity concentration bound is explicitly proved; otherwise they are normalized support profiles.

Table 1: Stability of the equation, gauges, and concentration bounds under limiting operations.

| **Operation** | **Equation** | **Boundedness** | **Pressure gauge** | **Mildness** | **Threshold** |
|---|---|---|---|---|---|
| Centered time shift | yes | yes after finite shift | shifted gauge | setup mildness | yes |
| Spatial recentering | covariant | yes | translated gauge | yes | yes |
| Parabolic rescaling | yes | normalized scale | scaled gauge | yes | yes |
| Tail recentering | quotient covariant | locally yes | tail gauge | setup closure | support profile or retained |
| Log-window normalization | normalized defect equation | normalized | quotient gauge | not automatic | support profile |
| Affine pressure quotient | modulo $\mathcal{C}_{\mathrm{aff}}$ | yes | affine removed | yes outside $\mathcal{C}_{\mathrm{aff}}$ | specified |
| Critical-tail blow-down | limiting equation | locally yes | scaled/quotient gauge | after bounded-origin realization | support profile |

**Theorem 1.11** (Stability under realized limiting operations)**.** *Every descendant, tail limit, concentrating successor, log-window normalization, critical-tail profile, and minimal-scale blow-down used in the proof of Theorem 1.14 is realized from the original blow-up sequence in the sense of Definition 1.9. More precisely, each such limit satisfies one of the following alternatives.*

(a) *It is a retained Seregin descendant of the original candidate, with the fixed descendant lower bound $\mathcal{V}_1 \geq \eta_*$.*

(b) *It is a normalized defect profile whose unnormalized defect measure has positive total mass on the selected window. If every normalized window descendant is excluded, then the original defect is excluded.*

(c) *It lies in a previously closed lower stratum, and the retained descendant heredity theorem applies.*

(d) *It generates a retained descendant or a normalized defect descendant satisfying* (a), (b), *or* (c).

*Consequently a contradiction obtained for a realized limiting profile is a contradiction for the original retained residual candidate, not merely for an auxiliary compactification limit. The theorem is used only through the concrete support-transfer statements proved for the individual compactifications below.*

*Proof.* It suffices to verify the assertion for the operations listed in Definition 1.9. Exact centered symmetries preserve the centered equation, local suitability, pressure gauges, and retained local velocity concentration; this is recorded in Lemma 8.2 and Theorem 8.6. Unreduced terminal recenterings before centering are the recenterings in the Seregin extraction theorem of the setup paper and are compact after the gauges in Proposition 1.19. Retained tail limits and concentrating successors are realized from the original sequence by diagonal lifting: Lemma 10.11 and Theorem 10.12 show that the limit remains in the normalized compactness class and keeps the fixed retained velocity threshold through its velocity part, up to the once-for-all threshold loss in Definition 3.4.

Normalized defect profiles are obtained by restricting a positive defect measure to a window and dividing by its mass. The mass of the chosen window may depend on the index and may tend to zero; this does not detach the normalized profile from the original sequence. It records a point in the support of the defect compactification. The renormalized-window heredity lemma

Lemma 9.52 below states explicitly that if all such support profiles are assigned to closed lower alternatives, then the defect measure itself has no surviving mass outside those alternatives.

Minimal-scale blow-downs are selected from retained or normalized defect windows; their centers and scales therefore come with the same defect measure or retained velocity lower bound. The pressure/gauge normalization for those blow-downs is treated in Lemmas 9.67 and 9.73 and Proposition 9.70. Thus every contradiction later in the proof is applied either to a retained descendant of the original profile or to a normalized support profile whose closure removes the corresponding defect. This proves the stated principle. □

**Lemma 1.12** (Support transfer in compact metrizable defect spaces)**.** *Let $X$ be a compact metrizable defect space and let*

$$\lambda_n = \frac{\mu_n \lfloor E_n}{\mu_n(E_n)} \qquad (\mu_n(E_n) > 0)$$

*be normalized realized defect measures converging weakly to $\lambda \in \mathcal{P}(X)$. Let $A \subset X$ be closed. If every normalized support profile in $\operatorname{supp}\lambda$ belongs to $A$, then the original normalized defect has no surviving mass in $X \setminus A$. Equivalently, if a positive amount of the normalized defect survives outside $A$, then there exists a support profile in $X \setminus A$.*

*Proof.* If positive normalized defect survives outside $A$, then $\lambda(X \setminus A) > 0$. Since $X$ is compact metrizable and $A$ is closed, $X \setminus A$ is open. By inner regularity choose a compact set $K \subset X \setminus A$ with $\lambda(K) > 0$. Then $K \cap \operatorname{supp}\lambda \neq \varnothing$, and every point of this intersection is a normalized support profile outside $A$, contradicting the support hypothesis. Hence $\lambda$ is supported on $A$, which is exactly the assertion that no normalized defect mass survives outside the closed excluded set. □

**Lemma 1.13** (Transfer of contradictions to the original blow-up sequence)**.** *Let $(V, P) \in \mathcal{R}(\mathcal{S}; I, J)$ be a retained residual candidate, and let $\mathcal{A}$ be a closed union of lower alternatives which is stable under the descendant operations used in this paper. Suppose that every limit realized from $(V, P)$ by the relevant construction has one of the following two properties:*

(i) *it is a retained Seregin descendant lying in $\mathcal{A}$;*
(ii) *it is a normalized support profile of a defect measure associated with the original sequence and lies in $\mathcal{A}$.*

*If $\mathcal{A}$ has already been excluded in the ordered residual decomposition, then the original candidate $(V, P)$ cannot exist.*

*Proof.* In case (i), retained descendant heredity Theorem 10.12 identifies the descendant as an admissible successor of the original residual candidate with the velocity threshold fixed in Definition 3.4. Since $\mathcal{A}$ is closed under the allowed limiting operations and is excluded, the retained descendant cannot occur.

In case (ii), the normalized support profile is obtained by restricting a positive defect measure to a window and renormalizing. If all such normalized support profiles lie in the closed excluded set $\mathcal{A}$, then regularity of the compactified defect measure implies that the defect measure has no mass outside $\mathcal{A}$; this is the support-transfer mechanism of Lemma 1.12, instantiated below for the individual compactifications. Hence the defect has no surviving support in the residual class after $\mathcal{A}$ is removed. In both cases the construction cannot produce a retained residual profile from $(V, P)$, contradicting the defining retained velocity concentration of the residual candidate. □

**Theorem 1.14** (Refined residual closure for the complete ordered decomposition)**.** *Relative to the inherited alternatives in Theorem 1.5 and Definition 1.6, one has*

$$\mathcal{C}_{\mathrm{ax}} = \mathcal{C}_{\mathrm{rot}} = \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}} = \mathcal{C}_{\mathrm{gen}} = \varnothing,$$
$$\mathcal{C}_{\log\mathrm{diff}} = \mathcal{C}_{\mathrm{Young}} = \mathcal{C}_{\mathrm{homcrit}} = \mathcal{C}_{\mathrm{logper}} = \mathcal{C}_{\mathrm{ap}} = \varnothing.$$

*Moreover,*

$$\mathcal{C}_{\mathrm{aff}} \cap \mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing,$$

*that is, no retained normalized residual representative remains in the affine/parasitic quotient stratum. Consequently*

$$\mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing.$$

*Proof.* Deferred. The proof is given in Section 12, after the terminal decomposition theorem and the exclusion of all reduced cases. The dependency order is recorded in Table 2. □

Table 2: Dependency table for the residual closure proof.

| **Result** | **Uses** | **Does not use** |
|---|---|---|
| Imported setup results | Setup paper only: local concentration, local Type I implication, normalized Seregin extraction, pressure atlas, lower classes, mildness, and conditional final assembly. | Paper IV residual closure. |
| R1 axisymmetric closure | Centered circulation equation, axis regularity, killed-strip comparison, and setup lower classes. | Terminal stratification assembly theorem. |
| R2 reduction | Rotational flux identities and finite unit-cylinder velocity concentration cover; closure is obtained only after terminal stratification. | R2 closure or terminal stratification conclusion. |
| R2 closure | R2 reduction, terminal stratification assembly, finite separated-family exclusion, and no atomic active profile. | Standalone rotational Liouville theorem. |
| R3 reduction | Hull realization, centered scale reselection, and retained velocity heredity; closure is obtained only after terminal stratification. | R3 closure or terminal stratification conclusion. |
| R3 closure | R3 reduction, terminal stratification assembly, sequence-$L^3$, and endpoint atomic contradiction. | Premature stationary-hull exclusion. |
| Concentration-set extraction | Local compactness, raw atlas defect measures, compact observer windows, and pressure atlas. | Endpoint Albritton–Barker theorem. |
| Diffuse-defect compactification | Concentration-set residual measure, observer compactification, and amenability of $\mathbb{R}^3 \rtimes \mathbb{R}$. | Generic closure. |
| Critical-tail closure | Bounded-origin realization, affine quotient, hidden-scale compactness, and coherent-tail alternatives. | Final residual exclusion. |
| Finite separated-profile exclusion | Diagonal lifting, finite directed graph recurrence, no atomic profile, and no infinite retained velocity concentration chain. | Terminal assembly theorem and global $L^3$ bound. |
| Atomic sequence-$L^3$ | Indecomposability and absence of diffuse, tail, separated, pressure-loss, and critical-tail alternatives. | Albritton–Barker endpoint conclusion. |

| **Result** | **Uses** | **Does not use** |
|---|---|---|
| Endpoint atomic contradiction | Bounded mild finite terminal shifts, sequence-$L^3$, and retained nonzero velocity concentration. | Residual decomposition. |
| Final assembly | All previous rows and the ordered decomposition. | Itself. |

**Corollary 1.15** (Residual input from the setup paper)**.** *For every normalized Seregin collection $\mathcal{S}$ generated by an admissible finite-energy Type I singularity in the setup paper, the setup remaining class is empty.*

*Proof deferred.* This is proved in Section 12, after Theorem 1.14 has been established and inserted into the setup paper's residual hypothesis. □

**Corollary 1.16** (Local pointwise Type I exclusion after residual closure)**.** *No finite-energy local pointwise Type I singularity can generate a normalized Seregin collection $\mathcal{S}$.*

*Proof deferred.* This is the final two-paper consequence recorded in Corollary 12.3; it is stated here only to identify the consequence of residual closure. □

1.3. **How to read the proof.** A reader interested only in the residual closure can navigate the paper as follows. The proof is read top-down by class.

(1) *Setup interface and Type-I implication* (Sections 1.2 and 1.4). Records the setup results used here as imported hypotheses and verifies that local pointwise Type-I singularities enter the admissible Seregin compactness setting.
(2) *Residual decomposition* (Definition 1.8). The ordered list of alternatives is
$$\mathcal{C}_{\mathrm{ax}},\ \mathcal{C}_{\mathrm{rot}},\ \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}},\ \mathcal{C}_{\mathrm{aff}},\ \mathcal{C}_{\log\mathrm{diff}},\quad \mathcal{C}_{\mathrm{Young}},\ \mathcal{C}_{\mathrm{homcrit}},\ \mathcal{C}_{\mathrm{logper}},\ \mathcal{C}_{\mathrm{ap}},\ \mathcal{C}_{\mathrm{gen}}.$$
The inherited list of allowed limiting operations is recorded in Theorem 1.11.
(3) *Class $\mathcal{C}_{\mathrm{ax}}$.* Centered angular-circulation equation (Lemma 4.1), one-dimensional killed radial absorption, axis-absorption Liouville theorem, closure (Theorem 4.4).
(4) *Class $\mathcal{C}_{\mathrm{rot}}$.* Coriolis Bernoulli identity, local flux dichotomy, reduction proposition (Proposition 5.8). Closure deferred until the terminal stratification assembly.
(5) *Class $\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$.* Stationary phase space, hull realization, centered scale reselection, reduction (Propositions 6.12 and 7.6). Closure deferred until the terminal stratification assembly.
(6) *Generic terminal residual.* Tail geometry and covariant observer calculus, asymptotic hull, recurrent-core rigidity, terminal concentration profiles, separated-profile theory, Albritton–Barker endpoint, generic exhaustion (Theorems 10.25 and 11.3).
(7) *Final assembly* (Section 12). Combines all class closures and the deferred lower-class arguments.

A reader who only wants to see the central technical core can read in this order: the asymptotic hull and recurrent-core rigidity, the terminal concentration-set extraction, the diffuse-defect compactification, the critical-tail compactification, and the minimal mesoscopic-scale reduction theorem (Theorem 9.81). Everything else is either tracking for the residual decomposition or reduction of the four analytic classes into this terminal part of the argument.

1.4. **Local Type I singularities enter the terminal concentration setting.** The purpose of this subsection is to make explicit the local reduction that is implicit in the two-paper proof architecture here. There are two logically different issues. The local pointwise Type I envelope gives Seregin compactness on every compact negative-time cylinder
$$B_R \times [-S, -\sigma], \qquad \sigma > 0,$$

but this compactness is strictly away from the terminal slice $t = 0$. It does not alone imply that the positive $L^3$-mass in $B_1 \times (-1, 0)$ survives away from the terminal layer $(-\sigma, 0)$. The latter is the terminal velocity no-escape condition used in the setup paper.

The implication below first records the admissible form used by the unified setup paper. It then uses the endpoint weak Serrin local Type I estimate of Albritton–Barker to derive the required scale-invariant $L^2$ bound on the terminal interval. Interpolation with the Type I envelope gives terminal velocity no-escape.

For this subsection, if $z_* = (x_*, T)$ and $r_n \downarrow 0$, write

$$u_n(x,t) := r_n u(x_* + r_n x, T + r_n^2 t), \qquad p_n(x,t) := r_n^2 p(x_* + r_n x, T + r_n^2 t).$$

Also write $Q_R := B_R(0) \times (-R^2, 0)$ and

$$Q_1(0,0) = B_1(0) \times (-1, 0).$$

**Definition 1.17** (Admissible local Type I concentration sequence)**.** Let $z_* = (x_*, T)$ satisfy the local Type I bound (1.10), and let $r_n \downarrow 0$. The corresponding rescaled velocities $u_n$ form an admissible local Type I concentration sequence if there are $\eta_v > 0$ such that

$$\iint_{B_1 \times (-1,0)} |u_n|^3 \, dx \, dt \geq \eta_v \qquad \text{for every } n, \tag{1.7}$$

and the terminal velocity no-escape condition holds:

$$\lim_{\sigma \downarrow 0} \sup_n \iint_{B_1 \times (-\sigma,0)} |u_n|^3 \, dx \, dt = 0. \tag{1.8}$$

Equivalently, after lowering the mass threshold, there are $\sigma_0 \in (0,1)$ and $\eta_1 > 0$ such that

$$\iint_{B_1 \times (-1,-\sigma_0)} |u_n|^3 \, dx \, dt \geq \eta_1 \qquad \text{for all } n. \tag{1.9}$$

**Lemma 1.18** (No-escape leaves positive mass away from the terminal layer)**.** *If $u_n$ is an admissible local Type I concentration sequence in the sense of Definition 1.17, then (1.9) holds for some $\sigma_0 \in (0,1)$ and some $\eta_1 > 0$.*

*Proof.* Choose $\sigma_0 \in (0,1)$ so small that

$$\sup_n \iint_{B_1 \times (-\sigma_0,0)} |u_n|^3 \, dx \, dt \leq \frac{1}{2} \eta_v,$$

which is possible by (1.8). Subtracting this terminal-layer contribution from (1.7) gives

$$\iint_{B_1 \times (-1,-\sigma_0)} |u_n|^3 \, dx \, dt \geq \frac{1}{2} \eta_v.$$

Take $\eta_1 = \eta_v / 2$. □

**Proposition 1.19** (Local Type I rescalings have Seregin compactness)**.** *Let $(u,p)$ be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0,T)$, let $z_* = (x_*, T)$ be a singular point, and assume that there exist $\rho > 0$ and $M < \infty$ such that*

$$\operatorname*{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \, \|u(t)\|_{L^\infty(B_\rho(x_*))} \leq M. \tag{1.10}$$

*Let $r_n \downarrow 0$, and let $(u_n, p_n)$ be the rescaled sequence above. Then the following hold.*

(i) *For every compact cylinder*

$$K = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0), \qquad R, S < \infty, \quad \sigma > 0,$$

*there exists* $n_0(K)$ *such that*

$$\|u_n\|_{L^\infty(K)} \le M\sigma^{-1/2}, \qquad n \ge n_0(K). \tag{1.11}$$

(ii) *On every compact cylinder* $K \Subset \mathbb{R}^3 \times (-\infty, 0)$ *there are time-dependent pressure gauges* $a_{n,K}(t)$ *and a constant* $C_K < \infty$ *such that, for* $n \ge n_0(K)$,

$$\|u_n\|_{L_t^\infty L_x^2(K)} + \|\nabla u_n\|_{L^2(K)} + \|p_n - a_{n,K}(t)\|_{L^{3/2}(K)} \le C_K. \tag{1.12}$$

*The constant* $C_K$ *may depend on the fixed Leray energy level of the original solution, but no concentration or no-escape assumption enters it.*

(iii) *After subtracting these local pressure gauges, the family* $(u_n, p_n)$ *is locally precompact in the standard Seregin topology on* $\mathbb{R}^3 \times (-\infty, 0)$*. Along every subsequence there is a further subsequence and an ancient suitable weak solution* $(U, Q)$ *such that, on every compact* $K \Subset \mathbb{R}^3 \times (-\infty, 0)$,

$$u_n \to U \quad \textit{strongly in } L^q(K), \qquad 1 \le q < \infty,$$

*and*

$$p_n - a_{n,K}(t) \rightharpoonup Q \quad \textit{weakly in } L^{3/2}(K).$$

*The limit is bounded and satisfies*

$$|U(x,t)| \le M(-t)^{-1/2}, \qquad t < 0.$$

(iv) *More generally, let* $\mathfrak{c} = (z_n)$, $z_n = (y_n, \tau_n)$*, be a parabolic recentering sequence for* $(u_n, p_n)$ *with* $\tau_n \le 0$*. Assume that* $\mathfrak{c}$ *remains in the local Type I window in the following precise sense: for every* $R, S < \infty$*, all sufficiently large* $n$ *satisfy*

$$r_n(|y_n| + R) < \rho, \qquad T + r_n^2(\tau_n - S) > T - \rho^2.$$

*If* $\mathfrak{c}$ *carries retained unit velocity concentration, then after passing to a subsequence it has no loss of local compactness in the sense of Definition 9.3, and its recentered fields converge locally to a bounded suitable terminal profile. Recenterings which do not satisfy this local-window condition are assigned by the compact-window-to-expanding-region protocol to the exterior, diffuse, pressure/noncompact, or critical-tail terminal alternatives rather than to the compact local profile class.*

*Proof.* Fix

$$K = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0).$$

For $n$ sufficiently large,

$$r_n R < \rho/2, \qquad r_n^2 S < \rho^2/2.$$

Hence, for a.e. $(x,t) \in K$,

$$x_* + r_n x \in B_\rho(x_*), \qquad T + r_n^2 t \in (T - \rho^2, T).$$

The local Type I bound (1.10) gives

$$|u_n(x,t)| = r_n |u(x_* + r_n x, T + r_n^2 t)| \le r_n M \left(T - (T + r_n^2 t)\right)^{-1/2} = M(-t)^{-1/2} \le M\sigma^{-1/2}.$$

This proves (1.11). In particular,

$$\operatorname*{ess\,sup}_{-S<t<-\sigma} \int_{B_R} |u_n(x,t)|^2 \, dx \le |B_R| M^2 \sigma^{-1},$$

and

$$\iint_K |u_n|^3 \, dx \, dt \le |K| M^3 \sigma^{-3/2}.$$

We next record the pressure estimate, where the nonlocal pressure contribution is localized in the local Type I compactness argument. We use the whole-space Leray pressure representative, so at each rescaled time

$$p_n(\cdot,t) = \mathcal{R}_i \mathcal{R}_j \big(u_{n,i} u_{n,j}\big)(\cdot,t)$$

modulo a function of time. Fix a cutoff $\chi_R \in C_c^\infty(B_{8R})$ with $\chi_R \equiv 1$ on $B_{4R}$, and write in $B_{4R}$

$$p_n = q_n + h_n, \qquad q_n = \mathcal{R}_i \mathcal{R}_j (\chi_R u_{n,i} u_{n,j}).$$

The local term satisfies

$$\|q_n\|_{L^{3/2}(B_{4R} \times [-S,-\sigma])} \le C_{R,S,\sigma,M}.$$

For the harmonic part choose the gauge $a_{n,K}(t) = h_n(0,t)$. The standard kernel difference estimate for the Riesz pressure kernel is as follows. If

$$K_{ij}(x) = c_3 \, \text{p.v.} \, \frac{3x_i x_j - \delta_{ij}|x|^2}{|x|^5}$$

is the Calderón–Zygmund pressure kernel, then for $|x| \le 2R$ and $|z| \ge 4R$,

$$|K_{ij}(x-z) - K_{ij}(-z)| \le C_R |z|^{-4}.$$

Decomposing the exterior region into dyadic annuli gives, for $x \in B_{2R}$,

$$|h_n(x,t) - a_{n,K}(t)| \le C_R \sum_{m \ge 2} 2^{-4m} \int_{2^m R < |z| < 2^{m+1} R} |u_n(z,t)|^2 \, dz.$$

For annuli whose physical image remains in $B_\rho(x_*)$, namely while $2^{m+1} R r_n \le \rho$, the local Type I bound and $-t \ge \sigma$ on $K$ give

$$\int_{2^m R < |z| < 2^{m+1} R} |u_n(z,t)|^2 \, dz \le C M^2 \sigma^{-1} (2^m R)^3.$$

Multiplication by $2^{-4m}$ leaves a summable $2^{-m}$ factor, so the corresponding part of the series is bounded by a geometric sum depending only on $R, M, \sigma$. For the remaining outer annuli one uses only the finite Leray energy of the original solution:

$$\int_{\mathbb{R}^3} |u_n(z,t)|^2 \, dz = r_n^{-1} \int_{\mathbb{R}^3} |u(y, T + r_n^2 t)|^2 \, dy.$$

If $m_n$ is the first index for which $2^{m_n} R r_n \gtrsim \rho$, then

$$\sum_{m \ge m_n} 2^{-4m} \int_{\mathbb{R}^3} |u_n(z,t)|^2 \, dz \le C_{\rho,R} \, r_n^3 \|u\|^2_{L^\infty(0,T;L^2)} \le C_{\rho,R,u}.$$

Thus $h_n - a_{n,K}$ is uniformly bounded on $B_{2R} \times [-S,-\sigma]$. Since this cylinder has finite measure,

$$\|h_n - a_{n,K}\|_{L^{3/2}(B_{2R} \times [-S,-\sigma])} \le C_K.$$

Together with the Calderón–Zygmund bound for $q_n$, this gives

$$\|p_n - a_{n,K}(t)\|_{L^{3/2}(B_{2R} \times [-S,-\sigma])} \le C_K.$$

Apply the local energy inequality for $(u_n, p_n)$ with a cutoff equal to one on $K$ and supported in a slightly larger compact cylinder $K' \Subset \mathbb{R}^3 \times (-\infty, 0)$. The already obtained local $L^2$, $L^3$, and pressure-gauge bounds control every term on the right-hand side, and therefore

$$\iint_K |\nabla u_n|^2 \, dx \, dt \le C_K.$$

This proves (1.12).

The Navier–Stokes equation now bounds $\partial_t u_n$ locally in a negative Sobolev space, for instance in $L^{3/2}_t H^{-2}_x(K)$. Aubin–Lions compactness and the uniform $L^\infty$ bound give strong local convergence of the velocity in every finite $L^q$, after passing to a subsequence. The pressure bound gives weak $L^{3/2}$ convergence after gauges. Passing to the limit in the distributional equations and in the local energy inequality gives an ancient suitable weak solution $(U, Q)$. The pointwise Type I bound passes to the limit, so $U$ is bounded on every compact negative-time cylinder. This proves (iii).

Finally, let $\mathfrak{c} = (y_n, \tau_n)$ be as in (iv), and set

$$u_n^{\mathfrak{c}}(y, s) = u_n(y + y_n, s + \tau_n), \qquad p_n^{\mathfrak{c}}(y, s) = p_n(y + y_n, s + \tau_n).$$

The local-window condition, together with $\tau_n \le 0$, is what is needed to repeat the preceding estimates on each fixed backward compact cylinder in the $(y, s)$-variables: the physical cylinders remain in $B_\rho(x_*) \times (T - \rho^2, T)$, the local Type I envelope gives the same $L^\infty$ bound on compact windows, and the pressure decomposition has the same local and finite-energy tail estimates. Thus the two failure modes in Definition 9.3, blow-up of the local dissipation measure or failure of local pressure-gauge control, cannot occur for such a retained local recentering sequence after subsequence extraction. The recentered fields therefore converge locally to a bounded suitable terminal profile. If the local-window condition fails, then the recentering leaves every bounded terminal window attached to the singular point, or its critical mass persists only through expanding regions; those are precisely the exterior and diffuse terminal alternatives rather than loss of local compactness inside the local Type I implication. □

1.5. **Pressure-projected terminal energy estimate.** The preceding compactness result is stated away from the terminal slice. The next estimate supplies the terminal-layer bound required for velocity no-escape. The local pressure projection removes the nonlocal pressure flux from the local energy inequality; the remaining estimate is an interior Caccioppoli propagation bound on balls contained in the local Type I window.

Let $v = u^{(r)}$, $\pi = p^{(r)}$, and $\Lambda = \rho/r$. After translating $z_*$ to $(0, 0)$, the local Type I bound gives

$$|v(x, s)| \le M(-s)^{-1/2} \qquad (x, s) \in B_\Lambda \times (-4, 0),$$

for all sufficiently small $r$. For $a \in \mathbb{R}^3$, $L \ge 1$, and $-4 < s < t < 0$, write

$$A_L(a, s) := \int_{B_L(a)} |v(x, s)|^2 \, dx, \qquad D_L(a; s, t) := \int_s^t \int_{B_L(a)} |\nabla v|^2 \, dx \, d\sigma.$$

**Lemma 1.20** (Local pressure-projected Caccioppoli inequality)**.** *There is a universal constant $C$ with the following property. Let $(v, \pi)$ be a suitable weak solution in $B_{8L}(a) \times (s_-, 0)$, $L \ge 1$, and assume*

$$|v(x, s)| \le M(-s)^{-1/2} \qquad \text{on } B_{8L}(a) \times (s_-, 0).$$

*If $s_0 \in (s_-, 0)$ is a Lebesgue time for $v$ and $\nabla v$, then for a.e. $t \in (s_0, 0)$,*

$$\begin{aligned} \int_{B_L(a)} |w(x, t)|^2 \, dx + D_L(a; s_0, t) &\le C A_{4L}(a, s_0) \\ &\quad + C \int_{s_0}^t \left( L^{-2} + M L^{-1} (-s)^{-1/2} \right) A_{4L}(a, s) \, ds. \end{aligned} \tag{1.13}$$

*The estimate is entirely local on $B_{8L}(a)$; no pressure tail from outside $B_{8L}(a)$ appears. Here $w = v + \nabla p_h$ is the local projected velocity defined in the proof below.*

*Proof.* The proof uses the standard local pressure-projection argument in the scale $L$ normalization. Let $B = B_{8L}(a)$. Denote by $E^*_B$ the Stokes pressure projection on $B$: for $F \in W^{-1,q}(B)$, $1 < q < \infty$, $\nabla E^*_B(F)$ is the gradient part of the stationary Stokes solution in $B$ with zero boundary velocity. It satisfies the scale-invariant estimates

$$\|\nabla E^*_B(\nabla \cdot F)\|_{L^q(B)} \le C_q \|F\|_{L^q(B)}, \qquad \|\nabla E^*_B(\Delta w)\|_{L^2(B)} \le C\|\nabla w\|_{L^2(B)}.$$

Set

$$\nabla p_h = -\nabla E^*_B(v), \qquad w := v + \nabla p_h.$$

The harmonic pressure $p_h$ is harmonic in $B$, and the projection estimates give

$$\|w\|_{L^2(B)} + \|\nabla p_h\|_{L^2(B)} \le C\|v\|_{L^2(B)}, \qquad \|\nabla w\|_{L^2(B_{6L}(a))} \le C\|\nabla v\|_{L^2(B)}.$$

The projected local energy inequality for $w$, obtained by applying the Stokes projection to the distributional Navier–Stokes equations and testing with $\zeta^2 w$, reads

$$\begin{aligned}
&\int |w(t)|^2\zeta^2 + \int_{s_0}^t \int |\nabla v|^2 \zeta^2 \\
&\quad \le C \int |w(s_0)|^2 \zeta^2 + C \int_{s_0}^t \int |w|^2 \left(|\partial_s \zeta| + |\Delta\zeta| + |\nabla\zeta|^2\right) \\
&\qquad + C \int_{s_0}^t \int |v|\,|w|^2\,|\nabla\zeta| + C \int_{s_0}^t \int |v|\,|\nabla p_h|^2\,|\nabla\zeta|.
\end{aligned} \tag{1.14}$$

Here $\zeta \in C^\infty_c(B_{4L}(a))$, $\zeta \equiv 1$ on $B_L(a)$, and

$$|\nabla\zeta| \le CL^{-1}, \qquad |\Delta\zeta| + |\partial_s\zeta| \le CL^{-2}.$$

The pressure-projection estimates allow $w$ and $\nabla p_h$ to be bounded by $v$ in $L^2(B_{4L}(a))$. Therefore the cutoff terms in (1.14) are bounded by

$$CL^{-2} \int_{s_0}^t A_{4L}(a,s)\,ds.$$

The two convective terms are bounded, using the Type I envelope, by

$$CL^{-1} \int_{s_0}^t \|v(s)\|_{L^\infty(B_{4L}(a))} A_{4L}(a,s)\,ds \le CML^{-1} \int_{s_0}^t (-s)^{-1/2} A_{4L}(a,s)\,ds.$$

The initial projected energy is controlled by

$$\int |w(s_0)|^2 \zeta^2\,dx \le CA_{4L}(a,s_0).$$

Combining the preceding bounds gives (1.13). □

**Definition 1.21** (Interior terminal energy estimate)**.** The local pointwise Type I window at $z_*$ is said to satisfy the interior terminal energy estimate if, for every fixed $R < \infty$, there are

$$r_R > 0, \qquad B_R < \infty,$$

such that every rescaling $v = u^{(r)}$, $0 < r < r_R$, with $\Lambda = \rho/r$, satisfies

$$\operatorname*{ess\,sup}_{-1<s<0} \int_{B_{8R}} |v(x,s)|^2\,dx + \int_{-1}^0 \int_{B_{8R}} |\nabla v|^2\,dx\,ds \le B_R. \tag{1.15}$$

Equivalently, the dependence on $A_{4L}$ in Lemma 1.20 can be stopped inside a fixed multiple of the target ball, uniformly as the boundary $\partial B_{\rho/r}$ of the local Type I window recedes to infinity.

**Proposition 1.22** (Pressure-projected reduction to an interior energy bound)**.** *Assume the local pointwise Type I bound (1.10) at $z_*$. If the interior energy estimate (1.15) holds with $R = 1$, then the velocity no-escape condition (1.8) holds for every positive local Type I concentration sequence at $z_*$. Consequently such a sequence is admissible in the sense of Definition 1.17.*

*Proof.* Let $u_n = u^{(r_n)}$ be any positive local Type I concentration sequence. By (1.15) with $R = 1$, after discarding finitely many indices,

$$\operatorname*{ess\,sup}_{-1<s<0} \int_{B_1} |u_n(x,s)|^2 \, dx \le B_1.$$

The local Type I envelope gives

$$\|u_n(s)\|_{L^\infty(B_1)} \le M(-s)^{-1/2}.$$

Therefore, for $0 < \sigma < 1$,

$$\begin{aligned}\int_{-\sigma}^0 \int_{B_1} |u_n|^3 \, dx \, ds &\le \int_{-\sigma}^0 \|u_n(s)\|_{L^\infty(B_1)} \int_{B_1} |u_n(x,s)|^2 \, dx \, ds \\ &\le 2MB_1\sigma^{1/2}.\end{aligned}$$

The finitely many discarded indices have absolutely continuous $|u_n|^3$-integrals on $B_1 \times (-1, 0)$, so taking the supremum over the whole sequence and then letting $\sigma \downarrow 0$ gives (1.8). The positive $Q_1$ mass hypothesis is the other part of Definition 1.17. □

*Remark* 1.23 (Pressure-projected reduction). Lemma 1.20 shows, independently of the Albritton–Barker endpoint input used below, that terminal no-escape reduces to an interior terminal energy estimate: after local pressure projection, the local energy inequality controls the projected velocity and the dissipation up to the larger-scale quantity $A_{4L}$. The estimate (1.15) prevents terminal-layer concentration entering from the boundary of the local Type I window. In the present paper this bound is supplied instead by the endpoint weak Serrin local Type I bounds in Theorem 1.24.

**Theorem 1.24** (Setup local Type-I implication)**.** *Let $(v, q)$ be a suitable weak solution in a parabolic cylinder*

$$Q_\rho(z_0) = B_\rho(x_0) \times (t_0 - \rho^2, t_0).$$

*Assume the local pointwise Type I envelope*

$$\operatorname*{ess\,sup}_{t_0-\rho^2<t<t_0} (t_0 - t)^{1/2} \|v(t)\|_{L^\infty(B_\rho(x_0))} \le M < \infty.$$

*Then the setup paper proves that the terminal local energy, pressure-gauge, and no-escape bounds stated in Proposition 1.19 hold on every smaller cylinder $Q_{\theta\rho}(z_0)$ with $0 < \theta < 1$, with constants depending only on $\rho, \theta, M$ and on the suitability constants of $(v, q)$ on $Q_\rho(z_0)$. In particular every rescaling sequence centered at a local pointwise Type I terminal singularity satisfies Definition 1.17.*

*Proof.* This theorem is imported verbatim from the companion setup paper; see Theorem 1.3 (2). Paper IV does not reprove the weak-Serrin-to-local-Type I estimate or the terminal no-escape estimate. Its only role here is to place first-generation terminal sequences inside the admissible local-window class and to supply the local energy, pressure, and no-escape bounds used by Definition 9.7. □

*Remark* 1.25 (Albritton–Barker as the underlying input). The setup proof of Theorem 1.24 relies on the Albritton–Barker endpoint weak-Serrin estimate, which gives finite scale-invariant $A, C, D, E$-quantities on every $Q_{\theta\rho}(z_0)$ under the local pointwise Type I envelope. The constants depend on local suitable-solution background quantities on $Q_\rho(z_0)$, with no circularity because the input envelope and the suitable-solution data are part of the hypothesis data of the cylinder $Q_\rho(z_0)$, while the conclusion is restricted to $Q_{\theta\rho}(z_0)$. The detailed weak-Serrin discussion belongs to the setup paper; it is recalled here only to identify the source of the constants used in Proposition 1.19.

**Theorem 1.26** (Local pointwise Type I implies terminal no-escape)**.** *Let $(u,p)$ be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0,T)$, let $z_* = (x_*, T)$, and assume that*

$$u \in L^\infty(0,T; L^2(\mathbb{R}^3)) \cap L^2(0,T; \dot{H}^1(\mathbb{R}^3)).$$

*Assume that $z_*$ satisfies the local pointwise Type I bound: there are $\rho > 0$ and $M < \infty$ such that*

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\, \|u(t)\|_{L^\infty(B_\rho(x_*))} \le M. \tag{1.16}$$

*For $0 < r < \rho$, set*

$$u^{(r)}(x,s) = r\, u(x_* + rx, T + r^2 s), \qquad p^{(r)}(x,s) = r^2 p(x_* + rx, T + r^2 s).$$

*Then for every $0 < R < \infty$ there are constants*

$$r_R \in (0, \rho/R), \qquad B_R < \infty,$$

*and time-dependent pressure gauges $a_{r,R}(s)$, depending on $R, M, \rho, T$ and the local suitable-solution background quantities of $(u,p)$ in $Q_\rho(z_*)$, such that the following terminal local energy estimate holds for every $0 < r < r_R$:*

$$\begin{aligned} \operatorname*{ess\,sup}_{-1<s<0} \int_{B_R} |u^{(r)}(x,s)|^2\,dx &+ \int_{-1}^0 \int_{B_R} |\nabla u^{(r)}|^2\,dx\,ds \\ &+ \int_{-1}^0 \int_{B_R} |p^{(r)}(x,s) - a_{r,R}(s)|^{3/2}\,dx\,ds \le B_R. \end{aligned} \tag{1.17}$$

*Consequently, for every $0 < \sigma < 1$ and every $0 < r < r_R$,*

$$\int_{-\sigma}^0 \int_{B_R} |u^{(r)}(x,s)|^3\,dx\,ds \le 2MB_R\sigma^{1/2}, \tag{1.18}$$

*and in particular*

$$\lim_{\sigma \downarrow 0} \sup_{0<r<r_R} \int_{-\sigma}^0 \int_{B_R} |u^{(r)}(x,s)|^3\,dx\,ds = 0. \tag{1.19}$$

*Proof.* Fix once and for all $0 < \theta < 1$, for instance $\theta = 1/2$, and put $L_R := \max\{R, 1\}$. By decreasing $r_R$, we may assume

$$B_{L_R r}(x_*) \times (T - L_R^2 r^2, T) \subset Q_{\theta\rho}(z_*) \qquad (0 < r < r_R).$$

The pointwise Type I bound implies the endpoint weak Serrin estimate on $Q_\rho(z_*)$. Indeed, after writing

$$f(t) = \|u(t)\|_{L^\infty(B_\rho(x_*))},$$

one has

$$|\{t \in (T-\rho^2, T) : f(t) > \alpha\}| \le M^2 \alpha^{-2}, \qquad \alpha > 0,$$

and therefore $u \in L_t^{2,\infty} L_x^\infty(Q_\rho(z_*))$. Applying Theorem 1.24 gives a finite constant $K_\theta$ such that $A, C, D, E \le K_\theta$ on all cylinders contained in $Q_{\theta\rho}(z_*)$.

Taking the cylinder $Q_{L_R r}(z_*)$ gives

$$(L_R r)^{-1} \operatorname*{ess\,sup}_{T-L_R^2 r^2<t<T} \int_{B_{L_R r}(x_*)} |u(x,t)|^2\,dx \le K_\theta,$$

and similarly the $D$- and $E$-bounds give

$$(L_R r)^{-1} \int_{T-L_R^2 r^2}^{T} \int_{B_{L_R r}(x_*)} |\nabla u|^2 \, dx \, dt \le K_\theta,$$

$$(L_R r)^{-2} \int_{T-L_R^2 r^2}^{T} \int_{B_{L_R r}(x_*)} |p - a_{r,R}^{\mathrm{phys}}(t)|^{3/2} \, dx \, dt \le K_\theta$$

for suitable time-dependent pressure gauges $a_{r,R}^{\mathrm{phys}}(t)$. Scaling these three estimates and restricting to

$$B_R \times (-1,0) \subset B_{L_R} \times (-L_R^2, 0)$$

gives (1.17) with

$$a_{r,R}(s) := r^2 a_{r,R}^{\mathrm{phys}}(T + r^2 s),$$

after enlarging the constant $B_R$ by a harmless factor depending on $R$ and $K_\theta$.

After translating $(x_*, T)$ to $(0,0)$, the rescaled fields also satisfy

$$|u^{(r)}(x,s)| \le M(-s)^{-1/2} \qquad (x,s) \in B_R \times (-1,0)$$

for all $r < r_R$, after decreasing $r_R$ if necessary. The terminal local energy estimate established above gives

$$\operatorname*{ess\,sup}_{-1<s<0} \int_{B_R} |u^{(r)}(x,s)|^2 \, dx \le B_R.$$

For $0 < \sigma < 1$, the local Type I bound above and the kinetic bound give

$$\begin{aligned} \int_{-\sigma}^{0} \int_{B_R} |u^{(r)}|^3 \, dx \, ds &\le \int_{-\sigma}^{0} \|u^{(r)}(s)\|_{L^\infty(B_R)} \int_{B_R} |u^{(r)}(x,s)|^2 \, dx \, ds \\ &\le M B_R \int_{-\sigma}^{0} (-s)^{-1/2} \, ds = 2 M B_R \sigma^{1/2}. \end{aligned}$$

This proves (1.18) and (1.19). □

**Corollary 1.27** (No diffuse terminal-layer mass)**.** *Under the hypotheses of Theorem 1.26, for every fixed $R < \infty$ there do not exist $r_n \downarrow 0$, $\sigma_n \downarrow 0$, and $\varepsilon_0 > 0$ such that*

$$r_n^{-2} \int_{T-\sigma_n r_n^2}^{T} \int_{B_{Rr_n}(x_*)} |u(x,t)|^3 \, dx \, dt \ge \varepsilon_0 \qquad \textit{for all } n.$$

*Proof.* For $u_n = u^{(r_n)}$, the left hand side is

$$\int_{-\sigma_n}^{0} \int_{B_R} |u_n(x,s)|^3 \, dx \, ds.$$

By (1.18), this is bounded by

$$2 M B_R \sigma_n^{1/2} \to 0,$$

which contradicts any fixed positive lower bound. □

**Corollary 1.28** (Local pointwise Type I concentration sequences are admissible)**.** *Let $(u,p)$ and $z_* = (x_*, T)$ satisfy the hypotheses of Theorem 1.26. Let $r_n \downarrow 0$ be any velocity concentration sequence satisfying*

$$\iint_{B_1 \times (-1,0)} |u_n|^3 \, dx \, dt \ge \eta_v > 0, \qquad u_n(x,t) = r_n u(x_* + r_n x, T + r_n^2 t).$$

*Then $u_n$ is an admissible local Type I concentration sequence in the sense of Definition 1.17.*

*Proof.* Apply Theorem 1.26 with $R = 1$. After discarding finitely many $n$, all $r_n < r_1$, and

$$\lim_{\sigma\downarrow 0}\sup_{n}\int_{-\sigma}^{0}\int_{B_1}|u_n|^3\,dx\,dt = 0.$$

The finitely many discarded indices do not affect the limit as $\sigma \downarrow 0$, because each fixed suitable weak solution has $|u_n|^3 \in L^1(B_1 \times (-1, 0))$. The positive mass hypothesis is the other condition in Definition 1.17. □

**Lemma 1.29** (Local Type I concentration enters the terminal dichotomy)**.** *Under the hypotheses of Proposition 1.19, let $r_n \downarrow 0$ be a velocity concentration sequence at $z_*$ satisfying*

$$\iint_{B_1\times(-1,0)}|u_n|^3\,dx\,dt \ge \eta_v > 0.$$

*Then $r_n$ is admissible by Corollary 1.28. Let $\sigma_0, \eta_1$ be given by Lemma 1.18. For the associated terminal extraction sequence $(V_N, P_N)$ generated by a Seregin limit of this sequence, choose the small-data threshold $\varepsilon_{\mathrm{sd}} \le \eta_1/2$. Then the origin extraction produces a bounded ancient Seregin profile with retained local velocity concentration away from the terminal layer. In particular the retained velocity concentration generated from the origin recentering has no loss of local compactness. Any retained local recentering which stays in the local Type I window has the same compactness property by Proposition 1.19; any concentration leaving that window is assigned to the exterior or diffuse terminal remainder. The paired terminal decomposition of Theorem 9.24, together with the mixed-profile exclusions and concentration-chain exclusions Theorems 9.102 and 10.21, reduces the retained possibilities to a single retained concentrating terminal profile or a finite separated family of retained concentrating terminal profiles.*

*Proof.* By Lemma 1.18,

$$\iint_{B_1\times(-1,-\sigma_0)}|u_n|^3\,dx\,dt \ge \eta_1 \qquad \text{for all } n.$$

The cylinder $B_1 \times [-1, -\sigma_0]$ is compactly contained in $\mathbb{R}^3 \times (-\infty, 0)$. Therefore Proposition 1.19 gives, after passing to a subsequence, strong local $L^3$ convergence on this cylinder:

$$u_n \to U \qquad \text{in } L^3(B_1 \times (-1, -\sigma_0)).$$

Consequently

$$\iint_{B_1\times(-1,-\sigma_0)}|U|^3\,dx\,dt \ge \eta_1.$$

Cover $B_1 \times (-1, -\sigma_0)$ by finitely many unit terminal cylinders strictly away from $t = 0$. At least one such cylinder carries a fixed positive velocity lower bound, and since the raw atlas defect measure dominates the velocity part, choosing $\varepsilon_{\mathrm{sd}} \le \eta_1/2$ after adjusting by this finite cover makes the corresponding origin-frame recentering raw unit-window active at threshold $\varepsilon_{\mathrm{sd}}$. This argument is precisely where no-escape is used: without (1.8), the lower bound could remain entirely in $B_1 \times (-\sigma, 0)$ for every fixed $\sigma > 0$.

The distinguished recentering satisfies the local-window condition in Proposition 1.19, so it is not in the local-compactness-failure class. Other retained local recenterings inside the same Type I window have the same compactness. Recenterings leaving the window are exterior or diffuse terminal remainders. The paired decomposition Theorem 9.24 records these remainders alongside the concentration set rather than treating them as mutually exclusive alternatives. The sole-source radiative cases are excluded by Theorem 9.27, mixed compact–diffuse or compact–noncompact coexistence is excluded by Theorem 9.102, and infinite chains of retained concentrating descendants are excluded by Theorem 10.21. Hence Corollary 9.25

leaves only a single retained concentrating terminal profile or a finite separated family of retained concentrating terminal profiles. □

**Theorem 1.30** (Local Type I singularities enter the terminal concentration setting)**.** *Let $(u, p)$ be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$, let $z_* = (x_*, T)$ be a singular point, and assume that $z_*$ satisfies the local Type I bound (1.10). Then there exists a bounded centered Type I Seregin profile with retained local velocity concentration generated from $(u, p, z_*)$. Consequently the local terminal concentration setting of Theorem 10.25 applies after the residual closure theorem is proved.*

*Proof.* By the local velocity concentration theorem in the setup paper, choose $r_n \downarrow 0$ and $\eta_v > 0$ with

$$\iint_{B_1 \times (-1,0)} |u_n|^3 \, dx \, dt \geq \eta_v, \qquad u_n(x, t) = r_n u(x_* + r_n x, T + r_n^2 t).$$

Theorem 1.26 and Corollary 1.28 show that this concentration sequence satisfies the terminal velocity no-escape condition required by the setup paper. Apply Lemma 1.29 to this sequence. The local Type I rescalings therefore produce at least one bounded terminal profile carrying retained local velocity concentration. After passing to the centered variables, the bound $|U(x, t)| \leq M(-t)^{-1/2}$ becomes the global centered Type I bound

$$|V(y, \tau)| \leq M,$$

so each retained profile generated by this implication is a bounded centered Type I Seregin profile.

If every such retained profile belongs to one of the previously closed lower classes, then the contradiction is supplied by the setup paper: small-amplitude and stationary $L^3$ profiles are excluded there, the uniformly tight class is excluded there, and the closed structured/decay class is excluded there. Otherwise choose a retained profile outside those lower classes. It is then a refined residual candidate, so Theorem 10.25 applies. In that residual setting, a finite separated family of retained concentrating terminal profiles is excluded by Theorem 10.24, infinite chains of retained concentrating descendants are excluded by Theorem 10.21, and mixed compact–diffuse tails are excluded by Theorem 9.102. The remaining single retained profile is the terminal case closed in Section 12. Thus the present theorem supplies the entry into the residual decomposition; the contradiction is the final two-paper corollary proved after the residual closure. □

**Corollary 1.31** (Local pointwise Type I exclusion after setup plus residual closure)**.** *Let $(u, p)$ be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$. Combining the residual closure Theorem 1.14 of this paper with the setup paper, the following holds: if $z_* = (x_*, T)$ is a singular point satisfying the local Type I bound (1.10), then the completed case closures give a contradiction. Equivalently, in the local Type I/local Type II dichotomy of the setup paper, every finite-energy singular point belongs to the local Type II alternative. The conclusion is the joint output of the two-paper chain; it is not asserted as a hidden hypothesis inside Paper IV alone.*

*Proof.* The entry assertion is Theorem 1.30. The contradiction is deferred to the final assembly, where the residual closure is proved and then inserted into the setup paper's conditional final theorem. The equivalence is the local Type I/local Type II dichotomy of the setup paper. □

## 2. Ordered residual profile decomposition

The following diagram records the ordered partition used in the paper. Each diamond is a stratification criterion. Each terminal branch is a possible singularity or candidate blow-up profile type selected by the criteria. The local Type II terminal alternative is the remaining

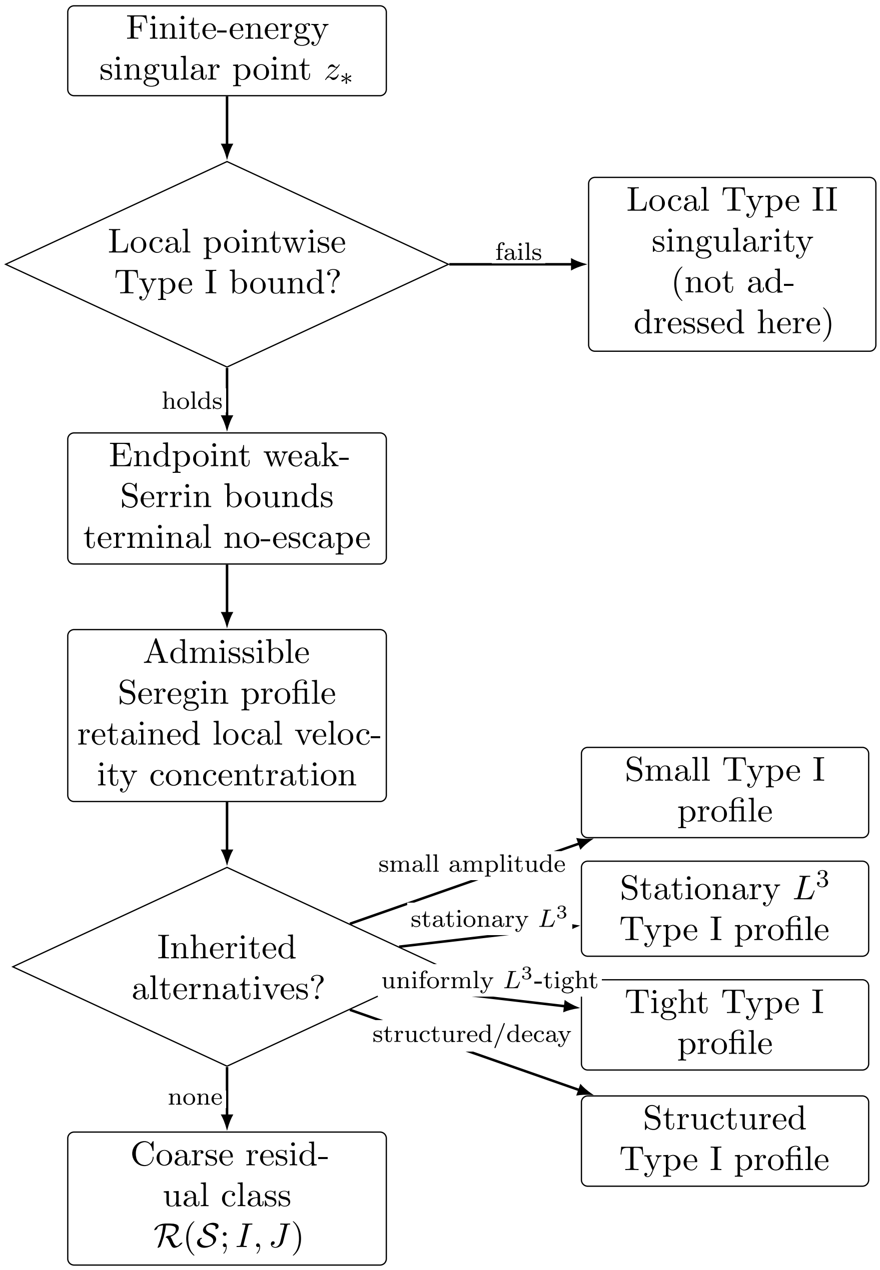


Figure 2. Entry diagram. Local pointwise Type I enters the admissible Seregin compactness setting through endpoint weak Serrin estimates and terminal no-escape. If the bound fails, the point belongs to the local Type II alternative, which is not addressed here.

regime after the local pointwise Type I case is excluded; every Type I terminal alternative is excluded either by the inherited alternatives or by the case-closure theorems proved below.

The loop from the last criterion back to the residual class is only a graphical way to indicate the ordered nature of the decomposition: if a lower stratum has not actually been removed, it is assigned to its corresponding terminal alternative rather than to $\mathcal{C}_{\mathrm{gen}}$. After the ordered removals

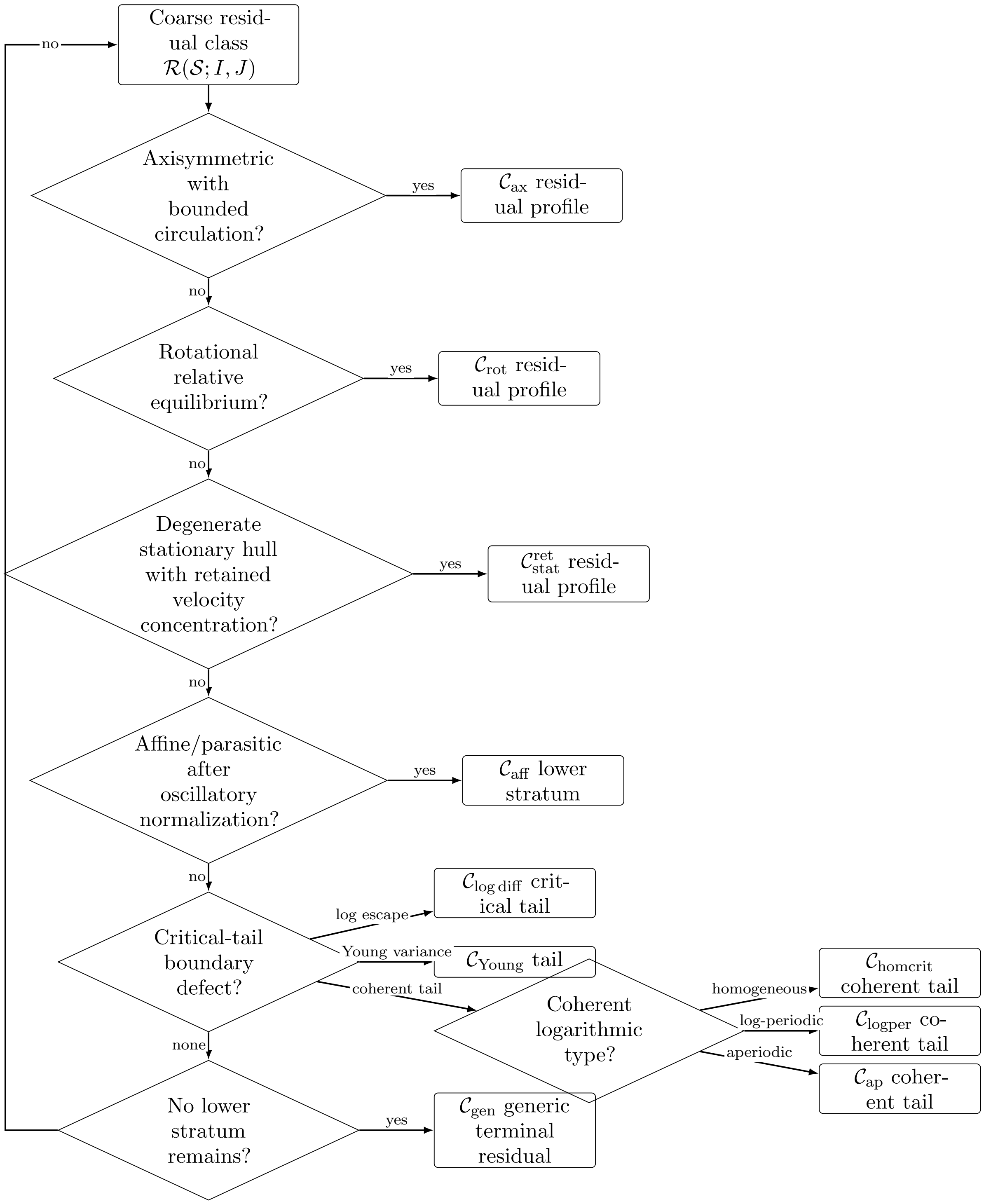


FIGURE 3. Residual stratification diagram. The final return arrow only records the ordered nature of the decomposition: if a lower stratum has not been removed, the profile is assigned to that terminal alternative; after all lower alternatives are removed, the remaining residual class is $\mathcal{C}_{\mathrm{gen}}$.

are performed, $\mathcal{C}_{\mathrm{gen}}$ is the remaining residual class by definition. For the table, write

$$\mathfrak{T}_\rho(u; z_*) := \operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\, \|u(t)\|_{L^\infty(B_\rho(x_*))}.$$

Table 3: Classification and closure table for the residual decomposition diagram. Each row gives the criterion selecting the terminal alternative, the mechanism that excludes it when it is a Type I alternative, and the statement used at that point. For the inherited alternatives, the displayed condition is a representative form of the inherited definition; the precise definitions and closures are those of the companion setup paper, as summarized in Theorem 1.3.

| **Terminal alternative** | **Criterion, exclusion mechanism, and reference** |
|---|---|
| Local Type II singularity | **Criterion.** For every $\rho > 0$, $\mathfrak{T}_\rho(u; z_*) = \infty$. **Role.** This is the complementary alternative left by the local Type I/Type II dichotomy after local pointwise Type I is excluded. **Link.** Corollary 1.31. |
| Small Type I profile | **Criterion.** Inherited small-amplitude class; in the setup normalization, $\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le \varepsilon_{\mathrm{sm}}$. **Exclusion.** Small-data/regularity closure recorded in the setup paper. **Link.** Definition 1.6. |
| Stationary $L^3$ Type I profile | **Criterion.** Inherited stationary class; representative condition $\partial_\tau V = 0$ and $V \in L^3(\mathbb{R}^3)$. **Exclusion.** Stationary $L^3$ Liouville/regularity closure recorded in the setup paper. **Link.** Definition 1.6. |
| Tight Type I profile | **Criterion.** Inherited tight class; representative condition $\lim_{R\to\infty} \sup_\tau \int_{\lvert y\rvert>R} \lvert V(y,\tau)\rvert^3\, dy = 0$. **Exclusion.** Endpoint $L^3$ rigidity for uniformly tight ancient dynamics. **Link.** Definition 1.6. |
| Structured Type I profile | **Criterion.** Inherited closed structured/decay class, $V \in \mathcal{C}^{\mathrm{setup}}_{\mathrm{struct/decay}}$. **Exclusion.** Closed structured/decay alternatives recorded in the setup paper. **Link.** Definition 1.6. |
| $\mathcal{C}_{\mathrm{ax}}$ residual profile | **Criterion.** Axisymmetric residual profile with bounded centered circulation $\Gamma(\rho, Z, \tau) = \rho V_\theta$ and $\|\Gamma\|_{L^\infty} < \infty$. **Exclusion.** Centered circulation equation plus axis-absorption Liouville theorem; the profile becomes no-swirl and falls into an earlier structured class. **Link.** Theorem 4.4. |
| $\mathcal{C}_{\mathrm{rot}}$ residual profile | **Criterion.** Rotational relative equilibrium $V(y,\tau) = Q(\tau)W(Q(\tau)^T y)$, $Q(\tau) = e^{\tau\Omega}$. **Reduction/closure.** Localized Coriolis-flux reduction sends the profile into lower classes already excluded by the setup paper or the terminal velocity-concentration decomposition; closure is only after terminal stratification assembly. **Link.** Corollary 10.27. |
| $\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$ residual profile | **Criterion.** Degenerate stationary-hull retained profile: along time translations $T_{s_k}V \to W$, where $W$ is stationary and $\mathcal{V}_1(W; 0) \ge \eta_*$. **Exclusion.** Scale reselection reduces to a stationary retained concentration profile, then terminal concentration excludes it. **Link.** Corollary 10.28. |
| $\mathcal{C}_{\mathrm{aff}}$ lower stratum | **Criterion.** The affine-normalized oscillation vanishes: $\Phi(S_c U) = 0$ for an admissible affine symmetry $S_c$. **Exclusion.** Oscillatory entry normalization forces retained residual profiles to keep positive non-affine oscillation after affine normalization. **Link.** Theorem 9.89. |
| $\mathcal{C}_{\log\mathrm{diff}}$ critical tail | **Criterion.** Logarithmic escape component in the realized critical-tail compactification: $\Lambda \ne 0$. **Exclusion.** Log-window descendant selection plus minimal mesoscopic-scale reduction and hidden-scale compactness. **Link.** Corollary 9.54. |
| $\mathcal{C}_{\mathrm{Young}}$ tail | **Criterion.** Tight annular critical-tail profile with non-Dirac Young measure: $\Lambda = 0$ and $\mathcal{Y}_{x,s} \ne \delta_{\bar{W}(x,s)}$ on a set of positive measure. **Exclusion.** No hidden-scale variance gives strong $L^3_{\mathrm{loc}}$ compactness, hence a Dirac Young measure. **Link.** Corollary 9.90. |

| **Terminal alternative** | **Criterion, exclusion mechanism, and reference** |
|---|---|
| $\mathcal{C}_{\mathrm{homcrit}}$ coherent tail | **Criterion.** Coherent $(-1)$-homogeneous tail: $W(r\omega, s) = r^{-1}A(\omega, s)$. **Exclusion.** Bounded-origin realization gives $W \in L^\infty_{\mathrm{loc}}$ near $r = 0$, forcing $A = 0$. **Link.** Corollary 9.100. |
| $\mathcal{C}_{\mathrm{logper}}$ coherent tail | **Criterion.** Nonzero log-periodic coherent tail: $Z(\rho + L, \omega, s) = Z(\rho, \omega, s)$, $L \neq 0$, where $Z = e^\rho W(e^\rho\omega, s)$. **Exclusion.** Bounded-origin realization gives $Z(\rho, \cdot, \cdot) \to 0$ as $\rho \to -\infty$; periodicity forces $Z \equiv 0$. **Link.** Corollary 9.100. |
| $\mathcal{C}_{\mathrm{ap}}$ coherent tail | **Criterion.** Nonzero aperiodic minimal logarithmic hull: $\mathcal{H}(Z)$ is minimal, nonperiodic, and nonzero. **Exclusion.** Bounded-origin realization puts 0 in the compact log-translation hull; minimality forces the hull to be $\{0\}$. **Link.** Corollary 9.100. |
| $\mathcal{C}_{\mathrm{gen}}$ generic terminal residual | **Criterion.** Ordered complement $$V \in \mathcal{R}(\mathcal{S}; I, J) \setminus \bigcup_{\mathcal{B} \in \mathfrak{B}_{\mathrm{lower}}} \mathcal{B},$$ where $\mathfrak{B}_{\mathrm{lower}}$ is the list of all preceding Type I alternatives. **Exclusion.** Paired concentration-set decomposition, finite separated-family exclusion, terminal indecomposability, sequence-$L^3$, parasitic-free mildness, and the Albritton–Barker endpoint Liouville theorem. **Link.** Theorem 11.3. |

## 3. Shared local notation

For $z_0 = (y_0, \tau_0)$ and $r > 0$, write

$$Q_r(z_0) := B_r(y_0) \times (\tau_0 - r^2, \tau_0).$$

**Definition 3.1** (Scale-invariant CKN concentration and defect measures)**.** For a centered profile $(V, P)$, define the velocity concentration

$$\mathcal{V}_r(V; z_0) := r^{-2} \iint_{Q_r(z_0)} |V|^3 \, dy \, d\tau, \tag{3.1}$$

and the quotient pressure concentration

$$D_r(P; z_0) := r^{-2} \inf_{c \in L^{3/2}(\tau_0 - r^2, \tau_0)} \iint_{Q_r(z_0)} |P(y, \tau) - c(\tau)|^{3/2} \, dy \, d\tau. \tag{3.2}$$

The scale-invariant local CKN quantity is

$$\mathcal{E}_r(V, P; z_0) := \mathcal{V}_r(V; z_0) + D_r(P; z_0). \tag{3.3}$$

Thus $\mathcal{E}_r$ is parabolically scale-invariant under the Navier–Stokes scaling.

If an explicit pressure atlas $\mathfrak{a}$ has been chosen, with local representative $a_{z_0,r}(\tau)$, we also write

$$\mathcal{E}^{\mathfrak{a}}_r(V, P; z_0) := r^{-2} \iint_{Q_r(z_0)} \left( |V|^3 + |P - a_{z_0,r}(\tau)|^{3/2} \right) dy \, d\tau. \tag{3.4}$$

Then

$$\mathcal{E}_r(V, P; z_0) \le \mathcal{E}^{\mathfrak{a}}_r(V, P; z_0).$$

Thresholds and regularity statements use the quotient quantity $\mathcal{E}_r$; compactness statements use a fixed setup pressure atlas when an explicit representative is needed.

For a terminal sequence $(V_n, P_n)$ with chosen pressure gauges $a_n(\tau)$, the atlas-dependent Radon defect measure is

$$d\mu_n := \left( |V_n|^3 + |P_n - a_n(\tau)|^{3/2} \right) dy \, d\tau. \tag{3.5}$$

The measure $\mu_n$ is not denoted by $\mathcal{E}_r$. On unit cylinders the factor $1^{-2}$ is invisible, so unit-scale atlas mass and unit-scale CKN concentration are compatible, with the atlas quantity dominating the quotient quantity. The velocity quantity $\mathcal{V}_r$ is the part used to certify nonzero retained limits under weak pressure convergence; the pressure-inclusive quantity $\mathcal{E}_r$ is used for CKN regularity and pressure-normalized smallness.

*Remark* 3.2 (Two-scale dictionary). We use two different normalizations. The quantity $\mathcal{E}_r$ is parabolic and CKN-critical, hence carries the factor $r^{-2}$. By contrast, the tail quantities $\mathcal{A}_n(z, r)$ are spatial/logarithmic averaged oscillations on fixed centered time windows; their natural normalization is $r^{-3}$. These two quantities are never identified.

**Definition 3.3** (Velocity retention at a level and unit CKN activity)**.** Fix $0 < \eta_*, \varepsilon_* < \infty$. For $\eta > 0$, a terminal profile $(U, \Pi)$ has retained local velocity concentration at level $\eta$ if, after the terminal normalization under consideration,

$$\mathcal{V}_1(U; 0) \geq \eta.$$

It has positive retained local velocity concentration if this holds for some $\eta > 0$. When no level is displayed, the setup-level retained descendant threshold is $\eta_*$. It has unit CKN activity if

$$\mathcal{E}_1(U, \Pi; 0) \geq \varepsilon_*.$$

Only retained velocity concentration is used to pass nontriviality to limits from approximating sequences. Pressure-inclusive CKN activity is used for local regularity, pressure compactness, and threshold smallness, and any pressure-only lower bound is treated through the pressure-gauge alternatives below.

**Definition 3.4** (Threshold hierarchy)**.** The constants used in the terminal concentration argument are chosen in the following order. The compact velocity lower bounds $\eta_0$ and $\eta_v$ are supplied by the imported setup results and the local concentration input of the companion setup paper, respectively. The local small-data threshold $\varepsilon_{\mathrm{sd}}$ is then fixed with

$$0 < \varepsilon_{\mathrm{sd}} \leq \min\{\eta_0, \eta_v, \varepsilon_{\mathrm{CKN}}\},$$

where $\varepsilon_{\mathrm{CKN}}$ denotes the universal CKN regularity threshold whenever that threshold is invoked. The retained descendant velocity threshold $\eta_*$ is chosen after the pressure gauges and compactness topology are fixed, with

$$0 < \eta_* \leq \frac{1}{2}\varepsilon_{\mathrm{sd}}.$$

The pressure-inclusive unit CKN activity threshold $\varepsilon_*$ is then chosen with

$$0 < \varepsilon_* \leq \frac{1}{2}\varepsilon_{\mathrm{sd}}.$$

Whenever a compactness or lower-semicontinuity passage loses a harmless fixed factor, the relevant threshold, $\eta_*$ for velocity retention or $\varepsilon_*$ for pressure-inclusive CKN activity, is replaced once and for all by a smaller positive value. No later argument chooses a new concentration threshold depending on a previously extracted iterated profile extraction.

**Definition 3.5** (Refined residual candidate)**.** A refined residual candidate is a bounded normalized Seregin ancient limit obtained from the admissible Seregin extraction theorem in the companion setup paper, which carries the positive compact local velocity concentration in (1.5), and which is not in any alternative already closed before the refined residual decomposition is entered. The definition is local: the only persistent lower bound attached to the candidate by the setup paper is the fixed compact velocity $L^3$ lower bound.

*Remark* 3.6 (No hidden global $L^3$ assertion). The terminal arguments below do not claim that a generic residual profile is globally $L^3$-tight or globally bounded in $L^\infty_\tau L^3_y$. The only nonlocal critical-norm conclusion used at the endpoint is the weaker sequence-$L^3$ completeness of retained indecomposable concentration profiles, and that conclusion is derived from local terminal alternatives rather than from an a priori global estimate on the original residual class.

3.1. **Notation, threshold, and gauge directories.** The following compact directories collect the persistent symbols, thresholds, and gauges used in the residual closure. They are not new definitions; each entry refers to the location where the symbol is first defined.

Velocity, pressure, and profile-space symbols.

- $(V, P)$: centered Seregin ancient profile in (1.2); first used in Section 1.
- $(U, Q)$: physical ancient representative when $(V, P)$ is treated under physical pullback (1.4); also used in $\mathcal{R}_z$-recentered form throughout Sections 8 and 9. The intended meaning is fixed locally in each section by the explicit definition of the immediately preceding rescaling.
- $(W, Q_W)$: blown-down, co-rotating, or tail profile, depending on the section; defined locally in Sections 5 and 9.3.
- $\mathcal{E}_r(V, P; z_0)$: quotient, scale-invariant parabolic CKN concentration, see Definition 3.1.
- $\mathcal{V}_r(V; z_0)$: velocity part of the scale-invariant local concentration; retained descendants and nonzero limiting profiles are certified with this quantity.
- $\mathcal{E}_1$: pressure-normalized unit CKN concentration, fixed at $r = 1$; used for local regularity, CKN smallness, and pressure estimates once the velocity lower bound has been identified.
- $\mu_n$: atlas-dependent Radon defect measure in (3.5); used for concentration-set and defect compactness, not as a scale-invariant quantity.
- $\mathcal{A}_n(z, r)$: spatial/logarithmic averaged affine-quotient oscillation with $r^{-3}$ normalization, distinct from $\mathcal{E}_r$.
- $\mathfrak{X}_M$: bounded terminal profile class with pressure gauges, Definition 8.7.
- $\mathfrak{X}_M^{\#}$: compact normalized descendant profile class, Definition 8.10.
- $\mathfrak{T}_M$: critical-tail compactification of triples $(\bar{W}, \mathcal{Y}, \Lambda)$, Definition 9.37.
- $\mathcal{C}_{\mathrm{aff}}$: affine/parasitic reduction/quotient stratum, Definition 8.16; not a usual nonempty/empty PDE class but an exclusion label, see Remark 8.17 and Theorem 9.89.
- $\mathrm{osc}_{3,\mathrm{sp}}$: spatial oscillation modulo time-dependent homogeneous gauges, Definition 9.57.
- $\mathrm{osc}_{3,\mathrm{ct}}$ and $\Phi$: constant-mode/non-affine retained oscillation, Definition 8.8 and Theorem 8.19.

Threshold ordering.

- $\eta_0$: compact velocity $L^3$ lower bound supplied by the base Seregin extraction Theorem 1.5.
- $\eta_v$: compact velocity lower bound from the local concentration input of the setup paper, used as initial mass for terminal extraction; see Definition 3.4.
- $\varepsilon_{\mathrm{sd}}$: local small-data threshold, $0 < \varepsilon_{\mathrm{sd}} \le \min\{\eta_0, \eta_v, \varepsilon_{\mathrm{CKN}}\}$, Definition 3.4.
- $\eta_*$: retained descendant velocity threshold, $0 < \eta_* \le \varepsilon_{\mathrm{sd}}/2$, used in Definition 3.3 and in concentration-set extraction.
- $\varepsilon_*$: unit CKN activity threshold, $0 < \varepsilon_* \le \varepsilon_{\mathrm{sd}}/2$, used only for pressure-inclusive CKN activity and smallness tracking.
- $\varepsilon_0$: minimal mesoscopic-scale oscillation threshold; depends only on the hidden-scale defect lower bound and on the finite-overlap constants of Lemma 9.59; first appears in Definition 9.58.
- $\eta_K$: compact oscillation lower bound on the recurrent concentration core, first introduced in Theorem 10.17.

The thresholds are fixed in the order $\eta_0, \eta_v \Rightarrow \varepsilon_{\mathrm{sd}} \Rightarrow (\eta_*, \varepsilon_*) \Rightarrow \varepsilon_0, \eta_K$, and no later argument chooses a new concentration threshold depending on a previously extracted iterated profile extraction. Phrases of the form "after lowering the threshold" refer either to the once-and-for-all reduction of $\eta_*$ or $\varepsilon_*$ recorded in Definition 3.4, or to a slight reduction of the mesoscopic threshold $\varepsilon_0$ recorded in Lemma 9.62; in either case the reduction does not disturb earlier choices.

Pressure and velocity gauges.

- $c(\tau)$ or $a_{z_0,r}(\tau)$: local pressure time gauge in (3.2) and (3.4); measurable in $\tau$; used because pressure is defined modulo functions of time.
- $c_n(s)$: homogeneous velocity gauge at the minimal mesoscopic scale, Definition 9.63; bounded; either compactifies or assigns the descendant to $\mathcal{C}_{\mathrm{aff}}$ by Lemma 9.74; no pointwise derivative $\partial_s c_n$ is used unless that compactness has already been obtained.
- $\beta(s) \cdot y + \alpha(s)$: affine parasitic pressure paired with a spatially homogeneous velocity mode; quotient notation introduced in Definition 8.16 and used only in the gauged formulation of the affine/parasitic stratum.
- $L_n(x, s) = b_n(s)\cdot x + a_n(s)$: scaled affine pressure at the minimal mesoscopic scale, Lemma 9.71; affine in $x$, measurable in $s$; discarded in the non-affine quotient and assigned to $\mathcal{C}_{\mathrm{aff}}$ when it does not vanish.
- $a_{n,K}(t)$: local Seregin pressure time gauge on a compact cylinder $K$, used by the local pressure-gauge compactness theorem inherited from the setup paper and recorded in Proposition 1.19.

Table 4: Glossary of residual symbols.

| **Symbol** | **Meaning** |
|---|---|
| $\mathcal{E}_r$ | Scale-invariant parabolic CKN concentration. |
| $D_r$ | Pressure quotient part of the CKN concentration, modulo time gauges. |
| $\mathcal{V}_r$ | Scale-invariant velocity concentration used for retained nonzero descendants. |
| $\mu_n$ | Raw atlas defect measure for concentration-set and compactness arguments. |
| $\mathcal{A}_n(z, r)$ | Spatial/log-tail averaged affine-quotient oscillation. |
| $\mathcal{H}_\infty$ | Full covariant spacetime asymptotic hull. |
| $\mathcal{C}_{\mathrm{ax}}$ | Axisymmetric bounded-circulation residual class. |
| $\mathcal{C}_{\mathrm{rot}}$ | Rotational relative-equilibrium residual class. |
| $\mathcal{C}_{\mathrm{stat}}^{\mathrm{ret}}$ | Stationary-hull residual class with retained local velocity concentration. |
| $\mathcal{C}_{\mathrm{aff}}$ | Affine/parasitic pressure residual quotient. |
| $\mathcal{C}_{\mathrm{log\,diff}}$ | Log-diffuse critical-tail residual alternative. |
| $\mathcal{C}_{\mathrm{Young}}$ | Young/variance critical-tail residual alternative. |
| $\mathcal{C}_{\mathrm{homcrit}}$ | Coherent homogeneous critical-tail alternative. |
| $\mathcal{C}_{\mathrm{logper}}$ | Coherent log-periodic critical-tail alternative. |
| $\mathcal{C}_{\mathrm{ap}}$ | Coherent aperiodic/minimal critical-tail alternative. |
| $\mathcal{C}_{\mathrm{gen}}$ | Generic terminal residual class. |
| Retained descendant | Descendant with inherited positive unit velocity concentration. |
| Normalized support profile / support profile | Normalized defect support profile, not necessarily retained at the original threshold. |
| First-generation recentering | Recentered window of the original setup terminal sequence; compact only under the local-window condition. |
| Ancient-profile recentering | Covariant recentering of an already extracted bounded ancient profile; compact by global ancient boundedness. |
| Local-window condition | Requirement that every fixed recentered cylinder undoes into the original local Type I setup cylinder. |
| Exterior/diffuse terminal branch | Positive residual mass outside active neighborhoods, below unit velocity thresholds after the reduction step. |

| Symbol | Meaning |
|---|---|
| Pressure/noncompact defect | Failure of local Seregin compactness or pressure gauge compactness under a selected recentering. |
| Critical-tail defect | Nonzero normalized annular/logarithmic residual profile visible only after critical-tail blow-down. |

## 4. Exclusion of the axisymmetric bounded-circulation residual class

In this section we prove the rigidity statement needed for the first refined residual class. The argument uses the scale-invariant angular circulation in centered variables. The essential point is that the centered Type I equation contains the Ornstein–Uhlenbeck drift. Thus the angular-circulation equation has an absorbing boundary at the symmetry axis together with a confining drift at spatial infinity. This gives a scalar Liouville theorem for bounded ancient solutions of the circulation equation. Related axisymmetric Liouville theorems for Navier–Stokes, in different settings, are proved in [6, 5].

Throughout the section we write cylindrical coordinates as $(\rho, \theta, Z)$, and an axisymmetric vector field as

$$V = V_\rho e_\rho + V_\theta e_\theta + V_Z e_Z.$$

The centered Navier–Stokes equation is

$$\partial_\tau V - \Delta V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V + (V \cdot \nabla)V + \nabla P = 0, \qquad \nabla \cdot V = 0. \tag{4.1}$$

For an axisymmetric solution define the centered angular circulation

$$\Gamma(\rho, Z, \tau) := \rho V_\theta(\rho, Z, \tau). \tag{4.2}$$

This is the same scale-invariant quantity as the physical circulation $rU_\theta$. If the physical representative is

$$U(x,t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)),$$

then

$$rU_\theta(r, z, t) = \rho V_\theta(\rho, Z, \tau), \qquad r = \sqrt{-t}\,\rho, \quad z = \sqrt{-t}\,Z.$$

**Lemma 4.1** (Centered circulation equation)**.** *Let $(V, P)$ be a smooth bounded axisymmetric solution of (4.1). Then $\Gamma = \rho V_\theta$ satisfies*

$$\partial_\tau \Gamma = \partial_\rho^2 \Gamma - \frac{1}{\rho} \partial_\rho \Gamma + \partial_Z^2 \Gamma - \left(V_\rho + \frac{\rho}{2}\right) \partial_\rho \Gamma - \left(V_Z + \frac{Z}{2}\right) \partial_Z \Gamma \tag{4.3}$$

*on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$. Moreover*

$$\Gamma(0, Z, \tau) = 0 \qquad \textit{for all } (Z, \tau) \in \mathbb{R} \times \mathbb{R}. \tag{4.4}$$

*Proof.* The angular component of (4.1) is

$$\partial_\tau V_\theta - \left(\Delta - \frac{1}{\rho^2}\right) V_\theta + \frac{1}{2}(\rho\partial_\rho + Z\partial_Z)V_\theta + \frac{1}{2} V_\theta + V_\rho \partial_\rho V_\theta + V_Z \partial_Z V_\theta + \frac{V_\rho V_\theta}{\rho} = 0.$$

Multiplying by $\rho$, using

$$\rho \left(\Delta - \frac{1}{\rho^2}\right) V_\theta = \partial_\rho^2 \Gamma - \frac{1}{\rho} \partial_\rho \Gamma + \partial_Z^2 \Gamma,$$

$$\rho \left[\frac{1}{2}(\rho\partial_\rho + Z\partial_Z)V_\theta + \frac{1}{2} V_\theta\right] = \frac{1}{2}(\rho\partial_\rho + Z\partial_Z)\Gamma,$$

and

$$\rho \left(V_\rho \partial_\rho V_\theta + V_Z \partial_Z V_\theta + \frac{V_\rho V_\theta}{\rho}\right) = V_\rho \partial_\rho \Gamma + V_Z \partial_Z \Gamma,$$

gives (4.3). Finally, smoothness of a bounded axisymmetric vector field at the axis implies $V_\theta(\rho, Z, \tau) = O(\rho)$ as $\rho \downarrow 0$, locally uniformly in $(Z, \tau)$. Hence $\Gamma = \rho V_\theta = O(\rho^2)$, which gives (4.4). ☐

We next record the scalar Liouville theorem used below. It is stated in a form that isolates precisely the PDE mechanism: bounded drift perturbations of the centered circulation operator cannot sustain a bounded ancient solution which vanishes on the symmetry axis.

**Lemma 4.2** (One-dimensional killed radial absorption)**.** *Fix $R > 1$, $M < \infty$, and a compact interval $K_\rho = [\delta, R-\delta] \Subset (0, R)$. Let $\rho_t$ be any weak solution on $(0, R)$, killed at $\{0, R\}$, of*

$$d\rho_t = -\left(\frac{1}{\rho_t} + \frac{\rho_t}{2} + \beta_t\right) dt + \sqrt{2}\, dW_t, \qquad |\beta_t| \le M,$$

*where $\beta_t$ may be adapted. If $\tau_\partial := \inf\{t : \rho_t \in \{0, R\}\}$, then*

$$\lim_{T\to\infty} \sup_{\rho_0 \in K_\rho} \mathbb{P}_{\rho_0}(\tau_\partial > T) = 0.$$

*More precisely, there are $T_0 = T_0(R, M, \delta) > 0$ and $\theta = \theta(R, M, \delta) > 0$ such that*

$$\inf_{\rho_0 \in K_\rho} \mathbb{P}_{\rho_0}(\tau_\partial \le T_0) \ge \theta.$$

*Proof.* Choose

$$\delta_0 := \min\{\delta/2,\ [4(M+2)]^{-1}\}.$$

We first show that every starting point in $K_\rho$ reaches the left boundary layer $(0, \delta_0]$ in a fixed time with a probability bounded from below. While $\rho_t \ge \delta_0$, the drift satisfies

$$-\left(\frac{1}{\rho_t} + \frac{\rho_t}{2} + \beta_t\right) \le M.$$

Fix $T_1 > 0$ and the deterministic linear path

$$\ell(t) := -\frac{R + MT_1 + 2\delta_0}{T_1}\, t, \qquad 0 \le t \le T_1.$$

On the Brownian tube event

$$E_1 := \left\{ \sup_{0 \le t \le T_1} \left| \sqrt{2}\, W_t - \ell(t) \right| \le \frac{\delta_0}{4} \right\},$$

the integral form of the SDE gives, for every $\rho_0 \in K_\rho$, that $\rho_t$ reaches $(0, \delta_0]$ before time $T_1$, unless it has already hit 0. The probability of $E_1$ is a strictly positive number

$$p_1 = p_1(R, M, \delta) > 0.$$

For completeness, this positivity follows by partitioning $[0, T_1]$ into finitely many subintervals and using the one-dimensional Dirichlet heat kernel lower bound in the strip of width $\delta_0/4$ around the line $\ell$.

It remains to absorb the process from the layer $(0, \delta_0]$. As long as the process stays in $(0, 2\delta_0]$,

$$-\left(\frac{1}{\rho_t} + \frac{\rho_t}{2} + \beta_t\right) \le M - \frac{1}{2\delta_0} \le -2.$$

Let $T_2 := \delta_0$. On the event

$$E_2 := \left\{ \sup_{0 \le t \le T_2} \left( \sqrt{2}\, W_t - \frac{t}{4} \right) \le \frac{\delta_0}{4} \right\},$$

which has probability $p_2 = p_2(M, \delta) > 0$, the same integral estimate gives

$$\rho_t \le \rho_0 - 2t + \frac{\delta_0}{4} + \frac{t}{4} \qquad \text{until hitting } 0.$$

Thus every initial point $\rho_0 \le \delta_0$ hits 0 before $T_2$, on $E_2$. Again $p_2 > 0$ is an explicit finite-time heat-kernel barrier estimate for Brownian motion below the affine boundary $\delta_0/4 + t/4$.

Combining the two blocks and using the strong Markov property gives

$$\inf_{\rho_0 \in K_\rho} \mathbb{P}_{\rho_0}(\tau_\partial \le T_1 + T_2) \ge p_1 p_2 =: \theta > 0.$$

With $T_0 := T_1 + T_2$, iteration gives

$$\sup_{\rho_0 \in K_\rho} \mathbb{P}_{\rho_0}(\tau_\partial > kT_0) \le (1-\theta)^k,$$

and the asserted decay follows. The constants are independent of the axial coordinate and of any truncation in that coordinate. ☐

**Lemma 4.3** (Axis-absorption Liouville theorem)**.** *Let $B = (B_\rho, B_Z)$ be a smooth vector field on $(0,\infty) \times \mathbb{R} \times \mathbb{R}$ satisfying*

$$\|B\|_{L^\infty} < \infty.$$

*Let $G \in C^{2,1}((0,\infty) \times \mathbb{R} \times \mathbb{R}) \cap L^\infty$ solve*

$$\partial_\tau G = \partial_\rho^2 G - \frac{1}{\rho}\partial_\rho G + \partial_Z^2 G - \left(B_\rho + \frac{\rho}{2}\right)\partial_\rho G - \left(B_Z + \frac{Z}{2}\right)\partial_Z G \tag{4.5}$$

*for all $\tau \in \mathbb{R}$, and assume*

$$\lim_{\rho \downarrow 0} G(\rho, Z, \tau) = 0 \qquad \text{locally uniformly in } (Z, \tau). \tag{4.6}$$

*Then $G \equiv 0$.*

*Proof.* Set $M_B = \|B\|_{L^\infty}$. For $R > 1$ define

$$h_R(\rho) := \frac{\displaystyle\int_0^{\min\{\rho,R\}} s \exp\left(\frac{s^2}{4} - M_B s\right) ds}{\displaystyle\int_0^R s \exp\left(\frac{s^2}{4} - M_B s\right) ds}. \tag{4.7}$$

Then $0 \le h_R \le 1$, $h_R(0) = 0$, $h_R(R) = 1$, and, on $0 < \rho < R$,

$$h_R'' + \left(M_B - \frac{\rho}{2} - \frac{1}{\rho}\right) h_R' = 0, \qquad h_R' > 0. \tag{4.8}$$

Let

$$\mathcal{L}_\tau f := \partial_\rho^2 f - \frac{1}{\rho}\partial_\rho f + \partial_Z^2 f - \left(B_\rho + \frac{\rho}{2}\right)\partial_\rho f - \left(B_Z + \frac{Z}{2}\right)\partial_Z f.$$

Since $B_\rho \ge -M_B$, (4.8) gives

$$\mathcal{L}_\tau h_R = h_R'' - \left(\frac{1}{\rho} + \frac{\rho}{2} + B_\rho\right) h_R' \le h_R'' + \left(M_B - \frac{\rho}{2} - \frac{1}{\rho}\right) h_R' = 0. \tag{4.9}$$

Thus $(\partial_\tau - \mathcal{L}_\tau) h_R \ge 0$.

We now compare $G$ to this barrier on the strip $\Omega_R = (0, R) \times \mathbb{R}$. Let $M_G = \|G\|_{L^\infty}$. We first record the Dirichlet strip decay estimate used by the comparison argument. If $q_{s,R}$ solves

$$\partial_\tau q_{s,R} = \mathcal{L}_\tau q_{s,R}, \qquad q_{s,R} = 0 \text{ on } \rho = 0, R, \qquad q_{s,R}(\rho, Z, s) = 1,$$

then, for every compact $K \Subset \Omega_R$ and every fixed $t$,

$$\lim_{s\to-\infty} \sup_{(\rho,Z)\in K} q_{s,R}(\rho,Z,t) = 0. \tag{4.10}$$

To prove (4.10), fix $K \Subset \Omega_R$, and choose $\delta > 0$ such that

$$K \subset [\delta, R-\delta] \times \mathbb{R}.$$

We use a one-dimensional absorption estimate for the $\rho$-component. Let $\rho_\tau$ be any weak solution of

$$d\rho_\tau = -\left(\frac{1}{\rho_\tau} + \frac{\rho_\tau}{2} + \beta_\tau\right) d\tau + \sqrt{2}\, dW_\tau, \qquad |\beta_\tau| \le M_B,$$

killed when it hits 0 or $R$. The coefficient $\beta_\tau$ may be any adapted process; in the application it is $B_\rho(\rho_\tau, Z_\tau, \tau)$. By Lemma 4.2, there are $T = T(R, M_B, \delta) > 0$ and $\theta = \theta(R, M_B, \delta) > 0$ such that

$$\inf_{\rho_0\in[\delta,R-\delta]} \mathbb{P}_{\rho_0}\{\rho_\tau \text{ hits } 0 \text{ or } R \text{ before time } T\} \ge \theta.$$

The parabolic maximum principle and the Feynman–Kac representation for the killed equation give

$$0 \le q_{s,R}(\rho,Z,t) \le \mathbb{P}_\rho\{\rho_\tau \text{ survives on } [s,t]\}.$$

Iterating the block estimate yields

$$\sup_{(\rho,Z)\in K} q_{s,R}(\rho,Z,t) \le (1-\theta)^{\lfloor (t-s)/T\rfloor},$$

which tends to 0 as $s \to -\infty$. This proves (4.10) directly on the infinite strip.

*Truncated Feynman–Kac justification.* The displayed killed-kernel and survival-probability statements are first established on the bounded box

$$\Omega_{\varepsilon,R,L} := (\varepsilon, R) \times (-L, L) \qquad (0 < \varepsilon < R < \infty,\ L > 0)$$

with Dirichlet boundary conditions on the entire boundary of $\Omega_{\varepsilon,R,L}$, where the standard killed parabolic theory provides a Feynman–Kac representation of the strip kernel $q_s^{(\varepsilon,R,L)}(\rho,Z,t)$ by the killed diffusion $(\rho_\tau, Z_\tau)$ with drift $(-(\rho^{-1} + \rho/2 + \beta_\tau), -(Z/2 + \gamma_\tau))$ and additive Brownian noise. The block survival estimate of Lemma 4.2 applies because the one-dimensional radial absorption in $\rho$ does not depend on the $Z$-component or on $\varepsilon$ or $L$ once $\delta \le \rho \le R - \delta$; hence

$$\sup_{(\rho,Z)\in K} q_s^{(\varepsilon,R,L)}(\rho,Z,t) \le (1-\theta)^{\lfloor (t-s)/T\rfloor}$$

holds uniformly in $\varepsilon$ and $L$ for any compact $K \Subset (\delta, R-\delta) \times [-L_0, L_0]$ and all $\varepsilon < \delta$, $L > L_0$. The truncated comparison $(W_R^\pm)_+ \le q_s^{(\varepsilon,R,L)}$ on $\Omega_{\varepsilon,R,L}$ then follows from the parabolic maximum principle applied to the truncated problem (the $\rho = \varepsilon$ boundary contributes nonpositively because both $G/M_G$ and $h_R$ are bounded by 1, so the truncated boundary corrector $\rho_{\rm tr}(\tau) := 2\sup\{|G/M_G| + h_R\} \cdot \mathbf{1}_{\rho=\varepsilon \text{ or } |Z|=L}$ is killed by the same Dirichlet kernel and tends to zero on every fixed compact $K$ as $L \to \infty$, $\varepsilon \downarrow 0$). Sending $L \to \infty$ recovers the half-strip $(\varepsilon, R) \times \mathbb{R}$ by monotone convergence of the killed semigroup; sending $\varepsilon \downarrow 0$ recovers $(0, R) \times \mathbb{R}$ using $h_R(0) = 0$ and the axis condition (4.6) on $G$; finally sending $R \to \infty$ gives the global statement (4.11) and the conclusion $G \equiv 0$ as in the next paragraph.

Fix $R > 1$, $s < t$, and set

$$W_R^+ := \frac{G}{M_G} - h_R, \qquad W_R^- := -\frac{G}{M_G} - h_R,$$

with the convention that the conclusion is immediate if $M_G = 0$. By (4.5) and (4.9),

$$(\partial_\tau - \mathcal{L}_\tau) W_R^\pm \le 0 \qquad \text{in } \Omega_R \times (s,t).$$

On $\rho = 0$, (4.6) and $h_R(0) = 0$ give $W_R^{\pm} \leq 0$. On $\rho = R$, $h_R(R) = 1$ and $|G| \leq M_G$ give again $W_R^{\pm} \leq 0$. The parabolic comparison principle with the Dirichlet strip kernel therefore yields, for $\tau = t$,

$$(W_R^{\pm})_+(\rho, Z, t) \leq q_{s,R}(\rho, Z, t) \qquad ((\rho, Z) \in \Omega_R).$$

Letting $s \to -\infty$ and using (4.10) gives $W_R^{\pm} \leq 0$ on compact subsets of $\Omega_R$. Since the compact subset is arbitrary,

$$|G(\rho, Z, t)| \leq M_G h_R(\rho) \qquad (0 < \rho < R, Z \in \mathbb{R},\ t \in \mathbb{R}). \tag{4.11}$$

Finally, for each fixed $\rho < \infty$, the denominator in (4.7) tends to infinity super-exponentially as $R \to \infty$, while the numerator remains fixed. Hence $h_R(\rho) \to 0$. Letting $R \to \infty$ in (4.11) gives $G(\rho, Z, t) = 0$ for every $\rho > 0$, $Z \in \mathbb{R}$, and $t \in \mathbb{R}$. The boundary value at $\rho = 0$ is already zero by assumption. Therefore $G \equiv 0$. □

**Theorem 4.4** (Exclusion of the axisymmetric bounded-circulation residual class)**.** *Let $\mathcal{C}_{\mathrm{ax}}(\mathcal{S}; I, J)$ be the axisymmetric bounded-circulation residual class in the refined residual decomposition. Then*

$$\mathcal{C}_{\mathrm{ax}}(\mathcal{S}; I, J) = \varnothing. \tag{4.12}$$

*Proof.* Assume, for contradiction, that $V \in \mathcal{C}_{\mathrm{ax}}(\mathcal{S}; I, J)$. By definition of $\mathcal{C}_{\mathrm{ax}}$, the associated physical ancient representative is axisymmetric and has bounded angular circulation. Equivalently, in centered variables,

$$\Gamma(\rho, Z, \tau) = \rho V_\theta(\rho, Z, \tau)$$

is bounded on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$. Since $V$ is a Seregin ancient limit, it is smooth and bounded in centered variables. Hence $B = (V_\rho, V_Z)$ is bounded.

By Lemma 4.1, $\Gamma$ satisfies (4.3) and vanishes at the axis. Applying Lemma 4.3 with $G = \Gamma$ and $B = (V_\rho, V_Z)$ gives

$$\Gamma \equiv 0.$$

Therefore $V_\theta \equiv 0$ for $\rho > 0$, and by smoothness also on the axis. Thus the associated ancient solution is axisymmetric without swirl. But the axisymmetric no-swirl class is one of the structured alternatives removed before the residual class is defined. This contradicts $V \in \mathcal{R}(\mathcal{S}; I, J)$, and therefore no such $V$ exists. □

*Remark* 4.5 (Why this does not assert the full bounded-swirl Liouville theorem). The proof uses the centered Type I equation, specifically the Ornstein–Uhlenbeck drift terms $-\rho\partial_\rho/2$ and $-Z\partial_Z/2$ in the scalar equation for $\Gamma$. It is therefore a Type I Seregin limit argument. It does not prove that every bounded ancient axisymmetric solution of the unrenormalized Navier–Stokes equations with bounded physical circulation is swirl-free. The latter lacks the confining centered drift and remains a different shell-transport problem.

## 5. The rotational relative-equilibrium residual class

We next isolate the second refined residual class. This class consists of centered Type I ancient limits whose time dependence is a rigid rotation of a fixed spatial profile. Passing to the co-rotating frame converts the centered Navier–Stokes equation into a stationary Leray profile equation with a Coriolis transport term. A whole-space Coriolis–Leray Liouville theorem for all bounded profiles is not used here. Instead, the Coriolis term is localized as an annular flux and is then treated by the local concentration analysis proved in Sections 9 and 10.

Throughout this section the centered equation is

$$\partial_\tau V - \Delta V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V + (V \cdot \nabla)V + \nabla P = 0, \qquad \nabla \cdot V = 0. \tag{5.1}$$

For $\Omega \in \mathfrak{so}(3)$, set

$$Q(\tau) := \exp(\tau\Omega) \in SO(3). \tag{5.2}$$

**Definition 5.1** (Rotational relative-equilibrium profile)**.** A bounded smooth ancient solution $(V, P)$ of (5.1) is a rotational relative equilibrium if there exist $\Omega \in \mathfrak{so}(3)$, a bounded smooth divergence-free vector field $W$, and a pressure profile $\Pi$, such that

$$V(y,\tau) = Q(\tau)W(Q(\tau)^T y), \qquad P(y,\tau) = \Pi(Q(\tau)^T y) \tag{5.3}$$

up to the usual time-dependent pressure gauge. The class $\mathcal{C}_{\mathrm{rot}}$ consists of such residual Seregin profiles, after the axisymmetric bounded-circulation class and the previously closed alternatives in the setup paper have been removed.

**Lemma 5.2** (Co-rotating stationary equation)**.** *Let $(V, P)$ be a rotational relative equilibrium. Then the co-rotating profile $(W, \Pi)$ satisfies*

$$-\Delta W + \frac{1}{2}y\cdot\nabla W + \frac{1}{2}W + (W\cdot\nabla)W + \Omega W - (\Omega y)\cdot\nabla W + \nabla\Pi = 0, \qquad \nabla\cdot W = 0. \tag{5.4}$$

*Equivalently,*

$$\mathbb{P}\Big[-\Delta W + \frac{1}{2}y\cdot\nabla W + \frac{1}{2}W + (W\cdot\nabla)W + \Omega W - (\Omega y)\cdot\nabla W\Big] = 0, \tag{5.5}$$

*where $\mathbb{P}$ is the whole-space Leray projector.*

*Proof.* Let $z = Q(\tau)^T y$. Since $Q'(\tau) = \Omega Q(\tau)$ and $Q(\tau)^T\Omega Q(\tau) = \Omega$, one has

$$\partial_\tau z = -\Omega z.$$

Therefore

$$\partial_\tau V(y,\tau) = Q(\tau)\Big(\Omega W(z) - (\Omega z)\cdot\nabla W(z)\Big).$$

The Laplacian, divergence, convection term, and pressure gradient are invariant under the orthogonal change of variables. Substitution into (5.1), followed by multiplication by $Q(\tau)^T$, gives (5.4). Applying the Leray projector gives (5.5). □

5.1. **The Bernoulli identity and the local Coriolis flux.** Write

$$F(y) := \frac{1}{2}y - \Omega y, \qquad A(y) := W(y) + F(y), \qquad \omega := \nabla\times W. \tag{5.6}$$

Since $\Omega$ is skew-symmetric, there is a unique vector $\varpi \in \mathbb{R}^3$ such that

$$\Omega y = \varpi \times y. \tag{5.7}$$

Then

$$\nabla\times F = -2\varpi. \tag{5.8}$$

**Lemma 5.3** (Coriolis Bernoulli identity)**.** *Let $(W, \Pi)$ be a smooth solution of (5.4). Define*

$$B := \frac{1}{2}|W|^2 + \Pi + F\cdot W. \tag{5.9}$$

*Then*

$$\nabla B + \nabla\times\omega - A\times\omega = 0, \tag{5.10}$$

*and consequently*

$$-\Delta B + A\cdot\nabla B = -|\omega|^2 - \omega\cdot(\nabla\times F) = -|\omega|^2 + 2\varpi\cdot\omega. \tag{5.11}$$

*Proof.* The vector identity

$$(W\cdot\nabla)W=\nabla\left(\frac{1}{2}|W|^2\right)-W\times\omega$$

and the identity

$$(F\cdot\nabla)W+(\nabla F)^TW=\nabla(F\cdot W)-F\times\omega$$

apply because $(\nabla F)^TW=\frac{1}{2}W+\Omega W$. Thus (5.4) can be rewritten as

$$-\Delta W+\nabla\left(\frac{1}{2}|W|^2+\Pi+F\cdot W\right)-(W+F)\times\omega=0.$$

Since $\nabla\cdot W=0$, one has $-\Delta W=\nabla\times\omega$, which gives (5.10).

Taking divergence in (5.10) gives

$$\Delta B=\nabla\cdot(A\times\omega),$$

because the divergence of a curl vanishes. Using

$$\nabla\cdot(A\times\omega)=\omega\cdot(\nabla\times A)-A\cdot(\nabla\times\omega),\qquad \nabla\times A=\omega+\nabla\times F,$$

we obtain

$$\Delta B=|\omega|^2+\omega\cdot(\nabla\times F)-A\cdot(\nabla\times\omega).$$

On the other hand, by (5.10),

$$A\cdot\nabla B=A\cdot(A\times\omega)-A\cdot(\nabla\times\omega)=-A\cdot(\nabla\times\omega).$$

Combining the last two identities gives (5.11). Finally, (5.8) follows from (5.7). ☐

**Lemma 5.4** (Localized Coriolis flux identity)**.** *Let $(W,\Pi)$ be as in Lemma 5.3, and let $\chi\in C_c^\infty(\mathbb{R}^3)$. Then*

$$\int_{\mathbb{R}^3}\chi\left(-\Delta B+A\cdot\nabla B+|\omega|^2\right)dy=-2\int_{\mathbb{R}^3}(W\times\varpi)\cdot\nabla\chi\,dy. \tag{5.12}$$

*In particular the Coriolis contribution is supported only where the cutoff varies.*

*Proof.* By (5.11),

$$-\Delta B+A\cdot\nabla B+|\omega|^2=2\varpi\cdot\omega.$$

Because $\varpi$ is constant and $\omega=\nabla\times W$,

$$\varpi\cdot\omega=\nabla\cdot(W\times\varpi).$$

Multiplying by $\chi$, integrating over $\mathbb{R}^3$, and integrating by parts gives

$$2\int\chi\,\varpi\cdot\omega=2\int\chi\,\nabla\cdot(W\times\varpi)=-2\int(W\times\varpi)\cdot\nabla\chi,$$

which proves (5.12). ☐

**Lemma 5.5** (Non-small Coriolis flux creates retained local velocity concentration)**.** *Let $\chi\in C_c^\infty(\mathbb{R}^3)$, let $A_\chi:=\operatorname{supp}\nabla\chi$, and put*

$$C_\chi:=\int_{A_\chi}|\nabla\chi|^{3/2}\,dy.$$

*Fix $T\in(0,1]$, and let*

$$\mathcal{T}_{\chi,T}:=\{(y,\tau):-T<\tau<0,\ Q(\tau)^Ty\in A_\chi\}$$

*be the swept annular tube in the original centered variables. Cover $\mathcal{T}_{\chi,T}$ by finitely many unit terminal cylinders*

$$\mathcal{T}_{\chi,T} \subset \bigcup_{m=1}^{N_{\chi,T}} Q_1(z_m).$$

*The number $N_{\chi,T}$ depends only on the compact annulus $A_\chi$, the time length $T$, and $\Omega$, and the cover is chosen with bounded overlap inside the normalized terminal domain under consideration. Let $\varepsilon_{\mathrm{sd}} > 0$ be the raw unit-window threshold used in Definition 9.3. If $\varpi \neq 0$ and*

$$\left| 2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi \, dy \right| \geq 2|\varpi| \, C_\chi^{2/3} \left( \frac{N_{\chi,T} \varepsilon_{\mathrm{sd}}}{T} \right)^{1/3}, \tag{5.13}$$

*then at least one cylinder $Q_1(z_m)$ is raw unit-window active at threshold $\varepsilon_{\mathrm{sd}}$ for the original centered solution $V$.*

*Proof.* Write

$$\delta_\chi := \left| 2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi \, dy \right|.$$

Hölder's inequality gives

$$\begin{aligned} \delta_\chi &\leq 2|\varpi| \int_{A_\chi} |W| \, |\nabla \chi| \, dy \\ &\leq 2|\varpi| \left( \int_{A_\chi} |W|^3 \, dy \right)^{1/3} C_\chi^{2/3}. \end{aligned}$$

Thus (5.13) implies

$$\int_{A_\chi} |W|^3 \, dy \geq \frac{N_{\chi,T} \varepsilon_{\mathrm{sd}}}{T}.$$

Since $V(y,\tau) = Q(\tau) W(Q(\tau)^T y)$ and $Q(\tau)$ is an isometry,

$$\begin{aligned} \iint_{\mathcal{T}_{\chi,T}} |V(y,\tau)|^3 \, dy \, d\tau &= \int_{-T}^{0} \int_{Q(\tau) A_\chi} |W(Q(\tau)^T y)|^3 \, dy \, d\tau \\ &= T \int_{A_\chi} |W(z)|^3 \, dz \geq N_{\chi,T} \varepsilon_{\mathrm{sd}}. \end{aligned}$$

Because the tube is covered by $N_{\chi,T}$ unit terminal cylinders with bounded overlap, one of them satisfies

$$\iint_{Q_1(z_m)} |V|^3 \, dy \, d\tau \geq \varepsilon_{\mathrm{sd}}.$$

The pressure contribution to the local unit defect measure is nonnegative after the allowed pressure gauge. Hence

$$\mu(Q_1(z_m)) \geq \varepsilon_{\mathrm{sd}}.$$

Thus the corresponding recentering sequence is raw unit-window active at threshold $\varepsilon_{\mathrm{sd}}$ in the sense of Definition 9.3. The retained nonzero descendant later obtained from this observer is certified by the displayed unit velocity lower bound, while the atlas pressure term is kept only in the raw defect measure. □

**Lemma 5.6** (Local Coriolis-flux dichotomy)**.** *Fix $\chi \in C_c^\infty(\mathbb{R}^3)$ and $T \in (0,1]$. Along every terminal extraction of a rotational relative equilibrium, the following priority alternatives are applied:*

- *(i) $\varpi \neq 0$, the flux across $A_\chi = \operatorname{supp} \nabla \chi$ satisfies the threshold (5.13), and the swept annular tube contains a unit-scale velocity concentration;*

*(ii) either $\varpi = 0$, or the threshold (5.13) fails; then the flux is a bounded compactly supported functional of $W$ on $A_\chi$, passes to terminal limits by local $L^3$ convergence, and creates no terminal alternative other than the single-profile, finite separated-profile, exterior-escape, diffuse-exterior-concentration, local-vanishing, or local-compactness-failure alternatives of Theorem 9.24.*

*Proof.* Alternative *(i)* is precisely Lemma 5.5. If $\varpi = 0$, then the flux vanishes identically. If $\varpi \neq 0$ and (5.13) fails, the flux is bounded by the right hand side of that threshold. In both cases the functional

$$W \longmapsto -2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi \, dy$$

depends only on $W$ on the compact annular set $A_\chi$. Under the local Seregin compactness supplied by the admissible setup hypotheses, velocities converge strongly in $L^3_{\mathrm{loc}}$; since $\nabla\chi \in L^{3/2}$ and $\varpi$ is fixed, the functional is continuous along every terminal subsequence:

$$\int (W_n \times \varpi) \cdot \nabla \chi \longrightarrow \int (W_\infty \times \varpi) \cdot \nabla \chi.$$

Thus a subthreshold flux is retained only as a local term in the limiting localized Bernoulli identity (5.12). It is not a new location of local velocity concentration. Every actual location of compact velocity $L^3$ concentration is still one of the terminal concentration-set alternatives listed in Theorem 9.24. □

*Remark* 5.7 (Why this is not a global Liouville argument). For $\Omega = 0$, the Coriolis flux in Lemma 5.4 vanishes. For $\Omega \neq 0$, the same lemma does not give a sign-definite maximum principle for $B$. It gives only the local dichotomy in Lemma 5.6: either a chosen annulus already contains a recentered local concentration, or the annular flux is a subthreshold compact functional which passes to the local terminal limit. No tail decay, whole-space $L^p$ bound, Gaussian energy estimate, or global pressure normalization is used.

### 5.2. **Reduction of the rotational relative-equilibrium class.**

**Proposition 5.8** (Rotational residual reduction)**.** *Every rotational relative-equilibrium residual either belongs to a previously closed lower class or produces a terminal concentration profile realized from the original blow-up sequence satisfying the hypotheses of the terminal stratification assembly theorem.*

*Proof.* Let $V \in \mathcal{C}_{\mathrm{rot}}(\mathcal{S}; I, J)$. By Definition 5.1 and Lemma 5.2, there are $\Omega \in \mathfrak{so}(3)$ and a bounded smooth divergence-free profile $(W, \Pi)$ satisfying the Coriolis–Leray equation (5.4). The centered representative $V$ is a normalized Seregin ancient limit, so by Theorem 1.5 it carries a fixed positive compact velocity $L^3$ concentration.

If $\Omega = 0$, then $V$ is stationary in the centered frame. It is reduced to the stationary setup class if it lies in the already closed stationary alternative, and otherwise to the stationary-hull residual branch handled by Proposition 6.12. We therefore assume $\Omega \neq 0$.

Choose a countable family of smooth cutoffs which is cofinal among compact annuli in the terminal variables. For each cutoff and each rational $T \in (0, 1]$, apply the local dichotomy Lemma 5.6. A superthreshold Coriolis flux produces a unit-scale velocity concentration and is therefore already part of the local concentration decomposition. A subthreshold Coriolis flux is a compactly supported functional of $W$, continuous under the local convergence supplied by the admissible setup compactness hypotheses, and hence remains inside the localized limiting equation; it carries no independent local velocity mass and creates no additional terminal alternative.

We now apply the paired terminal concentration-set/remainder decomposition to the inherited compact density. Loss of local compactness, local vanishing, exterior escape, and diffuse exterior concentration cannot be the sole locations of this density by Theorem 9.27. By the preceding paragraph, the Coriolis flux has already led either to a local concentration or to a compact local term in the selected terminal equation. Thus, after local concentration extraction, the retained density is located either in a finite separated family of retained velocity concentration profiles or in a single retained velocity concentration terminal profile, together with any residual diffuse, noncompact, or critical-tail remainder selected by the terminal extraction procedure. This gives a terminal input realized from the original blow-up sequence for the terminal stratification theorem; no R2 closure is used in this reduction step. □

**Proposition 5.9** (Already-closed integrable subcase). *Let $V$ be a rotational relative equilibrium with co-rotating profile $W$. If*

$$W \in L^3(\mathbb{R}^3), \tag{5.14}$$

*then $V \equiv 0$. More generally, the same conclusion holds under any condition which places the physical representative in $L^\infty_t L^3_x$ on its ancient time interval.*

*Proof.* Rotations and the Navier–Stokes critical scaling preserve the $L^3$ norm. Thus $W \in L^3$ implies

$$\sup_{\tau\in\mathbb{R}} \|V(\cdot,\tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Equivalently, the physical representative lies in the critical class $L^\infty_t L^3_x$. The endpoint $L^3$ class is one of the already closed alternatives used in the setup paper, by the Escauriaza–Seregin–Šverák endpoint regularity theorem [3] and the stationary Nečas–Růžička–Šverák Liouville theorem [7]. Therefore $V \equiv 0$. A zero profile has zero local velocity concentration, so no such profile can belong to the residual nonvanishing class. □

## 6. The degenerate stationary-hull residual class

The third refined residual class consists of Type I ancient limits whose time-translation hull contains a stationary profile with retained local velocity concentration lying on a non-isolated family of stationary centered profiles. The role of the stationary-family condition is to separate genuinely degenerate stationary profile families from isolated stationary profiles and from the symmetry-generated classes already removed in $\mathcal{C}_{\mathrm{ax}}$ and $\mathcal{C}_{\mathrm{rot}}$.

The retained-velocity concentration condition is essential. The mere presence of a stationary-family element in the hull of a residual ancient solution is insufficient. A stationary element may occur as a limit without retained local velocity concentration and therefore need not retain the local velocity concentration inherited from the original singular point. Such a hull element is not placed in a class that is to be closed by a stationary-profile exclusion. Therefore the class below is defined using *retained-velocity occurrence*: the stationary-family profile must itself occur as a Seregin limit carrying a positive local velocity concentration.

Throughout this section the centered equation is

$$\partial_\tau V - \Delta V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V + (V\cdot\nabla)V + \nabla P = 0, \qquad \nabla\cdot V = 0. \tag{6.1}$$

Its projected form is

$$\partial_\tau V = \mathcal{F}(V), \qquad \mathcal{F}(W) := \mathbb{P}\left[\Delta W - \frac{1}{2}(W + y\cdot\nabla W) - (W\cdot\nabla)W\right], \tag{6.2}$$

where $\mathbb{P}$ is the whole-space Leray projector.

6.1. **Stationary families and retained-velocity hull limits.**

**Definition 6.1** (Admissible stationary phase space)**.** An admissible stationary phase space is a Banach space $\mathfrak{B}$ of solenoidal vector fields on $\mathbb{R}^3$ such that:

(i) $\mathfrak{B} \hookrightarrow C^{2,\alpha}_{\mathrm{loc}}(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$ continuously for some $\alpha \in (0,1)$;
(ii) the map $\mathcal{F}$ in (6.2) is well-defined on an open subset $\mathcal{U} \subset \mathfrak{B}$ and maps $\mathcal{U}$ into a Banach space $\mathfrak{Y}$;
(iii) $\mathcal{F} : \mathcal{U} \to \mathfrak{Y}$ is $C^1$.

**Definition 6.2** (Nontrivial stationary family)**.** Let $W \in \mathcal{U} \subset \mathfrak{B}$ satisfy $\mathcal{F}(W) = 0$. We say that $W$ lies on an admissible nontrivial stationary family if there exist $\varepsilon > 0$ and a $C^1$ curve

$$a \mapsto W_a \in \mathcal{U}, \qquad |a| < \varepsilon,$$

such that

$$W_0 = W, \qquad \mathcal{F}(W_a) = 0 \quad (|a| < \varepsilon), \tag{6.3}$$

and such that the family is not locally generated only by the action of the rigid rotation group. Equivalently, after possibly reparametrizing the curve, one may assume that

$$\dot{W}_0 \notin \{\Omega W - (\Omega y) \cdot \nabla W : \Omega \in \mathfrak{so}(3)\} \tag{6.4}$$

whenever the tangent $\dot{W}_0$ exists and is nonzero.

**Definition 6.3** (Retained stationary-hull velocity concentration)**.** Let $(W, \Pi)$ be a stationary centered profile. We say that $(W, \Pi)$ is retained-stationary concentrating if it has retained velocity concentration in the normalized unit sense: there is a recentering/rescaling realized from the original blow-up sequence $\mathscr{T}$ such that

$$\mathcal{V}_1\big(\mathscr{T}W; 0\big) \geq \eta_*. \tag{6.5}$$

Equivalently, before the final unit normalization one may require $\mathcal{V}_r(W; z_0) \geq \eta_*$ for some parabolic cylinder $Q_r(z_0)$. Unnormalized large-cylinder mass is not a retained velocity concentration criterion. It has stationary-hull CKN activity if the same realized normalization also satisfies

$$\mathcal{E}_1(\mathscr{T}W, \mathscr{T}\Pi; 0) \geq \varepsilon_*.$$

Only the velocity condition (6.5) certifies that the stationary-hull profile is nonzero. The accompanying CKN activity condition is used only for CKN smallness and pressure estimates.

**Definition 6.4** (Degenerate stationary-hull residual class with retained velocity concentration)**.** Fix an admissible stationary phase space $\mathfrak{B}$. The degenerate stationary-hull residual class with retained local velocity concentration

$$\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}(\mathcal{S}; I, J; \mathfrak{B})$$

consists of those $V \in \mathcal{R}(\mathcal{S}; I, J)$, after the ordered removal of $\mathcal{C}_{\mathrm{ax}}$ and $\mathcal{C}_{\mathrm{rot}}$, for which there exists a sequence $s_k \in \mathbb{R}$ and a stationary profile $(W, \Pi)$ such that

$$(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k(\cdot)) \longrightarrow (W, \Pi) \tag{6.6}$$

locally smoothly for the velocity and locally in $L^{3/2}$ for the pressure modulo admissible gauges, and such that:

(i) $W \in \mathfrak{B}$ and $\mathcal{F}(W) = 0$;
(ii) $W$ lies on an admissible nontrivial stationary family in the sense of Definition 6.2;
(iii) $(W, \Pi)$ is retained-stationary concentrating in the sense of Definition 6.3;
(iv) $(W, \Pi)$ is not in any previously closed stationary, structured, axisymmetric bounded-circulation, or rotational relative-equilibrium class.

In the ordered residual decomposition one keeps this class of stationary-hull limits with retained velocity concentration as the degenerate stationary-hull alternative. The hull cases without retained local velocity concentration are assigned to the generic terminal residual class, where they are handled by the terminal concentration-set decomposition rather than by a stationary-profile exclusion.

*Remark* 6.5 (Why retained velocity concentration is necessary). The condition (6.5) prevents the following logical error. A nonstationary ancient solution may have a stationary profile in its hull with zero local velocity concentration on the relevant compact cylinders. Excluding that stationary profile would not exclude the original ancient solution. The stationary-profile exclusion used below applies only to stationary profiles that themselves occur as nontrivial Seregin limits with positive local concentration.

6.2. **Hull limits are again Seregin limits.** The next lemma is the basic compatibility between time-translation hulls in centered variables and rescaling sequences in physical variables. It is the reason that a stationary profile in the hull which retains local concentration may be treated as a blow-up profile of the original solution, not merely as a limit inside the already extracted ancient solution.

**Lemma 6.6** (Scale reselection for centered hull limits)**.** *Let $(V, P)$ be a centered Type I Seregin limit obtained from a finite-energy suitable weak solution $(u, p)$ at a singular point $z_* = (x_*, t_*)$. Suppose that, for some sequence $s_k \in \mathbb{R}$,*

$$(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k(\cdot)) \to (W, \Pi)$$

*locally in the Seregin topology. Then, after passing to a diagonal subsequence and reselecting the physical blow-up scales, $(W, \Pi)$ is itself a centered Type I Seregin limit of $(u, p)$ at $z_*$. If the convergence to $(W, \Pi)$ retains a positive local velocity lower bound, then $(W, \Pi)$ is a nontrivial Seregin limit with the same type of inherited nonvanishing condition.*

*Proof.* Let $\lambda_n \downarrow 0$ be a sequence of scales producing $(V, P)$. In centered variables, a time translation by $s$ is a change of the physical blow-up scale by a factor depending only on $s$. With the convention used in (6.1), write this adjusted scale as

$$\lambda_{n,s} := e^{-s/2}\lambda_n. \tag{6.7}$$

Indeed, writing $t = -e^{-\tau}$, a shift $\tau \mapsto \tau + s$ corresponds to multiplying the physical length scale by $e^{-s/2}$, so that

$$V_{\lambda e^{-s/2}}(y, \tau) = V_\lambda(y, \tau + s)$$

with the corresponding pressure gauge. Thus the rescaled fields built from $(u, p)$ at scale $\lambda_{n,s}$ are the same as the fields at scale $\lambda_n$, translated by $s$ in centered time.

Fix an exhaustion of compact cylinders $K_m \subset \mathbb{R}^3 \times \mathbb{R}$. For each $k$, first require

$$\lambda_{n(k)} \le k^{-1} e^{s_k/2}$$

when $s_k < 0$, and otherwise require simply $\lambda_{n(k)} \le k^{-1}$. Then $\lambda_{n(k),s_k} = e^{-s_k/2}\lambda_{n(k)} \to 0$. Increasing $n(k)$ if needed, arrange that the rescaled fields at scale $\lambda_{n(k)}$, shifted by $s_k$, are within $1/k$ of $(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k)$ on $K_k$ in the local Seregin topology. This is possible because the original sequence converges locally to $(V, P)$.

By the assumed hull convergence, the shifted limit $(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k)$ converges to $(W, \Pi)$ on each fixed compact cylinder. The diagonal sequence of physical rescalings with scales $\lambda_{n(k),s_k}$ therefore converges locally to $(W, \Pi)$. Hence $(W, \Pi)$ is again a centered Type I Seregin limit at $z_*$. The assertion about nonvanishing follows from strong local $L^3$ convergence of the velocity and the retained velocity lower bound; pressure compactness is used only to keep the limiting pressure gauge admissible. □

6.3. **Elementary consequences of stationary-family degeneracy.** The stationary-family condition has two immediate PDE consequences. First, a differentiable nontrivial family produces a nontrivial element of the kernel of the linearized stationary Leray operator. Second, if an ancient trajectory is constrained to a stationary family, then it is stationary.

**Lemma 6.7** (Stationary family implies linearized degeneracy)**.** *Let $a \mapsto W_a \in \mathfrak{B}$ be a $C^1$ stationary family satisfying (6.3). Then*

$$D\mathcal{F}(W_0)\dot{W}_0 = 0. \tag{6.8}$$

*In particular, if $\dot{W}_0 \neq 0$ modulo infinitesimal rotations, the linearized stationary Leray operator has a genuine non-symmetry kernel element.*

*Proof.* Differentiate $\mathcal{F}(W_a) = 0$ at $a = 0$. Since $\mathcal{F}$ is $C^1$ on $\mathfrak{B}$, this gives (6.8). The final statement is the nontriviality condition in Definition 6.2. □

**Lemma 6.8** (Trajectories constrained to a stationary family are stationary)**.** *Let $a \mapsto W_a \in \mathfrak{B}$ be a $C^1$ stationary family with $\dot{W}_a \neq 0$ on an interval. Suppose that a smooth solution of (6.2) has the form*

$$V(\tau) = W_{a(\tau)} \tag{6.9}$$

*for a $C^1$ function $a(\tau)$. Then $a$ is constant, and hence $V$ is stationary.*

*Proof.* Since $\mathcal{F}(W_a) = 0$ for every $a$, equation (6.2) gives

$$a'(\tau)\dot{W}_{a(\tau)} = \partial_\tau V(\tau) = \mathcal{F}(W_{a(\tau)}) = 0.$$

Because $\dot{W}_{a(\tau)} \neq 0$, we obtain $a'(\tau) = 0$. Thus $a$ is constant on the interval. □

**Corollary 6.9** (Exact stationary-family subcase)**.** *If an element of the residual class is itself an exact trajectory on an admissible stationary family, then it is a stationary centered profile. Hence it is excluded whenever it satisfies the stationary $L^3$ Liouville criterion, or more generally whenever the stationary profile belongs to one of the previously closed stationary or structured classes.*

*Proof.* Let $V(\tau) = W_{a(\tau)}$ be such an exact stationary-family trajectory. By Lemma 6.8, $a'(\tau) = 0$ on every interval on which the stationary-family parametrization is nondegenerate. Hence $a$ is constant and $V$ is a single stationary centered profile. If that stationary profile satisfies the stationary $L^3$ Liouville criterion or lies in one of the previously closed structured or ordered lower residual classes, the corresponding prior exclusion removes it from the residual class. □

6.4. **Stationary-hull profiles which occur as Seregin limits.** The preceding lemma removes the possible internal dynamics along a stationary family: exact stationary-family motion is stationary by Lemma 6.8. The remaining question is attainability by blow-up: whether a degenerate stationary-family profile can occur as a Seregin limit of a finite-energy suitable weak solution while retaining local velocity concentration. The following statement gives the precise Seregin-limit occurrence exclusion required to close the degenerate stationary-hull class with retained local velocity concentration. Its proof is deferred until the terminal stratification assembly theorem has been established.

**Proposition 6.10** (R3 degenerate stationary-family reduction target)**.** *Fix an admissible stationary phase space $\mathfrak{B}$. Let $(W, \Pi)$ be a stationary centered profile satisfying:*

*(i) $W \in \mathfrak{B}$, $\mathcal{F}(W) = 0$, and $W$ lies on an admissible nontrivial stationary family;*
*(ii) $(W, \Pi)$ is a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;*

*(iii)* $(W,\Pi)$ *is retained-stationary concentrating in the sense of Definition 6.3;*
*(iv)* $(W,\Pi)$ *is not in the small, stationary* $L^3$*, tight, closed structured/decay, axisymmetric bounded-circulation, or rotational relative-equilibrium classes already removed earlier in the stratification.*

*Then no such* $(W,\Pi)$ *exists after the terminal stratification assembly is inserted.*

*Proof deferred.* The proof is given after the terminal stratification assembly theorem in Corollary 10.28. The present statement records the exact R3 occurrence input that must be proved; no R3 exclusion is used before the terminal assembly theorem is available. □

*Remark* 6.11 (Interpretation of the Seregin limit occurrence theorem)*.* Proposition 6.10 excludes occurrence as a Type I Seregin limit. It does not assert that degenerate stationary profiles cannot exist as elliptic solutions of the stationary centered equation. It asserts only that such profiles cannot occur as Type I blow-up limits with retained local velocity concentration after all previously closed classes have been removed. This is the formulation required for the residual problem: the elliptic family may exist, but it cannot be reached by the local blow-up procedure coming from a finite-energy suitable weak solution.

### 6.5. **Reduction of the degenerate stationary-hull concentration class.**

**Proposition 6.12** (R3 stationary-hull reduction)**.** *Every retained stationary-hull residual profile either belongs to one of the stationary, tight, or structured classes already excluded by the setup paper, or it produces a stationary terminal concentration profile realized from the original blow-up sequence and satisfying the hypotheses of the terminal stratification assembly theorem.*

*Proof.* Let $V \in \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}(\mathcal{S}; I, J; \mathfrak{B})$. By Definition 6.4, there is a stationary profile $(W,\Pi)$ in the centered time-translation hull of $(V,P)$, and $(W,\Pi)$ is retained-stationary concentrating. The profile $W$ belongs to $\mathfrak{B}$, satisfies $\mathcal{F}(W) = 0$, and lies on an admissible nontrivial stationary family. Moreover, by the ordered definition of $\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$, it is not in any of the previously closed lower classes.

By Lemma 6.6, the hull profile $(W,\Pi)$ is itself a centered Type I Seregin limit of the original finite-energy suitable weak solution at the same singular point, after reselecting scales. Since local velocity concentration is retained, $(W,\Pi)$ satisfies every condition in Proposition 6.10, except that the actual contradiction is deferred to the terminal stratification assembly theorem. Thus the R3 branch has been reduced to a stationary terminal concentration profile realized from the original blow-up sequence; no R3 closure is used in this reduction step. □

**Proposition 6.13** (Already-closed integrable stationary-family subcase)**.** *Let* $(W,\Pi)$ *be a stationary centered profile lying on an admissible stationary family. If*

$$W \in L^3(\mathbb{R}^3), \tag{6.10}$$

*then* $W \equiv 0$*. Hence no retained-velocity hull profile satisfying* (6.10) *can occur in* $\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$*.*

*Proof.* The stationary physical representative associated with $W$ is a backward self-similar Navier–Stokes solution. The assumption $W \in L^3(\mathbb{R}^3)$ places it in the stationary $L^3$ class already excluded by the classical Nečas–Růžička–Šverák Liouville theorem [7], equivalently by the endpoint ancient Liouville theorem already recorded in the setup paper and cited below as Theorem 10.9. Hence $W \equiv 0$. A zero profile has zero local velocity concentration, so it cannot be retained-stationary concentrating. □

*Remark* 6.14 (Closure after terminal stratification assembly)*.* The preceding argument reduces the stationary-hull class with retained local velocity concentration from a time-translation hull question to the question whether a stationary centered profile in a nontrivial stationary family can occur as a Seregin limit. Proposition 6.10 is supplied only after the terminal stratification assembly theorem has been proved.

*Remark* 6.15 (Residual decomposition statement). For the residual decomposition, the third line is reduced as follows and closed only after the terminal stratification assembly theorem:

$$\begin{aligned}&\text{stationary-hull residual with retained velocity concentration}\\ &\xrightarrow{\text{stationary-family Seregin limit reduction}} \text{terminal stationary concentration profile}\\ &\xrightarrow{\text{terminal closure}} \varnothing.\end{aligned}$$

The precise class is the *degenerate stationary-hull residual class with retained local velocity concentration*.

## 7. Reduction of degenerate stationary-hull profiles with retained velocity concentration

We record the terminal concentration input needed for the degenerate stationary-hull class. The actual closure is deferred until after the terminal stratification theorem. The argument is not specific to the existence of a non-isolated stationary family. Once a stationary-family element occurs as a Type I Seregin limit carrying retained local velocity concentration, it is a stationary terminal concentration profile. Such profiles are later excluded by the same terminal stratification assembly and separated-profile exclusion used for the final residual class. No global coercive estimate for the original solution is used.

Throughout this section, a centered Type I profile satisfies

$$\partial_\tau U - \Delta U + \frac{1}{2}y \cdot \nabla U + \frac{1}{2}U + (U \cdot \nabla)U + \nabla \Pi = 0, \qquad \nabla \cdot U = 0 \tag{7.1}$$

on $\mathbb{R}^3 \times \mathbb{R}$, with the usual local suitability and pressure-gauge conventions.

### 7.1. Deferred terminal stratification corollary for R3.

We record the following specialization of the terminal stratification assembly theorem. It is stated here exactly in the form needed for the stationary-hull closure with retained local velocity concentration as a *reduction target*; the substantive proof is the terminal stratification proof later in the paper. Following the recommendation that only one actual R3 closure theorem appear after terminal assembly, this statement is recorded as a proposition rather than a theorem.

**Proposition 7.1** (R3 reduction target). *Let $(U, \Pi)$ be a bounded centered Type I Seregin profile satisfying local suitability and carrying a retained local velocity concentration, i.e. for some admissible recentering/rescaling $\mathscr{T}_{z_0,r}$,*

$$\mathcal{V}_1\big(\mathscr{T}_{z_0,r}U; 0\big) \geq \eta_*. \tag{7.2}$$

*Assume that $(U, \Pi)$ is outside the previously closed small-amplitude, stationary $L^3$, tight, closed structured/decay, axisymmetric bounded-circulation, and rotational relative-equilibrium classes. Then the following hold.*

- *(i) Every terminal extraction of $(U, \Pi)$ has the paired decomposition of Theorem 9.24: a retained velocity concentration set together with a residual remainder which is locally vanishing, diffuse, or noncompact after neighborhoods of the concentration set are removed.*
- *(ii) Local vanishing, exterior escape, diffuse exterior concentration, and loss of local compactness cannot be the sole locations of the retained compact velocity concentration (7.2).*
- *(iii) Mixed compact–diffuse and compact–noncompact residual profiles, and infinite chains of retained concentrating descendants, are impossible in the refined residual class.*
- *(iv) Finite separated families of retained velocity concentration profiles are impossible.*

(v) *Any remaining single retained velocity concentration profile is terminally indecomposable: it has no retained concentrating tail limit, no diffuse exterior concentration, and no loss of local compactness along escaping recenterings.*

(vi) *Every retained terminally indecomposable concentration profile has a backward sequence of finite critical norm: there exist $\tau_k \to -\infty$ such that*

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty. \tag{7.3}$$

(vii) *Retained terminal concentration profiles are either affine/parasitic lower-stratum profiles or have parasitic-free finite-shift mild physical pullbacks. Therefore the Albritton–Barker endpoint Liouville theorem applies to any normalized residual profile satisfying (7.3).*

*Proof deferred.* This is the R3 specialization of Theorem 10.25. It is stated here only as a reduction target for the stationary-hull section; its proof is the terminal stratification proof in Sections 9 and 10. □

*Remark* 7.2 (Local nature of the theorem). The theorem above is a local velocity concentration statement. It uses the local suitable-weak-solution compactness and pressure normalization developed from the Caffarelli–Kohn–Nirenberg theory [2] together with the imported setup results: local positive concentration and Type I Seregin compactness. It then uses paired terminal profile decomposition, sole-source exclusions, critical-tail compactification with realized critical-tail rigidity, recurrence for concentration chains, and the finite separated-profile theorem. It does not assert or require a global a priori bound for the original physical solution.

For reference, we record the stationary specialization of the atomic sequence-$L^3$ lemma used after the terminal stratification assembly theorem.

**Lemma 7.3** (Indecomposable local profiles have backward sequence-$L^3$ control)**.** *Let $(U, \Pi)$ be a bounded smooth ancient solution of (7.1) which belongs to the normalized parasitic-free compact state class $\mathfrak{X}^{\#}_M$ imported from the setup construction. Suppose that $(U, \Pi)$ has retained local velocity concentration and is terminally indecomposable: it has no retained concentrating tail limit, no diffuse exterior concentration in the parabolic CKN sense, and no escaping recentering sequence along which local compactness, suitability, or pressure control fails. Then there exists a sequence $\tau_k \to -\infty$ such that*

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty. \tag{7.4}$$

*Proof.* This is the stationary-profile specialization of Lemma 10.6. The hypotheses listed here are the terminal indecomposability hypotheses used there: retained unit velocity concentration, no retained concentrating tail limit, no diffuse exterior concentration, and no escaping recentering with local compactness or pressure-gauge failure. Applying Lemma 10.6 gives (7.4). No separate exterior activity level is introduced in this stationary specialization; the only exterior activity selection is the velocity quantity $m_n^{\mathrm{vel}}$ in Lemma 10.6, and pressure-inclusive CKN activity is not used to certify nontriviality of a tail profile. □

### 7.2. **Deferred stationary concentration statement.**

*Remark* 7.4 (Deferred R3 stationary consequence). Let $(W, \Pi)$ be a stationary solution of the centered equation,

$$-\Delta W + \frac{1}{2} y \cdot \nabla W + \frac{1}{2} W + (W \cdot \nabla)W + \nabla \Pi = 0, \qquad \nabla \cdot W = 0. \tag{7.5}$$

Assume that:

(i) $(W, \Pi)$ is obtained as a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
(ii) $(W, \Pi)$ has retained local velocity concentration in the sense of (7.2);
(iii) $(W, \Pi)$ is outside all previously closed lower classes listed in Proposition 7.1.

Then no such $(W, \Pi)$ exists. This statement is the stationary case of the R3 closure after Theorem 10.25; it follows from Corollary 10.28 and is recorded here only to name the exact stationary concentration statement used by the stationary-hull reduction.

### 7.3. **Deferred degenerate stationary-hull closure statement.**

*Remark* 7.5 (Deferred R3 degenerate-family consequence). Fix an admissible stationary phase space $\mathfrak{B}$. There is no stationary centered profile $(W, \Pi)$ satisfying all of the following conditions:

(i) $W \in \mathfrak{B}$, $W$ solves the stationary centered equation (7.5), and $W$ lies on an admissible nontrivial stationary family;
(ii) $(W, \Pi)$ is a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
(iii) $(W, \Pi)$ has retained local velocity concentration;
(iv) $(W, \Pi)$ is outside the previously closed lower classes.

This statement is recorded as a deferred R3 consequence; it follows from the post-terminal R3 closure Corollary 10.28. The stationary-family condition in (i) is additional structure; once the terminal stratification assembly theorem is available, the stationary concentration argument applies to these profiles in particular.

**Proposition 7.6** (Stationary-hull reduction to terminal stratification)**.** *Let $\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}(\mathcal{S}; I, J; \mathfrak{B})$ be the degenerate stationary-hull residual class with retained local velocity concentration. Every element of this class produces a stationary terminal concentration profile realized from the original blow-up sequence and satisfying the hypotheses of Theorem 10.25.*

*Proof.* Let $V \in \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}(\mathcal{S}; I, J; \mathfrak{B})$. By the definition of the degenerate stationary-hull residual class with retained local velocity concentration, the centered translation hull of $(V, P)$ contains a stationary profile $(W, \Pi)$ with retained local velocity concentration which lies in an admissible nontrivial stationary family and is outside the previously closed lower classes. The scale-reselection lemma for centered hull limits shows that $(W, \Pi)$ is itself a centered Type I Seregin limit of the original finite-energy suitable weak solution at the same singular point.

Thus $(W, \Pi)$ satisfies the hypotheses of the terminal stratification assembly theorem in the stationary case. The actual emptiness conclusion is recorded after that terminal stratification assembly theorem as Corollary 10.28. □

*Remark* 7.7 (How this enters the residual decomposition). The third residual line is reduced as follows and closed only after the terminal stratification theorem:

$$\begin{gathered}\text{degenerate stationary-hull residual with retained velocity concentration}\\ \Longrightarrow \text{stationary profile with retained velocity concentration} \Longrightarrow \varnothing.\end{gathered}$$

The second implication is proved by the terminal stratification assembly: decomposable profiles produce either an excluded separated concentration family, diffuse exterior concentration, or loss of local compactness; indecomposable profiles have backward sequence-$L^3$ control and are ruled out by the endpoint ancient Liouville theorem.

## 8. Tail geometry of the generic terminal residual

We now pass to the generic terminal residual class. The preliminary point is that the centered equation is not invariant under ordinary fixed spatial translations. The correct far-field recenterings are self-similar covariant translations.

**Definition 8.1** (Covariant translations)**.** For $a \in \mathbb{R}^3$, define

$$(T_aV)(y,\tau) := V(y + e^{\tau/2}a, \tau), \qquad (T_aP)(y,\tau) := P(y + e^{\tau/2}a, \tau).$$

**Lemma 8.2** (Covariant translation invariance)**.** *If $(V, P)$ solves (1.2), then $(T_aV, T_aP)$ also solves (1.2).*

*Proof.* Set $z = y + e^{\tau/2}a$. Then

$$\partial_\tau(T_aV)(y,\tau) = (\partial_\tau V)(z,\tau) + \frac{1}{2}e^{\tau/2}a \cdot \nabla V(z,\tau).$$

Moreover,

$$\frac{1}{2}y \cdot \nabla(T_aV)(y,\tau) = \frac{1}{2}z \cdot \nabla V(z,\tau) - \frac{1}{2}e^{\tau/2}a \cdot \nabla V(z,\tau).$$

The two additional transport terms cancel in $\partial_\tau T_aV + \frac{1}{2}y \cdot \nabla T_aV$. The Laplacian, the zeroth-order term, the convection term, the pressure gradient, and the divergence constraint are translation covariant. Substitution into (1.2) proves the claim. □

*Remark* 8.3 (Fixed translates create a spurious drift)*.* If one sets $V_n(y,\tau) = V(y + x_n, \tau)$ with fixed $x_n$, then the centered equation acquires the additional transport term

$$-\frac{1}{2}x_n \cdot \nabla V_n.$$

Thus fixed translates do not define a compact invariant far-field hull for the renormalized dynamics.

**Definition 8.4** (Spacetime covariant action)**.** For $s \in \mathbb{R}$, define the centered time translation

$$(\Theta_sV)(y,\tau) := V(y, \tau + s), \qquad (\Theta_sP)(y,\tau) := P(y, \tau + s).$$

For a tail center $x \in \mathbb{R}^3$ observed at centered time $\sigma$, define the covariant tail recentering

$$\mathscr{T}_{(x,\sigma)}V(y,s) := V(y + e^{s/2}x, \sigma + s), \qquad \mathscr{T}_{(x,\sigma)}P(y,s) := P(y + e^{s/2}x, \sigma + s).$$

Equivalently,

$$\mathscr{T}_{(x,\sigma)} = \Theta_\sigma T_{e^{-\sigma/2}x}.$$

Thus $\mathscr{T}_{(x,\sigma)}$ is a composition of exact symmetries of the centered equation. Ordinary parabolic recenterings are used below only for local terminal extraction before one has passed to a centered ancient profile; all escaping tail limits of centered profiles use $\mathscr{T}_{(x,\sigma)}$.

**Lemma 8.5** (Semidirect covariance of the centered action)**.** *The centered time translations and spatial covariant translations satisfy*

$$\Theta_s T_a \Theta_{-s} = T_{e^{s/2}a}, \qquad s \in \mathbb{R}, \ a \in \mathbb{R}^3.$$

*Equivalently, if $G = \mathbb{R}^3 \rtimes \mathbb{R}$ acts by*

$$\mathscr{G}_{(a,s)} := \Theta_s T_a,$$

*then the group law in this convention is*

$$(a,s) \cdot (b,t) = (e^{-t/2}a + b, \ s + t), \qquad \mathscr{G}_{(a,s)}\mathscr{G}_{(b,t)} = \mathscr{G}_{(a,s)\cdot(b,t)}. \tag{8.1}$$

*In particular the subgroup $\{T_a : a \in \mathbb{R}^3\}$ is normal in the full spacetime covariant action, and the subspace of observables invariant under all spatial covariant translations is invariant under every $\Theta_s$.*

*Proof.* For a profile $V$,

$$(\Theta_s T_a \Theta_{-s} V)(y,\tau) = V(y + e^{(\tau+s)/2}a, \tau) = (T_{e^{s/2}a}V)(y,\tau).$$

The pressure transforms identically. The asserted invariance of the spatially invariant observable subspace is the same identity written on the induced $L^2$-representation. The group law (8.1) follows from

$$\Theta_s T_a \Theta_t T_b = \Theta_{s+t} T_{e^{-t/2}a+b}.$$

□

**Theorem 8.6** (Covariant observer calculus). *Let $Z := \mathbb{R}^3 \times \mathbb{R}$ denote the centered observer space, with points $z = (\xi, \sigma)$. For $a \in \mathbb{R}^3$ and $\sigma \in \mathbb{R}$,*

$$(T_a U)(y,\tau) = U(y + e^{\tau/2}a, \tau), \qquad (\Theta_\sigma U)(y,\tau) = U(y, \tau+\sigma).$$

*Then*

$$\Theta_\sigma T_a = T_{e^{\sigma/2}a}\Theta_\sigma, \qquad \Theta_\sigma T_a \Theta_{-\sigma} = T_{e^{\sigma/2}a}.$$

*For $z = (\xi,\sigma)$, define the covariant observer*

$$\mathcal{R}_z U := \Theta_\sigma T_{e^{-\sigma/2}\xi} U.$$

*Equivalently,*

$$(\mathcal{R}_{(\xi,\sigma)}U)(y,s) = U(y + e^{s/2}\xi, \sigma + s).$$

*The covariant tube pulled back from $Q_r(0,0)$ is*

$$\mathcal{Q}_r^{\mathrm{cov}}(\xi,\sigma) := \left\{ (Y,\tau) : |\tau - \sigma| < r^2,\ |Y - e^{(\tau-\sigma)/2}\xi| < r \right\}.$$

*The map*

$$(y,s) \mapsto (Y,\tau) = (y + e^{s/2}\xi, \sigma + s)$$

*is a bijection from $Q_r(0,0)$ onto $\mathcal{Q}_r^{\mathrm{cov}}(\xi,\sigma)$ with spacetime Jacobian one. The same formulas apply to the pressure, modulo the admissible time-dependent gauges.*

*Proof.* The semidirect identities are Lemma 8.5; written without conjugation,

$$(\Theta_\sigma T_a U)(y,\tau) = U(y + e^{(\tau+\sigma)/2}a, \tau+\sigma) = (T_{e^{\sigma/2}a}\Theta_\sigma U)(y,\tau).$$

Taking $a = e^{-\sigma/2}\xi$ gives

$$(\Theta_\sigma T_{e^{-\sigma/2}\xi}U)(y,s) = U(y + e^{(s+\sigma)/2}e^{-\sigma/2}\xi, \sigma+s) = U(y + e^{s/2}\xi, \sigma+s).$$

If $Y = y + e^{s/2}\xi$ and $\tau = \sigma + s$, then

$$y = Y - e^{(\tau-\sigma)/2}\xi, \qquad s = \tau - \sigma,$$

so $|s| < r^2$ and $|y| < r$ are equivalent to the defining inequalities for $\mathcal{Q}_r^{\mathrm{cov}}(\xi,\sigma)$. The transformation is a time-dependent spatial translation, hence has Jacobian one. □

**Definition 8.7** (Terminal profile class and local topology). For $M < \infty$, let $\mathfrak{X}_M$ be the class of pairs $(U,\Pi)$, modulo addition of arbitrary functions of time to $\Pi$, such that:

(i) $U$ is a smooth bounded ancient solution of (1.2) on $\mathbb{R}^3 \times \mathbb{R}$, with $\|U\|_{L^\infty} \le M$;
(ii) $(U,\Pi)$ is locally suitable on every compact terminal cylinder;
(iii) on every compact cylinder the pressure has an admissible $L^{3/2}$-representative after subtracting a time-dependent gauge.

The topology on $\mathfrak{X}_M$ is

$$U_n \to U \quad \text{in } C^\infty_{\mathrm{loc}}, \qquad \Pi_n - c_n(\tau) \rightharpoonup \Pi \quad \text{in } L^{3/2}_{\mathrm{loc}}$$

for suitable time-dependent gauges $c_n$. All profile limits, tail descendants, concentrating descendant hulls, and recurrent-core limits below are taken in this topology.

**Definition 8.8** (Observer oscillation and spatial oscillation)**.** Let $Q = B \times I$ be a bounded covariant terminal cylinder. For a velocity field $U$, define the constant-in-time oscillation functional

$$\mathrm{osc}_{3,\mathrm{ct}}(U;Q) := \inf_{c\in\mathbb{R}^3} \iint_Q |U-c|^3 \, dy\, d\tau.$$

Define also the spatial oscillation modulo time-dependent homogeneous gauges

$$\mathrm{osc}_{3,\mathrm{sp}}(U;Q) := \inf_{c\in L^3(I;\mathbb{R}^3)} \iint_Q |U(y,\tau)-c(\tau)|^3 \, dy\, d\tau.$$

Thus $\mathrm{osc}_{3,\mathrm{ct}}$ detects nonconstant time-dependent homogeneous modes, while $\mathrm{osc}_{3,\mathrm{sp}}$ is the quotient seminorm used to remove spatially homogeneous parasitic modes. A covariant observer $z \in Z$ is $\eta$-oscillatory for $U$ if

$$\mathrm{osc}_{3,\mathrm{ct}}\big(U; \mathcal{Q}_1^{\mathrm{cov}}(z)\big) \ge \eta.$$

On each compact observer window we fix a uniformly locally finite family of observer centers $\{z_k\} \subset Z$, subordinate to a bounded-overlap covariant unit-cylinder cover. The corresponding local oscillation measure on observer-center sets $E \subset Z$ is

$$\mu_U^{\mathrm{osc}}(E) := \sum_{z_k\in E} \mathrm{osc}_{3,\mathrm{ct}}\big(U; \mathcal{Q}_1^{\mathrm{cov}}(z_k)\big).$$

For a sequence $U_n$ we write $\mu_n^{\mathrm{osc}}$ for this measure. All uses of $\mu_n^{\mathrm{osc}}$ are on compact observer-center windows or on specified escaping windows after restriction, so the bounded overlap gives uniform local finiteness. Replacing the cover by another bounded-overlap covariant unit cover changes only harmless constants and never changes the alternatives “positive threshold,” “vanishing threshold,” or “diffuse defect.” The concentration-set extraction uses $\mathrm{osc}_{3,\mathrm{ct}}$, because after parasitic-free normalization it detects genuine non-affine oscillation. Mesoscopic cylinders use $\mathrm{osc}_{3,\mathrm{sp}}$, equivalently the functional $\mathcal{A}_n(z,r)$ of Definition 9.57, because the minimal-scale quotient must ignore time-dependent homogeneous gauges. Thus the concentration-set construction and the mesoscopic affine quotient are kept distinct.

**Lemma 8.9** (Local compactness and continuity of the profile transformations)**.** *Let $(U_n,\Pi_n) \subset \mathfrak{X}_M$ be a sequence with the local suitability and pressure-gauge bounds supplied by the setup paper on every fixed compact cylinder. Then every subsequence has a further subsequence converging in the topology of $\mathfrak{X}_M$. Moreover the actions $T_a$, $\Theta_s$, the covariant tail recenterings $\mathscr{T}_{(x,\sigma)}$ are continuous on $\mathfrak{X}_M$. The affine maps satisfy the bound-tracking continuity*

$$S_c : \mathfrak{X}_M \longrightarrow \mathfrak{X}_{M+|c|},$$

*again with the pressure interpreted modulo the corresponding transformed time gauges.*

*Proof.* On each compact cylinder $K$, the setup local suitability, dissipation, strong local $L^3$-velocity compactness, and pressure-gauge compactness give a Seregin subsequential limit in the local topology of $\mathfrak{X}_M$. Diagonalizing over the countable compact exhaustion gives the asserted subsequence. With the setup pressure-atlas bounds and local suitability fixed, interior parabolic estimates for (1.2) upgrade the velocity convergence to $C^\infty_{\mathrm{loc}}$; the pressure representatives converge weakly in $L^{3/2}_{\mathrm{loc}}$ after the local gauges. The formulas defining $T_a$, $\Theta_s$, $\mathscr{T}_{(x,\sigma)}$, and $S_c$ are smooth changes of variables and affine pressure changes on each fixed compact cylinder. The first three preserve the $L^\infty$ bound $M$. The affine map $S_c$ subtracts the constant vector $c$, so its velocity bound is at most $M + |c|$. All four maps preserve the local topology and transform admissible gauges into admissible gauges. No global spatial norm or global pressure representative is used. $\square$

**Definition 8.10** (Compact normalized descendant profile class)**.** Let $\mathfrak{X}_M^\# \subset \mathfrak{X}_M$ denote the sequential closure, in the local topology of $\mathfrak{X}_M$, of all normalized Seregin profiles from the setup paper and all profiles obtained from them by the limiting operations used in this paper: centered time translations, covariant translations, covariant tail recenterings, terminal profile limits, concentrating descendant limits, recurrent-core limits, and the canonical affine normalization whenever the affine parameter is bounded. The closure is always taken with the local pressure gauges supplied by the normalized Seregin compactness theorem.

**Lemma 8.11** (Compactness and continuity on the normalized profile class)**.** *The space $\mathfrak{X}_M^\#$ is compact in the local topology inherited from $\mathfrak{X}_M$. The maps $T_a$, $\Theta_s$, and the covariant recenterings used to form descendants act continuously on $\mathfrak{X}_M^\#$. The affine map satisfies the bound-tracking continuity*

$$S_c : \mathfrak{X}_M^\# \longrightarrow \mathfrak{X}_{M+|c|}^\#.$$

*Proof.* By definition, every sequence in $\mathfrak{X}_M^\#$ is a diagonal limit of normalized Seregin descendants with the same local $L^\infty$, local suitability, pressure-gauge, and interior smooth bounds on each fixed compact cylinder. Applying Lemma 8.9 on an exhaustion of compact cylinders gives a further subsequential limit in $\mathfrak{X}_M$. The closure defining $\mathfrak{X}_M^\#$ contains that limit, so $\mathfrak{X}_M^\#$ is sequentially compact; the local topology is metrizable on bounded sets by the standard countable exhaustion, hence compact.

Continuity of the limiting operations is the continuity already proved in Lemma 8.9, restricted to the closed normalized profile class. The affine map changes only the velocity bound and the explicit affine pressure gauge, so it maps $\mathfrak{X}_M^\#$ continuously into $\mathfrak{X}_{M+|c|}^\#$. $\square$

**Definition 8.12** (Affine homogeneous symmetry)**.** For $c \in \mathbb{R}^3$, define

$$(S_cV)(y,\tau) := V(y-2c,\tau) - c, \qquad (S_cP)(y,\tau) := P(y-2c,\tau) + \frac{1}{2}c\cdot y.$$

**Lemma 8.13** (Affine renormalized Galilean symmetry)**.** *If $(V,P)$ solves (1.2), then $(S_cV, S_cP)$ also solves (1.2).*

*Proof.* Set $U(y,\tau) = V(y-2c,\tau) - c$, $\Pi(y,\tau) = P(y-2c,\tau) + \frac{1}{2}c\cdot y$, and $z = y - 2c$. Then

$$\partial_\tau U = (\partial_\tau V)(z,\tau), \qquad \Delta U = (\Delta V)(z,\tau), \qquad \nabla U = (\nabla V)(z,\tau).$$

The nonlinear term is

$$(U\cdot\nabla)U = (V-c)\cdot\nabla V = (V\cdot\nabla)V - c\cdot\nabla V,$$

where all quantities on the right are evaluated at $(z,\tau)$. The drift term satisfies

$$\frac{1}{2}y\cdot\nabla U = \frac{1}{2}(z+2c)\cdot\nabla V = \frac{1}{2}z\cdot\nabla V + c\cdot\nabla V.$$

The two $c\cdot\nabla V$ terms cancel. Finally,

$$\frac{1}{2}U + \nabla\Pi = \frac{1}{2}(V-c) + \nabla P + \frac{1}{2}c = \frac{1}{2}V + \nabla P.$$

Substitution into (1.2) at the point $(z,\tau)$ gives the equation for $(U,\Pi)$. The divergence condition is unchanged by translation and subtracting a constant vector. $\square$

**Definition 8.14** (Pressure gauges and affine quotient)**.** On every ordinary local cylinder $Q$, pressure representatives are identified modulo time functions:

$$P_Q \sim P_Q + c_Q(\tau).$$

At the affine/parasitic quotient level, and only there, one also identifies pressure representatives modulo the affine spatial functions paired with spatially homogeneous parasitic velocities:

$$P_Q \sim P_Q + b_Q(\tau) \cdot y + c_Q(\tau).$$

The affine term is allowed exactly when it corresponds to a removable homogeneous mode in the velocity equation, as in Definition 8.16. Generic terminal residual branches are always considered after this quotient has either been fixed by the canonical parasitic-free representative or assigned to $\mathcal{C}_{\mathrm{aff}}$.

**Lemma 8.15** (Pressure atlas compatibility)**.** *Let $Q, Q'$ be overlapping local cylinders in a normalized Seregin extraction. Before affine quotienting, admissible pressure representatives satisfy*

$$P_Q - P_{Q'} = c_{Q,Q'}(\tau)$$

*on $Q \cap Q'$. At the affine/parasitic quotient level they satisfy*

$$P_Q - P_{Q'} = b_{Q,Q'}(\tau) \cdot y + c_{Q,Q'}(\tau)$$

*with the affine term paired with the corresponding removable homogeneous velocity mode. The velocity equation is unchanged after the time-gauge identification, and after the affine identification it is interpreted in the quotient equation with the parasitic correction removed or assigned to $\mathcal{C}_{\mathrm{aff}}$.*

*Proof.* The time-gauge statement is the setup pressure atlas compatibility imported in Theorem 1.3. The affine statement follows by substituting a spatially homogeneous velocity $b(\tau)$ into the centered equation: the only pressure gradient it can create is $-(\partial_\tau b + \frac{1}{2}b)$, hence an affine pressure in $y$, modulo a time function. This is the quotient convention in Definition 8.14. All terminal limiting operations below choose their pressure gauges before taking compactness limits, and if the affine term is nontrivial the profile is assigned to $\mathcal{C}_{\mathrm{aff}}$. $\square$

**Definition 8.16** (Affine/parasitic lower stratum and residual normalization)**.** The affine/parasitic lower stratum $\mathcal{C}_{\mathrm{aff}}$ consists of the spatially homogeneous modes which are invisible to the spatial quotient seminorm $\mathrm{osc}_{3,\mathrm{sp}}$. It has two equivalent realizations.

(i) A genuine centered profile belongs to $\mathcal{C}_{\mathrm{aff}}$ if, after one affine homogeneous symmetry $S_c$, its velocity is spatially constant. In the parasitic-free representative this is the zero profile.

(ii) A quotient limit belongs to $\mathcal{C}_{\mathrm{aff}}$ if it is represented by a locally bounded spatially homogeneous velocity $b(\tau)$. In the weak centered equation this mode is paired with the affine pressure functional

$$\big(-\partial_\tau b(\tau) - \tfrac{1}{2}b(\tau)\big) \cdot y$$

in $\mathcal{D}'_\tau \mathcal{D}'_y$. If the mode is realized as a smooth suitable profile, this affine pressure is an ordinary local pressure; if it arises only as a measurable homogeneous gauge $c_n(\tau)$, it is used only in the quotient weak formulation and never as an independent $L^{3/2}_{\mathrm{loc}}$ pressure.

In the ordered residual decomposition this stratum is part of the previously closed lower classes, not part of $\mathcal{C}_{\mathrm{gen}}$. Equivalently, every generic residual profile is considered after the canonical affine normalization which removes its spatially homogeneous mild mode. Retained non-affine oscillation in the residual class is therefore always measured after this normalization.

*Remark* 8.17 (How affine constants are used). When a compact recurrent tail core is shown below to consist only of spatially homogeneous parasitic modes, the contradiction is not that an unnormalized nonzero constant has zero local $L^3$-mass. The contradiction is that these modes have already been assigned to the closed lower stratum in Definition 8.16; after the canonical affine normalization they are the zero profile and cannot represent a generic residual profile carrying retained velocity concentration or non-affine oscillation.

**Lemma 8.18** (Homogeneous gauges generate only the affine/parasitic stratum). *Let $c_n : I \to \mathbb{R}^3$ be bounded in $L^\infty(I)$, and let*

$$C_n(y,\tau) := c_n(\tau).$$

*Every local weak limit, time translate, or normalized noncompactness limit of $(C_n)$ has vanishing spatial quotient oscillation on every compact cylinder and belongs to $\mathcal{C}_{\mathrm{aff}}$. Conversely, after the canonical parasitic-free normalization, no nonzero spatially homogeneous mode remains in the residual profile class.*

*Proof.* For every compact cylinder $Q$,

$$\inf_{c(\tau)} \iint_Q |C_n(y,\tau) - c(\tau)|^3 \, dy \, d\tau = 0,$$

with the choice $c(\tau) = c_n(\tau)$. The same identity is stable under local weak limits and time recenterings. Hence every such limit has zero spatial quotient oscillation on every compact cylinder and is, by Definition 8.16, an affine/parasitic quotient mode. If a homogeneous mode is realized as a smooth centered solution, substituting $U = b(\tau)$ into (1.2) gives

$$\partial_\tau b + \frac{1}{2} b + \nabla P = 0,$$

so the pressure is affine in $y$ up to a time gauge, exactly as in the definition of $\mathcal{C}_{\mathrm{aff}}$. If $b$ is only a bounded measurable gauge, the same identity is understood distributionally and only in the quotient formulation. The residual representative is defined after removal of this quotient direction; therefore a nonzero retained effect of $b$ is assigned to the closed lower stratum, while the parasitic-free residual profile has no such mode. $\square$

**Theorem 8.19** (Closed non-affine oscillation functional). *Let $Q \Subset \mathbb{R}^3 \times \mathbb{R}$ be a bounded connected parabolic cylinder. For $U \in \mathfrak{X}_M$, define*

$$d_Q(U) := \mathrm{osc}_{3,\mathrm{ct}}(U;Q)^{1/3}.$$

*Then*

$$|d_Q(U) - d_Q(V)| \le \|U - V\|_{L^3(Q)}.$$

*In particular $d_Q$ is continuous under the local topology of $\mathfrak{X}_M$. Moreover,*

$$d_Q(U) = 0 \quad \Longleftrightarrow \quad U \text{ is spatially and temporally constant on } Q.$$

*Fix once and for all a nested exhaustion*

$$Q_1 \Subset Q_2 \Subset \cdots, \qquad \bigcup_{j=1}^\infty Q_j = \mathbb{R}^3 \times \mathbb{R},$$

*by bounded connected parabolic cylinders. Define*

$$\Phi(U) := \sum_{j=1}^\infty 2^{-j} \min\{1, d_{Q_j}(U)\}.$$

*Then $\Phi$ is continuous on $\mathfrak{X}_M$, and*

$$\Phi(U) = 0 \quad \Longleftrightarrow \quad U \text{ is globally constant.}$$

*Consequently, for every $\eta > 0$, the non-affine oscillation set*

$$\mathfrak{A}_\eta := \{(U,\Pi) \in \mathfrak{X}_M : \Phi(U) \geq \eta\}$$

*is closed in the local profile topology.*

*Proof.* For every $c \in \mathbb{R}^3$, the triangle inequality in $L^3(Q)$ gives

$$\|U - c\|_{L^3(Q)} \leq \|U - V\|_{L^3(Q)} + \|V - c\|_{L^3(Q)}.$$

Taking the infimum over $c$ gives

$$d_Q(U) \leq \|U - V\|_{L^3(Q)} + d_Q(V).$$

Interchanging $U$ and $V$ proves the Lipschitz estimate. Local smooth convergence implies local $L^3$-convergence, so $d_Q$ is continuous in the topology of $\mathfrak{X}_M$.

If $d_Q(U) = 0$, choose $c_k \in \mathbb{R}^3$ such that $\|U - c_k\|_{L^3(Q)} \to 0$. The sequence $c_k$ is bounded; otherwise

$$\|U - c_k\|_{L^3(Q)} \geq |c_k|\,|Q|^{1/3} - \|U\|_{L^3(Q)} \to \infty,$$

which is impossible. Passing to a subsequence, $c_k \to c$, and hence $U = c$ almost everywhere on $Q$. Since $U$ is smooth, $U \equiv c$ pointwise on $Q$. The converse is immediate.

Each summand in the definition of $\Phi$ is continuous and bounded by $2^{-j}$. The Weierstrass test gives uniform convergence of the defining series, hence continuity of $\Phi$. If $\Phi(U) = 0$, then $d_{Q_j}(U) = 0$ for every $j$, so $U$ is constant on every $Q_j$. The cylinders are nested and overlap, so the constants agree and $U$ is globally constant. The converse is again immediate. Closedness of $\mathfrak{A}_\eta$ follows from continuity of $\Phi$. □

**Corollary 8.20** (Compact normalized non-affine oscillation sets)**.** *For every $\eta > 0$,*

$$\mathfrak{A}^\#_\eta := \{(U,\Pi) \in \mathfrak{X}^\#_M : \Phi(U) \geq \eta\}$$

*is compact in the normalized profile topology.*

*Proof.* By Lemma 8.11, $\mathfrak{X}^\#_M$ is compact. By Theorem 8.19, $\Phi$ is continuous in the local topology. Therefore $\mathfrak{A}^\#_\eta$ is a closed subset of a compact space. □

**Theorem 8.21** (Affine normalization dichotomy)**.** *Let $S_c$ be the affine homogeneous symmetry of Definition 8.12, and suppose the ordered lower strata include all constant centered profiles as in Definition 8.16. For every $(U,\Pi) \in \mathfrak{X}_M$, exactly one of the following alternatives holds:*

- (i) $\mathrm{osc}_{3,\mathrm{sp}}(U;Q) = 0$ *on every compact cylinder $Q$. Then $U$ is a spatially homogeneous quotient mode, and $(U,\Pi)$ belongs to the affine/parasitic lower stratum $\mathcal{C}_{\mathrm{aff}}$.*
- (ii) *For the parasitic-free representative, after an admissible affine symmetry $S_c$, one has $\Phi(S_cU) > 0$. Equivalently, for some compact cylinder $Q$,* $\mathrm{osc}_{3,\mathrm{ct}}(S_cU;Q) > 0$.

*In the retained normalized residual space, alternative* (i) *has already been removed by the canonical affine/parasitic normalization, so only alternative* (ii) *can remain. Moreover, if $K \subset \mathfrak{X}_M$ is compact and*

$$K \cap \{(U,\Pi) : \Phi(U) = 0\} = \varnothing,$$

*then there exists $\eta_K > 0$ such that*

$$\Phi(U) \geq \eta_K \qquad \textit{for every } (U,\Pi) \in K.$$

*Proof.* If $\mathrm{osc}_{3,\mathrm{sp}}(U;Q) = 0$ on every compact cylinder, then on each time slice $U$ is spatially constant in the distributional sense. The local constants are compatible on overlapping cylinders, so $U(y,\tau) = b(\tau)$ locally in time for a measurable bounded function $b$. By Definition 8.16, this is precisely the affine/parasitic quotient stratum; if it is realized by a smooth profile, the corresponding affine pressure is the one described there, and otherwise it is used only in the quotient weak formulation.

If the spatial oscillation does not vanish identically, $U$ is not a spatially homogeneous quotient mode. After choosing the canonical parasitic-free representative and applying any admissible affine symmetry $S_c$, the resulting velocity is not globally constant. Hence, by Theorem 8.19, $\Phi(S_cU) > 0$. Conversely, if $\Phi(S_cU) = 0$, then $S_cU$ is globally constant, and the original profile is in $\mathcal{C}_{\mathrm{aff}}$. This proves the dichotomy and the retained normalized statement. For the compact statement, $\Phi$ is continuous and strictly positive on $K$, so it attains a positive minimum

$$\eta_K := \min_{(U,\Pi)\in K} \Phi(U) > 0.$$

□

**Definition 8.22** (Escaping covariant hull)**.** Let $V$ be a bounded smooth ancient solution of (1.2). Define

$$\mathcal{H}_\infty(V) := \left\{W :\ \exists(a_n, s_n) \in G,\ |(a_n, s_n)|_{\mathrm{cov}} \to \infty, \mathscr{G}_{(a_n,s_n)}V \to W \text{ in } C^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R})\right\}.$$

Here $G = \mathbb{R}^3 \rtimes \mathbb{R}$ is the covariant spacetime group in Lemma 8.5, and any proper left-invariant metric may be used for $|(a, s)|_{\mathrm{cov}}$. Thus the recurrent tail hull is invariant under the full covariant action, not merely under ordinary spatial translations.

**Theorem 8.23** (Compactness of the normalized escaping covariant hull)**.** *Let $(V, P) \in \mathfrak{X}^\#_M$ be represented after canonical affine/parasitic normalization. Assume that every escaping covariant sequence for $(V, P)$ has a compact subsequence in the normalized local topology with the setup pressure atlas fixed; if this assumption fails, the sequence is assigned instead to the local-compactness or pressure-gauge-loss terminal alternative. In the compactness branch, $\mathcal{H}_\infty(V)$ is nonempty and compact in $C^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R})$. Every $W \in \mathcal{H}_\infty(V)$ is a bounded smooth ancient solution of (1.2) with $\|W\|_{L^\infty} \le M$.*

*Proof.* For any escaping covariant sequence $(a_n, s_n)$, the transformed solutions $\mathscr{G}_{(a_n,s_n)}V$ solve the same equation and have the same $L^\infty$ bound by Lemmas 8.2 and 8.5. The $L^\infty$ bound alone is not used to obtain a time modulus or derivative bound. By the compactness-branch assumption, the transformed profiles have a subsequence converging in the normalized pressure-atlas topology, hence locally smoothly in velocity. Passing to the limit in the equation gives another bounded ancient solution. Nonemptiness follows by applying this compactness branch to any escaping sequence in $G$.

It remains to record closedness of the escaping hull. Suppose

$$W_j \in \mathcal{H}_\infty(V), \qquad W_j \to W \quad \text{in } C^\infty_{\mathrm{loc}}.$$

For each $j$, choose an escaping sequence $g_{j,k} \in G$ with $\mathscr{G}_{g_{j,k}}V \to W_j$. Choose $k(j)$ so large that

$$|g_{j,k(j)}|_{\mathrm{cov}} \ge j, \qquad d_j\left(\mathscr{G}_{g_{j,k(j)}}V, W_j\right) \le j^{-1},$$

where $d_j$ is a metric for $C^j$-convergence on $\overline{B_j} \times [-j, j]$. Then

$$\mathscr{G}_{g_{j,k(j)}}V \to W \quad \text{in } C^\infty_{\mathrm{loc}}, \qquad |g_{j,k(j)}|_{\mathrm{cov}} \to \infty.$$

Hence $W \in \mathcal{H}_\infty(V)$. This proves closedness.

To see sequential precompactness of the whole hull, let $W_j \in \mathcal{H}_\infty(V)$ be arbitrary. For each $j$, choose an escaping sequence $g_{j,k}$ with $\mathscr{G}_{g_{j,k}}V \to W_j$, and choose $k(j)$ so that

$$|g_{j,k(j)}|_{\mathrm{cov}} \ge j, \qquad d_j\left(\mathscr{G}_{g_{j,k(j)}}V, W_j\right) \le j^{-1}.$$

The sequence $\mathscr{G}_{g_{j,k(j)}}V$ is an escaping transformed sequence in the compactness branch, so it has a locally smooth convergent subsequence. Along that subsequence, $W_j$ converges to the same limit. The closedness argument places the limit in $\mathcal{H}_\infty(V)$. Thus every sequence in the hull has a convergent subsequence with limit still in the hull; metrizability of the local smooth topology gives compactness. $\square$

**Lemma 8.24** (Full covariant invariance of the escaping covariant hull)**.** *For every $g \in G$,*

$$\mathscr{G}_g\mathcal{H}_\infty(V) = \mathcal{H}_\infty(V).$$

*In particular, if $W \in \mathcal{H}_\infty(V)$, then both $T_aW$ and $\Theta_sW$ belong to $\mathcal{H}_\infty(V)$ for all $a \in \mathbb{R}^3$ and $s \in \mathbb{R}$.*

*Proof.* Let $W = \lim_n \mathscr{G}_{g_n}V$, with $|g_n|_{\mathrm{cov}} \to \infty$. Then

$$\mathscr{G}_gW = \lim_n \mathscr{G}_g\mathscr{G}_{g_n}V = \lim_n \mathscr{G}_{gg_n}V.$$

Since the metric on $G$ is proper and left-invariant, $|gg_n|_{\mathrm{cov}} \to \infty$. Hence $\mathscr{G}_gW \in \mathcal{H}_\infty(V)$. Applying the same argument to $g^{-1}$ gives equality. $\square$

**Definition 8.25** (Tail alternatives)**.** Let $\mathcal{C}$ denote the set of constant vector fields. Define the following classes only in the compact normalized hull branch, after the local-compactness and pressure-gauge-loss alternatives identified in Theorem 8.23:

$$\mathcal{C}_{\mathrm{gen}_{\text{tail-const}}} := \{V \in \mathcal{C}_{\mathrm{gen}} : \mathcal{H}_\infty(V) \subset \mathcal{C}\},$$

and

$$\mathcal{C}_{\mathrm{gen}_{\text{tail-prof}}} := \{V \in \mathcal{C}_{\mathrm{gen}} : \mathcal{H}_\infty(V) \text{ contains a nonconstant profile}\}.$$

**Proposition 8.26** (Tail dichotomy and tail-zero normalization)**.** *Every $V \in \mathcal{C}_{\mathrm{gen}}$ belongs to $\mathcal{C}_{\mathrm{gen}_{\text{tail-const}}} \cup \mathcal{C}_{\mathrm{gen}_{\text{tail-prof}}}$. If $\mathcal{H}_\infty(V) = \{C_\infty\}$ is a singleton constant hull, then*

$$U := S_{C_\infty}V$$

*solves (1.2) and satisfies*

$$\mathcal{H}_\infty(U) = \{0\}.$$

*Proof.* The dichotomy is the definition. Since $T_a$ and $S_c$ commute,

$$T_a(S_{C_\infty}V) = S_{C_\infty}(T_aV).$$

Thus every far-field limit of $S_{C_\infty}V$ is

$$S_{C_\infty}(C_\infty) = 0.$$

The equation is preserved by Lemma 8.13. $\square$

**Definition 8.27** (Tail-minimal recurrent core)**.** A nonempty compact set $\mathcal{M} \subset \mathcal{H}_\infty(V)$ is a tail-minimal recurrent core if it is invariant under all covariant translations $T_a$, invariant under all time translations

$$(\Theta_sW)(y,\tau) := W(y,\tau+s),$$

and contains no proper nonempty compact subset with the same two invariance properties.

**Lemma 8.28** (Pressure gradients in invariant local averages)**.** *Let $\mathcal{M}$ be a compact set of smooth bounded solutions of (1.2), compact in the locally smooth topology and invariant under the covariant translations $T_a$ and the time translations $\Theta_s$. Let $\mu$ be a Borel probability measure on $\mathcal{M}$ invariant under both actions. For each $W \in \mathcal{M}$, let $P_W$ be any smooth local pressure near $(0,0)$ satisfying (1.2). Put*

$$u_i(W) := W_i(0,0), \qquad p_i(W) := \partial_iP_W(0,0), \qquad b_{ij}(W) := u_i(W)u_j(W).$$

*Let $\Pi_0$ denote the orthogonal projection in $L^2(\mathcal{M},\mu)$ onto the closed subspace of functions invariant under all spatial covariant translations $T_a$, and set*

$$u_i^0 := \Pi_0 u_i, \qquad p_i^0 := \Pi_0 p_i.$$

*Then the pressure contribution in the recurrent-core energy average is gauge-independent and satisfies*

$$\int_{\mathcal{M}} u_i(W) p_i(W)\, d\mu(W) = \int_{\mathcal{M}} u_i^0(W) p_i^0(W)\, d\mu(W). \tag{8.2}$$

*Equivalently, the nonzero spatial-frequency part of the pressure gradient does no work in invariant average:*

$$\int_{\mathcal{M}} u_i(W)\big(p_i(W) - p_i^0(W)\big)\, d\mu(W) = 0. \tag{8.3}$$

*Proof.* The additive pressure gauge does not enter, because only $\nabla P_W$ appears. The pressure gradient is fixed by the velocity through the equation

$$\nabla P_W = \Delta W - \frac{1}{2}(W + y\cdot\nabla W) - \partial_\tau W - (W\cdot\nabla)W.$$

Thus $W \mapsto p_i(W)$ is a continuous bounded local observable on $\mathcal{M}$. The local pressure reconstruction step in the imported setup results supplies the local pressure representatives; the present argument uses only their gradients, so the additive gauges never enter.

Let $U_a$ be the unitary representation of spatial covariant translations on $L^2(\mathcal{M},\mu)$,

$$(U_a F)(W) := F(T_a W).$$

Let $D_1, D_2, D_3$ be its skew-adjoint infinitesimal generators. For a smooth local observable $F$, $D_k F$ is obtained by differentiating $F(T_{he_k}W)$ at $h = 0$. In particular, at $\tau = 0$,

$$D_k u_i(W) = \partial_k W_i(0,0).$$

Hence

$$D_i u_i = 0 \qquad \text{in } L^2(\mathcal{M},\mu), \tag{8.4}$$

because each $W$ is divergence-free.

Taking the spatial divergence of (1.2) gives the local pressure Poisson equation

$$-\Delta_y P_W = \partial_i\partial_j(W_i W_j). \tag{8.5}$$

Indeed, the divergence of $\partial_\tau W$, $\Delta W$, and $W$ is zero; the divergence of $y\cdot\nabla W$ is $\partial_i(y_j\partial_j W_i) = \partial_j W_j + y_j\partial_j\partial_i W_i = 0$; and $(W\cdot\nabla)W_i = \partial_j(W_j W_i)$. Applying the infinitesimal translation generators at $(0,0)$ gives

$$D_i p_i = -D_i D_j b_{ij}. \tag{8.6}$$

The pressure gradient is curl-free, hence

$$D_k p_i = D_i p_k. \tag{8.7}$$

Let

$$A := -\sum_{k=1}^{3} D_k^2.$$

By the spectral theorem for the commuting unitary group $U_a$, $A$ is self-adjoint and nonnegative, and $\Pi_0$ is the spectral projection of $A$ at the eigenvalue 0. Define the nonzero spatial part

$$q_i := p_i - p_i^0 = (I - \Pi_0)p_i.$$

Equations (8.6) and (8.7) imply, on the nonzero spectral subspace,

$$q_k = D_k A^{-1}(I-\Pi_0)D_iD_jb_{ij}$$

in the standard regularized sense. Explicitly, with

$$q_{k,\varepsilon} := D_k(\varepsilon + A)^{-1}(I-\Pi_0)D_iD_jb_{ij}, \qquad \varepsilon > 0,$$

the joint spectral resolution of $D_1, D_2, D_3$ gives

$$q_{k,\varepsilon} \longrightarrow q_k \quad \text{in } L^2(\mathcal{M},\mu) \quad \text{as } \varepsilon \downarrow 0.$$

The convergence is precisely the multiplier convergence

$$\frac{\xi_k\xi_i\xi_j}{\varepsilon + |\xi|^2} \longrightarrow \frac{\xi_k\xi_i\xi_j}{|\xi|^2} \qquad (\xi \neq 0),$$

together with $(I-\Pi_0)$, which removes the point $\xi = 0$. The curl-free relation (8.7) says that the nonzero spatial spectral part of $p$ is parallel to $\xi$, and (8.6) fixes its longitudinal coefficient; there is therefore no additional harmonic component.

For each $\varepsilon > 0$, skew-adjointness of $D_k$ and (8.4) give

$$\begin{aligned}\int_{\mathcal{M}} u_k q_{k,\varepsilon}\, d\mu &= \int_{\mathcal{M}} u_k D_k(\varepsilon + A)^{-1}(I-\Pi_0)D_iD_jb_{ij}\, d\mu \\ &= -\int_{\mathcal{M}} (D_ku_k)(\varepsilon + A)^{-1}(I-\Pi_0)D_iD_jb_{ij}\, d\mu = 0.\end{aligned}$$

Passing to the limit in $L^2$ yields

$$\int_{\mathcal{M}} u_k q_k\, d\mu = 0,$$

which is (8.3). Since $p_i^0$ lies in the range of $\Pi_0$ and $u_i - u_i^0$ is orthogonal to that range,

$$\int_{\mathcal{M}} (u_i - u_i^0)p_i^0\, d\mu = 0.$$

Combining this with (8.3) proves (8.2). □

**Theorem 8.29** (Rigidity of recurrent profile tail cores)**.** *Let $\mathcal{M}$ be a compact tail-minimal recurrent core for (1.2) arising from the normalized Seregin setup in the companion setup paper, with the affine zero mode interpreted by the affine symmetry Lemma 8.13. Equivalently, after applying one fixed affine normalization $S_c$, the spatially homogeneous part of the core satisfies the bounded mild centered formulation recorded in the imported setup results. Then every $W \in \mathcal{M}$ is spatially and temporally constant. More precisely, there exists $m \in \mathbb{R}^3$ such that*

$$W(y,\tau) \equiv m \qquad \text{for every } W \in \mathcal{M}.$$

*After applying the affine symmetry $S_m$, the core is reduced to the zero solution.*

*Proof.* The group generated by covariant translations and time translations is amenable on the compact invariant set $\mathcal{M}$. Choose an invariant Borel probability measure $\mu$ with full support on a minimal component, and write

$$\langle F \rangle := \int_{\mathcal{M}} F(W)\, d\mu(W)$$

for local observables. Let $P_W$ be a local pressure associated to $W$. As in Lemma 8.28, set

$$u_i(W) := W_i(0,0), \qquad p_i(W) := \partial_i P_W(0,0),$$

and let $\Pi_0$ be the orthogonal projection onto observables invariant under all spatial covariant translations. Write

$$u_i^0 := \Pi_0 u_i, \qquad p_i^0 := \Pi_0 p_i.$$

Invariance gives

$$\langle \partial_\tau F(W)(0,0)\rangle = 0, \qquad \langle \partial_j F(W)(0,0)\rangle = 0$$

for every smooth local observable $F$. These identities follow first for difference quotients generated by the group action and then by smoothness.

Evaluate (1.2) at $(0,0)$. The drift term vanishes at $y=0$, which gives

$$\partial_\tau W(0,0) - \Delta W(0,0) + \frac{1}{2}W(0,0) + (W(0,0)\cdot\nabla)W(0,0) + \nabla P_W(0,0) = 0.$$

Project this identity onto the spatially invariant subspace. The spatial derivative terms vanish under $\Pi_0$: the Laplacian is $D_jD_ju_i$, and the nonlinear term is $D_j(u_ju_i)$ because $D_ju_j=0$. Time translations do not literally commute with the covariant spatial translations; by Lemma 8.5 they normalize them:

$$\Theta_s T_a \Theta_{-s} = T_{e^{s/2}a}.$$

Hence the subspace of observables invariant under all $T_a$ is invariant under $\Theta_s$, and the orthogonal projection $\Pi_0$ commutes with the time-translation representation. Therefore $\Pi_0\partial_\tau u_i = \partial_\tau u_i^0$. Thus

$$\partial_\tau u_i^0 + \frac{1}{2}u_i^0 + p_i^0 = 0 \qquad \text{in } L^2(\mathcal{M},\mu). \tag{8.8}$$

Taking the inner product of (8.8) with $u_i^0$, summing in $i$, and using invariance under time translations,

$$\left\langle u_i^0 \partial_\tau u_i^0\right\rangle = \frac{1}{2}\left\langle \partial_\tau |u^0|^2\right\rangle = 0,$$

gives

$$\left\langle u_i^0 p_i^0\right\rangle = -\frac{1}{2}\left\langle |u^0|^2\right\rangle. \tag{8.9}$$

Testing the equation against $W$, evaluating at $(0,0)$, and averaging, the time term is

$$\langle W\cdot\partial_\tau W\rangle = \frac{1}{2}\left\langle \partial_\tau |W|^2\right\rangle = 0.$$

The diffusion term gives

$$-W\cdot\Delta W = |\nabla W|^2 - \operatorname{div}(W\cdot\nabla W),$$

and therefore

$$-\langle W\cdot\Delta W\rangle = \left\langle |\nabla W|^2\right\rangle.$$

The nonlinear term is a divergence:

$$W\cdot(W\cdot\nabla)W = \frac{1}{2}W\cdot\nabla|W|^2 = \frac{1}{2}\operatorname{div}(|W|^2W),$$

so its average is zero. Decompose

$$p_i = p_i^0 + (p_i - p_i^0).$$

The nonzero spatial part of the pressure gradient does no work in invariant average by Lemma 8.28, and the remaining spatial zero mode is evaluated by (8.9). Hence

$$\langle u_i p_i\rangle = \left\langle u_i^0 p_i^0\right\rangle = -\frac{1}{2}\left\langle |u^0|^2\right\rangle.$$

Thus

$$\left\langle |\nabla W|^2\right\rangle + \frac{1}{2}\left\langle |W|^2\right\rangle - \frac{1}{2}\left\langle |u^0|^2\right\rangle = 0. \tag{8.10}$$

Both terms are nonnegative. The second nonnegative quantity is

$$\left\langle |u|^2\right\rangle - \left\langle |u^0|^2\right\rangle = \left\langle |u-u^0|^2\right\rangle,$$

because $\Pi_0$ is an orthogonal projection in $L^2(\mathcal{M},\mu)$. Therefore

$$\langle|\nabla W|^2\rangle = 0, \qquad \langle|u-u^0|^2\rangle = 0.$$

Since $\mu$ has full support and the observables are continuous in the local smooth topology, $\nabla W(0,0) = 0$ for every $W \in \mathcal{M}$. Invariance moves $(0,0)$ to any $(y,\tau)$, so $\nabla W(y,\tau) = 0$ everywhere. Thus every element of $\mathcal{M}$ is spatially constant:

$$W(y,\tau) = a_W(\tau).$$

It remains only to identify the allowed spatially homogeneous zero mode. This is not a pressure-gauge assertion. By the normalization in the setup paper quoted in the statement, there is one fixed affine normalization $S_c$ for which the spatially homogeneous representatives satisfy the bounded mild centered formulation recorded in the imported setup results. For a spatially constant mild centered solution $B(\tau)$, the nonlinear Duhamel term in the mild formulation vanishes, and the Ornstein–Uhlenbeck–Stokes semigroup acts on constants by

$$S(\tau-\sigma)B(\sigma) = e^{-(\tau-\sigma)/2}B(\sigma).$$

Thus

$$B(\tau) = e^{-(\tau-\sigma)/2}B(\sigma) \qquad (\sigma < \tau).$$

Letting $\sigma \to -\infty$ and using boundedness of the ancient trajectory gives $B(\tau) = 0$ for every $\tau$. Undoing the fixed affine normalization gives a single vector $m = c$ and

$$W(y,\tau) \equiv m \qquad (W \in \mathcal{M}).$$

The affine symmetry $S_m$ then maps the core to the zero solution. □

## 9. Terminal concentration profiles and separated centers

The terminal part of the proof is local in parabolic cylinders. It does not require a global $L^3$ bound, a global CKN budget, a finite critical-mass budget, or a global nonlinear profile decomposition for the original residual profile. Everything is formulated on fixed compact terminal cylinders after applying the recentering sequence under consideration.

**Definition 9.1** (Parabolic recentering sequence)**.** Let $(V_n,P_n)$ be a terminal extraction sequence on expanding domains $D_n \uparrow \mathbb{R}^3\times\mathbb{R}$, with local suitability and pressure-gauge control. A parabolic recentering sequence is a sequence

$$\mathfrak{c} = (z_n)_{n\ge 1}, \qquad z_n = (y_n,\tau_n) \in D_n.$$

Two recentering sequences are equivalent if

$$\sup_n \left(|y_n - y_n'| + |\tau_n - \tau_n'|^{1/2}\right) < \infty,$$

and asymptotically separated if the same parabolic distance tends to infinity. The recentered fields associated to $\mathfrak{c}$ are denoted

$$V_n^{\mathfrak{c}}(y,s) := V_n(y+y_n, s+\tau_n), \qquad P_n^{\mathfrak{c}}(y,s) := P_n(y+y_n,s+\tau_n) - a_n^{\mathfrak{c}}(s),$$

where $a_n^{\mathfrak{c}}$ is an admissible time-dependent local pressure gauge. All convergence assertions for a recentering sequence are made on fixed compact sets in the variables $(y,s)$.

**Definition 9.2** (Combined parabolic-covariant descendant recentering)**.** Let $g_n = (y_n,\tau_n)$ be an ordinary parabolic recentering sequence for a terminal extraction $(V_n,P_n)$, and let $\zeta_n = (\xi_n,\sigma_n)$ be a covariant tail recentering sequence in the centered limit generated by $g_n$. The combined parabolic-covariant descendant recentering is denoted

$$d_n = (g_n;\xi_n,\sigma_n)$$

and acts on the original terminal sequence by

$$V_n^d(y,s) := V_n(y + y_n + e^{s/2}\xi_n,\ s + \tau_n + \sigma_n),$$

with the pressure pulled back by the same formula and then adjusted by an admissible time-dependent gauge. Its relative distance from $g_n$ is

$$|\xi_n| + |\sigma_n|^{1/2}.$$

Thus $d_n$ is separated from $g_n$ exactly when this relative distance tends to infinity. This definition is used only to lift descendants of a centered profile back to the original terminal extraction; ordinary first-generation terminal recenterings remain the ordinary parabolic recenterings of Definition 9.1.

**Definition 9.3** (Raw atlas defect measure and raw unit-window active recenterings)**.** After choosing local pressure gauges $a_n(\tau)$, define

$$d\mu_n := \Big(|V_n|^3 + |P_n - a_n(\tau)|^{3/2}\Big)\, dy\, d\tau, \qquad d\nu_n := |\nabla V_n|^2\, dy\, d\tau.$$

The measure $\mu_n$ is the atlas-dependent Radon measure from Definition 3.1. It is raw, gauge-dependent, and used on fixed unit observer cylinders for local compactness and concentration-set extraction. It is not a scale-invariant CKN measure. Scale-invariant threshold and regularity statements at arbitrary radius are written with $\mathcal{V}_r$, $D_r$, or $\mathcal{E}_r$. The raw mass condition below is only a selection criterion for observer windows. It never certifies that a limiting profile is nonzero: retained nonzero profiles are certified by $\mathcal{V}_1$ or by the explicitly stated non-affine oscillation functional. If the selected raw mass is pressure-only, it is treated by Lemma 9.5. The local mass of a recentering sequence $\mathfrak{c} = (z_n)$ is

$$m(\mathfrak{c}) := \limsup_{n\to\infty} \mu_n(Q_1(z_n)).$$

For a combined parabolic-covariant descendant recentering $d_n = (g_n; \xi_n, \sigma_n)$, the same notation means the mass of the pulled back defect measure in the combined coordinates:

$$m(\mathfrak{d}) := \limsup_{n\to\infty} \mu_n^{\mathfrak{d}}(Q_1(0,0)),$$

where $\mu_n^{\mathfrak{d}}$ is obtained from $\mu_n$ by the coordinate change in Definition 9.2 and by the admissible pressure gauges used in that recentered frame. Thus first-generation recentering sequences and lifted covariant descendants use the same local unit-cylinder criterion, expressed in their own local coordinates. For a fixed raw unit-window threshold $\varepsilon_{\mathrm{sd}} > 0$, the recentering sequence is raw unit-window active at threshold $\varepsilon_{\mathrm{sd}}$ if $m(\mathfrak{c}) \ge \varepsilon_{\mathrm{sd}}$, and locally vanishing at unit scale otherwise. It has loss of local compactness if

$$\limsup_{n\to\infty} \nu_n(Q_2(z_n)) = \infty$$

or if the local pressure gauges fail to be uniformly controlled in $L^{3/2}(Q_2(z_n))$.

**Lemma 9.4** (Gauge compatibility and retained-velocity semicontinuity)**.** *Let $(U_n, \Pi_n) \to (U, \Pi)$ in the topology of $\mathfrak{X}_M$ on a fixed compact cylinder containing $Q_1(0,0)$. After choosing the pressure gauges used in that convergence, the following hold.*

(i) *The pressure gauges are compatible with centered time shifts, covariant translations, covariant tail recenterings, affine normalization, and diagonal extraction of descendant profiles: applying any of these operations only changes the gauge by another admissible function of time.*

(ii) *The local CKN quantity is lower semicontinuous in the only direction used for pressure-inclusive limits:*

$$\mathcal{E}_1(U,\Pi;0) \le \liminf_{n\to\infty} \mathcal{E}_1(U_n,\Pi_n;0).$$

*Consequently, if the limit profile has $\mathcal{E}_1(U,\Pi;0) \ge \eta$, then all sufficiently far approximating profiles have CKN mass at least any prescribed $\eta' < \eta$.*

(iii) *If retained nontriviality is carried by the velocity part, then it is closed under the same convergence:*

$$\liminf_{n\to\infty} \mathcal{V}_1(U_n;0) \ge \eta \quad \Longrightarrow \quad \mathcal{V}_1(U;0) \ge \eta.$$

*In applications where the lower bound is first obtained for velocity alone and then converted into a CKN lower bound, the descendant threshold is chosen below the resulting positive constant as in Definition 3.4.*

*Proof.* Time shifts and covariant translations compose the pressure with a smooth local change of variables; affine normalization adds the explicit linear pressure $\frac{1}{2}c \cdot y$; and every local pressure is defined only up to a time-dependent function. Thus admissible gauges remain admissible under all operations listed in (i). For (ii), strong local convergence of $U_n$ gives

$$\iint_{Q_1} |U_n|^3 \to \iint_{Q_1} |U|^3.$$

For the pressure part, pass to the quotient space

$$L^{3/2}(Q_1)\big/L^{3/2}((-1,0);\{\text{spatial constants}\}).$$

The quotient norm is convex and weakly lower semicontinuous; equivalently, after choosing any admissible atlas gauges realizing the local convergence,

$$D_1(\Pi;0) \le \liminf_{n\to\infty} D_1(\Pi_n;0).$$

Adding this to the strong $L^3$ convergence of the velocity gives the claimed lower semicontinuity direction for $\mathcal{E}_1$. The velocity-only assertion follows directly from strong $L^3(Q_1)$ convergence. The statement is entirely local on $Q_1$. □

**Lemma 9.5** (No pressure-only retained active limit)**.** *Let $(U_n,\Pi_n) \to (U,\Pi)$ in the local Seregin topology on $Q_2(0,0)$, after passing to compatible pressure gauges whenever such gauges are available. Assume that, on the cylinders where pressure compactness is invoked, the pressure representatives are the normalized whole-space Riesz pressures modulo functions of time and modulo the affine/parasitic quotient of Definition 8.14. If $U = 0$ on $Q_2(0,0)$, then*

$$D_1(\Pi;0) = 0.$$

*Consequently, suppose*

$$\liminf_{n\to\infty} \mathcal{E}_1^{\mathfrak{a}}(U_n,\Pi_n;0) \ge \eta > 0 \qquad \textit{but} \qquad \mathcal{V}_1(U;0) = 0.$$

*Then the velocity part of the approximating concentration vanishes in the limit, and the remaining lower bound is pressure-only. At least one of the following ordered alternatives holds; when more than one applies, the proof uses the first applicable alternative:*

(i) *the pressure gauges or pressure masses are not strongly compact in the local $L^{3/2}$ quotient topology on $Q_2(0,0)$, so the recentering belongs to the pressure-loss part of the noncompact terminal alternative;*

(ii) *the limiting pressure defect is affine in space modulo functions of time and is assigned to the affine/parasitic quotient stratum $\mathcal{C}_{\mathrm{aff}}$;*

(iii) *the local Riesz part on a slightly larger cylinder carries the pressure lower bound, and hence yields a positive velocity lower bound on a finite family of neighboring unit cylinders.*

*Consequently no pressure-only lower bound is used below to certify a retained nonzero terminal profile. Retained descendants are certified by $\mathcal{V}_1$, or by the non-affine oscillation functional $\Phi$, and pressure-inclusive $\mathcal{E}_1$ is used only after one of the three alternatives above has been resolved.*

*Proof.* If $U = 0$ on $Q_2$, then the limiting centered equation gives, in distributions on $Q_2$,

$$\nabla\Pi = -\partial_\tau U + \Delta U - \frac{1}{2}y\cdot\nabla U - \frac{1}{2}U - (U\cdot\nabla)U = 0.$$

Thus $\Pi = c(\tau)$ on $Q_2$, and its quotient pressure concentration on $Q_1$ vanishes:

$$D_1(\Pi;0) = 0.$$

If the gauges are not compact, or if the pressure representatives are only weakly compact and retain a nonzero $L^{3/2}$ defect in the local quotient, this is alternative (i) by definition of the pressure-loss part of the noncompact terminal alternative. Assume therefore that, after subtracting the chosen time gauges, the pressures are strongly compact in $L^{3/2}_{\mathrm{loc}}$ modulo spatial constants. Since $\mathcal{V}_1(U;0) = 0$ and $U_n \to U$ strongly in $L^3(Q_1)$, the velocity contribution to $\mathcal{E}_1^{\mathfrak{a}}(U_n,\Pi_n;0)$ tends to zero. Any remaining positive lower bound is therefore represented by the strong $L^{3/2}$ quotient pressure limit modulo gauges.

If $U = 0$ on $Q_2$, the first paragraph shows that this quotient pressure limit has zero $D_1$-mass, a contradiction. Hence a genuine pressure lower bound with compact pressure gauges can only survive through either an affine/parasitic pressure mode, through a local Riesz part generated by velocity products on the finite enlargement used by the pressure representation, or through failure of the chosen normalized whole-space pressure representative to remain compact in the local quotient.

On $Q_2$, the pressure satisfies the local decomposition

$$\Pi_n - a_n(\tau) = \mathcal{R}_i\mathcal{R}_j(U_{n,i}U_{n,j}) + h_n,$$

where $h_n$ is harmonic in the spatial variables on the smaller cylinder, after the standard cutoff used in the setup pressure atlas. If the velocity products on the finite enlargement used by this cutoff decomposition carry a positive $L^{3/2}$ lower bound, then a finite unit-cylinder cover gives (iii). Otherwise the Riesz source tends to zero strongly in $L^{3/2}$ on that enlargement, and the Riesz part of the pressure converges strongly to zero in $L^{3/2}(Q_1)$. Any remaining nonzero pressure defect is therefore harmonic in space on the unit cylinder. Its affine part is the parasitic pressure mode allowed in $\mathcal{C}_{\mathrm{aff}}$, giving (ii). If the harmonic representative is not affine after quotienting by time functions, then it is not an admissible compact pressure-only remnant under the normalized whole-space Riesz representation modulo time and affine gauges assumed in the lemma. It is therefore recorded as pressure-gauge or pressure-representation failure, namely alternative (i). This proves that a retained terminal profile is never certified by pressure alone. □

**Lemma 9.6** (Local pressure decomposition and pressure-only alternatives)**.** *Let $(u_n, p_n)$ be locally suitable on a parabolic cylinder $Q_4 = B_4 \times (-16, 0)$ with $\|u_n\|_{L^\infty(Q_4)} \le M$ and pressure gauges $a_n(t)$ compatible with Definition 8.14. Fix a compactly supported cutoff $\chi \in C_c^\infty(B_3)$ with $\chi \equiv 1$ on $B_2$. Then on $B_2$*

$$p_n - a_n(t) = p_n^{\mathrm{loc}} + h_n, \qquad p_n^{\mathrm{loc}} := \mathcal{R}_i\mathcal{R}_j(\chi u_{n,i}u_{n,j}), \tag{9.1}$$

*where $h_n$ is harmonic in the spatial variables on $B_2$ for almost every $t \in (-4, 0)$ and depends only on the values of $(u_n, p_n - a_n)$ on $Q_4 \setminus Q_2$ (the boundary data of the cutoff representation). The following four properties hold.*

(i) Local Riesz strong continuity. *If $u_n \to 0$ strongly in $L^3(Q_4)$, then $p_n^{\mathrm{loc}} \to 0$ strongly in $L^{3/2}(Q_2)$, with $\|p_n^{\mathrm{loc}}\|_{L^{3/2}(Q_2)} \le C\|u_n\|^2_{L^3(Q_4)}$.*

(ii) Harmonic moment compactness. *For every multi-index $\alpha$ with $|\alpha| \le K$ and every Schwartz test function $\psi$ compactly supported in $B_1$,*

$$t \mapsto \int (\partial^\alpha h_n)(x,t)\,\psi(x)\,dx$$

*is bounded uniformly in $n$ by a constant depending only on $\|u_n\|_{L^3(Q_4)}$, $\|p_n - a_n\|_{L^{3/2}(Q_4)}$, $K$, and $\psi$.*

(iii) Affine quotient record. *The quotient of $h_n$ by spatial affine functions*

$$h_n(x,t) \sim h_n(x,t) + (\alpha_n(t) + \beta_n(t)\cdot x)$$

*is bounded in any local norm by the same data; the quotient $[h_n] \in D_1$ is the affine quotient of Definition 8.14. If the affine part of $h_n$ is the only carrier of a positive pressure lower bound, the corresponding recentering belongs to the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$.*

(iv) Non-affine harmonic record. *If $[h_n]$ has a nonzero limit $[h]$ in the quotient $D_1$, then either $[h]$ is purely affine (case* (ii) *of Lemma 9.5), or the non-affine harmonic remainder is not used as retained activity. If a nonzero local Riesz part is present on the finite enlargement, it gives neighboring velocity activity by* (i)*. If no such Riesz part is present, the non-affine harmonic remainder is recorded as failure of the normalized whole-space pressure representative, modulo time and affine gauges, to be compact in the local quotient; this is the pressure/noncompact alternative.*

*The conclusion of* (iv) *is the precise mechanism behind the reduction of the pressure-only case: only the local Riesz component is allowed to produce neighboring velocity activity. A genuinely non-affine harmonic quotient is not converted into a velocity source by itself; outside the affine quotient it is treated as pressure-representation or pressure-compactness failure.*

*Proof.* The decomposition (9.1) is the standard local pressure cutoff: writing

$$-\Delta(p_n - a_n) = \partial_i\partial_j(u_{n,i}u_{n,j})$$

on $Q_4$ and applying the Riesz operators $\mathcal{R}_i\mathcal{R}_j$ to the cutoff right-hand side $\partial_i\partial_j(\chi u_{n,i}u_{n,j})$ gives a local representative of the velocity-product part of the pressure, and the difference $h_n = p_n - a_n - p_n^{\mathrm{loc}}$ is then a harmonic function in space on $B_2$ at almost every time. This is the same construction used in the setup paper for the local pressure atlas (Theorem 1.3 (3)).

(i) The boundedness of Riesz transforms on $L^{3/2}$ gives

$$\|p_n^{\mathrm{loc}}\|_{L^{3/2}(Q_2)} \le C\|\chi u_{n,i}u_{n,j}\|_{L^{3/2}(Q_4)} \le C\|u_n\|^2_{L^3(Q_4)}.$$

Strong $L^3$-vanishing of $u_n$ on $Q_4$ thus forces strong $L^{3/2}$-vanishing of $p_n^{\mathrm{loc}}$ on $Q_2$.

(ii) The standard interior estimates for harmonic functions in $B_2$ bound every spatial derivative of $h_n$ on $B_1$ by an $L^{3/2}$-norm of $h_n$ on $B_2$, which in turn is bounded by $\|p_n - a_n\|_{L^{3/2}(Q_4)} + \|p_n^{\mathrm{loc}}\|_{L^{3/2}(Q_2)}$. The displayed bound on time-integrated moments follows.

(iii) The affine subspace of harmonic functions on $B_2$ is finite-dimensional; the projection onto it is bounded for any local norm. The quotient $[h_n]$ is therefore bounded. By the affine pressure quotient fixed in Definition 8.14, the affine part of $h_n$ is the parasitic pressure mode allowed in $\mathcal{C}_{\mathrm{aff}}$; a recentering with positive lower bound carried only by this mode is the affine/parasitic recentering and is recorded in $\mathcal{C}_{\mathrm{aff}}$.

(iv) Assume the limit $[h]$ is non-affine in the quotient $D_1$. A harmonic function may be non-affine while still having no local Poisson source in $B_2$; therefore this fact alone is not used to produce a velocity lower bound. If the localized Riesz component $p_n^{\mathrm{loc}}$ has a nonzero lower bound on the finite enlargement entering the cutoff, (i) and boundedness of the Riesz transforms give a neighboring unit cylinder with positive velocity-product mass, hence the velocity alternative of Lemma 9.5. If $p_n^{\mathrm{loc}} \to 0$ strongly and $[h]$ is non-affine, then the surviving pressure lower bound is neither affine/parasitic nor represented by the normalized whole-space Riesz pressure modulo the allowed gauges. It is therefore assigned to the pressure-representation or pressure-compactness failure alternative, not to retained velocity concentration. ☐

**Definition 9.7** (Local-window condition for first-generation recenterings)**.** Let $(V_n, P_n)$ be a first-generation terminal sequence obtained by undoing the setup paper local Type I rescaling on a fixed physical cylinder $\mathcal{Q}_{\mathrm{loc}}$. A parabolic recentering sequence $\mathfrak{c} = (z_n)$, $z_n = (y_n, \tau_n)$, satisfies the *local-window condition* if, for every compact cylinder

$$K_{R,S} := B_R(0) \times [-S, S] \Subset \mathbb{R}^3 \times \mathbb{R},$$

the cylinder obtained by writing $K_{R,S}$ in the $\mathfrak{c}$-centered coordinates and then undoing the setup rescaling is contained in $\mathcal{Q}_{\mathrm{loc}}$ for all sufficiently large $n$. Equivalently, every fixed compact cylinder in the recentered variables lies in the part of the first-generation extraction where the setup pressure atlas, local energy inequality, and endpoint Type I bounds are available.

**Lemma 9.8** (Local-window compactness for first-generation recenterings)**.** *Let $(V_n, P_n)$ be a first-generation admissible terminal sequence from the setup extraction, and let $\mathfrak{c} = (z_n)$ satisfy the local-window condition of Definition 9.7. If*

$$\liminf_{n\to\infty} \mathcal{V}_1(V_n; z_n) > 0,$$

*then the recentered sequence admits a subsequence and admissible pressure gauges for which*

$$(V_n^{\mathfrak{c}}, P_n^{\mathfrak{c}}) \longrightarrow (U, \Pi)$$

*locally in the Seregin topology. The limit is locally suitable, pressure normalized, bounded on compact terminal windows by the imported setup bounds, and nonzero:*

$$\mathcal{V}_1(U; 0) > 0$$

*after lowering the velocity threshold once.*

*Proof.* Fix $R, S < \infty$. By the local-window condition, the physical image of $K_{2R,2S}$ in the $\mathfrak{c}$-frame lies inside the setup local Type I cylinder for all large $n$. The imported setup results therefore gives the local energy, dissipation, velocity $L^3$, and pressure-gauge bounds on this physical image, uniformly in $n$, after the pressure gauges supplied by the setup atlas. Rescaling back to the $\mathfrak{c}$-coordinates gives the standard Seregin compactness bounds on $K_{R,S}$: strong local $L^3$ compactness for the velocity, weak local $L^{3/2}$ compactness for pressure modulo time functions, and stability of the local energy inequality.

Diagonalizing over $R, S \in \mathbb{N}$ gives a locally suitable terminal limit. The retained lower bound is carried by the velocity part, so strong $L^3_{\mathrm{loc}}$ convergence gives a positive $\mathcal{V}_1$ lower bound in the limit after the fixed threshold loss. No compactness is asserted outside the local-window region. ☐

**Lemma 9.9** (Covariant compactness or pressure-atlas loss for descendants)**.** *Let $(U, \Pi) \in \mathfrak{X}_M^{\#}$ be represented after canonical affine/parasitic normalization. For every covariant recentering sequence $g_n \in G$, either the transforms*

$$\mathscr{G}_{g_n}(U, \Pi)$$

*have no compact closure in the normalized local topology with the setup pressure atlas fixed, in which case the sequence is assigned to the local-compactness or pressure-gauge-loss terminal alternative, or they admit a locally compact pressure-gauge subsequence in the Seregin topology. In the compactness branch, if*

$$\liminf_{n\to\infty} \mathcal{V}_1(\mathscr{G}_{g_n} U; 0) > 0,$$

*then the limiting descendant is nonzero.*

*Proof.* The covariant action is an exact symmetry of the centered equation by Lemma 8.5. Hence the transformed velocities remain bounded by $\|U\|_{L^\infty}$ and solve the same centered equation on all of $\mathbb{R}^3 \times \mathbb{R}$. This boundedness alone is not used as a compactness theorem. If the setup pressure-atlas representatives and local compactness bounds fail to have a compact subsequence, this is precisely the noncompact/pressure-gauge terminal alternative. Otherwise the transformed profiles lie in a compact subset of $\mathfrak{X}_M^\#$, so a diagonal subsequence converges in the normalized local topology to a bounded ancient suitable descendant. Strong local $L^3$ convergence transfers any velocity lower bound to the limit. □

**Lemma 9.10** (Escaping first-generation recenterings are assigned to terminal alternatives)**.** *Let $(V_n, P_n)$ be a first-generation terminal sequence and let $\mathfrak{c} = (z_n)$ carry retained unit velocity concentration:*

$$\liminf_{n\to\infty} \mathcal{V}_1(V_n; z_n) > 0.$$

*If $\mathfrak{c}$ fails the local-window condition, then it is not handled by local compactness. Instead it is assigned, according to the compact-window-to-expanding-region protocol of Theorem 9.17, to one of the terminal exterior alternatives: an exterior retained velocity profile, a diffuse defect, a pressure/noncompact defect, or a critical-tail defect.*

*Proof.* Failure of the local-window condition means that some fixed recentered compact cylinder leaves the physical setup cylinder after undoing the first-generation scaling. The imported local Seregin compactness theorem is therefore not available on that cylinder, so the recentering cannot be treated by Lemma 9.8. The retained unit velocity concentration gives a positive residual measure in the corresponding selected expanding terminal region. Applying Theorem 9.17 to that region selects one of the protocol's priority alternatives: a selected unit observer with velocity concentration, a residual measure escaping to the observer boundary, a failure of local compactness or pressure-gauge compactness, or a nonzero normalized critical-tail scale. These are reduction alternatives, not compactness conclusions. □

**Corollary 9.11** (Local compactness or alternative selection for retained recenterings)**.** *Let $(V_n, P_n)$ be a terminal sequence obtained either as a first-generation setup sequence or as a descendant of an already extracted bounded ancient profile. Any recentering sequence carrying retained unit velocity concentration is of exactly one of the following types.*

(i) *It is first-generation and satisfies the local-window condition; then Lemma 9.8 gives a nonzero locally compact Seregin limit.*
(ii) *It acts on an already extracted normalized ancient profile and its pressure-atlas orbit has compact closure; then Lemma 9.9 gives a nonzero locally compact descendant.*
(iii) *It is either first-generation and fails the local-window condition, or it is an ancient-profile recentering whose pressure-atlas orbit has no compact closure; then Lemmas 9.9 and 9.10 assigns it to an exterior, diffuse, pressure/noncompact, or critical-tail terminal branch.*

*Thus retained unit velocity concentration under an arbitrary recentering is never used by itself as a compactness hypothesis.*

*Proof.* First-generation recenterings either satisfy or fail the local-window condition. The two cases are handled by Lemmas 9.8 and 9.10. Recenterings of extracted ancient profiles act on a bounded ancient profile defined on the full centered ancient space. They are handled by Lemma 9.9: compact pressure-atlas orbits give locally compact descendants, while noncompact or pressure-gauge-loss orbits are assigned to the corresponding terminal alternative. These cases exhaust the recenterings used in the terminal extraction arguments. □

**Theorem 9.12** (Local concentration-set extraction)**.** *Let $Z = \mathbb{R}^3 \times \mathbb{R}$ be the centered covariant observer space, and let $K_m \Subset Z$, $m \geq 1$, be a compact exhaustion. Let $\mu_n$ be the raw atlas defect measures of a terminal sequence, with local finite mass on compact observer windows:*

$$\sup_n \mu_n(K_m^+) < \infty \qquad (m \geq 1),$$

*where $K_m^+$ is a compact enlargement such that every covariant unit cylinder $\mathcal{Q}_1^{\mathrm{cov}}(z)$, $z \in K_m$, lies in $K_m^+$. The compact-window concentration sets are*

$$\widetilde{A}^{\eta}_{n,m} := \{z \in K_m : \mu_n(\mathcal{Q}_1^{\mathrm{cov}}(z)) \geq \eta\}.$$

*For the Hausdorff–Fell extraction we use the equivalent smoothed version below; thresholds are adjusted only by fixed cutoff constants. Fix $\chi \in C_c^\infty(Q_1(0,0))$, $0 \leq \chi \leq 1$, with $\chi \equiv 1$ on $Q_{1/2}(0,0)$. For $z \in Z$, put*

$$\chi_z(Y,\tau) := \chi\big(\mathcal{R}_z^{-1}(Y,\tau)\big), \qquad M_n(z) := \int \chi_z \, d\mu_n.$$

*For $\eta > 0$, define the compact-window concentration set*

$$A^{\eta}_{n,m} := \{z \in K_m : M_n(z) \geq \eta\}.$$

*After passing to a subsequence, for every $m$ and every rational $\eta > 0$, the compact sets $A^{\eta}_{n,m}$ converge in the Hausdorff–Fell topology to compact sets $A^{\eta}_m \subset K_m$. Moreover, for every open neighborhood $O \subset K_m$ of $A^{\eta}_m$,*

$$\limsup_{n\to\infty} \sup_{z \in K_m \setminus O} M_n(z) \leq \eta.$$

*Thus outside every neighborhood of the concentration set $A^{\eta}_m$, no $\eta$-concentrating covariant unit tube remains in the compact observer window $K_m$. The concentration set is the closed locus $A^{\eta}_m$, not a finite or countable enumeration of tubes.*

*Proof.* For fixed $n$, the map $z \mapsto \chi_z$ is continuous in $C_c$ on $K_m$, and all supports are contained in the compact enlargement $K_m^+$. Since $\mu_n$ is a Radon measure finite on $K_m^+$, $M_n$ is continuous on $K_m$. Therefore $A^{\eta}_{n,m}$ is compact.

The hyperspace of compact subsets of the compact metric space $K_m$ is compact in the Hausdorff–Fell topology. Hence, for each fixed pair $(m, \eta)$, a subsequence exists along which $A^{\eta}_{n,m}$ converges. A diagonal argument over $m \in \mathbb{N}$ and $\eta \in \mathbb{Q}_+$ gives simultaneous convergence.

Let $O \subset K_m$ be an open neighborhood of $A^{\eta}_m$. If the stated residual bound failed, there would exist $\delta > 0$, a subsequence $n_k$, and points $z_{n_k} \in K_m \setminus O$ such that

$$M_{n_k}(z_{n_k}) \geq \eta + \delta.$$

After replacing $O$ by a slightly smaller open neighborhood of $A^{\eta}_m$, we may assume $K_m \setminus O$ is compact. Passing to a further subsequence,

$$z_{n_k} \to z_* \in K_m \setminus O.$$

For all large $k$,

$$z_{n_k} \in A^{\eta+\delta/2}_{n_k,m}.$$

Hausdorff–Fell convergence gives $z_* \in A^{\eta+\delta/2}_m$. Since $A^{\eta+\delta/2}_m \subset A^{\eta}_m \subset O$, this contradicts $z_* \in K_m \setminus O$. The residual estimate follows. All bounds used here are local on $K_m^+$; no global CKN budget or countability of concentrating profiles is assumed. $\square$

**Lemma 9.13** (Closed-active-locus reduction)**.** *Let $A^\eta_m \subset K_m$ be the compact concentration loci produced by Theorem 9.12 for a rational threshold $\eta \in (0, \varepsilon_{\mathrm{CKN}}/2)$ and a compact observer window $K_m \Subset Z$. Then $A^\eta_m$ is a compact subset of $K_m$ for which the following reduction holds. For every open neighborhood $O \Supset A^\eta_m$ in $K_m$ there is $N(O) \in \mathbb{N}$ such that for all $n \geq N(O)$ the following hold simultaneously.*

(1) Sole carrier of compact retained concentration. *Every covariant unit cylinder $\mathcal{Q}^{\mathrm{cov}}_1(z) \subset K_m$ with center $z \in K_m \setminus O$ carries less than $\eta$ defect mass: $M_n(z) < \eta$.*
(2) Locality of the residual. *The residual measure $\mu_n\lfloor(K_m \setminus O)$ satisfies $\sup_{z\in K_m\setminus O} M_n(z) < \eta$; equivalently, no covariant unit cylinder centered in $K_m \setminus O$ is $\eta$-concentrating.*
(3) No additional compact concentration locus. *Any further Hausdorff–Fell limit point of compact $\eta$-concentration sets in $K_m$ is contained in $A^\eta_m$; the compact concentration locus is the unique maximal closed locus of compact retained concentration in $K_m$.*

*Consequently, the only candidate compact carriers of retained terminal concentration on $K_m$ at threshold $\eta$ are the points of $A^\eta_m$, and the trichotomy alternatives of Definition 9.15 (vanishing, diffuse, or noncompact residual) describe the entirety of the residual after a $\rho$-neighborhood of $A^\eta_m$ is removed.*

*Proof.* Compactness of $A^\eta_m$ is recorded in Theorem 9.12. Statement (1)–(2) is the displayed residual estimate of that theorem applied with neighborhood $O$ of $A^\eta_m$. Statement (3) follows because every other compact $\eta$-concentration limit $A' \subset K_m$ along the same extraction is, by Hausdorff–Fell limit-point characterization, contained in $A^\eta_m$: otherwise a point $z' \in A' \setminus A^\eta_m$ would carry a covariant unit defect mass $\geq \eta$ and could not be evicted from $K_m \setminus O$ for $O$ any open neighborhood of $A^\eta_m$, contradicting (1). The trichotomy then exhausts the residual on $K_m \setminus U_\rho(A^\eta_m)$ by definition of Definition 9.15. □

**Lemma 9.14** (Closed active loci reduce to single, finite separated, or chain alternatives)**.** *Let $A^\eta_m \subset K_m$ be a nonempty compact active locus from Theorem 9.12. For each $z \in A^\eta_m$, choose centers $z_n \in A^{\eta'}_{n,m}$, with $\eta' < \eta$, converging to $z$ in the Hausdorff–Fell realization and extract the corresponding locally compact retained velocity profile whenever the local-window/ancient-profile compactness hypotheses of Corollary 9.11 hold. Declare two realized centers preliminarily related if their realizing recentering sequences have bounded covariant parabolic distance and converge, after the admissible pressure gauges and after passing to a common subsequence, to the same retained terminal profile class. The non-separated local profile equivalence relation is the transitive closure of this preliminary relation. Then the compact active locus contributes to the terminal assembly in exactly one of the following priority ways.*

(i) *All realized centers in $A^\eta_m$ lie in one equivalence class; the compact active locus represents a single retained profile class.*
(ii) *There are at least two separated equivalence classes; choosing one realizing sequence from each of finitely many such classes gives a finite separated family of retained velocity concentration profiles in the sense of Definition 9.103.*
(iii) *After every finite separated subfamily has been selected, a further retained active descendant remains separated from the selected family. Iterating the realized descendant construction and using Lemma 10.11 and Theorem 10.12 produces an infinite retained concentration chain.*

*Thus a continuum active set is not a separate terminal alternative: after quotienting by non-separated local profile equivalence, it enters the terminal assembly as a single profile class, a finite separated family, or an infinite retained chain.*

*Proof.* The relation is an equivalence relation by construction. If all realized centers are equivalent, the active locus has one non-separated local profile class, proving (i).

If not, choose two inequivalent realized center sequences. Their covariant distance cannot stay bounded along a subsequence that realizes the same local profile class, because then they would be related by the preliminary relation. Thus, after passing to the realizing subsequences defining the inequivalent classes, they are separated in the sense of Definition 9.103. The same argument selects any finite number of separated inequivalent classes, giving (ii) whenever the process stops after finitely many selections.

If the process does not stop, construct inductively a sequence of retained active descendants, each separated from the finite family already chosen. The diagonal lifting lemma realizes each descendant as a recentering of the original terminal extraction, and descendant heredity keeps the profile in the same normalized retained residual class unless it has already fallen into a closed lower stratum. Removing the closed lower strata, the inductive sequence is an infinite retained concentration chain. This is (iii). □

**Definition 9.15** (Residual measure after concentration-set removal)**.** Let $U_\rho(A^\eta_{n,m})$ be the $\rho$-neighborhood of the concentration set in the compact observer window $K_m$. The residual measure after deleting positive-threshold concentration neighborhoods is

$$\mu^{\mathrm{rem}}_{n,m,\eta,\rho} := \mu_n \lfloor \big(K_m \setminus U_\rho(A^\eta_{n,m})\big)\,.$$

The terminal trichotomy uses this restricted measure: vanishing means $\mu^{\mathrm{rem}}_{n,m,\eta,\rho}(K_m) \to 0$ along the selected extraction; diffuse defect means positive residual mass remains while no unit cylinder exceeds the concentration threshold after deletion; noncompact defect means that an escaping residual recentering loses local compactness or pressure-gauge control under its own normalization.

**Definition 9.16** (Canonical countable family of expanding terminal regions)**.** The *canonical countable family* $\mathcal{E} \subset 2^Z$ is the closure under finite intersection of the family of regions of one of the following elementary types:

(i) exterior balls $\{|y| > N\}$ with $N \in \mathbb{N}$;
(ii) dyadic spatial annuli $\{2^k < |y| < 2^{k+1}\}$ with $k \in \mathbb{Z}$;
(iii) time slabs $\{|\tau - q| < \ell\}$ with $q \in \mathbb{Q}$ and $\ell \in \mathbb{Q}_{>0}$;
(iv) logarithmic annuli $\{2^k < |y - y_0| < 2^{k+1}\}$ with $y_0 \in \mathbb{Q}^3$ and $k \in \mathbb{Z}$,

together with their images under finitely many covariant observer shifts $T_a$ with $a \in \mathbb{Q}^3 \rtimes \mathbb{Q}$ and parabolic rescalings $\Theta_s$ with $s \in \mathbb{Q}$. Equivalently, $\mathcal{E}$ is the set of all finite intersections of the rationally parametrized images of (i)–(iv). This family is countable.

Every concrete exterior or tail window used in the residual proof is of one of the following two forms: (a) an element of $\mathcal{E}$ up to covariant action; (b) a multiplicative or additive scale-shift of an element of $\mathcal{E}$ by an index-dependent parameter $R_n \to \infty$ or $R_n \downarrow 0$. In case (a) the inclusion in $\mathcal{E}$ is direct; in case (b) the relevant scale-shifted region is one of the canonical dyadic annuli or logarithmic annuli of $\mathcal{E}$ after substitution $R_n \mapsto 2^{k(n)}$ for an integer index $k(n)$ chosen within a bounded distance from $\log_2 R_n$, at the cost of an absolute multiplicative loss in the captured mass.

For convenience we write $E^{(j)}_n = g^{(j)}_n \cdot E^{(j)}$, where $E^{(j)} \in \mathcal{E}$ and $g^{(j)}_n$ is the index-dependent covariant shift or scale change selecting the $j$-th window at time $n$.

**Theorem 9.17** (Compact-window to expanding-region terminal protocol)**.** *Let $K_m \uparrow Z$ be the compact observer exhaustion used in Theorem 9.12, and let $\mu_n$ be the raw terminal defect measures. Let $\mathcal{E}$ be the canonical countable family of expanding terminal regions of Definition 9.16, indexed by $j \in \mathbb{N}$, and write $E^{(j)}_n \subset Z$ for its members along the extraction. After one diagonal subsequence, for every $j$, every rational $\eta > 0$, and every rational $\rho > 0$, and for every CKN threshold $\eta \le \varepsilon_{\mathrm{CKN}}/2$, exactly one of the following* priority alternatives *occurs in the order listed* *for the concentration-deleted measures*

$$\mu_n \lfloor \big(E_n^{(j)} \setminus U_\rho(A_{n,m}^\eta)\big).$$

(a) Expanding-region vanishing. *For every $m$,*

$$\mu_n\big(E_n^{(j)} \cap K_m \setminus U_\rho(A_{n,m}^\eta)\big) \longrightarrow 0,$$

*and the remaining mass outside $K_m$ tends to zero as $m \to \infty$ along the selected protocol.*

(b) Compact subthreshold regular residual. *A positive amount of residual mass remains in some fixed compact observer window $K_m$ after deleting $U_\rho(A_{n,m}^\eta)$, every unit cylinder inside the residual region has CKN activity $\mathcal{E}_1 < \eta$, pressure gauges are compact, and no escape outside $K_m$ takes place.*

(c) Boundary diffuse defect. *A positive amount of residual mass leaves every fixed $K_m$; normalized restrictions converge on $\overline{Z}$ to a nonzero probability measure supported on $\overline{Z} \setminus Z$.*

(d) Noncompact or pressure-gauge defect. *There exists a recentering sequence $z_n \in E_n^{(j)}$ carrying positive velocity or atlas CKN mass for which the local Seregin compactness or pressure-gauge compactness required in Definition 9.3 fails.*

(e) Critical-tail defect. *No unit observer carries a retained velocity threshold and no compactness failure occurs, but there are scales $R_n \to \infty$ or $R_n \downarrow 0$, according to the observer convention, for which the normalized residual measure on the associated annular/logarithmic windows has nonzero limiting mass. This is the critical-tail compactification input of Definition 9.37.*

*The priority order is:* (a) *preempts the others; failing* (a)*, the alternatives* (b)–(e) *are intrinsically distinct after deleting unit cylinders above the CKN threshold, with* (b) *being the only compact-window alternative in which no unit cylinder has CKN activity above $\eta$. By Lemma 9.18, alternative* (b) *carries no retained terminal velocity concentration and is removed from the residual decomposition before any later use; the terminal stratification theorem Theorem 10.25 therefore lists only the alternatives* (c)–(e) *together with the four compact retained alternatives proved later.*

*Proof.* The extraction in Theorem 9.12 is already diagonal over the countable pairs $(m, \eta)$. Refine it over the countable selected regions $E_n^{(j)}$ and rational deletion radii $\rho$; since $\mathcal{E}$ is itself countable by Definition 9.16, the resulting diagonal extraction is a countable refinement. If alternative (a) fails, then for some $j, \eta, \rho$ either a positive amount of mass remains in some fixed compact window after deleting $U_\rho(A_{n,m}^\eta)$, or positive mass escapes every fixed compact window.

In the first case, if a unit observer inside the residual set has velocity activity above the velocity threshold, it is an active velocity recentering and its observer cylinder is raw-measure active by Lemma 9.21; but every raw-measure active observer at threshold $\eta$ has been deleted by construction, so this case does not occur after deletion. If the local compactness or pressure gauges fail along the selected observer sequence, this is (d). If compactness holds but the lower bound is pressure-only, Lemma 9.5 places it either in (d), to the affine/parasitic quotient, or to a neighboring velocity observer. The remaining sub-alternative is (b): every unit cylinder in $K_m \setminus U_\rho(A_{n,m}^\eta)$ has CKN activity $\mathcal{E}_1 < \eta$, pressure gauges are compact, and the residual mass is contained in $K_m$ for some fixed $m$ (no escape).

In the second case, normalize the residual restrictions outside $K_m$ and use weak compactness of probability measures on $\overline{Z}$. The limit has no mass on any compact subset of $Z$, hence is supported on the boundary and gives (c). Finally suppose that the residual mass is invisible at every fixed unit observer but is not lost in the compactification. Cover the selected tail region by dyadic annular/logarithmic windows $\mathcal{A}_{n,k}$ with bounded overlap, ordered by increasing $|k|$ and then by sign. Choose $k(n)$ to be the first index for which

$$\mu_n\big(\mathcal{A}_{n,k(n)} \setminus U_\rho(A^\eta_{n,m})\big) \geq \frac{1}{2} \sup_k \mu_n\big(\mathcal{A}_{n,k} \setminus U_\rho(A^\eta_{n,m})\big) > 0.$$

Normalize the residual measure on this selected window. By Lemma 9.19, after passing to a further subsequence either the normalized log-radius measures have a nonzero vague limit on the logarithmic compactification (case (e)), or a fixed multiplicative annulus captures a positive fraction of the mass at macroscopic scales $R_n$, again giving case (e) via the macroscopic critical-tail input of Definition 9.37. This selection is local in the chosen windows and uses no global mass budget. □

**Lemma 9.18** (Compact subthreshold regular residual cannot carry retained velocity concentration)**.** *Let the residual measure $\mu^{\mathrm{rem}}_{n,m,\eta,\rho}$ of Definition 9.15 fall into alternative* (b) *of Theorem 9.17 at the threshold $\eta \leq \varepsilon_{\mathrm{CKN}}/2$, where $\varepsilon_{\mathrm{CKN}}$ is the CKN smallness constant of Theorem 1.3 (1). Then the corresponding portion of the residual measure is supported on a relatively compact set of $\varepsilon_{\mathrm{CKN}}$-regular points of the underlying suitable solutions, and the following two conclusions hold.*

(i) Local regularity. *The compact-window residual mass is carried by velocity fields which are uniformly bounded and locally smooth on a slight enlargement of the selected residual region; in particular it cannot accumulate retained terminal velocity concentration at any compact observer.*

(ii) Disjointness from singular cores. *After alternative* (b) *has been recorded and removed, no surviving residual measure with retained velocity concentration intersects the same compact region. Equivalently, alternative* (b) *cannot coexist with a retained singular concentration core inside the same compact observer window after the CKN threshold has been chosen below $\varepsilon_{\mathrm{CKN}}$.*

*Proof.* Fix the compact window $K_m$ and the deletion radius $\rho$ provided by the alternative. Let $R_m = \mathrm{dist}(K_m, \partial K_{m+1}) > 0$. Cover $K_m$ by a finite collection of unit covariant cylinders $\mathcal{Q}^{\mathrm{cov}}_1(z_i)$, $i = 1, \dots, N_m$, arranged so that the slightly enlarged cylinders $\mathcal{Q}^{\mathrm{cov}}_{1+\delta}(z_i)$ with $\delta \in (0, R_m/4)$ still lie in $K_{m+1} \setminus U_{\rho/2}(A^\eta_{n,m})$ for all sufficiently large $n$, using that the CKN-threshold locus has been deleted at radius $\rho$ and the cylinders are chosen at distance $\rho/2$ from it.

By the compact subthreshold hypothesis, every such cylinder satisfies

$$\mathcal{E}_1(U_n, \Pi_n; z_i) < \eta \leq \varepsilon_{\mathrm{CKN}}/2.$$

The pressure gauge attached to $(U_n, \Pi_n)$ on $K_{m+1}$ is compact in the relevant atlas (Definition 9.3). By the imported $\varepsilon$-regularity criterion of Caffarelli–Kohn–Nirenberg, recorded in Theorem 1.3 (1), each cylinder $\mathcal{Q}^{\mathrm{cov}}_1(z_i)$ is $\varepsilon_{\mathrm{CKN}}$-regular for the underlying suitable solution $(U_n, \Pi_n)$ along the diagonal extraction: the velocity is bounded by an absolute constant on $\mathcal{Q}^{\mathrm{cov}}_{1-\delta}(z_i)$ and is locally smooth there. Taking the finite union over $i$ and over the at most countably many windows in alternative (b) gives a uniformly bounded smooth representative of the velocity on a slight enlargement of the residual support, proving (i).

For (ii), a retained singular concentration core inside the same compact window has, by definition of the CKN threshold below $\varepsilon_{\mathrm{CKN}}$, at least one unit cylinder with CKN activity exceeding $\eta$. All such cylinders are removed by passing to the concentration-deleted measure at radius $\rho$. The compact subthreshold alternative is recorded only on the complement of these cylinders. Hence the compact subthreshold residual and any retained singular core are disjointly supported on the compact window after deletion. In particular the compact subthreshold residual carries no retained terminal velocity concentration, and is removed from the residual decomposition before the terminal stratification theorem Theorem 10.25 is applied. □

**Lemma 9.19** (Annular tight/log-escape dichotomy)**.** *Let $\lambda_n$ be normalized residual probability measures on dyadic annuli or logarithmic windows in escaping tail regions, with $\lambda_n(\mathbb{R}^3 \setminus \{0\}) = 1$ and supports tending to infinity or to zero in the radial sense. Then after passing to a subsequence exactly one of the following alternatives occurs.*

(1) Macroscopic annular tightness. *There exist macroscopic scales $R_n$ (with $R_n \to \infty$ or $R_n \downarrow 0$ according to the observer convention) and constants $0 < a < b < \infty$ and $\delta > 0$ such that*
$$\lambda_n\big(\{aR_n < |x| < bR_n\}\big) \geq \delta \qquad \textit{for all sufficiently large } n.$$
(2) Logarithmic escape. *The pushforwards $\Lambda_n := (\log|\cdot|)_*\lambda_n$ on $\mathbb{R}$ have a nonzero vague limit $\Lambda$ on the two-point compactification $\overline{\mathbb{R}}_{\log} = [-\infty, +\infty]$, with $\Lambda$ charging at least one of the points $\pm\infty$ or having no exponentially decaying mass on every fixed bounded log-window.*

*Proof.* For each $n$ and each $L > 0$, set
$$\alpha_n(L) := \lambda_n\big(\{e^{-L} \leq |x| \leq e^L\}\big) = \Lambda_n([-L, L]).$$
Since $\alpha_n(L)$ is non-decreasing in $L$ and bounded above by 1, the limits $\alpha(L) := \liminf_n \alpha_n(L)$ and $\bar{\alpha}(L) := \limsup_n \alpha_n(L)$ exist and are non-decreasing in $L$. Pass to a subsequence so that $\alpha_n(L) \to \alpha_\infty(L)$ for every rational $L$, with $\alpha_\infty$ non-decreasing and bounded by 1; set
$$m := \sup_L \alpha_\infty(L) = \lim_{L\to\infty} \alpha_\infty(L) \in [0, 1].$$
If $m < 1$, a positive amount $1 - m$ of mass escapes every fixed bounded log-window in the limit; hence $\Lambda_n$ restricted to $\overline{\mathbb{R}}_{\log}$ has weak-$*$ limit points charging at least one of $\pm\infty$, and we are in alternative (2).

If $m = 1$ but $\alpha_\infty$ is not eventually constant on every bounded window, the same conclusion holds in the form "no exponentially decaying mass on every fixed bounded log-window," i.e. (2).

Otherwise $m = 1$ and $\alpha_\infty$ is eventually constant: there is $L_0 > 0$ with $\alpha_\infty(L_0) = 1$, so the $\lambda_n$ are tight in the log-radius variable. Cover $[-L_0, L_0]$ by finitely many sub-intervals $I_k = [\log a_k, \log b_k]$ of length $\leq 1$ with $0 < a_k < b_k < \infty$. At least one of these intervals carries mass at least $1/(2N)$ along a further subsequence, where $N$ is the number of intervals; that is,
$$\lambda_n\big(\{a_k < |x| < b_k\}\big) \geq \frac{1}{2N}$$
for all sufficiently large $n$. Since the original measures live on escaping windows, the supports translate by index-dependent macroscopic scales $R_n$; after factoring out this translation, the captured fixed annulus is $\{a_kR_n < |x| < b_kR_n\}$, giving alternative (1). $\square$

*Remark* 9.20 (How the dichotomy enters the protocol)*.* The dichotomy Lemma 9.19 is invoked exactly at the last step of the proof of Theorem 9.17: the normalized residual measure on the selected dyadic annular window is either macroscopically annular tight, in which case it gives the macroscopic critical-tail input of Definition 9.37, or it has nonzero logarithmic escape, in which case it yields the log-defect measure $\Lambda$ used in the critical-tail compactification of Definition 9.37. In both cases alternative (e) of the protocol is realized.

**Lemma 9.21** (Oscillatory observers are raw-measure active)**.** *Let $\mu_n$ be the raw atlas defect measure used in Theorem 9.12, so that $\mu_n$ dominates the velocity measure $|U_n|^3\,dy\,d\tau$. If a covariant unit observer $z$ satisfies*
$$\mathrm{osc}_{3,\mathrm{ct}}(U_n; \mathcal{Q}_1^{\mathrm{cov}}(z)) \geq \eta,$$
*then the same observer is raw-measure active at threshold $\eta$:*
$$\mu_n(\mathcal{Q}_1^{\mathrm{cov}}(z)) \geq \eta.$$

*Consequently, after the raw atlas concentration sets of Theorem 9.12 are removed at every positive threshold, no positive constant-in-time oscillatory unit observer remains outside those sets.*

*Proof.* By definition of the constant-in-time oscillation,

$$\operatorname{osc}_{3,\mathrm{ct}}(U_n; \mathcal{Q}_1^{\mathrm{cov}}(z)) = \inf_{c\in\mathbb{R}^3} \iint_{\mathcal{Q}_1^{\mathrm{cov}}(z)} |U_n - c|^3 \le \iint_{\mathcal{Q}_1^{\mathrm{cov}}(z)} |U_n|^3.$$

The raw atlas defect measure contains the velocity contribution, hence

$$\iint_{\mathcal{Q}_1^{\mathrm{cov}}(z)} |U_n|^3 \le \mu_n(\mathcal{Q}_1^{\mathrm{cov}}(z)).$$

Thus oscillation implies raw-measure activity with the same threshold. The last statement follows by applying the local concentration-set extraction theorem to all rational positive thresholds and then lowering thresholds by harmless fixed cover and cutoff constants when the smoothed bump convention is used. □

**Lemma 9.22** (Spatial quotient defect measures are locally CKN-controlled)**.** *On the same bounded-overlap covariant unit-cylinder cover, define the spatial quotient oscillation measure*

$$\nu_U^{\mathrm{sp}}(E) := \sum_{z_k\in E} \operatorname{osc}_{3,\mathrm{sp}}\big(U; \mathcal{Q}_1^{\mathrm{cov}}(z_k)\big).$$

*Then $\nu_U^{\mathrm{sp}}$ is locally finite on compact observer windows and is dominated, up to the fixed cover multiplicity, by the local velocity part of the defect measure. In particular, every compactness argument in Theorem 9.29 applies without change to $\nu_U^{\mathrm{sp}}$ after concentration-set removal.*

*Proof.* For each unit covariant cylinder,

$$\operatorname{osc}_{3,\mathrm{sp}}\big(U; \mathcal{Q}_1^{\mathrm{cov}}(z_k)\big) \le \iint_{\mathcal{Q}_1^{\mathrm{cov}}(z_k)} |U|^3.$$

Summing over a bounded-overlap cover gives local finiteness and domination by the local velocity contribution, hence by the raw atlas defect measure. The proof of Theorem 9.29 uses only positivity, local finiteness, bounded-overlap domination, and weak compactness of normalized restrictions on the observer compactification. These properties hold for $\nu_U^{\mathrm{sp}}$ exactly as they hold for the constant-in-time oscillation measure $\mu_U^{\mathrm{osc}}$. □

**Lemma 9.23** (Separation of recentered cylinders)**.** *Let $\mathfrak{c}_i = (z_{i,n})$ and $\mathfrak{c}_j = (z_{j,n})$ be asymptotically separated recentering sequences. For every fixed $R < \infty$, the cylinder $Q_R(z_{j,n})$, written in the coordinates of $\mathfrak{c}_i$, leaves every compact subset of the recentered variables. Consequently no positive compact velocity $L^3$ concentration assigned to $\mathfrak{c}_j$ is visible on any fixed compact cylinder of the $\mathfrak{c}_i$-frame.*

*Proof.* In the $\mathfrak{c}_i$-coordinates the center of $Q_R(z_{j,n})$ is

$$\big(y_{j,n} - y_{i,n},\ \tau_{j,n} - \tau_{i,n}\big).$$

The parabolic distance of this point from the origin tends to infinity by asymptotic separation. Hence, for every fixed compact terminal cylinder $K$, one has

$$K \cap \big(Q_R(z_{j,n}) - z_{i,n}\big) = \varnothing$$

for all sufficiently large $n$. This is a purely local statement about the relative positions of the two recentering sequences. It does not estimate the solution on all of space and does not use a global norm. □

**Theorem 9.24** (Paired terminal concentration-profile decomposition)**.** *Every terminal concentration sequence admits a paired decomposition*

$$\textit{retained velocity concentration set} \quad + \quad \textit{residual remainder.}$$

*More precisely, after passing to a subsequence, the compact observer-window concentration sets $A_m^\eta$ of Theorem 9.12 exist simultaneously for every $m$ and every rational $\eta > 0$. Every observer sequence that remains in these sets at a fixed positive velocity threshold and has no loss of local compactness generates a bounded retained terminal profile after local pressure gauges. Outside every neighborhood of the concentration set in each fixed compact observer window, the residual raw unit-window mass is below the chosen threshold in the precise concentration-deleted sense of $\mu^{\rm rem}_{n,m,\eta,\rho}$ from Definition 9.15. Applying the compact-window to expanding-region protocol Theorem 9.17, the compact subthreshold regular residual is removed first by Lemma 9.18. The surviving residual remainder is then of one of the following priority types:*

$$\mathcal{R}_{\rm van}, \qquad \mathcal{R}_{\rm diff}, \qquad \mathcal{R}_{\rm noncomp}, \qquad \mathcal{R}_{\rm crit}.$$

*Here $\mathcal{R}_{\rm van}$ means the concentration-deleted residual measures vanish on compact observer windows and then on the selected expanding regions; $\mathcal{R}_{\rm diff}$ means positive concentration-deleted residual mass persists at the boundary of the observer compactification while every fixed unit cylinder is below threshold; and $\mathcal{R}_{\rm noncomp}$ means that some escaping residual recentering loses the local Seregin compactness or pressure-gauge control; and $\mathcal{R}_{\rm crit}$ means that residual mass is first visible only after annular/logarithmic critical-tail normalization. Thus mixed configurations such as one compact concentration profile plus diffuse exterior remainder are part of the decomposition; they are not silently identified with the pure diffuse stratum.*

*Proof.* Apply the local concentration-set extraction theorem, Theorem 9.12, on the compact observer exhaustion $\{K_m\}$. It produces closed concentration sets $A_m^\eta$ and the residual estimate

$$\limsup_{n\to\infty} \sup_{z\in K_m\setminus O} M_n(z) \le \eta$$

outside every neighborhood $O$ of $A_m^\eta$. This is the precise replacement for a maximal finite or countable tube family: concentrating centers are recorded by closed loci, and the residual is defined by restricting away from their neighborhoods.

If an observer sequence stays in a positive-threshold concentration set and has no loss of the local compactness specified in Definition 9.3, then Corollary 9.11 gives a bounded retained terminal profile: the diagonal compactness is supplied there either by the first-generation local-window compactness lemma or by covariant compactness for bounded ancient profiles. If such an observer sequence loses local dissipation or pressure-gauge control, the corresponding remainder is $\mathcal{R}_{\rm noncomp}$.

It remains to classify what is outside the concentration sets. Apply the compact-window-to-expanding-region protocol Theorem 9.17 to the selected exterior and tail regions used in the terminal extraction. Its vanishing alternative is $\mathcal{R}_{\rm van}$. Its boundary-measure alternative is $\mathcal{R}_{\rm diff}$. Its compactness or pressure-gauge failure alternative is $\mathcal{R}_{\rm noncomp}$, while its critical-scale alternative is recorded for the later critical-tail branch. These residual alternatives are mutually exclusive once the concentration sets and selected protocol are fixed, and they exhaust the residual possibilities. No step asserts that a diffuse or exterior remainder cannot coexist with compact concentration profiles. ☐

**Corollary 9.25** (Persistent local velocity concentration forces retained velocity concentration)**.** *If a terminal sequence satisfies*

$$\liminf_{n\to\infty} \iint_{Q_1(0,0)} |V_n|^3 \, dy \, d\tau \ge \eta_0 > 0$$

*with $\eta_0 \geq \varepsilon_{\rm sd}$, then the terminal decomposition contains a local concentration profile unless the distinguished observer sequence has loss of local compactness. In the generic residual after local vanishing and local compactness failure are excluded, the persistent compact velocity concentration is therefore carried by the retained velocity concentration set of Theorem 9.24; the locus may contain one, finitely many, infinitely many, or continuum many concentrating observer classes, and the residual remainder may still be diffuse until the mixed compact–diffuse stratum is excluded.*

*Proof.* The raw atlas defect measure $\mu_n$ dominates the velocity contribution. Hence the origin recentering sequence has $\mu$-mass at least $\eta_0$, and is raw unit-window active at threshold $\varepsilon_{\rm sd}$ unless it has loss of local compactness. If it has no loss of local compactness, concentration-set extraction places its observer class in a positive-threshold concentration set $A^\eta_m$ of Theorem 9.24. □

**Lemma 9.26** (Local vanishing and exterior alternatives do not carry compact concentration)**.** *Let a terminal extraction inherit the positive compact velocity $L^3$ concentration supplied by the imported setup results. Then the retained compact velocity concentration cannot be carried solely by local vanishing, exterior escape, or diffuse exterior concentration.*

*Proof.* The assertion is local. By the local no-escape and concentration input in the imported setup results, failure of local vanishing at a singular terminal point gives a positive local velocity concentration sequence. The setup paper carries this through the Type I Seregin extraction as the compact velocity lower bound (1.5). Cover the fixed compact cylinder $K$ in (1.5) by finitely many unit terminal cylinders. At least one cylinder in the cover carries a positive velocity lower bound, say $\eta_K > 0$. Since $\mu_n$ dominates $|V_n|^3\,dy\,d\tau$, choosing $\varepsilon_{\rm sd} \leq \eta_K$ makes that unit cylinder raw unit-window active at threshold $\varepsilon_{\rm sd}$. Thus the distinguished compact concentration cannot be locally vanishing.

Local vanishing after removal of raw unit-window active recentering sequences means that the residual profile has raw unit-window mass below $\varepsilon_{\rm sd}$ on each fixed compact terminal cylinder; it therefore cannot contain the fixed compact velocity lower bound. An exterior escaping concentration is visible only after its own recentering; relative to the distinguished compact terminal cylinder it escapes every bounded window. By Lemma 9.23, it cannot be the sole location of a lower bound that remains in the distinguished compact window. Finally, diffuse exterior concentration is defined by persistence through expanding regions while all fixed unit cylinders are locally vanishing. If such a sequence contained the compact lower-bound threshold from the imported setup results, the positive concentration input in that setting would select a unit terminal cylinder with mass at least the velocity threshold, contradicting local vanishing. Thus none of these alternatives can be the unique location of the retained compact velocity concentration. □

**Theorem 9.27** (Local exclusion of noncompact and nonconcentrating terminal alternatives)**.** *For terminal sequences generated by the Seregin setup theorem in the companion setup paper, loss of local compactness, local vanishing, exterior escape, and diffuse exterior concentration are not possible sole locations of the retained compact velocity concentration.*

*Proof.* Retained recentering sequences with loss of local compactness are removed by Corollary 9.11. Local vanishing, exterior escape, and diffuse exterior concentration are removed by Lemma 9.26. The proof uses only the positive concentration input and the local compactness hypotheses in the imported setup results, and fixed compact terminal cylinders. □

**Definition 9.28** (Observer compactification)**.** We fix a metrizable compactification $\overline{Z}$ of the observer space $Z = \mathbb{R}^3 \times \mathbb{R}$ with the following properties.

(i) Compact observer balls of finite covariant radius embed continuously.
(ii) The covariant observer maps generated by $T_a$ and $\Theta_s$ extend continuously to $\overline{Z}$.

(iii) Unit-cylinder concentration evaluation maps are upper semicontinuous after the fixed smoothing convention of Theorem 9.12.
(iv) The normalized restrictions of locally finite defect measures are tight as probability measures on $\overline{Z}$.

Such a compactification is obtained by embedding $Z$ into the product of compact intervals generated by a countable separating family of bounded continuous observer functions, including the smoothed unit-cylinder evaluation functions and the coordinate cutoffs for a compact exhaustion, and then taking the closure.

**Theorem 9.29** (Diffuse-defect compactness). *Let $\mu_n^{\rm osc}$ be the residual constant-in-time oscillation measures of Definition 8.8, after removing neighborhoods of all positive-threshold concentration sets from Theorem 9.12. By Lemma 9.21, this removal also removes all positive-threshold oscillatory unit observers on compact windows. Assume there are observer regions $E_n \subset Z$, escaping every compact subset of $Z$, and a number $m_0 > 0$ such that*

$$\mu_n^{\rm osc}(E_n) \geq m_0,$$

*while no unit observer in $E_n$ carries a positive oscillation threshold:*

$$\sup_{z \in E_n} \operatorname{osc}_{3,\rm ct}(U_n; \mathcal{Q}_1^{\rm cov}(z)) \longrightarrow 0.$$

*Let $\overline{Z}$ be the compact metrizable observer compactification of Definition 9.28, and define probability measures*

$$\lambda_n := \frac{\mu_n^{\rm osc} \lfloor E_n}{\mu_n^{\rm osc}(E_n)} \qquad \text{on } \overline{Z}.$$

*After passing to a subsequence,*

$$\lambda_n \rightharpoonup \lambda$$

*weakly as probability measures on $\overline{Z}$. The limit satisfies*

$$\lambda(\overline{Z}) = 1, \qquad \lambda(B) = 0 \quad \text{for every } B \Subset Z.$$

*Thus the limit is a nonzero diffuse-defect profile supported at the boundary of the observer compactification, not a local concentration profile.*

*Proof.* The measures $\lambda_n$ are probability measures on the compact metric space $\overline{Z}$, so weak compactness gives a convergent subsequence. Testing against the constant function 1 preserves total mass and gives $\lambda(\overline{Z}) = 1$.

Let $B \Subset Z$. Since $E_n$ escapes every compact subset of $Z$, one has $B \cap E_n = \varnothing$ for all sufficiently large $n$. Hence $\lambda_n(B) = 0$ eventually, and the portmanteau theorem gives $\lambda(B) = 0$. The additional unit-scale diffuseness assumption gives the stronger moving-window estimate

$$\sup_{z \in E_n} \lambda_n\big(B_Z(z,1)\big) \leq \frac{C}{m_0} \sup_{\zeta \in E_n} \operatorname{osc}_{3,\rm ct}(U_n; \mathcal{Q}_1^{\rm cov}(\zeta)) \longrightarrow 0.$$

Here $B_Z(z,1)$ is any fixed unit ball for a proper metric on $Z$, and $C$ is the bounded-overlap constant of the observer cover. The restriction $\lambda_n = \mu_n^{\rm osc} \lfloor E_n / \mu_n^{\rm osc}(E_n)$ is why only cover centers lying in $E_n$ enter the estimate. Therefore no positive unit-scale local oscillation profile is hidden inside the escaping windows. The proof is local on the chosen observer windows and uses no global critical-mass or energy budget. □

**Lemma 9.30** (Support transfer for the diffuse-defect compactification). *Let $X_{\rm def}$ be the compact metrizable diffuse-defect space of Theorem 9.32, and let*

$$\lambda_n = \frac{\mu_n^{\rm osc} \lfloor E_n}{\mu_n^{\rm osc}(E_n)}$$

*be the normalized realized diffuse-defect measures from Theorem 9.29, viewed in $X_{\mathrm{def}}$. If $\lambda_n \rightharpoonup \lambda$ and $A \subset X_{\mathrm{def}}$ is a closed union of already excluded diffuse alternatives, then support of $\lambda$ in $A$ implies that no diffuse defect mass survives outside $A$. Conversely, surviving normalized diffuse mass outside $A$ produces a sequence-realized support profile in $X_{\mathrm{def}} \setminus A$.*

*Proof.* This is Lemma 1.12 with $X = X_{\mathrm{def}}$, $\mu_n = \mu_n^{\mathrm{osc}}$, and the escaping observer regions $E_n$. The construction theorem makes $X_{\mathrm{def}}$ compact metrizable and keeps the limit inside the closed sequence-realized subset, so the support point obtained outside $A$ is a realized diffuse-defect profile. □

**Definition 9.31** (Defect observables)**.** Let $M \geq 1$ be the velocity bound of the underlying admissible class. Fix the following six countable families of bounded continuous observables, each defined on diffuse-defect candidates produced by Theorem 9.29.

(1) *Observer test functions.* A countable dense family $\{\varphi_j\}_{j\in\mathbb{N}} \subset C(\overline{Z})$, separating probability measures on the compact metrizable observer compactification $\overline{Z}$ of Definition 9.28.
(2) *Smoothed unit-cylinder activity observables.* For each rational $z \in Z$ and each $\eta \in \mathbb{Q}_{>0}$, the smooth bounded functional $\Psi_{z,\eta}$ recording the convolved CKN activity $\mathcal{E}_1$ of a defect representative on the unit covariant cylinder $\mathcal{Q}_1^{\mathrm{cov}}(z)$, thresholded by a smooth cutoff at $\eta$. This family is countable.
(3) *Compact-exhaustion cutoff observables.* For each $m \in \mathbb{N}$, a smooth cutoff $\zeta_m$ supported in $K_{m+1}$ and equal to 1 on $K_m$, recording the mass that the defect measure places on the compact exhaustion of Theorem 9.12.
(4) *Local velocity observables.* For each compact cylinder $K_m$ and each finite list $F_1, \dots, F_N$ of polynomials of total degree $\leq m$ in the variables $W, \nabla W, \dots, \nabla^m W$, the localized observable $\int F_j(W, \nabla W, \dots)\zeta_m \, dy \, d\tau$.
(5) *Pressure-gradient observables modulo gauges.* For each compactly supported divergence-free test field $\Psi \in C_c^\infty(K_m, \mathbb{R}^3)$ with rational coefficients in a countable dense subspace, the bounded observable
$$\mathcal{P}_\Psi(W, P) = \langle \nabla P, \Psi \rangle,$$
computed for representatives modulo the time-only and affine pressure gauges of Definition 8.14.
(6) *Log-annular observables.* For each rational $\rho_0$ and $A > 0$, the cutoffs $\chi_{\rho_0, A}$ of Definition 9.40 applied to the log-radial pushforward of any defect representative.

The combined family is countable; let $\{\mathcal{O}_k\}_{k\in\mathbb{N}}$ enumerate it after fixing a uniform bound $\|\mathcal{O}_k\|_\infty \leq C_k$.

**Theorem 9.32** (Construction of the diffuse-defect compactification)**.** *There exists a compact metrizable space $X_{\mathrm{def}}$, a continuous action $\sigma \colon G \times X_{\mathrm{def}} \to X_{\mathrm{def}}$ of $G = \mathbb{R}^3 \rtimes \mathbb{R}$, and a closed $G$-invariant subset*
$$X_{\mathrm{def}}^{\mathrm{anc}} \subset X_{\mathrm{def}},$$
*such that every normalized diffuse-defect sequence generated by the original terminal sequence has a subsequential limit in $X_{\mathrm{def}}^{\mathrm{anc}}$.*

*Proof.* Let $\{\mathcal{O}_k\}_{k\in\mathbb{N}}$ be the observable family of Definition 9.31 with uniform bound $\|\mathcal{O}_k\|_\infty \leq C_k$. Define the embedding
$$\iota \colon \omega \longmapsto \big(\mathcal{O}_k(\omega)\big)_{k\in\mathbb{N}} \in \prod_{k\in\mathbb{N}} [-C_k, C_k],$$
where $\omega$ ranges over the diffuse-defect candidates produced by Theorem 9.29. By Tychonoff's theorem, the product space
$$\mathcal{X}_0 := \prod_{k\in\mathbb{N}} [-C_k, C_k]$$

is compact and metrizable (the metric $d(x,y)=\sum_k 2^{-k}\frac{|x_k-y_k|}{1+|x_k-y_k|}$ generates the product topology). Set

$$X_{\mathrm{def}} := \overline{\iota(\text{all defect candidates})} \subset \mathcal{X}_0.$$

$X_{\mathrm{def}}$ is compact metrizable as a closed subset of a compact metrizable space.

The covariant action: for each $g=(a,s)\in G$, the pullback of every observable $\mathcal{O}_k$ by the covariant observer map $T_a Z_s$ is again an observable in the chosen family (Definition 9.31 is closed under the rational $G$-action and is dense enough that the closure under $G$ lies in the same product space after a permutation of coordinates). Hence the map $\sigma_g\colon \iota(\omega)\mapsto\iota(g\cdot\omega)$ extends continuously to $X_{\mathrm{def}}$, and $(g,x)\mapsto\sigma_g(x)$ is jointly continuous because the rational orbit on the countable family of observables is dense in the natural $G$-orbit map and convergence in $X_{\mathrm{def}}$ is coordinatewise.

The sequence-realized class:

$$X_{\mathrm{def}}^{\mathrm{anc}} := \overline{\iota(\text{sequence-realized defect profiles})} \subset X_{\mathrm{def}},$$

where "sequence-realized" means realized from the original blow-up sequence in the sense of Definition 1.9. This gives a closed subset of the compact product, and the diagonal argument below verifies that this closure is still represented by limits of realized states rather than by an abstract compactification artifact.

It remains to check $G$-invariance of $X_{\mathrm{def}}^{\mathrm{anc}}$ and the diagonal-closure property. $G$-invariance follows because the covariant observer maps belong to the allowed realized operations. For the diagonal-closure property, suppose $\omega_j\in X_{\mathrm{def}}^{\mathrm{anc}}$ and $\omega_j\to\omega$ in $X_{\mathrm{def}}$. By definition of the closure each $\omega_j$ is a limit point of a sequence $\iota(\omega_j^{(n)})$ with $\omega_j^{(n)}$ realized from the original blow-up sequence. Choose indices $n=n(j)$ so large that $d(\iota(\omega_j),\iota(\omega_j^{(n(j))}))<2^{-j}$. Then $\iota(\omega_j^{(n(j))})\to\iota(\omega)$, and the $\omega_j^{(n(j))}$ are realized. Hence $\omega$ is in the closure of the realized states, i.e. $\omega\in X_{\mathrm{def}}^{\mathrm{anc}}$.

Finally, every normalized diffuse-defect sequence generated by the original terminal sequence is a sequence $\omega_n$ of realized states. Their images $\iota(\omega_n)$ lie in the compact metrizable space $X_{\mathrm{def}}$ and have a convergent subsequence; the limit lies in $X_{\mathrm{def}}^{\mathrm{anc}}$ by the closure argument above. □

**Definition 9.33** (Diffuse-defect profile class)**.** The compact metrizable space $X_{\mathrm{def}}=\mathfrak{D}_M$ and its closed $G$-invariant sequence-realized subset $X_{\mathrm{def}}^{\mathrm{anc}}=\mathfrak{D}_M^{\mathrm{anc}}$ are those constructed in Theorem 9.32. The diffuse-defect mass functional is normalized to be identically one on $\mathfrak{D}_M$. The covariant observer group $G=\mathbb{R}^3\rtimes\mathbb{R}$ acts on $\mathfrak{D}_M$ continuously through the action $\sigma$ of Theorem 9.32.

**Proposition 9.34** (Amenable averaging protocol for realized diffuse defects)**.** *Let $G=\mathbb{R}^3\rtimes\mathbb{R}$ act continuously on the compact metrizable diffuse-defect compactification $X_{\mathrm{def}}=\mathfrak{D}_M$. Let $\nu_0$ be a Borel probability measure supported on the closed sequence-realized class $\mathfrak{D}_M^{\mathrm{anc}}$. Then there exists a $G$-invariant Borel probability measure $\nu$ on $X_{\mathrm{def}}$, obtained as a weak-* limit of Følner averages*

$$\nu_\alpha := \frac{1}{|F_\alpha|}\int_{F_\alpha} g_\#\nu_0\,dg,$$

*where $(F_\alpha)$ is a left Følner net in $G$. Moreover $\nu(\mathfrak{D}_M^{\mathrm{anc}})=1$, so the invariant diffuse defect remains sequence-realized.*

*Proof.* The semidirect group $G=\mathbb{R}^3\rtimes\mathbb{R}$ is solvable and hence amenable; choose a left Følner net $(F_\alpha)$ for a left Haar measure. The action map $g\mapsto g_\#\nu_0$ is weak-* continuous because the action on $\mathfrak{D}_M$ is continuous. Therefore each $\nu_\alpha$ is a Borel probability measure. Compactness of $X_{\mathrm{def}}$ implies weak-* compactness of $\mathcal{P}(X_{\mathrm{def}})$, so a subnet converges to a probability measure $\nu$.

Let $h \in G$ and let $F \in C(X_{\mathrm{def}})$. Then

$$\int F\, d(h_\# \nu_\alpha) - \int F\, d\nu_\alpha = \frac{1}{|F_\alpha|} \int_{F_\alpha} \left[ \int F(hg\omega)\, d\nu_0(\omega) - \int F(g\omega)\, d\nu_0(\omega) \right] dg.$$

The absolute value is bounded by

$$2\|F\|_{L^\infty} \frac{|hF_\alpha \triangle F_\alpha|}{|F_\alpha|},$$

which tends to zero by the Følner property. Passing to the weak-* limit gives $h_\# \nu = \nu$. Since $\mathfrak{D}_M^{\mathrm{anc}}$ is closed and $G$-invariant, each averaged measure $\nu_\alpha$ is supported on $\mathfrak{D}_M^{\mathrm{anc}}$. Weak-* limits preserve support on closed sets, so $\nu(\mathfrak{D}_M^{\mathrm{anc}}) = 1$. □

**Theorem 9.35** (Recurrent diffuse-defect measure)**.** *The covariant observer action of $G$ is continuous on $\mathfrak{D}_M$. Every nonzero diffuse-defect profile $\lambda \in \mathfrak{D}_M$ generates a $G$-invariant Borel probability measure $\mathbb{P}$ on $\mathfrak{D}_M$. The normalized diffuse-defect mass remains one under this averaging.*

*Proof.* The continuity of the covariant observer action is the conclusion of the construction theorem Theorem 9.32. In that construction, the action is first defined on realized defect profiles by the covariant observer maps and then extended continuously to the compact product closure of the countable observable family. Apply Proposition 9.34 to $\nu_0 = \delta_\lambda$. The resulting invariant probability measure $\mathbb{P}$ is supported on the closed sequence-realized diffuse-defect class. Since the diffuse-defect mass is identically one on $\mathfrak{D}_M$, its average against $\mathbb{P}$ is also one. □

**Theorem 9.36** (Diffuse-defect trichotomy)**.** *Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$ be a normalized terminal residual sequence with $\|U_n\|_{L^\infty} \le M$. Suppose that, after removing the concentration sets at every positive affine-normalized threshold, there are escaping observer windows $E_n$ such that*

$$\mu_n^{\mathrm{osc}}(E_n) \ge m_0 > 0.$$

*Then, after passing to a subsequence, the residual diffuse defect falls into one of the following ordered alternatives:*

- D1. Regenerated concentration. *There exist covariant observers $z_n$ and a threshold $\eta > 0$ such that*
$$\mathrm{osc}_{3,\mathrm{ct}}(U_n; \mathcal{Q}_1^{\mathrm{cov}}(z_n)) \ge \eta.$$
*This produces a retained concentrating descendant.*
- D2. Affine/parasitic collapse. *Every velocity tangent associated with the diffuse-defect hull is locally constant after the admissible affine normalization; the defect belongs to the affine/parasitic lower stratum of Definition 8.16.*
- D3. Critical diffuse tail. *There is a nonzero recurrent diffuse-defect profile supported on $\overline{Z} \setminus Z$, carrying normalized diffuse oscillation mass one, annihilating every fixed unit concentration observable, and not contained in the affine/parasitic lower stratum.*

*Equivalently, residual diffuse defects are contained in the union of the regenerated concentration alternative, the affine/parasitic stratum, and the critical diffuse-tail class $\mathcal{C}_{\mathrm{critdiff}}$.*

*Proof.* If D1 occurs, the asserted trichotomy holds. Assume D1 does not occur. Then, for every fixed $\eta > 0$, all sufficiently large $n$ satisfy

$$\sup_{z \in E_n} \mathrm{osc}_{3,\mathrm{ct}}(U_n; \mathcal{Q}_1^{\mathrm{cov}}(z)) < \eta.$$

The hypotheses of Theorem 9.29 apply to $\mu_n^{\mathrm{osc}} \lfloor E_n$, and hence produce a diffuse-defect profile $\lambda$ with total mass one and zero mass on each compact subset of $Z$. By Theorem 9.35, the orbit closure of $\lambda$ carries a $G$-invariant probability measure $\mathbb{P}$.

If every local velocity tangent in the support of $\mathbb{P}$, after the canonical affine normalization, is locally constant, then Definition 8.16 places the defect in the affine lower stratum. This is D2. If not, the invariant diffuse boundary profile has positive normalized spatial quotient oscillation mass, no retained unit-scale velocity concentration observable, and non-affine velocity content. This is precisely the critical diffuse-tail alternative D3. The alternatives are ordered by first extracting all regenerated concentration and then all affine/parasitic collapse; after those removals, the remaining nonzero diffuse profile is $\mathcal{C}_{\mathrm{critdiff}}$. □

**Definition 9.37** (Critical-tail compactification)**.** The class $\mathcal{C}_{\mathrm{critdiff}}$ consists of recurrent diffuse-defect profiles produced by Theorem 9.36 which are supported on the boundary of the covariant observer compactification, carry normalized spatial quotient oscillation mass one, annihilate every fixed unit concentration observable, and are not affine/parasitic.

The critical-tail compactification $\mathfrak{T}_M$ is the compact profile class of triples

$$\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda)$$

generated by such critical diffuse profiles after macroscopic blow-down. Here $\bar{W}$ is the barycentric macroscopic velocity, $\mathcal{Y}_{x,s}$ is the Young measure recording unresolved bounded macroscopic oscillation, and $\Lambda$ is the log-scale diffuse measure recording mass escaping every fixed multiplicative annulus. The coherent subspace $\mathfrak{T}_{\mathrm{coh}}$ is the special case

$$\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}, \qquad \Lambda = 0.$$

The purely log-diffuse subspace is denoted $\mathfrak{T}_{\log\mathrm{diff}}$, and the oscillatory Young-tail subspace is denoted $\mathfrak{T}_{\mathrm{Young}}$. A sequence

$$\mathcal{T}_n = (\bar{W}_n, \mathcal{Y}_n, \Lambda_n)$$

with pressure representatives $Q_n$ in the time/affine quotient converges to $\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda)$, with pressure representative $Q$, in $\mathfrak{T}_M$ if:

(i) $\bar{W}_n \rightharpoonup^* \bar{W}$ in $L^\infty_{\mathrm{loc}}$ on macroscopic annular cylinders;

(ii) the Young measures converge against every bounded continuous velocity observable with compact support in the macroscopic variables: for every $\psi \in C_c^\infty$ supported in a macroscopic annular cylinder and every $F \in C_b(\mathbb{R}^3)$,

$$\iint \psi(x,s) \int F(\xi)\, d\mathcal{Y}_{n,x,s}(\xi)\, dx\, ds \longrightarrow \iint \psi(x,s) \int F(\xi)\, d\mathcal{Y}_{x,s}(\xi)\, dx\, ds;$$

(iii) the log-scale defect measures $\Lambda_n$ converge vaguely on the logarithmic annular compactification;

(iv) pressure gradients converge distributionally after the time-dependent gauges and affine pressure quotients fixed in Definition 8.14;

(v) the affine quotient data converge in the finite-dimensional local quotient topology; and

(vi) when the profiles are sequence-realized, the limiting profile remains in the closed sequence-realized support class generated by the original terminal sequence; equivalently, the associated support data converge in the diffuse-defect state space $X^{\mathrm{anc}}_{\mathrm{def}}$.

For coherent elements, the corresponding macroscopic tangent pair $(W, Q)$ belongs to $\mathcal{C}_{\mathrm{cohcrit}}$ and satisfies on annular cylinders $\{0 < r_0 < |x| < r_1 < \infty\} \times I$

$$\partial_s W + \frac{1}{2} x \cdot \nabla W + \frac{1}{2} W + \nabla Q = 0, \qquad \nabla \cdot W = 0$$

in distributions, with nonzero spatial quotient oscillation and recurrence under the induced logarithmic dilation flow.

**Theorem 9.38** (Compactness of the critical-tail compactification)**.** *Every bounded sequence-realized family of critical-tail profiles has a convergent subsequence in $\mathfrak{T}_M$. If all profiles in the sequence are sequence-realized, then the limit is sequence-realized.*

*Proof.* The barycentric velocities are bounded in $L^\infty_{\mathrm{loc}}$, so Banach–Alaoglu gives local weak-$*$ subsequential convergence. The fundamental compactness theorem for Young measures gives convergence of the oscillation measures against bounded continuous velocity observables. The logarithmic annular compactification is metrizable on the countable family of windows used in the extraction, hence Prokhorov compactness gives a vaguely convergent subsequence of the log-scale measures. The pressure components are recorded only through distributional gradients after the time and affine gauges; local pressure compactness modulo these gauges gives distributional subsequential limits. The affine quotient data are finite-dimensional on each fixed time window, and the quotient topology was chosen so that bounded sequences have subsequential limits. Diagonalizing over the countable annular exhaustion gives convergence in all components listed in Definition 9.37. For sequence-realized sequences, choose for each approximating profile a realizing subsequence from the original terminal sequence and diagonalize over the annular exhaustion and the countable observables in the topology above. The diagonal realizing sequence converges to the same limit, so the limiting critical-tail profile remains sequence-realized. $\square$

**Lemma 9.39** (Support transfer for the critical-tail compactification)**.** *Let $\mathfrak{T}_M$ be the compact metrizable critical-tail compactification and let*

$$\lambda_n = \frac{\mu_n \lfloor E_n}{\mu_n(E_n)}$$

*be normalized realized critical-tail defect measures, where $\mu_n(E_n) > 0$. If $\lambda_n \rightharpoonup \lambda \in \mathcal{P}(\mathfrak{T}_M)$ and $A \subset \mathfrak{T}_M$ is closed, then support of $\lambda$ in $A$ removes all surviving critical-tail defect mass outside $A$. If positive normalized critical-tail mass survives outside $A$, then a sequence-realized support profile in $\mathfrak{T}_M \setminus A$ exists.*

*Proof.* This is Lemma 1.12 with $X = \mathfrak{T}_M$. The compactness theorem above ensures that subsequential limits of realized critical-tail profiles remain sequence-realized, so the support point outside the closed set is not an abstract compactification artifact. $\square$

**Definition 9.40** (Log-annular compactification)**.** Set $\rho = \log|x|$ and write $\overline{\mathbb{R}}_{\log} = [-\infty, +\infty]$ for the two-point compactification of the log-radius axis. Fix the countable family of log-window cutoffs

$$\chi_{\rho_0,A}(\rho) := \chi\Big(\tfrac{\rho-\rho_0}{A}\Big), \qquad \rho_0 \in \mathbb{Q},\ A \in \mathbb{Q}_{>0},$$

where $\chi$ is the fixed compactly supported even cutoff of Section 9.1. The log-annular space $\overline{\mathbb{R}}_{\log}$ is endowed with the metrizable vague topology generated by these cutoffs. A sequence $\Lambda_n$ of finite Borel measures on $\overline{\mathbb{R}}_{\log}$ converges to $\Lambda$ vaguely if

$$\int \chi_{\rho_0,A}(\rho)\, d\Lambda_n(\rho) \longrightarrow \int \chi_{\rho_0,A}(\rho)\, d\Lambda(\rho)$$

for every cutoff in the countable family above and additionally for the test functions

$$\rho \mapsto \chi(\rho/A), \quad \rho \mapsto \chi((\rho \mp A)/1)$$

witnessing the masses at $\pm\infty$.

**Definition 9.41** (Young-measure convergence at the macroscopic critical scale)**.** Let $\overline{B_M} \subset \mathbb{R}^3$ be the closed velocity ball of radius $M$. A sequence of Young measures $\mathcal{Y}_n$ on the macroscopic spacetime annulus $\{a < |x| < b\} \times I$ with values in $\overline{B_M}$ converges to $\mathcal{Y}$ if for every $\psi \in C_c^\infty(\{a < |x| < b\} \times I)$ and every $F \in C(\overline{B_M})$,

$$\iint \psi(x,s) \int F(\xi)\, d\mathcal{Y}_{n,x,s}(\xi)\, dx\, ds \longrightarrow \iint \psi(x,s) \int F(\xi)\, d\mathcal{Y}_{x,s}(\xi)\, dx\, ds.$$

This is the fundamental theorem of Young measures applied to the bounded sequence of macroscopic representatives $W_n$, with associated Young measures $\mathcal{Y}_{n,x,s}$ recording the unresolved bounded oscillations at $(x,s)$. Equivalently, $\mathcal{Y}_n \to \mathcal{Y}$ in the $\sigma(L^\infty, L^1)$-topology induced by tensor test functions.

**Definition 9.42** (Distributional pressure-gradient convergence at critical scale)**.** A sequence of pressure profiles $Q_n$, each defined on the same macroscopic annular cylinder up to the affine and time-only gauges of Definition 8.14, converges *annularly distributionally* to $Q$, written $\nabla Q_n \to \nabla Q$ in $\mathcal{D}'_{\mathrm{ann}}$, if for every compactly supported divergence-free test vector field $\Psi \in C_c^\infty(\{a < |x| < b\} \times I, \mathbb{R}^3)$

$$\langle \nabla Q_n, \Psi \rangle = -\langle Q_n, \nabla \cdot \Psi \rangle \to 0$$

and for every general compactly supported $\Phi \in C_c^\infty(\{a < |x| < b\} \times I, \mathbb{R}^3)$ one has

$$\langle \nabla Q_n, \Phi \rangle \to \langle \nabla Q, \Phi \rangle,$$

where the left-hand sides are computed for representatives of $Q_n$ chosen inside the affine quotient.

**Lemma 9.43** (Critical-tail Young balance with pressure and viscous defects)**.** *Let $V_n \in L^\infty$ be uniformly bounded by $M$ and satisfy the exact critical-tail blow-down equation*

$$\partial_s V_n - R_n^{-2} \Delta V_n + \frac{1}{2} x \cdot \nabla V_n + \frac{1}{2} V_n + R_n^{-1} V_n \cdot \nabla V_n + \nabla Q_n = 0, \qquad \nabla \cdot V_n = 0 \tag{9.2}$$

*on a fixed macroscopic annular cylinder $\{a < |x| < b\} \times I$, with $R_n \to \infty$, affine pressure quotient fixed, and pressure gradients satisfying the gauge of Definition 9.42. Let $(\bar{W}, \mathcal{Y}, \Lambda)$ be the limiting critical-tail profile. After passing to the subsequence defining $\mathfrak{T}_M$, the pressure and viscous pairings*

$$\mathscr{P}_{n,F}[\psi] := \iint \psi(x,s)\, \nabla_\xi F(V_n(x,s)) \cdot \nabla Q_n(x,s)\, dx\, ds$$

$$\mathscr{D}_{n,F}[\psi] := -R_n^{-2} \iint \psi(x,s)\, \nabla_x V_n(x,s)^T \nabla_\xi^2 F(V_n(x,s)) \nabla_x V_n(x,s)\, dx\, ds$$

*converge for every $\psi \in C_c^\infty$ and every $F \in C^2(\overline{B_M})$ in the countable dense test class used to define the critical-tail topology. Denote the limits by $\mathscr{P}_F[\psi]$ and $\mathscr{D}_F[\psi]$. Then the Young measure satisfies the exact weak balance*

$$\begin{aligned} &\iint \left( \partial_s \psi + \tfrac{1}{2} x \cdot \nabla_x \psi + \tfrac{3}{2} \psi \right) \int F(\xi)\, d\mathcal{Y}_{x,s}(\xi)\, dx\, ds \\ &\qquad = \frac{1}{2} \iint \psi \int \xi \cdot \nabla_\xi F(\xi)\, d\mathcal{Y}_{x,s}(\xi)\, dx\, ds + \mathscr{P}_F[\psi] + \mathscr{D}_F[\psi]. \end{aligned} \tag{9.3}$$

*with the usual approximation from $C^2$ to bounded continuous observables on $\overline{B_M}$ whenever the velocity derivatives and the two defect pairings are represented by uniform limits. In particular, if $F$ is homogeneous of degree $p$ and $\mathscr{P}_F[\psi] = \mathscr{D}_F[\psi] = 0$, then*

$$\iint \left( \partial_s \psi + \tfrac{1}{2} x \cdot \nabla_x \psi + \tfrac{3-p}{2} \psi \right) \int F(\xi)\, d\mathcal{Y}_{x,s}(\xi)\, dx\, ds = 0.$$

*The log-scale component $\Lambda$, which records the covariant positions of the selected annular windows rather than a pressure-renormalized velocity observable, is invariant under the induced log-translation $\rho \mapsto \rho + \frac{1}{2} s$ on $\overline{\mathbb{R}}_{\log}$.*

*Proof.* Test (9.2) against $\psi(x,s)F'(V_n(x,s))$ for $F \in C^2(\overline{B_M})$ and $\psi \in C_c^\infty$. Each term is integrated by parts inside the chosen macroscopic annular cylinder where $\psi$ has compact support. The viscous term contributes

$$R_n^{-2}\iint F'(V_n)\Delta V_n\,\psi\,dx\,ds = -R_n^{-2}\iint\Big[\psi\,\nabla_x V_n^T\nabla_\xi^2 F(V_n)\nabla_x V_n + \nabla_\xi F(V_n)\nabla_x V_n\cdot\nabla\psi\Big]\,dx\,ds.$$

Since $F'$ and $F''$ are bounded on $\overline{B_M}$, $V_n$ is bounded by $M$, and the local energy bound from Lemma 9.66 gives $R_n^{-2}\iint|\nabla V_n|^2\psi\,dx\,ds \le C$, the Hessian part has a subsequential distributional limit on the countable test family; this is $\mathscr{D}_F[\psi]$. The remaining first-derivative part equals $-R_n^{-2}\iint\nabla_x F(V_n)\cdot\nabla\psi$, which after integration by parts is $R_n^{-2}\iint F(V_n)\Delta\psi = O(R_n^{-2})$ and tends to 0.

The convective term, after the standard chain-rule manipulation, equals

$$R_n^{-1}\iint F'(V_n)(V_n\cdot\nabla V_n)\psi\,dx\,ds = R_n^{-1}\iint(V_n\cdot\nabla)F(V_n)\,\psi\,dx\,ds = -R_n^{-1}\iint F(V_n)V_n\cdot\nabla\psi\,dx\,ds,$$

using $\nabla\cdot V_n = 0$. Since $F$ and $V_n$ are bounded, this term is $O(R_n^{-1})$ and tends to 0.

The drift term $\frac12 x\cdot\nabla V_n + \frac12 V_n$ tested against $F'(V_n)\psi$ gives, after the chain rule and integration by parts in $x$,

$$\iint\Big(\tfrac12 x\cdot\nabla F(V_n)+\tfrac12 F'(V_n)V_n\Big)\psi\,dx\,ds = -\iint F(V_n)\Big(\tfrac12\nabla\cdot(x\psi)\Big)\,dx\,ds+\iint \tfrac12 F'(V_n)V_n\,\psi\,dx\,ds,$$

where the boundary term vanishes by compact support of $\psi$. Using $\frac12\nabla\cdot(x\psi) = \frac32\psi + \frac12 x\cdot\nabla\psi$ gives

$$-\iint F(V_n)\Big(\tfrac32\psi + \tfrac12 x\cdot\nabla\psi\Big)\,dx\,ds + \iint \tfrac12 F'(V_n)V_n\,\psi\,dx\,ds.$$

Combining with the remaining $\partial_s$ term, which gives $-\iint F(V_n)\partial_s\psi\,dx\,ds$ after integration by parts in time, and keeping the pressure term explicitly, we obtain

$$\iint F(V_n)\Big(\partial_s\psi+\tfrac12 x\cdot\nabla\psi+\tfrac32\psi\Big)\,dx\,ds-\iint \tfrac12 F'(V_n)V_n\,\psi\,dx\,ds = \iint \psi\,F'(V_n)\cdot\nabla Q_n\,dx\,ds+o(1).$$

Choosing $F(\xi) = \xi\cdot e_k$ recovers the linear barycenter equation; choosing more general $F$ and passing to the Young-measure limit using Definition 9.41 gives the exact weak identity

$$\begin{aligned}\iint\Big(\partial_s\psi + \tfrac12 x\cdot\nabla\psi + \tfrac32\psi\Big)\int F(\xi)\,d\mathcal{Y}_{x,s}(\xi)\,dx\,ds \\ = \iint \tfrac12\psi\int F'(\xi)\xi\,d\mathcal{Y}_{x,s}(\xi)\,dx\,ds + \mathscr{P}_F[\psi] + \mathscr{D}_F[\psi].\end{aligned}$$

This is (9.3). The homogeneous-observable special case follows from Euler's identity $\xi\cdot\nabla_\xi F = pF$ whenever the pressure and viscous defect pairings vanish for that observable. If either pairing does not vanish, the nonzero defect is retained as pressure-gradient or Young/viscous defect data in $\mathfrak{T}_M$ and is assigned to the pressure/noncompact, affine quotient, or Young critical-tail alternatives rather than silently discarded.

The same calculation applied to the radial pushforward $\rho\mapsto\log|x|$ of $V_n$, or directly to the dyadic annular $\Lambda_n$, gives the invariance of $\Lambda$ under the induced log-translation $\rho\mapsto\rho+\frac12 s$. $\square$

9.1. **Log-annular virial identities for recurrent critical defects.** Throughout this subsection let $\chi\in C_c^\infty((-2,2))$ satisfy $0\le\chi\le1$ and $\chi\equiv1$ on $(-1,1)$. For $\rho\in\mathbb{R}$ and $A\ge1$, set

$$\phi_{\rho,A}(y) := \chi\bigg(\frac{\log|y|-\rho}{A}\bigg), \qquad R := e^\rho.$$

The cutoff is used only on annuli avoiding the origin. When differentiating with respect to $\rho$, $A$ is kept fixed.

**Lemma 9.44** (Log-annular oscillation virial identity)**.** *Let $(U,\Pi) \in \mathfrak{X}_M$ be a smooth bounded solution of (1.2). Fix $A \geq 1$, $\rho \in \mathbb{R}$, and put $\phi = \phi_{\rho,A}$. Define the weighted affine gauge*

$$c_{\rho,A}(\tau) := \frac{\int \phi(y) U(y,\tau)\, dy}{\int \phi(y)\, dy}, \qquad V_{\rho,A}(y,\tau) := U(y,\tau) - c_{\rho,A}(\tau),$$

*so that*

$$\int \phi V_{\rho,A}\, dy = 0.$$

*Set*

$$E_{\rho,A}(\tau) := \int \phi |V_{\rho,A}|^2\, dy, \qquad \mathcal{V}_{\rho,A}(\tau) := R^{-1} E_{\rho,A}(\tau),$$
$$D_{\rho,A}(\tau) := \int \phi |\nabla U|^2\, dy, \qquad \mathcal{D}_{\rho,A}(\tau) := R^{-1} D_{\rho,A}(\tau).$$

*Then $\mathcal{V}_{\rho,A}$ is locally absolutely continuous in $\tau$, and for a.e. $\tau$, after subtracting an arbitrary pressure gauge $\pi(\tau)$, one has*

$$\begin{aligned} \partial_\tau \mathcal{V}_{\rho,A} + 2\mathcal{D}_{\rho,A} + \frac{1}{2}\partial_\rho \mathcal{V}_{\rho,A} = R^{-1} &\int (\Delta\phi)|V_{\rho,A}|^2\, dy \\ &+ R^{-1} \int |V_{\rho,A}|^2 U \cdot \nabla\phi\, dy \\ &+ 2R^{-1} \int (\Pi - \pi(\tau)) V_{\rho,A} \cdot \nabla\phi\, dy. \end{aligned} \tag{9.4}$$

*Moreover*

$$|\nabla \phi_{\rho,A}(y)| \leq \frac{C_\chi}{A|y|}, \qquad |\Delta \phi_{\rho,A}(y)| \leq \frac{C_\chi}{A|y|^2} + \frac{C_\chi}{A^2|y|^2}, \tag{9.5}$$

*and*

$$y \cdot \nabla \phi_{\rho,A}(y) = \frac{1}{A}\chi'\left(\frac{\log|y| - \rho}{A}\right), \qquad \partial_\rho \phi_{\rho,A} = -y \cdot \nabla \phi_{\rho,A}.$$

*Proof.* Since $c_{\rho,A}(\tau)$ is the weighted least-squares affine gauge,

$$\int \phi V_{\rho,A}\, dy = 0.$$

Hence the derivative of the gauge drops out:

$$\frac{d}{d\tau} \int \phi |U - c_{\rho,A}|^2 = 2\int \phi V_{\rho,A} \cdot \partial_\tau U.$$

Substitute the centered equation

$$\partial_\tau U = \Delta U - (U \cdot \nabla)U - \frac{1}{2} y \cdot \nabla U - \frac{1}{2} U - \nabla \Pi.$$

The diffusion term gives

$$2\int \phi V \cdot \Delta U = -2\int \phi |\nabla U|^2 + \int (\Delta\phi)|V|^2.$$

The convective term gives, using $U \cdot \nabla U = U \cdot \nabla V$ and $\nabla \cdot U = 0$,

$$-2\int \phi V \cdot (U \cdot \nabla U) = -\int \phi U \cdot \nabla |V|^2 = \int |V|^2 U \cdot \nabla\phi.$$

The centered drift and linear terms give

$$-\int \phi V\cdot(y\cdot\nabla U)-\int \phi V\cdot U=-\frac{1}{2}\int \phi y\cdot\nabla|V|^2-\int\phi|V|^2$$
$$=\frac{1}{2}\int y\cdot\nabla\phi|V|^2+\frac{1}{2}\int\phi|V|^2.$$

The terms involving $c_{\rho,A}$ vanish because $\int\phi V=0$. Finally, for any scalar gauge $\pi(\tau)$,

$$-2\int\phi V\cdot\nabla\Pi=2\int(\Pi-\pi(\tau))V\cdot\nabla\phi,$$

because $\nabla\cdot V=0$ and

$$\int\pi(\tau)V\cdot\nabla\phi\,dy=\pi(\tau)\int\nabla\cdot(\phi V)\,dy=0.$$

Combining these identities yields

$$\partial_\tau E_{\rho,A}+2D_{\rho,A}=\int(\Delta\phi)|V|^2+\int|V|^2U\cdot\nabla\phi+2\int(\Pi-\pi)V\cdot\nabla\phi$$
$$+\frac{1}{2}\int\phi|V|^2+\frac{1}{2}\int y\cdot\nabla\phi|V|^2.$$

Since

$$\partial_\rho\phi_{\rho,A}=-y\cdot\nabla\phi_{\rho,A}$$

and the weighted least-squares condition also removes the $\partial_\rho c_{\rho,A}$ term,

$$\partial_\rho E_{\rho,A}=\int\partial_\rho\phi_{\rho,A}|V|^2=-\int y\cdot\nabla\phi_{\rho,A}|V|^2.$$

Therefore

$$\frac{1}{2}R^{-1}\left(\int\phi|V|^2+\int y\cdot\nabla\phi|V|^2\right)=-\frac{1}{2}\partial_\rho(R^{-1}E_{\rho,A})=-\frac{1}{2}\partial_\rho\mathcal{V}_{\rho,A}.$$

Thus, after multiplying the energy identity by $R^{-1}$, the centered drift and zeroth-order contribution is $-\frac{1}{2}\partial_\rho\mathcal{V}_{\rho,A}$. Moving it to the left gives (9.4). The cutoff bounds follow by direct differentiation in polar coordinates. □

*Remark* 9.45 (Virial normalization). The normalization $R^{-1}\int\phi|U-c|^2$ is the correct scale for an actual spatial tail of size $|y|^{-1}$. It is not automatically a compact observable for an arbitrary bounded residual profile. In the critical-tail reduction below, one uses it either on selected scale-critical defect windows or as an observable on the compact recurrent defect hull after the required scale-critical virial tightness has been verified. Without that compactness, the virial identity remains valid for smooth annular cutoffs but cannot by itself close the critical-tail alternatives.

**Lemma 9.46** (Realized log-window virial compactness)**.** *Let $\mathcal{D}\in\mathfrak{T}_M$ be a recurrent critical-tail defect generated by a normalized terminal sequence realized from the original blow-up sequence after concentration-set removal and affine normalization. Then, along the selected defect subsequence and after the admissible local pressure gauges of Proposition 1.19, the annular virial observables $\mathcal{V}_{\rho,A}$, the local dissipation $\mathcal{D}_{\rho,A}$, and the right-hand side flux contributions*

$$R^{-1}\int(\Delta\phi)|V|^2,\qquad R^{-1}\int|V|^2U\cdot\nabla\phi,\qquad 2R^{-1}\int(\Pi-\pi)V\cdot\nabla\phi$$

*appearing in (9.4) are compact in the local distributional topology and define continuous observables on the closed hull of $\mathcal{D}$ under the joint time-translation and log-translation flow. Equivalently, the hypothesis on legitimate compact observability invoked in Lemma 9.47 is automatic for realized* *recurrent defects, after either ordinary minimal mesoscopic-scale reduction or membership in the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$.*

*Proof.* Each ingredient is bounded by the uniform $L^\infty$ bound on the realizing sequence and by the local pressure-gauge compactness inherited through Proposition 1.19. Specifically, on every fixed log-annular cylinder $\Sigma_{\rho,A} = \{e^{\rho-A} \le |y| \le e^{\rho+A}\} \times [s_0, s_1]$, the cutoff $\phi$ and its derivatives are supported away from the spatial origin and satisfy the scaled bounds $|\nabla\phi| \lesssim |y|^{-1}$, $|\Delta\phi| \lesssim |y|^{-2}$ from (9.5). Combined with $\|U_n\|_{L^\infty(\Sigma_{\rho,A})} \le M$, the integrals $\int \phi|V_{\rho,A}|^2$, $\int(\Delta\phi)|V|^2$, and $\int |V|^2 U \cdot \nabla\phi$ are uniformly bounded. The pressure flux term $\int(\Pi - \pi)V \cdot \nabla\phi$ involves only the cutoff gradient, supported away from the origin; using the local pressure gauge subtraction $\Pi - \pi$ recorded in Lemma 1.20 and Proposition 9.70, the corresponding integrals are uniformly bounded in $L^1$. Local dissipation $\int \phi|\nabla U|^2$ is uniformly bounded along the realizing sequence by the local energy inequality of the suitable weak formulation.

Continuity in the local distributional topology is the standard fact that each integral is a continuous bilinear pairing of the $L^\infty$ field against a smooth compactly supported test function, with the pressure gauge chosen to make $\Pi - \pi$ locally $L^{3/2}$-compact, see Lemma 9.71. The joint time-translation and log-translation flow extends these observables to the recurrent hull by density. If at any stage the homogeneous gauge sequence loses compactness, Lemma 9.74 assigns the descendant to $\mathcal{C}_{\mathrm{aff}}$ and the assertion is recorded as the affine alternative. In the complementary case the observability hypothesis of Lemma 9.47 is verified, and the reduction provided by the minimal mesoscopic-scale theorem Theorem 9.81 closes the diagram. □

**Lemma 9.47** (Recurrent virial equality case and reduction)**.** *Let $\mathcal{D} \in \mathfrak{T}_M$ be a recurrent critical-tail defect generated by a bounded normalized terminal sequence after concentration-set removal and affine normalization. By Lemma 9.46 the log-annular virials in Lemma 9.44 are legitimate compact observables on the recurrent hull of $\mathcal{D}$ in the sense of Remark 9.45, after either minimal mesoscopic-scale reduction or assignment to $\mathcal{C}_{\mathrm{aff}}$. Let $\mathbb{P}$ be an invariant probability measure on that hull for the joint time-translation and log-translation flow. Assume also that the averaged pressure gauges are chosen by a local annular pressure decomposition, so that the three right-hand side terms in (9.4) have limits as defect observables. Then the following conclusions hold.*

(i) Zero-flux equality case. *If*

$$\lim_{A\to\infty} \int \left| R^{-1} \int (\Delta\phi)|V|^2 + R^{-1} \int |V|^2 U \cdot \nabla\phi + 2R^{-1} \int (\Pi - \pi)V \cdot \nabla\phi \right| d\mathbb{P} = 0, \tag{9.6}$$

*then every local tangent in the support of $\mathbb{P}$ has zero spatial gradient. Consequently*

$$\mathcal{D} \in \mathcal{C}_{\mathrm{aff}}.$$

(ii) Tight Dirac flux reduction. *Suppose (9.6) fails, and suppose the nonzero limiting flux is neither concentrated on a positive-threshold unit concentrating observer nor lost to logarithmic infinity. Suppose moreover that the annular blow-downs selected by this nonzero flux generate a Dirac Young measure. Then the selected annular blow-downs converge strongly in $L^3_{\mathrm{loc}}$ on macroscopic annular cylinders to a nonzero coherent critical tail. Hence*

$$\mathcal{D} \in \mathcal{C}_{\mathrm{cohcrit}}.$$

*The minimal mesoscopic-scale reduction theorem proved below supplies the tight-Dirac alternative for realized recurrent defects. Combined with that theorem, the preceding alternatives give the inclusion*

$$\mathcal{D} \in \mathcal{C}_{\mathrm{conc}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{cohcrit}}.$$

*Proof.* Integrate (9.4) against the invariant measure $\mathbb{P}$. The averaged $\partial_\tau \mathcal{V}_{\rho,A}$ term vanishes by time-translation invariance. The averaged $\partial_\rho \mathcal{V}_{\rho,A}$ term vanishes by log-translation invariance. Therefore

$$2 \int \mathcal{D}_{\rho,A}\, d\mathbb{P} = \int \mathcal{F}_{\rho,A}\, d\mathbb{P},$$

where $\mathcal{F}_{\rho,A}$ denotes the sum of the three right-hand side flux terms in (9.4). Under (9.6) and by nonnegativity of the dissipation,

$$\lim_{A\to\infty} \int \mathcal{D}_{\rho,A}\, d\mathbb{P} = 0.$$

Choose a countable exhausting family of ordinary unit covariant cylinders. For each cylinder, choose $A$ large enough that the corresponding cutoff is identically one on a fixed enlargement. The preceding limit then implies that, for $\mathbb{P}$-a.e. profile, the local dissipation $\int_Q |\nabla U|^2$ vanishes on every cylinder in the exhausting family. Hence $\nabla U = 0$ distributionally on every compact cylinder. Thus every local tangent is spatially constant. By the affine normalization dichotomy, this is precisely the affine/parasitic lower stratum $\mathcal{C}_{\mathrm{aff}}$.

We now prove the tight Dirac flux reduction. Since the averaged flux does not vanish and is not concentrated on a unit concentrating observer, there are logarithmic annuli with nonzero normalized flux but with no positive-threshold unit constant-in-time oscillatory subcylinder. Since the flux is not lost to logarithmic infinity, these annuli may be selected with a tight macroscopic scale $R_n \to \infty$. Passing to the corresponding macroscopic variables gives bounded blown-down fields

$$\widetilde{U}_n := \mathcal{R}_{(y_n,\sigma_n)} U_n, \qquad W_n(x,s) = \widetilde{U}_n(R_n x, s) - c_n(s)$$

on every fixed annular cylinder, with pressure and affine gauges normalized by the exact blow-down definitions of Definition 9.84. Terminal compactness gives a weak-$*$ $L^\infty_{\mathrm{loc}}$ limit and an associated Young measure. By the Dirac hypothesis on the selected flux, the Young measure is $\delta_W$ almost everywhere. Uniform boundedness then upgrades convergence to strong $L^3_{\mathrm{loc}}$ convergence on annular cylinders. Passing to the limit in the exact ungauged equation (9.22), the viscous and nonlinear terms vanish by the factors $R_n^{-2}$ and $R_n^{-1}$, and the normalized pressure gradients converge in annular distributions. The limit therefore satisfies

$$\partial_s W + \frac{1}{2} x \cdot \nabla W + \frac{1}{2} W + \nabla Q = 0, \qquad \nabla \cdot W = 0,$$

on macroscopic annular cylinders. The selected flux is nonzero, so the limit has nonzero spatial quotient oscillation. Thus $(W, Q) \in \mathcal{C}_{\mathrm{cohcrit}}$. □

9.2. **Logarithmic escape as a hidden-scale obstruction.** The logarithmic escape alternative is not ruled out below by a separate global tail estimate. The point of the next reduction is narrower and more useful: once tight annular descendants have no hidden-scale variance, a nonzero logarithmic escape component cannot survive. Thus the log-diffuse alternative and the Young alternative are both reduced to the same minimal mesoscopic-scale theorem.

**Definition 9.48** (Log-diffuse recurrent critical defect)**.** A realized recurrent critical-tail defect is log-diffuse if, after projecting the normalized diffuse oscillation measures to the logarithmic radial variable $\rho = \log|y|$, their log marginals $\lambda_n$ satisfy

$$\sup_{\rho_0\in\mathbb{R}} \lambda_n([\rho_0 - A, \rho_0 + A]) \longrightarrow 0 \qquad \text{for every fixed } A < \infty.$$

Equivalently, the critical-tail compactification has a nonzero logarithmic escape component $\Lambda$. This alternative is denoted by $\mathcal{C}_{\log\mathrm{diff}}$.

**Definition 9.49** (Log-window normalized support profile)**.** Let $\lambda_n$ be the normalized logarithmic defect marginals of a log-diffuse critical defect. A log-window normalized support profile, also called a log-window support profile, is obtained by choosing numbers $A \geq 1$, centers $\rho_n$, and intervals

$$I_n = [\rho_n - A, \rho_n + A]$$

with $\lambda_n(I_n) > 0$, restricting the defect measures to $I_n$, and renormalizing:

$$\widetilde{\lambda}_n := \frac{\lambda_n \lfloor I_n}{\lambda_n(I_n)}.$$

After applying the logarithmic recentering $\rho \mapsto \rho - \rho_n$, the renormalized measures are tight in the fixed interval $[-A, A]$. Any critical-tail profile obtained as a subsequential limit of these normalized restrictions is called a log-window normalized support profile of the original log-diffuse defect. It is a retained descendant only if a separate unit velocity lower bound is proved.

**Lemma 9.50** (Nonzero logarithmic escape has tight log-window support profiles)**.** *Let $\mathcal{D} \in \mathcal{C}_{\log \mathrm{diff}}$ be a nonzero realized recurrent critical-tail defect after concentration-set removal and affine normalization. Then, for some fixed $A \geq 1$, $\mathcal{D}$ has a nonzero log-window normalized support profile. This profile is a tight macroscopic annular critical-tail realization and retains normalized spatial quotient oscillation mass.*

*Proof.* The logarithmic marginals $\lambda_n$ are probability measures. Fix any $A \geq 1$. Cover $\mathbb{R}_\rho$ by the intervals

$$I_{k,A} := [2Ak - A, 2Ak + A], \qquad k \in \mathbb{Z}.$$

For each $n$, at least one interval has positive $\lambda_n$-mass. Choose one such interval and denote it by $I_n = [\rho_n - A, \rho_n + A]$. Restricting to $I_n$, renormalizing, and translating $\rho_n$ to the origin gives probability measures supported in the fixed compact interval $[-A, A]$. Compactness of the observer boundary profile class and the local Seregin compactness of $\mathfrak{X}_M$ give a subsequential limit. By construction, the limiting profile has total normalized defect mass one, hence is nonzero.

The logarithmic recentering fixes the selected multiplicative annulus, so the resulting critical-tail realization is tight in logarithmic scale. The original diffuse defect was measured after affine normalization by the oscillation defect measure; the normalized restriction preserves total oscillation mass equal to one. Hence the normalized support profile carries nonzero spatial quotient oscillation. No positive lower bound for the unnormalized mass $\lambda_n(I_n)$ is required, because the profile is a normalized boundary profile. □

**Lemma 9.51** (Support transfer for log-window defect compactification)**.** *Let $X_{\log}$ be the compact metrizable log-window defect space obtained by restricting the critical-tail compactification $\mathfrak{T}_M$ to normalized logarithmic windows, and let*

$$\lambda_n = \frac{\Lambda_n \lfloor I_n}{\Lambda_n(I_n)} \qquad (\Lambda_n(I_n) > 0)$$

*be normalized realized log-window restrictions. If $\lambda_n \rightharpoonup \lambda \in \mathcal{P}(X_{\log})$ and $A \subset X_{\log}$ is closed, then support of $\lambda$ in $A$ removes all surviving log-window defect mass outside $A$; any positive surviving mass outside $A$ gives a normalized log-window support profile outside $A$.*

*Proof.* Apply Lemma 1.12 in the compact metrizable space $X_{\log}$. The normalization by $\Lambda_n(I_n)$ is exactly the realized support-profile normalization; the size of $\Lambda_n(I_n)$ is irrelevant after passing to probability measures. □

**Lemma 9.52** (Renormalized log-window heredity)**.** *Let $\mathcal{D} \in \mathcal{C}_{\log \mathrm{diff}}$ be a log-diffuse defect realized from the original blow-up sequence, and let $\mathcal{F} \subset \mathfrak{T}_M$ be a closed union of alternatives which is closed under retained descendants and normalized support profiles. Suppose that every nonzero* *log-window normalized support profile of $\mathcal{D}$ belongs to $\mathcal{F}$. Then the original log-diffuse defect itself has no surviving mass outside $\mathcal{F}$. In particular, if $\mathcal{F}$ has already been excluded by the lower-stratum closure theorems, then $\mathcal{D}$ is impossible.*

*Proof.* It remains to verify that the selected logarithmic windows remain tied to the original defect even if their unnormalized mass tends to zero. Let $\Lambda$ be the compactified logarithmic defect measure associated with $\mathcal{D}$. If $\Lambda(\mathfrak{T}_M \setminus \mathcal{F}) > 0$, the concrete support-transfer lemma Lemma 9.51 gives a nonzero log-window normalized support profile outside $\mathcal{F}$. This contradicts the hypothesis that every nonzero log-window normalized support profile lies in $\mathcal{F}$. Hence $\Lambda$ is supported on $\mathcal{F}$. Since $\mathcal{F}$ is closed under retained descendants and normalized support profiles and is already excluded, the original defect has no admissible surviving normalized support profile. The fact that $\lambda_n(I_n)$ may tend to zero does not affect the argument: the normalized restriction records a support profile of the original probability measure, and a positive original defect measure cannot be supported entirely on excluded closed alternatives. □

**Theorem 9.53** (Logarithmic escape reduces to hidden-scale compactness)**.** *Assume Theorem 9.83. Then every nonzero log-diffuse critical-tail defect generated after concentration-set removal and affine normalization has a nonzero normalized support profile in*

$$\mathcal{C}_{\mathrm{conc}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{cohcrit}}.$$

*Consequently, after the concentration, affine/parasitic, and coherent critical-tail alternatives have been removed, no log-diffuse alternative remains.*

*Proof.* Let $\mathcal{D} \in \mathcal{C}_{\log\mathrm{diff}}$ be nonzero. By Lemma 9.50, nonzero tight log-window normalized support profiles exist. Let $\widetilde{\mathcal{D}}$ be any such profile. If this support profile regenerates a positive-threshold unit concentrating observer, then it belongs to $\mathcal{C}_{\mathrm{conc}}$. If its spatial quotient oscillation vanishes, then Theorem 8.21 places it in $\mathcal{C}_{\mathrm{aff}}$. It remains to consider the case in which it is neither concentrating nor affine.

Because $\widetilde{\mathcal{D}}$ was obtained by restriction to a fixed logarithmic window, its logarithmic escape component is zero. The only remaining possible obstruction in the critical-tail compactification is a non-Dirac annular Young defect. By Theorem 9.83, every tight annular blow-down generated by the descendant has no hidden-scale variance. Hence Lemma 9.56 makes the Young measure Dirac. The coherent extraction Corollary 9.86 then gives a nonzero coherent critical tail; thus $\widetilde{\mathcal{D}} \in \mathcal{C}_{\mathrm{cohcrit}}$.

In all cases the log-window normalized support profile lies in the closed excluded set

$$\mathcal{F} := \mathcal{C}_{\mathrm{conc}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{cohcrit}}.$$

By Lemma 9.52, if every log-window normalized support profile of the original log-diffuse defect lies in $\mathcal{F}$, then the original defect has no surviving mass outside $\mathcal{F}$. Once the concentration, affine/parasitic, and coherent critical-tail alternatives are removed, this makes the original log-diffuse defect impossible. □

**Corollary 9.54** (Log-diffuse alternative excluded by minimal mesoscopic scale reduction)**.** *Inside the retained residual decomposition,*

$$\mathcal{C}_{\log\mathrm{diff}} = \varnothing.$$

*Equivalently, after affine/parasitic lower-stratum removal, logarithmic hidden escape is excluded by the same hidden-scale compactness theorem that removes the Young alternative.*

*Proof.* By Theorem 9.81, Theorem 9.83 applies to all tight annular descendants. Hence Theorem 9.53 reduces every nonzero log-diffuse defect to a descendant in $\mathcal{C}_{\mathrm{conc}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{cohcrit}}$. The concentration alternative has been removed by terminal concentration-set extraction, the affine stratum by Theorem 9.89, and the coherent critical-tail alternative by Corollary 9.100. Hence no log-diffuse alternative remains. □

### 9.3. Hidden-scale Young variance.

**Definition 9.55** (Hidden-scale variance)**.** Let $W_n$ be a macroscopic critical-tail blow-down on a compact annular cylinder

$$K \Subset \{0 < |x| < \infty\} \times I.$$

For $0 < \delta < 1$, set

$$\mathfrak{H}_K(W_n, \delta) := \sup_{\substack{|h|<\delta \\ |\theta|<\delta}} \|W_n(\cdot + h, \cdot + \theta) - W_n\|_{L^3(K)},$$

where the translated functions are evaluated on a fixed slightly larger annular cylinder containing all $(x+h, s+\theta)$ with $(x,s) \in K$ and $|h| + |\theta| < \delta$. The sequence has no hidden-scale variance on $K$ if

$$\lim_{\delta \downarrow 0} \limsup_{n\to\infty} \mathfrak{H}_K(W_n, \delta) = 0.$$

It has no hidden-scale variance if this holds on every compact macroscopic annular cylinder.

**Lemma 9.56** (Hidden-scale variance is the Young obstruction)**.** *Let $W_n$ be uniformly bounded in $L^\infty_{\mathrm{loc}}$ on macroscopic annular cylinders and suppose*

$$W_n \rightharpoonup^* \bar{W} \qquad \text{in } L^\infty_{\mathrm{loc}}.$$

*On a compact annular cylinder $K$, the following are equivalent:*

(i) *$W_n \to \bar{W}$ strongly in $L^3(K)$;*

(ii) *the generated Young measure is Dirac on $K$:*

$$\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)} \qquad \text{for a.e. } (x,s) \in K;$$

(iii) *$W_n$ has no hidden-scale variance on $K$.*

*Consequently a realized Young critical tail is precisely a failure of the hidden-scale variance condition on some compact annular cylinder.*

*Proof.* The uniform $L^\infty(K)$ bound gives a uniform $L^3(K)$ bound. The Fréchet–Kolmogorov compactness criterion on $L^3(K)$, applied after extending the functions to a fixed slightly larger cylinder, says that this bounded family is relatively compact in $L^3(K)$ if and only if the translation modulus in Definition 9.55 tends to zero. Thus (iii) is equivalent to relative compactness in $L^3(K)$. Any strong subsequential limit must agree with the weak-* limit $\bar{W}$, so relative compactness is equivalent to strong convergence to $\bar{W}$. This proves (i)⇔(iii).

Strong $L^3(K)$ convergence implies convergence in measure, hence the associated Young measure is $\delta_{\bar{W}}$. Conversely, if the generated Young measure is Dirac and $\{W_n\}$ is uniformly bounded in $L^\infty(K)$, then $W_n \to \bar{W}$ in measure on $K$. Uniform boundedness gives uniform integrability of $|W_n - \bar{W}|^3$, so Vitali convergence yields strong $L^3(K)$ convergence. Hence (i)⇔(ii). □

**Definition 9.57** (Multiscale affine oscillation)**.** Let $(U_n, \Pi_n)$ be a terminal sequence. For a covariant observer $z = (\xi, \sigma)$ and $r \geq 1$, define the critical-tail cylinder

$$\mathcal{Q}^{\mathrm{cov}}_{r,\mathrm{ct}}(z) := \mathcal{R}_z\big(B_r(0) \times (-1,1)\big).$$

Define

$$\mathcal{A}_n(z,r) := r^{-3}\, \mathrm{osc}_{3,\mathrm{sp}}\big(\mathcal{R}_z U_n; B_r(0) \times (-1,1)\big).$$

The exponent 3 is the spatial dimension; the centered time interval is kept at fixed size because critical-tail blow-downs are spatial/logarithmic blow-downs, not parabolic rescalings. Allowing the minimizing constant to depend on time removes the harmless spatially homogeneous affine mode. Unit concentration-set removal is formulated with $\mathrm{osc}_{3,\mathrm{ct}}$; after the affine/parasitic quotient has been applied, the mesoscopic residual analysis is formulated with $\mathrm{osc}_{3,\mathrm{sp}}$, and $\sup_z \mathcal{A}_n(z,1)$ is below the retained spatial-oscillation threshold on the residual exterior window, after the fixed cover constants are absorbed into the threshold.

**Definition 9.58** (Minimal mesoscopic oscillation scale)**.** Let $R_n \to \infty$ be the macroscopic scale of a tight critical-tail blow-down. A minimal mesoscopic oscillation scale is a sequence of covariant centers $z_n$, radii $r_n$, and a number $\varepsilon_0 > 0$ such that

$$1 \ll r_n \ll R_n, \qquad \mathcal{A}_n(z_n, r_n) \geq \varepsilon_0,$$

and, after replacing $r_n$ by a dyadic comparable scale if necessary,

$$\sup_z \sup_{1 \leq r \leq r_n/2} \mathcal{A}_n(z,r) < \varepsilon_0/2$$

on the selected residual exterior window. Thus all smaller retained subscales are affine-flat, while the scale $r_n$ is the first scale at which non-affine oscillation is visible.

**Lemma 9.59** (Finite-overlap control of affine oscillation)**.** *Let $I \Subset \mathbb{R}$, let $f \in L^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times I)$ with $\|f\|_{L^\infty} \leq M$, and let*

$$B_1(a_0), B_1(a_1), \ldots, B_1(a_N)$$

*be a finite chain of unit balls such that*

$$|B_1(a_j) \cap B_1(a_{j+1})| \geq \kappa > 0 \qquad (j = 0, \ldots, N-1).$$

*Assume that for each $j$ there exists $c_j \in L^\infty(I;\mathbb{R}^3)$, $|c_j| \leq M$, such that*

$$\int_I \int_{B_1(a_j)} |f(y,s) - c_j(s)|^3 \, dy \, ds \leq \delta.$$

*Then, for $0 \leq j,k \leq N$,*

$$\int_I |c_j(s) - c_k(s)|^3 \, ds \leq C(N,\kappa)\delta, \tag{9.7}$$

*and*

$$\int_I \int_{B_1(a_j)} |f(y,s) - c_k(s)|^3 \, dy \, ds \leq C(N,\kappa)\delta. \tag{9.8}$$

*Consequently, if all unit cylinders in a bounded chain have affine-normalized oscillation $o(1)$, then all their local affine gauges agree up to $o(1)$ in $L^3(I)$.*

*Proof.* For neighboring indices put $E_j = B_1(a_j) \cap B_1(a_{j+1})$. Since $|E_j| \geq \kappa$, the pointwise inequality

$$|c_j(s) - c_{j+1}(s)|^3 \leq 4\kappa^{-1} \int_{E_j} \big( |f(y,s) - c_j(s)|^3 + |f(y,s) - c_{j+1}(s)|^3 \big) \, dy$$

holds for a.e. $s$. Integrating in $s$ gives

$$\int_I |c_j - c_{j+1}|^3 \, ds \leq C_\kappa \delta.$$

Summing along the chain and using

$$\left| \sum_{\ell=j}^{k-1} (c_\ell - c_{\ell+1}) \right|^3 \leq |j-k|^2 \sum_{\ell=j}^{k-1} |c_\ell - c_{\ell+1}|^3$$

proves (9.7). Finally,

$$|f - c_k|^3 \le 4|f - c_j|^3 + 4|c_j - c_k|^3$$

on $B_1(a_j) \times I$, and (9.8) follows from (9.7). $\square$

**Lemma 9.60** (Spacetime finite-overlap translation control)**.** *Let $K \Subset \mathbb{R}^3 \times I$, let $(h_n, \theta_n) \to 0$, and let $f_n$ be uniformly bounded in $L^\infty$ on a fixed neighborhood of*

$$K \cup \big(K + (h_n, \theta_n)\big).$$

*Assume that every unit covariant cylinder in a finite spacetime chain covering the segment joining $(x, s) \in K$ to $(x + h_n, s + \theta_n)$ has spatial quotient oscillation $o(1)$. Then there are spatially homogeneous functions $a_n, b_n \in L^\infty(I; \mathbb{R}^3)$ such that*

$$\|f_n - a_n\|_{L^3(K)} + \|f_n(\cdot + h_n, \cdot + \theta_n) - b_n\|_{L^3(K)} = o(1),$$

*and*

$$\|f_n(\cdot + h_n, \cdot + \theta_n) - f_n - (b_n - a_n)\|_{L^3(K)} = o(1).$$

*Consequently, if the translated difference has a positive $L^3(K)$ lower bound while no unit concentrating observer is present, then that lower bound is carried by the spatially homogeneous time-dependent mode $b_n - a_n$. In the ordered residual decomposition this descendant belongs to the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$.*

*Proof.* Cover $K$ and $K + (h_n, \theta_n)$ by finitely many unit spacetime cylinders contained in the fixed neighborhood. Refine the cover so that any two successive cylinders in the chain have spatial overlap bounded below and time overlap bounded below by constants depending only on $K$. On each cylinder choose a near-minimizing spatially homogeneous gauge. Applying Lemma 9.59 on each time-overlap interval and summing along the finite chain shows that all gauges on the chain agree in $L^3$ on their common time intervals up to $o(1)$. Gluing the finitely many local gauges by this overlap relation gives global-in-space functions $a_n(s)$ on the part of the chain covering $K$ and $b_n(s)$ on the part covering $K + (h_n, \theta_n)$. The first displayed estimate follows from the local small oscillation and finite overlap. Subtracting the two approximations gives the second displayed estimate.

If the translated difference remains bounded below, then $\|b_n - a_n\|_{L^3(\mathrm{proj}_s K)}$ stays bounded below. The corresponding descendant has zero $\mathrm{osc}_{3,\mathrm{sp}}$ on every compact cylinder but a nonzero homogeneous time-dependent component. That is the affine/parasitic lower stratum in the ordered profile decomposition. $\square$

**Lemma 9.61** (Translation defect produces concentration, affine collapse, or a mesoscopic radius)**.** *Let $K \Subset \{0 < |x| < \infty\} \times I$, let*

$$\widetilde{U}_n := \mathcal{R}_{(y_n, \sigma_n)} U_n, \qquad W_n(x, s) = \widetilde{U}_n(R_n x, s) - c_n(s), \qquad R_n \to \infty,$$

*be a bounded macroscopic blow-down, and suppose that for some $(h_n, \theta_n) \to 0$ and $\eta_0 > 0$,*

$$\|W_n(\cdot + h_n, \cdot + \theta_n) - W_n\|_{L^3(K)} \ge \eta_0.$$

*Set*

$$\ell_n := R_n\big(|h_n| + |\theta_n|^{1/2}\big).$$

*The observer metric used here is the fixed parabolic metric on covariant observer space,*

$$d_{\mathrm{obs}}\big((h, \theta), (0, 0)\big) \simeq |h| + |\theta|^{1/2}.$$

*Multiplication by $R_n$ converts this macroscopic observer displacement into the corresponding centered spatial/logarithmic tail scale. This is an observer blow-down scale, not a Navier–Stokes parabolic rescaling of the centered time variable. Then, after passing to a subsequence, exactly one of the following alternatives is selected by the ordered priority rule.*

(i) *A positive-threshold unit concentrating observer is regenerated.*
(ii) *The defect is carried only by a spatially homogeneous time-dependent affine mode; equivalently the descendant is assigned to* $\mathcal{C}_{\mathrm{aff}}$.
(iii) $1 \ll \ell_n \ll R_n$, *and there are covariant centers* $z_n$, *radii* $r_n \sim \ell_n$, *and* $\varepsilon_0 = \varepsilon_0(\eta_0, K) > 0$ *such that*

$$\mathcal{A}_n(z_n, r_n) \geq \varepsilon_0.$$

*Proof.* Since $(h_n, \theta_n) \to 0$, $\ell_n/R_n \to 0$. First assume $\ell_n$ is bounded. The two points compared in the original centered variables are then connected, on the compact set $K$, by a uniformly finite chain of covariant unit cylinders. If a unit cylinder in that chain has a positive constant-in-time oscillation lower bound, alternative (i) holds. Otherwise the spatial quotient oscillation on every cylinder in the chain tends to zero. By Lemma 9.60, the displayed translation defect can persist only through a spatially homogeneous time-dependent mode. This is alternative (ii). Thus, if neither (i) nor (ii) occurs, $\ell_n \to \infty$.

In that remaining case $1 \ll \ell_n \ll R_n$. Cover $K$ and its translated copy by finitely many balls of radius comparable to $|h_n| + |\theta_n|^{1/2}$ in macroscopic variables. Pulling this cover back to the centered variables gives covariant cylinders of radius comparable to $\ell_n$. If all such cylinders had spatial quotient oscillation below a sufficiently small $\varepsilon_0 = \varepsilon_0(\eta_0, K)$, then Lemma 9.60, after rescaling the finite overlap constants, would again force the translated fields to differ by only an affine/parasitic time mode. This is alternative (ii), contrary to the present case. Therefore one such cylinder has $\mathcal{A}_n(z_n, r_n) \geq \varepsilon_0$ with $r_n \sim \ell_n$, proving (iii). $\square$

**Lemma 9.62** (Failure of hidden-scale compactness produces a minimal mesoscopic scale)**.** *Let* $\widetilde{U}_n = \mathcal{R}_{(y_n,\sigma_n)} U_n$ *and*

$$W_n(x, s) = \widetilde{U}_n(R_n x, s) - c_n(s)$$

*be a tight macroscopic annular blow-down after concentration-set removal. If* $W_n$ *fails the hidden-scale variance condition on some compact annular cylinder* $K$, *then either a positive-threshold unit concentrating observer is regenerated, the affine stratum* $\mathcal{C}_{\mathrm{aff}}$ *is reached by a descendant, or there is a minimal mesoscopic oscillation scale in the sense of Definition 9.58.*

*Proof.* Failure of hidden-scale variance gives a compact annular cylinder $K$, a number $\eta_0 > 0$, and translations $(h_n, \theta_n) \to 0$ such that, after passing to a subsequence,

$$\|W_n(\cdot + h_n, \cdot + \theta_n) - W_n\|_{L^3(K)} \geq \eta_0.$$

By Lemma 9.61, the translation defect either regenerates a unit concentrating observer, reaches $\mathcal{C}_{\mathrm{aff}}$, or gives centers $z_n$, radii $\rho_n$ with $1 \ll \rho_n \ll R_n$, and $\varepsilon_0 > 0$ such that

$$\mathcal{A}_n(z_n, \rho_n) \geq \varepsilon_0.$$

In the last case, let $r_n$ be the smallest dyadic radius in $[1, C\rho_n]$ for which

$$\sup_z \mathcal{A}_n(z, r_n) \geq \varepsilon_0$$

on the selected residual exterior window. Unit concentration-set removal excludes $r_n = O(1)$, so $r_n \to \infty$. Since $r_n \lesssim \rho_n$ and $\rho_n/R_n \to 0$, one has $r_n/R_n \to 0$. By dyadic minimality, all smaller scales up to $r_n/2$ have oscillation below $\varepsilon_0/2$, after slightly lowering $\varepsilon_0$. This is a minimal mesoscopic oscillation scale. $\square$

**Definition 9.63** (Minimal-scale spatial blow-down)**.** Let $z_n$ and $r_n$ be a minimal mesoscopic oscillation scale. Write

$$U_n^{z_n} := \mathcal{R}_{z_n} U_n, \qquad \Pi_n^{z_n} := \mathcal{R}_{z_n} \Pi_n.$$

Choose measurable velocity gauges $c_n : (-T, T) \to \mathbb{R}^3$, with $|c_n| \le M$, which are minimizing, or within $o(1)$ of minimizing, the affine oscillation on each compact time interval. The minimal-scale spatial blow-down is

$$V_n(x, s) := U_n^{z_n}(r_n x, s), \qquad W_n(x, s) := V_n(x, s) - c_n(s).$$

Thus $V_n$ is the ungauged blow-down and $W_n$ is the affine-normalized velocity used to measure non-affine oscillation. The associated scaled pressure is

$$Q_n(x, s) := r_n^{-1} \Pi_n^{z_n}(r_n x, s),$$

modulo the admissible addition of affine functions of $x$ and arbitrary functions of $s$. Affine pressure terms are always paired with the corresponding spatially homogeneous velocity gauge in the weak formulation; no pointwise time derivative of $c_n$ is required. This is not the Navier–Stokes parabolic scaling. It is the spatial/logarithmic tail blow-down adapted to the centered residual equation. The pressure normalization is chosen so that $\nabla_x Q_n$ is the rescaled pressure-gradient term appearing in the centered blow-down equation.

*Remark* 9.64 (First-bad pressure and gauge convention). Throughout the minimal mesoscopic-scale argument, all distributional equations, pressure Poisson equations, and local energy inequalities are written for the ungauged field $V_n$ and its genuine scaled pressure $Q_n$. The field $W_n = V_n - c_n(s)$ is used only for quotient quantities: spatial oscillation, Young measures, and variance after the homogeneous gauge sequence has either compactified in time or has been assigned to the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$. Thus an "affine pressure paired with $c_n$" is quotient notation. It is not asserted to be an $L^{3/2}_{\mathrm{loc}}$ pressure, and no pointwise derivative $\partial_s c_n$ is used unless compactness of $c_n$ has already been obtained.

**Lemma 9.65** (Minimal-scale blow-down equation)**.** *Let $V_n, W_n, Q_n, c_n$ be the minimal-scale blow-downs of Definition 9.63. On every compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$, $V_n$ and $W_n$ are divergence-free, $\|V_n\|_{L^\infty(K)} \le M$, $\|W_n\|_{L^\infty(K)} \le 2M$, and the ungauged field satisfies*

$$\partial_s V_n - r_n^{-2} \Delta V_n + \frac{1}{2} x \cdot \nabla V_n + \frac{1}{2} V_n + r_n^{-1} (V_n \cdot \nabla) V_n + \nabla Q_n = 0 \tag{9.9}$$

*in distributions. The affine-normalized field $W_n = V_n - c_n(s)$ is used for oscillation and Young-measure compactness. Its equation is understood only in the quotient sense made precise in Lemma 9.67; no pointwise derivative of $c_n$ and no $L^{3/2}_{\mathrm{loc}}$ affine pressure for $W_n$ is assumed. Moreover*

$$-\Delta Q_n = r_n^{-1} \partial_i \partial_j \big(V_{n,i} V_{n,j}\big) \tag{9.10}$$

*in distributions, after the same gauges.*

*Proof.* The covariant recentering $\mathcal{R}_{z_n}$ is an exact symmetry of the centered equation, so $(U_n^{z_n}, \Pi_n^{z_n})$ solves (1.2). Since centered time is not rescaled,

$$\partial_s V_n = (\partial_\tau U_n^{z_n})(r_n x, s), \qquad \Delta_y U_n^{z_n} = r_n^{-2} \Delta_x V_n,$$
$$y \cdot \nabla_y U_n^{z_n} = x \cdot \nabla_x V_n, \qquad (U_n^{z_n} \cdot \nabla_y) U_n^{z_n} = r_n^{-1} (V_n \cdot \nabla_x) V_n.$$

If $Q_n = r_n^{-1} \Pi_n^{z_n}(r_n x, s)$, then

$$\nabla_x Q_n = (\nabla_y \Pi_n^{z_n})(r_n x, s).$$

Substituting these identities gives the scaled equation for $V_n$. We do not treat $W_n = V_n - c_n(s)$ as carrying an ordinary pressure unless the gauges have additional time regularity. Instead, all PDE identities below are first written for the ungauged field $V_n$, and the passage to $W_n$ is made only after quotienting out spatially homogeneous modes. Taking the divergence of the ungauged scaled equation gives (9.10); the linear drift has zero divergence contribution because $V_n$ is divergence-free. The $L^\infty$ bounds follow from $\|U_n\|_{L^\infty} \le M$ and $|c_n| \le M$. $\square$

**Lemma 9.66** (Local energy inequality under spatial/logarithmic blow-down)**.** *Let* $(U_n, \Pi_n)$ *be locally suitable centered profiles realized from the original blow-up sequence in the sense of Definition 1.9, and let*

$$V_n(x,s) = U_n^{z_n}(r_n x, s), \qquad Q_n(x,s) = r_n^{-1}\Pi_n^{z_n}(r_n x, s),$$

*be the minimal-scale spatial/logarithmic blow-downs of Definition 9.63. Then* $(V_n, Q_n)$ *satisfy the local energy inequality associated to* (9.9) *on every relatively compact spacetime domain* $K \Subset \mathbb{R}^3 \times \mathbb{R}$*: for every nonnegative* $\phi \in C_c^\infty(K)$*,*

$$\begin{aligned} r_n^{-2}\iint |\nabla V_n|^2 \phi \, dx\, ds \le \iint \frac{|V_n|^2}{2}\Big(\partial_s \phi + r_n^{-2}\Delta\phi + \frac{1}{2}x\cdot\nabla\phi + \frac{1}{2}\phi\Big)\, dx\, ds \\ + r_n^{-1}\iint \frac{|V_n|^2}{2} V_n \cdot \nabla\phi \, dx\, ds + \iint Q_n V_n \cdot \nabla \phi \, dx\, ds. \end{aligned} \tag{9.11}$$

*In particular,*

$$r_n^{-2}\iint |\nabla V_n|^2 \phi\, dx\, ds \le C(M, \|\phi\|_{C^2}, |K|, r_n) \tag{9.12}$$

*with a finite constant uniform in* $n$*, and the right-hand side of* (9.11) *converges, as* $n \to \infty$*, to the distributional drift–pressure identity for the limiting bounded profile.*

*Proof.* The setup paper local energy inequality (recalled inside the imported results Theorem 1.3 (3)) states that for every locally suitable centered profile $(U, \Pi)$ and every nonnegative $\Phi \in C_c^\infty(\mathbb{R}^3 \times \mathbb{R})$,

$$\iint |\nabla_y U|^2 \Phi \, dy\, d\tau \le \iint \frac{|U|^2}{2}\big(\partial_\tau\Phi + \Delta_y\Phi + \tfrac{1}{2}y\cdot\nabla_y\Phi + \tfrac{1}{2}\Phi\big)\, dy\, d\tau + \iint \Big(\tfrac{|U|^2}{2} + \Pi\Big) U\cdot\nabla_y\Phi\, dy\, d\tau. \tag{9.13}$$

This identity is preserved by the exact covariant recentering $\mathcal{R}_{z_n}$, so it applies to $U_n^{z_n}$ and $\Pi_n^{z_n}$ with the same constants.

Apply (9.13) to $U_n^{z_n}, \Pi_n^{z_n}$ with the test function

$$\Phi(y,\tau) := r_n^{-3}\,\phi(y/r_n, \tau)$$

where $\phi \in C_c^\infty(K)$ is fixed. Change variables $y = r_n x$, keeping $\tau = s$ (centered time is not rescaled). Then

$$\iint |\nabla_y U_n^{z_n}|^2 \Phi\, dy\, d\tau = \iint |\nabla_y U_n^{z_n}(r_n x, s)|^2 r_n^{-3}\phi(x,s)\, r_n^3\, dx\, ds = r_n^{-2}\iint |\nabla_x V_n|^2 \phi\, dx\, ds,$$

because $|\nabla_y U_n^{z_n}|^2(r_n x, s) = r_n^{-2}|\nabla_x V_n|^2(x,s)$. Similarly,

$$\iint \frac{|U_n^{z_n}|^2}{2}\partial_\tau\Phi\, dy\, d\tau = \iint \frac{|V_n|^2}{2}\partial_s\phi\, dx\, ds,$$

$$\iint \frac{|U_n^{z_n}|^2}{2}\Delta_y\Phi\, dy\, d\tau = r_n^{-2}\iint \frac{|V_n|^2}{2}\Delta_x\phi\, dx\, ds,$$

$$\iint \frac{|U_n^{z_n}|^2}{2}\tfrac{1}{2}y\cdot\nabla_y\Phi\, dy\, d\tau = \iint \frac{|V_n|^2}{2}\tfrac{1}{2}x\cdot\nabla_x\phi\, dx\, ds,$$

$$\iint \frac{|U_n^{z_n}|^2}{2}\tfrac{1}{2}\Phi\, dy\, d\tau = \iint \frac{|V_n|^2}{2}\tfrac{1}{2}\phi\, dx\, ds.$$

For the convective and pressure-flux terms,

$$\iint \Big(\tfrac{|U_n^{z_n}|^2}{2} + \Pi_n^{z_n}\Big)U_n^{z_n}\cdot\nabla_y\Phi\, dy\, d\tau = r_n^{-1}\iint \tfrac{|V_n|^2}{2}V_n\cdot\nabla_x\phi\, dx\, ds + \iint Q_n V_n\cdot\nabla_x\phi\, dx\, ds,$$

using $Q_n = r_n^{-1}\Pi_n^{z_n}(r_n x, s)$, $\nabla_y \Phi = r_n^{-4}\nabla_x \phi(\cdot/r_n, \cdot)$, and $dy = r_n^3 dx$. Substituting all changes into (9.13) gives (9.11).

The uniform bound (9.12) follows from the $L^\infty$ bounds $|V_n| \le M$, the local $L^{3/2}$-pressure compactness in Lemma 9.73, Hölder's inequality on the compact support of $\phi$, and the explicit dependence of the right-hand side on $\|\phi\|_{C^2}$, $|K|$, and $r_n$. Since $r_n \to \infty$, the viscous and convective terms carry the vanishing factors $r_n^{-2}$ and $r_n^{-1}$. After passing to the pressure-gauge subsequence supplied by Lemma 9.73 and to the bounded-velocity subsequence, the remaining drift and pressure-flux terms pass to their distributional limits. This is the local energy identity used in the proof of Lemma 9.78. □

**Lemma 9.67** (Minimal-scale quotient formulation)**.** *Let $V_n, W_n, c_n, Q_n$ be as in Lemma 9.65. The minimal-scale analysis uses the following quotient formulation.*

(i) *The distributional equation and local energy inequality are always applied to the ungauged field $V_n$ and its pressure $Q_n$.*
(ii) *The affine-normalized field $W_n = V_n - c_n(s)$ is used only for oscillation functionals and Young measures, which are invariant under spatially homogeneous shifts after the gauge sequence is either compact or assigned to $\mathcal{C}_{\mathrm{aff}}$.*
(iii) *If $c_n$ is not compact in $L^3_{\mathrm{loc}}$ in time, the homogeneous gauge sequence itself generates an affine/parasitic descendant. If $c_n \to c$ strongly in $L^3_{\mathrm{loc}}$, all Young-measure variances and strong compactness statements pass from $V_n$ to $W_n$ by deterministic translation of the generated Young measures.*

*Thus the symbol "affine pressure paired with $c_n$" means a quotient functional, not an asserted $L^{3/2}_{\mathrm{loc}}$ pressure unless the time regularity of $c_n$ has been separately proved.*

*Proof.* Statement (i) is the construction in Lemma 9.65. The local energy inequality is stable under the minimal-scale spatial blow-down for $V_n$, because $V_n$ is obtained from an actual centered solution by an exact covariant recentering and a spatial scaling; no subtraction of $c_n$ is involved.

For (ii) and (iii), observe that subtracting a spatially homogeneous function does not change any $\mathrm{osc}_{3,\mathrm{sp}}$-type seminorm in which the same class of homogeneous gauges is minimized over. If the sequence $c_n$ is not compact in $L^3_{\mathrm{loc}}$, Lemma 9.74 places the resulting homogeneous descendant in $\mathcal{C}_{\mathrm{aff}}$. If $c_n \to c$ strongly, then Lemma 9.75 identifies the Young measure of $W_n$ with the deterministic translate of the Young measure of $V_n$, and the variance densities agree. Hence all compactness and variance conclusions used for $W_n$ follow from identities proved for $V_n$, without requiring $\partial_s c_n$ to be a function. □

**Lemma 9.68** (Covariance of the whole-space Riesz pressure)**.** *At the finite-energy Leray level one may choose the standard whole-space pressure representative*

$$p(\cdot, t) = \mathcal{R}_i \mathcal{R}_j (u_i u_j)(\cdot, t)$$

*modulo a function of time. This representative is preserved, modulo the corresponding time gauge, by physical Type I scaling, the centered self-similar change of variables, centered time translation, spatial translation, and the covariant translations used in this paper. Local pressure representatives are restrictions of this whole-space representative, followed only by the admissible addition of functions of time.*

*Proof.* Let $K_{ij}$ be the Calderón–Zygmund pressure kernel. It is homogeneous of degree $-3$:

$$K_{ij}(\lambda x) = \lambda^{-3} K_{ij}(x), \qquad \lambda > 0.$$

Therefore, if

$$u^\lambda(x,t) = \lambda u(x_0 + \lambda x, t_0 + \lambda^2 t), \qquad p^\lambda(x,t) = \lambda^2 p(x_0 + \lambda x, t_0 + \lambda^2 t),$$

then

$$p^{\lambda}(\cdot,t) = \mathcal{R}_i\mathcal{R}_j(u_i^{\lambda}u_j^{\lambda})(\cdot,t)$$

modulo the scaled time gauge. Spatial translations commute with convolution by $K_{ij}$, and time translations only relabel the time variable. The centered change of variables is a time-dependent parabolic scaling with the pressure factor $P(y,\tau) = (-t)p(x,t)$; hence the same homogeneity gives the centered Riesz representative at each fixed centered time. The covariant translations used below are compositions of these exact spatial/time changes and the admissible pressure gauges recorded in the centered equation. Since all additional gauges are functions of time, their gradients vanish and their restrictions to compact cylinders are precisely the local pressure gauges used in the Seregin topology. □

**Lemma 9.69** (Harmonic remainders in normalized pressure limits). *Let $(U_n, \Pi_n)$ be a sequence of normalized profiles obtained from the whole-space representative in Lemma 9.68, before applying any homogeneous velocity quotient. Suppose that $U_n \to U$ locally smoothly and that, after time-dependent gauges, $\Pi_n \to \Pi$ locally in $L^{3/2}$ or weakly in $L^{3/2}$. Then*

$$\Pi(\cdot,s) - \mathcal{R}_i\mathcal{R}_j(U_iU_j)(\cdot,s)$$

*is a time-dependent constant. After the affine/parasitic quotient is applied, the only additional admissible harmonic pressure direction is a linear function $\beta(s)\cdot y$; this linear part is the pressure paired with the spatially homogeneous parasitic velocity mode and is absent in the parasitic-free representative.*

*Proof.* Before the affine quotient, the pressure of each approximating profile is the whole-space Riesz representative, modulo a time gauge. Since the velocities are uniformly bounded in $L^\infty$, these representatives have uniformly bounded BMO seminorms on $\mathbb{R}^3$:

$$\|\mathcal{R}_i\mathcal{R}_j(U_{n,i}U_{n,j})(\cdot,s)\|_{\mathrm{BMO}} \le CM^2.$$

The BMO bound is invariant under the exact scalings and translations in Lemma 9.68. Hence, after passing to a subsequence at each fixed time interval and using a diagonal extraction, the whole-space representatives converge weakly-* in local BMO-duality to a whole-space BMO pressure $P^*$, modulo time gauges. The local $L^{3/2}$-pressure limit $\Pi$ agrees with $P^*$ on each compact cylinder, up to the same time-dependent gauge, because both are local limits of the same normalized representatives. Passing to the limit in the pressure Poisson equation gives

$$-\Delta P^*(\cdot,s) = \partial_i\partial_j(U_iU_j)(\cdot,s)$$

in distributions. Thus the difference

$$h(\cdot,s) := P^*(\cdot,s) - \mathcal{R}_i\mathcal{R}_j(U_iU_j)(\cdot,s)$$

is harmonic in space and belongs to $\mathrm{BMO}(\mathbb{R}^3)$ modulo an additive constant. An entire harmonic BMO function is constant: indeed, for each ball $B_R(x)$, the interior estimate for harmonic functions and the John–Nirenberg control of mean oscillation give

$$|\nabla h(x,s)| \le CR^{-1}\|h(\cdot,s)\|_{\mathrm{BMO}(\mathbb{R}^3)}$$

with $C$ independent of $R$; letting $R\to\infty$ yields $\nabla h(\cdot,s) = 0$. Since $\Pi$ and $P^*$ differ only by the local time gauge, before affine quotienting the difference between $\Pi$ and $\mathcal{R}_i\mathcal{R}_j(U_iU_j)$ is spatially constant.

Equivalently, if a non-affine harmonic component were present on some compact ball and time interval, elliptic regularity would give a point where a nonzero spatial derivative of order at least one remains. Testing against compactly supported functions with zero spatial mean, and with zero first moments when isolating the affine quotient, would detect this component.

Such a component cannot arise from the uniformly BMO whole-space Riesz representatives plus time-only gauges described in Lemma 9.68.

The affine/parasitic normalization is the only subsequent operation that changes the velocity by a spatially homogeneous function. For a smooth homogeneous mode $b(s)$, substitution into the centered equation gives the affine pressure gradient

$$\nabla_y\big[(-\partial_s b - \tfrac{1}{2}b)\cdot y\big].$$

For merely bounded measurable homogeneous gauges this expression is used only as the quotient functional specified in Definition 8.16, Remark 9.64, and Lemma 9.67. Thus the normalized profile class contains no non-affine harmonic pressure remainders: only time gauges and, inside $\mathcal{C}_{\mathrm{aff}}$, the explicit affine parasitic pressure direction are allowed. □

**Proposition 9.70** (Normalized pressure representation modulo affine parasitic modes)**.** *Every normalized Seregin, terminal, covariant descendant, minimal-scale blow-down, or concentrating descendant profile used in the residual profile class is represented by a pressure of the form*

$$\Pi(\cdot,s) = \mathcal{R}_i\mathcal{R}_j(U_iU_j)(\cdot,s) + \beta(s)\cdot y + \alpha(s), \tag{9.14}$$

*where $\alpha$ is an arbitrary time gauge and $\beta(s)\cdot y$ is the affine pressure paired with the spatially homogeneous parasitic velocity mode. Outside the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$, the representative is taken with $\beta = 0$.*

*Proof.* The whole-space Riesz representative and its covariance under the exact operations are given by Lemma 9.68. The exclusion of non-affine harmonic pressure remainders in normalized limits is Lemma 9.69. The only additional pressure direction created by affine/parasitic normalization is the explicit linear pressure paired with the homogeneous velocity gauge. Hence (9.14) is the complete normalized pressure representation. If $\beta \neq 0$ has a retained effect, the associated profile is assigned to $\mathcal{C}_{\mathrm{aff}}$; in the parasitic-free representative used outside $\mathcal{C}_{\mathrm{aff}}$, one sets $\beta = 0$. □

**Lemma 9.71** (Scaled pressure compactness modulo affine gauges)**.** *Let $(U_n,\Pi_n)$ be bounded centered solutions with $\|U_n\|_{L^\infty} \le M$, let $z_n$ be covariant centers, let $r_n \to \infty$, and set*

$$\widetilde{U}_n(y,s) := (\mathcal{R}_{z_n}U_n)(y,s), \qquad \widetilde{\Pi}_n(y,s) := (\mathcal{R}_{z_n}\Pi_n)(y,s).$$

*For*

$$\widetilde{Q}_n(x,s) := r_n^{-1}\widetilde{\Pi}_n(r_nx,s)$$

*and every compact cylinder $K = B_R \times I \Subset \mathbb{R}^3 \times \mathbb{R}$, there are affine functions*

$$L_n(x,s) = b_n(s)\cdot x + a_n(s)$$

*such that*

$$\widetilde{Q}_n - L_n \to 0 \qquad \text{strongly in } L^{3/2}(K). \tag{9.15}$$

*The same statement holds for the minimal-scale pressures $Q_n$ in the quotient sense of Lemma 9.67. If the affine parts $L_n$ carry a nonzero retained descendant, then that descendant lies in the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$. Otherwise the affine parts may be discarded in the affine-normalized quotient, and the non-affine pressure converges strongly to zero.*

*Proof.* The covariant recentering is an exact symmetry, so it suffices to work with a single bounded centered solution $(U,\Pi)$. For a.e. $s$, the pressure Poisson equation is

$$-\Delta\Pi(\cdot,s) = \partial_i\partial_j(U_iU_j)(\cdot,s)$$

in $\mathbb{R}^3$. Let

$$\Pi^{\mathrm{loc}}(\cdot,s) := \mathcal{R}_i\mathcal{R}_j(U_iU_j)(\cdot,s)$$

be the whole-space Calderón–Zygmund representative. Since $\|U\|_{L^\infty} \le M$,

$$\|\Pi^{\mathrm{loc}}(\cdot, s)\|_{\mathrm{BMO}(\mathbb{R}^3)} \le CM^2$$

uniformly in $s$. Hence, by the John–Nirenberg inequality, for every ball $B_\rho$,

$$\inf_{a\in\mathbb{R}} \|\Pi^{\mathrm{loc}}(\cdot, s) - a\|_{L^{3/2}(B_\rho)} \le CM^2|B_\rho|^{2/3}.$$

After the scaling $x \mapsto r_n x$, this gives

$$\inf_{a_n(s)} \|r_n^{-1}\Pi^{\mathrm{loc}}(r_n\cdot, s) - a_n(s)\|_{L^{3/2}(B_R)} \le C_{R,M} r_n^{-1} \to 0.$$

The bound is uniform for $s \in I$, hence after choosing the measurable minimizing averages $a_n(s)$ one also has

$$\|r_n^{-1}\Pi^{\mathrm{loc}}(r_n\cdot, s) - a_n(s)\|_{L^{3/2}(B_R\times I)} \le C_{R,M,I} r_n^{-1} \to 0.$$

By Proposition 9.70, the only remaining pressure component in the normalized profile class is

$$\beta(s) \cdot y + \alpha(s).$$

After scaling,

$$r_n^{-1}\big(\beta(s) \cdot r_n x + \alpha(s)\big) = \beta(s) \cdot x + r_n^{-1}\alpha(s),$$

which is an affine function of $x$. Taking this function as $L_n$, the remaining pressure is the scaled Leray representative already estimated above, and it tends to zero strongly in $L^{3/2}(B_R \times I)$. This proves (9.15). The minimal-scale pressure is used only up to the affine quotient described in Lemma 9.67; in that quotient the same estimate applies. □

**Lemma 9.72** (Pressure–velocity product convergence)**.** *Let $K \Subset \mathbb{R}^3 \times \mathbb{R}$. Suppose*

$$Q_n \to Q \quad \textit{strongly in } L^{3/2}(K), \qquad W_n \rightharpoonup^* W \quad \textit{in } L^\infty(K),$$

*with* $\sup_n \|W_n\|_{L^\infty(K)} < \infty$*. Then*

$$Q_n W_n \to QW \qquad \textit{in } \mathcal{D}'(K).$$

*Proof.* Let $\varphi \in C_c^\infty(K;\mathbb{R}^3)$. Then

$$\iint_K (Q_n W_n - QW) \cdot \varphi = \iint_K (Q_n - Q)W_n \cdot \varphi + \iint_K Q(W_n - W) \cdot \varphi.$$

The first term tends to zero by the strong $L^{3/2}$ convergence of $Q_n$ and the uniform $L^\infty$ bound on $W_n$. Since $Q\varphi \in L^1(K)$, the second term tends to zero by weak-* convergence in $L^\infty(K)$. □

**Lemma 9.73** (Pressure compactness at a minimal mesoscopic scale)**.** *Let $Q_n$ be as in Lemma 9.65. Assume in addition the* normalized whole-space Riesz-pressure hypothesis*: each $Q_n$ is the whole-space Riesz-pressure representative of the minimal-scale rescaled solution, normalized modulo the affine pressure gauges and modulo time-only functions in the sense of Definition 8.14 and the local pressure atlas imported in Theorem 1.3 (3); equivalently, each $Q_n$ is the canonical normalized representative produced by Lemma 9.6 after subtracting affine gauges. For every compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$, exactly one of the following holds:*

(i) *the affine pressure/velocity gauges produce a retained descendant in $\mathcal{C}_{\mathrm{aff}}$;*

(ii) *after discarding those affine gauges in the affine-normalized quotient, there exists a pressure limit $Q$ such that*

$$Q_n \to Q \qquad \textit{strongly in } L^{3/2}(K).$$

*In fact, in the non-affine quotient one may take $Q = 0$. Consequently, if $W_n \rightharpoonup^* \bar{W}$ in $L^\infty_{\mathrm{loc}}$, then*

$$Q_n W_n \rightharpoonup Q\bar{W} \qquad \textit{in distributions on } K.$$

*Proof.* This is Lemma 9.71 applied to the minimal-scale scaling. If the affine gauges retain nonzero descendant concentration, we are in alternative (i). Otherwise the affine gauges are precisely the quotient directions removed by affine normalization, and the remaining pressure converges strongly to 0 in $L^{3/2}(K)$. Strong convergence of the pressure and weak-* convergence of the uniformly bounded velocity give the product convergence in distributions by Lemma 9.72. □

**Lemma 9.74** (Homogeneous gauges: compactness modulo the affine stratum)**.** *Let $c_n : I \to \mathbb{R}^3$ be uniformly bounded in $L^\infty(I)$ on every compact interval $I \Subset \mathbb{R}$. Then, after passing to a subsequence, exactly one of the following alternatives is used in the ordered minimal-scale argument:*

(i) *$c_n \to c$ strongly in $L^3_{\mathrm{loc}}(\mathbb{R};\mathbb{R}^3)$;*
(ii) *the spatially homogeneous sequence*

$$C_n(y,s) := c_n(s)$$

*has a noncompact homogeneous quotient defect. This defect has zero spatial quotient oscillation on every compact cylinder and is assigned to the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$.*

*Proof.* If $c_n$ has a strongly convergent subsequence in $L^3_{\mathrm{loc}}$, we pass to that subsequence and are in alternative (i). If no such subsequence exists, then the family is not relatively compact in $L^3(I)$ for some compact interval $I \Subset \mathbb{R}$. Since the $L^\infty(I)$ bound gives uniform boundedness and tightness on this fixed interval, the only possible failure in the Fréchet–Kolmogorov criterion is time-translation equicontinuity. Hence there are translations $\theta_n \to 0$ and $\eta > 0$ such that

$$\|c_n(\cdot + \theta_n) - c_n\|_{L^3(I)} \geq \eta.$$

The associated spatially homogeneous profiles $C_n(y,s) = c_n(s)$ have zero spatial quotient oscillation on every compact cylinder. The displayed lower bound records noncompactness only in the homogeneous time-dependent quotient; it does not assert the existence of a strongly convergent nonzero homogeneous descendant. By the affine/parasitic routing convention and Lemma 8.18, this quotient defect belongs to $\mathcal{C}_{\mathrm{aff}}$. This is alternative (ii). □

**Lemma 9.75** (Variance is invariant under homogeneous velocity gauges)**.** *Let $V_n$ be uniformly bounded in $L^\infty_{\mathrm{loc}}$, let $c_n \to c$ strongly in $L^3_{\mathrm{loc}}(\mathbb{R})$, and set*

$$W_n(x,s) := V_n(x,s) - c_n(s).$$

*If $V_n$ generates a Young measure $\nu_{x,s}$ with barycenter $\bar{V}$, then $W_n$ generates the translated Young measure*

$$\widetilde{\nu}_{x,s}(A) = \nu_{x,s}(A + c(s))$$

*with barycenter $\bar{W} = \bar{V} - c(s)$. The variance densities agree:*

$$\langle |\xi - \bar{V}(x,s)|^2 \rangle_{\nu_{x,s}} = \langle |\xi - \bar{W}(x,s)|^2 \rangle_{\widetilde{\nu}_{x,s}}.$$

*Consequently any distributional variance inequality proved for $V_n$ passes unchanged to the affine-normalized fields $W_n$.*

*Proof.* After passing to a subsequence, $c_n(s) \to c(s)$ a.e. and in measure on every compact time interval. For every continuous compactly supported function $F(\xi)$ and every compactly supported test function $\varphi(x,s)$,

$$\iint \varphi F(W_n)\,dx\,ds = \iint \varphi F(V_n - c_n(s))\,dx\,ds \to \iint \varphi \int F(\xi - c(s))\,d\nu_{x,s}(\xi)\,dx\,ds.$$

This is precisely the translated Young measure. Translation by a deterministic vector preserves centered second moments, giving the variance identity. The distributional inequality involves only this variance density and therefore is unchanged. □

**Lemma 9.76** (Support transfer for the Young variance defect)**.** *Let $K \Subset \mathbb{R}^3 \times \mathbb{R}$ be a compact macroscopic cylinder and let $X_Y(K) = K \times \mathcal{P}(\overline{B_M})$, where $\overline{B_M}$ is the compact velocity ball fixed by the local $L^\infty$ bound. Let*

$$\lambda_n = \frac{\mu_n^Y \lfloor E_n}{\mu_n^Y(E_n)} \qquad (\mu_n^Y(E_n) > 0)$$

*be normalized realized Young-measure variance defects on $X_Y(K)$. If $\lambda_n \rightharpoonup \lambda$ and $A \subset X_Y(K)$ is closed, then support of $\lambda$ in $A$ removes all surviving Young variance defect outside $A$; positive surviving variance outside $A$ produces a normalized Young support profile outside $A$.*

*Proof.* The space $X_Y(K)$ is compact metrizable because $K$ is compact and $\mathcal{P}(\overline{B_M})$ is compact metrizable in the weak topology. The conclusion is Lemma 1.12 applied to this space. This is the only measure-theoretic transfer used when the hidden-scale Young defect is removed after the variance inequality. □

**Lemma 9.77** (Renormalization for the coherent barycenter equation)**.** *Let $Z \in L^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R})$, $Q \in L^{3/2}_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R})$, and $\nabla \cdot Z = 0$. Suppose*

$$\partial_s Z + \frac{1}{2} x \cdot \nabla Z + \frac{1}{2} Z + \nabla Q = 0$$

*in distributions. Then*

$$\partial_s \frac{|Z|^2}{2} + \frac{1}{2} x \cdot \nabla \frac{|Z|^2}{2} + \frac{1}{2}|Z|^2 + \operatorname{div}(QZ) = 0$$

*in distributions.*

*Proof.* Let $\varrho_\varepsilon$ be a standard space–time mollifier and write $Z_\varepsilon = \varrho_\varepsilon * Z$, $Q_\varepsilon = \varrho_\varepsilon * Q$ on a compact cylinder $K$, after extending across a slightly larger compact cylinder. Since $x/2$ is smooth and Lipschitz on $K$,

$$\varrho_\varepsilon * ((x/2) \cdot \nabla Z) - (x/2) \cdot \nabla Z_\varepsilon \to 0 \qquad \text{in } L^p(K)$$

for every finite $p$. The mollified equation is therefore

$$\partial_s Z_\varepsilon + \frac{1}{2} x \cdot \nabla Z_\varepsilon + \frac{1}{2} Z_\varepsilon + \nabla Q_\varepsilon = R_\varepsilon, \qquad R_\varepsilon \to 0 \quad \text{in } L^p_{\mathrm{loc}}.$$

Multiplying by $Z_\varepsilon$ gives

$$\partial_s \frac{|Z_\varepsilon|^2}{2} + \frac{1}{2} x \cdot \nabla \frac{|Z_\varepsilon|^2}{2} + \frac{1}{2}|Z_\varepsilon|^2 + Z_\varepsilon \cdot \nabla Q_\varepsilon = R_\varepsilon \cdot Z_\varepsilon.$$

Because $\nabla \cdot Z = 0$, also $\nabla \cdot Z_\varepsilon = 0$, and hence

$$Z_\varepsilon \cdot \nabla Q_\varepsilon = \operatorname{div}(Q_\varepsilon Z_\varepsilon).$$

Now $Z_\varepsilon \to Z$ in $L^p_{\mathrm{loc}}$ for every finite $p$, while $Z \in L^\infty_{\mathrm{loc}}$, and $Q_\varepsilon \to Q$ in $L^{3/2}_{\mathrm{loc}}$. Thus $Q_\varepsilon Z_\varepsilon \to QZ$ in distributions, and $R_\varepsilon \cdot Z_\varepsilon \to 0$ in distributions. Letting $\varepsilon \downarrow 0$ gives the claimed renormalized identity. □

**Lemma 9.78** (Barycenter equation and variance inequality)**.** *Let $V_n, W_n, Q_n, c_n$ be a minimal-scale blow-down. Then either the sequence has a retained affine/parasitic descendant in $\mathcal{C}_{\mathrm{aff}}$, or, after passing to a subsequence, $c_n \to c$ strongly in $L^3_{\mathrm{loc}}$,*

$$V_n \rightharpoonup^* \bar{V}, \qquad W_n \rightharpoonup^* \bar{W} \qquad \text{in } L^\infty_{\mathrm{loc}},$$

*and $V_n, W_n$ generate Young measures $\nu_{x,s}$ and $\widetilde{\nu}_{x,s}$, respectively. Set*

$$E(x,s) := \langle |\xi|^2 \rangle_{\widetilde{\nu}_{x,s}}, \qquad \mathcal{V}(x,s) := E(x,s) - |\bar{W}(x,s)|^2.$$

*Then $\bar V$ satisfies the coherent transport equation with pressure $Q$. The affine-normalized barycenter $\bar W = \bar V - c(s)$ satisfies the same equation only as a quotient modulo spatially homogeneous modes; after passing to the parasitic-free representative outside $\mathcal{C}_{\mathrm{aff}}$, this quotient equation is represented as*

$$\partial_s \bar W + \frac{1}{2}x\cdot\nabla\bar W + \frac{1}{2}\bar W + \nabla Q = 0, \qquad \nabla\cdot\bar W = 0, \tag{9.16}$$

*and the nonnegative variance density $\mathcal{V} \in L^\infty_{\mathrm{loc}}$ satisfies*

$$\partial_s\mathcal{V} + \frac{1}{2}x\cdot\nabla\mathcal{V} + \mathcal{V} \le 0 \tag{9.17}$$

*in distributions.*

*Proof.* By Lemma 9.74, either the homogeneous gauge defect belongs to $\mathcal{C}_{\mathrm{aff}}$, or $c_n \to c$ strongly in $L^3_{\mathrm{loc}}$ after passing to a subsequence. We work in the second case. The weak-* compactness of $V_n, W_n$ and generation of Young measures are standard consequences of the uniform $L^\infty_{\mathrm{loc}}$ bound. Testing (9.9) against a compactly supported divergence-free vector field, the viscous term is bounded by

$$Cr_n^{-2}\|W_n\|_{L^\infty},$$

and the nonlinear term by

$$Cr_n^{-1}\|V_n\|^2_{L^\infty}.$$

Both tend to zero. The pressure compactness in Lemma 9.73 gives either a retained descendant in $\mathcal{C}_{\mathrm{aff}}$, or the pressure gradient limit in the affine-normalized quotient. Thus $\bar V$ satisfies the coherent transport equation. Since $\bar W = \bar V - c(s)$, passing from $\bar V$ to $\bar W$ changes only the spatially homogeneous quotient component. We do not differentiate $c$ as a function. Instead, Lemma 9.67 says that this is an affine/parasitic quotient direction. If it has retained contribution, the profile is in $\mathcal{C}_{\mathrm{aff}}$; otherwise the canonical parasitic-free representative removes it, and the representative satisfies (9.16).

We next pass to the local energy inequality for the ungauged fields $V_n$. The appropriate form is the scaled suitable inequality of Lemma 9.66, applied to nonnegative compactly supported test functions. This avoids using a formal pointwise multiplication of the scaled equation by $V_n$. There is no affine pressure work term in this inequality; the homogeneous gauge has not been subtracted. In (9.11), the viscous diffusion and nonlinear-flux terms vanish in the limit because of the factors $r_n^{-2}$ and $r_n^{-1}$, while the dissipation term is nonnegative and may be dropped. The strong pressure convergence gives

$$\mathrm{div}(Q_nV_n) \to \mathrm{div}(Q\bar V).$$

Therefore

$$\partial_s\frac{E_V}{2} + \frac{1}{2}x\cdot\nabla\frac{E_V}{2} + \frac{1}{2}E_V + \mathrm{div}(Q\bar V) \le 0$$

in distributions, where

$$E_V(x,s) := \langle|\xi|^2\rangle_{\nu_{x,s}}$$

is the second moment of the Young measure generated by $V_n$. On the other hand, applying Lemma 9.77 to the barycenter equation for $\bar V$ gives

$$\partial_s\frac{|\bar V|^2}{2} + \frac{1}{2}x\cdot\nabla\frac{|\bar V|^2}{2} + \frac{1}{2}|\bar V|^2 + \mathrm{div}(Q\bar V) = 0.$$

Subtracting gives the variance inequality for the variance of $V_n$. By Lemma 9.75, the same variance density is $\mathcal{V}$, the variance of $W_n$. This proves (9.17). □

**Lemma 9.79** (Bounded ancient variance Liouville theorem)**.** *Let $F \in L^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R})$ satisfy*

$$0 \le F \le C < \infty$$

*and*

$$\partial_s F + \frac{1}{2} x \cdot \nabla F + F \le 0$$

*in distributions on $\mathbb{R}^3 \times \mathbb{R}$. Then $F \equiv 0$.*

*Proof. Step 1: renormalization.* Set $G(x,s) := e^s F(x,s)$, so that $0 \le G \le Ce^s$ and, for every smooth nonnegative compactly supported test function $\psi$,

$$\langle G, (-\partial_s - \tfrac{1}{2}x \cdot \nabla - \tfrac{3}{2})\psi\rangle = \langle F, (-\partial_s - \tfrac{1}{2}x \cdot \nabla - \tfrac{1}{2})(e^s\psi)\rangle = \langle F, e^s(-\partial_s - \tfrac{1}{2}x \cdot \nabla - \tfrac{1}{2})\psi - e^s\psi\rangle.$$

Adding the two contributions reduces to $\langle F, e^s(-\partial_s - \frac{1}{2}x \cdot \nabla - \frac{3}{2})\psi\rangle$, and the distributional hypothesis $\partial_s F + \frac{1}{2}x \cdot \nabla F + F \le 0$ applied to $e^s\psi \ge 0$ gives

$$\partial_s G + \frac{1}{2} x \cdot \nabla G \le 0 \qquad \text{in distributions on } \mathbb{R}^3 \times \mathbb{R}. \tag{9.18}$$

*Step 2: backward adjoint semigroup.* The forward characteristic flow of $\widetilde{L} := \partial_s + \frac{1}{2}x \cdot \nabla$ starting from $(x_0, s_0)$ is $x(\sigma) = e^{(\sigma - s_0)/2}x_0$. Define, for $T \ge 0$, the backward adjoint operator on test functions

$$(\mathcal{S}_T\psi)(x,s) := e^{-3T/2}\psi\big(e^{-T/2}x,\ s - T\big).$$

The prefactor $e^{-3T/2}$ is the Jacobian of $x \mapsto e^{-T/2}x$ on $\mathbb{R}^3$, so $\int \mathcal{S}_T\psi \, dx\, ds = \int \psi \, dx\, ds$ and the family $\{\mathcal{S}_T\}_{T \ge 0}$ is the adjoint of the forward characteristic flow on $L^1(\mathbb{R}^3 \times \mathbb{R})$. In particular, by the substitution $x = e^{T/2}x'$, $s = s' + T$ (Jacobian $e^{3T/2}$),

$$\langle G, \mathcal{S}_T\psi\rangle = \iint G(e^{T/2}x', s' + T)\, \psi(x', s')\, dx'\, ds'. \tag{9.19}$$

*Step 3: monotonicity in $T$.* Fix $\psi \ge 0$ smooth and compactly supported. A direct computation using the chain rule shows that $\frac{d}{dT}\mathcal{S}_T\psi = -\widetilde{L}^*(\mathcal{S}_T\psi)$ where $\widetilde{L}^* = -\partial_s - \frac{1}{2}x \cdot \nabla - \frac{3}{2}$; equivalently,

$$\frac{d}{dT}\langle G, \mathcal{S}_T\psi\rangle = \langle G, -\widetilde{L}^*(\mathcal{S}_T\psi)\rangle = \langle \widetilde{L}G, \mathcal{S}_T\psi\rangle.$$

Since $\mathcal{S}_T\psi \ge 0$ and $\widetilde{L}G \le 0$ distributionally by (9.18),

$$\frac{d}{dT}\langle G, \mathcal{S}_T\psi\rangle \le 0. \tag{9.20}$$

*Step 4: Lebesgue limit and conclusion.* Pick a standard nonnegative mollifier $\psi_0 \in C^\infty_c(\mathbb{R}^3 \times \mathbb{R})$ of unit total mass, fix $(x_1, s_1) \in \mathbb{R}^3 \times \mathbb{R}$ and $T \ge 0$, and set $(x_0, s_0) := (e^{-T/2}x_1, s_1 - T)$. Apply (9.20) to the rescaled test function $\psi_\varepsilon(x,s) = \varepsilon^{-4}\psi_0\big((x - x_0)/\varepsilon, (s - s_0)/\varepsilon\big)$: since $T \mapsto \langle G, \mathcal{S}_T\psi_\varepsilon\rangle$ is nonincreasing on $[0, \infty)$,

$$\langle G, \mathcal{S}_T\psi_\varepsilon\rangle \le \langle G, \psi_\varepsilon\rangle.$$

By (9.19),

$$\langle G, \mathcal{S}_T\psi_\varepsilon\rangle = \iint G(e^{T/2}x', s' + T)\, \psi_\varepsilon(x', s')\, dx'\, ds'$$

which approximates $G(e^{T/2}x_0, s_0 + T) = G(x_1, s_1)$ at the rate dictated by Lebesgue differentiation. Similarly $\langle G, \psi_\varepsilon\rangle \to G(x_0, s_0)$ as $\varepsilon \downarrow 0$ at common Lebesgue points. Therefore, at every common Lebesgue point of $G$,

$$G(x_1, s_1) \le G(e^{-T/2}x_1, s_1 - T).$$

Translating back through $G = e^s F$,

$$e^{s_1}F(x_1, s_1) \le e^{s_1 - T}F(e^{-T/2}x_1, s_1 - T), \tag{9.21}$$

hence $F(x_1, s_1) \le e^{-T} F(e^{-T/2} x_1, s_1 - T) \le Ce^{-T}$ using the hypothesis $F \le C$ a.e. Letting $T \to \infty$ gives $F(x_1, s_1) = 0$ at every Lebesgue point, hence almost everywhere. The distributional inequality is satisfied by the zero a.e. representative, so $F \equiv 0$. □

**Lemma 9.80** (Minimal-scale blow-downs are strongly compact)**.** *Every minimal-scale blow-down $W_n$ either has a retained affine/parasitic descendant in $\mathcal{C}_{\mathrm{aff}}$, or has, after passing to a subsequence, a limit $\bar{W}$ such that*

$$W_n \to \bar{W} \qquad \textit{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R}).$$

*The limit satisfies* (9.16)*.*

*Proof.* By Lemma 9.78, either the affine/parasitic alternative has occurred, or the Young-measure variance $\mathcal{V}$ satisfies the hypotheses of Lemma 9.79. Hence $\mathcal{V} \equiv 0$. Thus

$$\widetilde{\nu}_{x,s} = \delta_{\bar{W}(x,s)} \qquad \text{for a.e. } (x, s).$$

Uniform $L^\infty_{\mathrm{loc}}$ boundedness then upgrades convergence in measure to strong convergence in $L^3_{\mathrm{loc}}$. The limit equation has already been proved in Lemma 9.78. □

**Theorem 9.81** (Minimal mesoscopic oscillation scale reduction)**.** *Every minimal mesoscopic oscillation scale generated by a realized recurrent critical-tail extraction after concentration-set removal and affine normalization has a descendant in*

$$\mathcal{C}_{\mathrm{conc}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{cohcrit}}.$$

*Proof.* Let $z_n, r_n$ be a minimal mesoscopic oscillation scale. If, before passing to the minimal-scale blow-down, some unit covariant observer in the selected residual window carries a positive constant-in-time oscillation threshold, then the concentration alternative is regenerated and the descendant lies in $\mathcal{C}_{\mathrm{conc}}$. It remains to consider the case in which unit concentration has been removed at all fixed positive thresholds.

Form the minimal-scale blow-down $W_n$ of Definition 9.63. By Lemma 9.80, either the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$ is reached, or, after passing to a subsequence,

$$W_n \to W \qquad \text{strongly in } L^3_{\mathrm{loc}},$$

and $(W, Q)$ satisfies the coherent critical-tail transport equation

$$\partial_s W + \frac{1}{2} x \cdot \nabla W + \frac{1}{2} W + \nabla Q = 0, \qquad \nabla \cdot W = 0.$$

The minimal-scale lower bound and strong convergence give

$$\mathrm{osc}_{3,\mathrm{sp}}\big(W; B_1(0) \times (-1, 1)\big) \ge \varepsilon_0.$$

Thus the limit is not spatially affine/parasitic on the unit critical-tail window. If this non-affine lower bound were lost in a further descendant, the descendant would be assigned to $\mathcal{C}_{\mathrm{aff}}$ by Theorem 8.21. Otherwise the strongly realized nonzero limit is a coherent critical-tail descendant, hence belongs to $\mathcal{C}_{\mathrm{cohcrit}}$. Therefore every minimal mesoscopic oscillation scale is assigned to $\mathcal{C}_{\mathrm{conc}}$, $\mathcal{C}_{\mathrm{aff}}$, or $\mathcal{C}_{\mathrm{cohcrit}}$, as claimed. □

*Remark* 9.82 (Critical-tail reduction theorem)*.* Theorem 9.81 is the critical-tail reduction theorem proved above. It is a local multiscale rigidity statement: oscillation first appearing at an intermediate scale $1 \ll r_n \ll R_n$ must either regenerate a concentrating observer, collapse into the affine/parasitic lower stratum, or produce a coherent critical tail. The log-diffuse alternative has been reduced above to the same hidden-scale compactness statement through tight log-window normalized support profiles.

**Theorem 9.83** (No hidden-scale variance from minimal mesoscopic scale reduction)**.** *Every tight macroscopic annular blow-down generated by a realized recurrent critical-tail extraction after concentration-set removal and affine normalization has no hidden-scale variance on compact macroscopic annular cylinders.*

*Proof.* Suppose that hidden-scale variance fails. By Lemma 9.62, either a unit concentrating observer is regenerated, the affine stratum is reached, or a minimal mesoscopic oscillation scale exists. The first alternative contradicts concentration-set removal. The affine alternative is removed by Theorem 9.89. In the third alternative, Theorem 9.81 gives a descendant in $\mathcal{C}_{\mathrm{conc}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{cohcrit}}$. The concentration case again contradicts concentration-set removal; the affine case is removed by Theorem 9.89; and the coherent critical-tail case is closed by Corollary 9.100. Thus all alternatives are impossible, and hidden-scale variance cannot fail. ☐

**Definition 9.84** (Macroscopic critical-tail blow-down map)**.** Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$, let $z_n = (y_n, \sigma_n)$ be covariant observer centers, and let $R_n \to \infty$ be a selected macroscopic scale. Put

$$\widetilde{U}_n := \mathcal{R}_{z_n} U_n, \qquad \widetilde{\Pi}_n := \mathcal{R}_{z_n} \Pi_n.$$

The ungauged spatial/logarithmic blow-down is

$$V_n(x, s) := \widetilde{U}_n(R_n x, s), \qquad \widehat{Q}_n(x, s) := R_n^{-1}\big(\widetilde{\Pi}_n(R_n x, s) - a_n(s)\big),$$

where $a_n(s)$ is the time-dependent pressure gauge. If an affine pressure component

$$L_n(y, s) = b_n(s) \cdot y + a_n(s)$$

is removed in the affine quotient, the corresponding quotient pressure is

$$Q_n(x, s) := R_n^{-1}\big(\widetilde{\Pi}_n(R_n x, s) - L_n(R_n x, s)\big).$$

Velocity oscillations are measured after a spatially homogeneous velocity gauge

$$W_n(x, s) := V_n(x, s) - c_n(s), \qquad |c_n(s)| \le M.$$

The map is not the Navier–Stokes parabolic scaling: centered time is not rescaled. The scaling identities is

| term | blow-down size |
|---|---|
| $\partial_s V_n$ | $1$ |
| $\Delta V_n$ | $R_n^{-2}$ |
| $(V_n \cdot \nabla) V_n$ | $R_n^{-1}$ |
| $\frac{1}{2} y \cdot \nabla_y U_n$ | $\frac{1}{2} x \cdot \nabla_x V_n$ |
| $\nabla_y \widetilde{\Pi}_n$ | $\nabla_x \widehat{Q}_n$. |

Thus $V_n$ satisfies the exact distributional equation

$$\partial_s V_n - R_n^{-2} \Delta V_n + \frac{1}{2} x \cdot \nabla V_n + \frac{1}{2} V_n + R_n^{-1} (V_n \cdot \nabla) V_n + \nabla \widehat{Q}_n = 0. \tag{9.22}$$

If $Q_n$ is used instead of $\widehat{Q}_n$, the equation is understood in the affine/parasitic quotient; the missing term is the spatially homogeneous force $b_n(s)$, which is either assigned to $\mathcal{C}_{\mathrm{aff}}$ or removed by the canonical affine normalization. The field $W_n$ is used for oscillation and Young-measure compactness, not as an ungauged Navier–Stokes solution unless the time regularity of $c_n$ has separately been obtained.

**Theorem 9.85** (Critical-tail compactification extraction)**.** *Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$ be a bounded normalized terminal residual sequence, $\|U_n\|_{L^\infty} \le M$. Assume that, after removal of all affine-normalized concentrating covariant observers, there remains a nonzero diffuse exterior oscillation defect. Then, after passing to a subsequence, the sequence generates a nonzero element*

$$\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda) \in \mathfrak{T}_M.$$

*Moreover, the barycentric velocity satisfies the linear critical-tail transport law*

$$\partial_s \bar{W} + \frac{1}{2} x \cdot \nabla \bar{W} + \frac{1}{2} \bar{W} + \nabla Q = 0, \qquad \nabla \cdot \bar{W} = 0$$

*in distributions on macroscopic annular cylinders, and the Young/log state* $(\mathcal{Y}, \Lambda)$ *satisfies the balance identity of Lemma 9.43; in particular the log-scale component satisfies the induced log-dilation balance, while nonlinear velocity observables retain the pressure and viscous defect pairings explicitly recorded in* (9.3)*.*

*Proof.* Let $\nu_n^{\mathrm{sp}}$ be the spatial quotient oscillation defect measure after concentration-set removal, as in Lemma 9.22. By the mixed-diffuse hypothesis there are escaping observer windows $E_n$ and $m_0 > 0$ such that

$$\nu_n^{\mathrm{sp}}(E_n) \geq m_0.$$

Normalize

$$\lambda_n := \frac{\nu_n^{\mathrm{sp}} \lfloor E_n}{\nu_n^{\mathrm{sp}}(E_n)}.$$

By Lemma 9.22 and the proof of Theorem 9.29, after passing to a subsequence these probability measures converge to a boundary diffuse-defect profile. Because all positive-threshold concentrating observers have been removed in the stronger $\mathrm{osc}_{3,\mathrm{ct}}$-sense, the limit gives no mass to ordinary unit concentrating observers in $Z$.

We now compactify the critical boundary without imposing annular tightness. Apply Prokhorov compactness first in the observer compactification and then in the logarithmic radial compactification. If the normalized defect is tight in a fixed multiplicative annulus, select macroscopic scales $R_n \to \infty$, observers $z_n = (y_n, \sigma_n)$, affine velocity gauges $c_n(s)$, and pressure gauges. On fixed annular cylinders use the exact blow-down map of Definition 9.84:

$$V_n(x, s) = \widetilde{U}_n(R_n x, s), \qquad W_n(x, s) = V_n(x, s) - c_n(s),$$

with $\widetilde{U}_n = \mathcal{R}_{z_n} U_n$. The uniform bound $\|U_n\|_{L^\infty} \leq M$ gives weak-* compactness of $W_n$ in $L^\infty_{\mathrm{loc}}$, and the fundamental theorem of Young measures gives, after passing to a further subsequence, a barycenter $\bar{W}$ and Young measure $\mathcal{Y}_{x,s}$. If the sequence is strongly compact in $L^3_{\mathrm{loc}}$, then $\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}$; otherwise $\mathcal{Y}$ records the unresolved oscillation defect. If the defect is not tight in any fixed logarithmic annulus, the escaping logarithmic mass is recorded by $\Lambda$. These alternatives are exhaustive and define an element of $\mathfrak{T}_M$. Since the normalized defect mass is one, $\mathcal{T} \neq 0$.

It remains to identify the equation satisfied by the barycenter. The ungauged fields $V_n$ satisfy the exact scaled equation (9.22). Passing from $V_n$ to $W_n = V_n - c_n(s)$ is used only to measure oscillation. We do not differentiate the homogeneous gauge $c_n(s)$. Instead, the exact equation is passed to the limit for $V_n$ and then projected to the affine/parasitic quotient. If the homogeneous component represented by $c_n$ is retained, it is assigned to $\mathcal{C}_{\mathrm{aff}}$; otherwise the canonical affine normalization removes it from the non-affine quotient barycenter. For every compactly supported test function, the viscous and nonlinear contributions are bounded respectively by

$$C R_n^{-2} \|V_n\|_{L^\infty}, \qquad C R_n^{-1} \|V_n\|_{L^\infty}^2,$$

and hence both tend to zero. The pressure gauges give distributional compactness of the remaining gradients on annuli. Passing to the weak-* barycentric limit of $V_n$, then applying the affine quotient just described, yields the displayed linear critical-tail transport equation for the non-affine barycenter $\bar{W}$. Applying Lemma 9.43 gives the exact Young/log balance: the log-scale component satisfies the induced log-dilation balance, while nonlinear velocity observables retain the pressure and viscous defect pairings displayed in (9.3). This construction is local on macroscopic annuli and uses no global critical norm, CKN budget, or energy estimate. □

**Corollary 9.86** (Coherent extraction under annular tightness and strong compactness)**.** *If the critical-tail defect $\mathcal{T} \in \mathfrak{T}_M$ produced by Theorem 9.85 satisfies $\Lambda = 0$ and $\mathcal{Y}_{x,s} = \delta_{W(x,s)}$ for a.e. $(x, s)$, then there exist macroscopic scales, covariant observers, affine gauges, and pressure gauges for which the blown-down fields converge strongly in $L^3_{\mathrm{loc}}$ and weak-$*$ in $L^\infty_{\mathrm{loc}}$ to a nonzero coherent critical tail $(W, Q) \in \mathcal{C}_{\mathrm{cohcrit}}$.*

*Proof.* This is the coherent subcase of Theorem 9.85. The condition $\Lambda = 0$ rules out loss of mass across logarithmic scale, and the Dirac-valued Young measure gives strong $L^3_{\mathrm{loc}}$ convergence by uniform $L^\infty$ boundedness. Nontriviality follows from the normalized affine oscillation mass. $\square$

**Theorem 9.87** (Coherent critical-tail classification)**.** *Let $(W, Q) \in \mathcal{C}_{\mathrm{cohcrit}}$ be a coherent critical tail. In logarithmic radial variables*

$$x = e^\rho \omega, \qquad Z(\rho, \omega, s) = e^\rho W(e^\rho \omega, s),$$

*its compact log-translation/time-translation hull belongs to exactly one of*

$$\mathcal{C}_{\mathrm{affine}}, \qquad \mathcal{C}_{\mathrm{homcrit}}, \qquad \mathcal{C}_{\mathrm{logper}}, \qquad \mathcal{C}_{\mathrm{ap}}.$$

*Here $\mathcal{C}_{\mathrm{homcrit}}$ is the homogeneous $(-1)$-tail class, $\mathcal{C}_{\mathrm{logper}}$ is the nonzero log-periodic tail class, and $\mathcal{C}_{\mathrm{ap}}$ is the remaining aperiodic minimal log-translation hull class.*

*Proof.* Let

$$\mathcal{H}(Z) := \overline{\{Z(\rho + a, \omega, s + b) : a, b \in \mathbb{R}\}}$$

be the compact logarithmic translation hull. If the affine-normalized oscillation vanishes on every compact log-cylinder, then the tail is affine/parasitic. Otherwise, if every element of $\mathcal{H}(Z)$ is independent of $\rho$, then $W$ is $(-1)$-homogeneous and belongs to $\mathcal{C}_{\mathrm{homcrit}}$. If the log-translation orbit has a nonzero period, then $W$ is discretely self-similar at infinity and belongs to $\mathcal{C}_{\mathrm{logper}}$. If neither alternative holds, compact-flow decomposition gives a minimal compact invariant subsystem with no nonzero log period; this is the aperiodic log-translation class $\mathcal{C}_{\mathrm{ap}}$. These alternatives exhaust compact log-translation hulls. $\square$

**Theorem 9.88** (Realized critical-tail rigidity by stratification)**.** *Let $\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda) \in \mathfrak{T}_M$ be a nonzero realized critical-tail profile generated by a bounded normalized residual Navier–Stokes terminal sequence after terminal concentration-set removal. Then the ordered stratification selects one of the following priority profile alternatives:*

(a) *$\mathcal{T}$ regenerates affine-normalized local concentration;*
(b) *$\mathcal{T}$ is affine/parasitic;*
(c) *$\mathcal{T}$ is log-diffuse;*
(d) *$\mathcal{T}$ is a Young critical tail;*
(e) *$\mathcal{T}$ is a coherent homogeneous critical tail;*
(f) *$\mathcal{T}$ is a coherent log-periodic critical tail;*
(g) *$\mathcal{T}$ is a coherent aperiodic critical tail.*

*In the ordered refined decomposition, the log-diffuse, Young, homogeneous, log-periodic, and aperiodic critical-tail alternatives are lower structured strata. Thus every nonzero realized critical-tail profile which remains after these lower strata have been removed either regenerates local concentration or collapses to the affine/parasitic lower stratum.*

*Proof.* If the covariant observer orbit closure of $\mathcal{T}$ intersects a constant-in-time concentration set $\{U : \mathrm{osc}_{3,\mathrm{ct}}(U; Q_1^{\mathrm{cov}}) \geq \eta\}$ for some $\eta > 0$, then alternative *(a)* holds. This case is removed first.

If every spatial quotient oscillation observable vanishes on the profile, then Theorem 8.21 and Definition 8.16 place the profile in the affine/parasitic lower stratum. This is alternative *(b)*. It remains to consider the case in which the profile has nonzero non-affine diffuse content and does not regenerate retained unit-scale velocity concentration.

By Theorem 9.85, the residual normalized diffuse mass is represented by the critical-tail compactification. Thus the boundary mass has only two mechanisms: mass escaping every bounded logarithmic annulus, recorded by $\Lambda$, and annular blow-down mass, recorded by $(\bar{W}, \mathcal{Y})$. If $\Lambda \neq 0$, the profile is log-diffuse, giving *(c)*. If $\Lambda = 0$ but $\mathcal{Y}_{x,s}$ is not a Dirac mass on a set of positive macroscopic measure, the profile is a Young critical tail, giving *(d)*.

The remaining case is coherent: $\Lambda = 0$ and $\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}$ a.e. Then Corollary 9.86 gives a coherent critical tail $(\bar{W}, Q) \in \mathcal{C}_{\text{cohcrit}}$. Applying Theorem 9.87 gives exactly one of the homogeneous, log-periodic, or aperiodic coherent alternatives, namely *(e)*, *(f)*, or *(g)*, unless the affine/parasitic case already occurred. The alternatives are mutually exclusive after the priority order above: regenerated concentration is removed first, then affine collapse, then log-diffuse mass, then Young oscillation, and finally the coherent log-translation classification. The final assertion follows from this ordered classification. □

9.4. **Complete closure of lower alternatives.** The critical-tail classification above is a classification theorem rather than an emptiness theorem. The complete residual decomposition is closed by local lower-stratum exclusions together with the minimal mesoscopic-scale theorem Theorem 9.81. That theorem first gives hidden-scale compactness through Theorem 9.83. The log-diffuse alternative is reduced to hidden-scale compactness by the log-window descendant selection of Theorem 9.53; the Young alternative is reduced to it by the Fréchet–Kolmogorov/Young-measure equivalence in Lemma 9.56. The critical-tail reduction theorem used in both reductions is Theorem 9.81. These are local statements about realized residual descendants; none is a global $L^\infty_t L^3_x$, finite-energy-tail, or global CKN-budget estimate.

**Theorem 9.89** (Oscillatory entry normalization)**.** *Let $(U, \Pi)$ be a retained residual profile after the earlier lower classes recorded in the companion setup paper have been removed. Then exactly one of the following holds:*

(i) *after canonical affine/parasitic normalization and an admissible affine symmetry $S_c$, the profile retains positive constant-in-time non-affine oscillation:*
$$\Phi(S_c U) > 0;$$
(ii) *all spatial oscillations $\operatorname{osc}_{3,\text{sp}}(U; Q)$ vanish on compact cylinders, and the profile belongs to the affine/parasitic quotient stratum $\mathcal{C}_{\text{aff}}$.*

*Consequently the retained normalized residual decomposition has*
$$\mathcal{C}_{\text{aff}} \cap \mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing.$$

*Proof.* The dichotomy is Theorem 8.21. If all spatial oscillations $\operatorname{osc}_{3,\text{sp}}(U; Q)$ vanish on compact cylinders, then the profile is a spatially homogeneous quotient mode and belongs to $\mathcal{C}_{\text{aff}}$. If this alternative is not available after lower-stratum removal, the parasitic-free representative has $\Phi(S_c U) > 0$ for an admissible affine symmetry $S_c$. A profile in $\mathcal{C}_{\text{aff}}$ therefore cannot carry retained normalized residual concentration, so $\mathcal{C}_{\text{aff}} \cap \mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing$. □

**Corollary 9.90** (Young alternative exclusion from minimal mesoscopic scale reduction)**.**
$$\mathcal{C}_{\text{Young}} = \varnothing.$$

*Proof.* Let a retained residual profile generate a nonzero Young critical-tail alternative. By Corollary 9.54, the log-diffuse alternative is already empty, so the defect has a tight macroscopic annular realization. Let $W_n$ be the corresponding annular blow-down and $\mathcal{Y}_{x,s}$ its Young measure. By Theorems 9.81 and 9.83, $W_n$ has no hidden-scale variance on compact annular cylinders. Hence Lemma 9.56 gives
$$\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)} \qquad \text{for a.e. } (x, s),$$
contradicting the definition of $\mathcal{C}_{\text{Young}}$. Therefore $\mathcal{C}_{\text{Young}} = \varnothing$. □

**Theorem 9.91** (Critical-tail rigidity from minimal mesoscopic scale reduction)**.** *Every nonzero recurrent critical-tail defect generated by a realized residual terminal sequence belongs to*

$$\mathcal{C}_{\mathrm{conc}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{cohcrit}}.$$

*Equivalently, after regenerated concentration, affine/parasitic collapse, and coherent critical tails have been removed, no recurrent critical-tail defect remains.*

*Proof.* Let $\mathcal{D} \in \mathfrak{T}_M$ be a nonzero recurrent critical-tail defect. By Theorem 9.88, the ordered priority alternative is one of: regenerated concentration, affine/parasitic collapse, log-diffuse defect, Young critical tail, or coherent critical tail. The first two alternatives are precisely $\mathcal{C}_{\mathrm{conc}}$ and $\mathcal{C}_{\mathrm{aff}}$. The log-diffuse alternative is empty by Corollary 9.54. The Young alternative is empty by Corollary 9.90. The only remaining nonempty critical-tail alternative is therefore the coherent alternative $\mathcal{C}_{\mathrm{cohcrit}}$. This proves the inclusion. Once $\mathcal{C}_{\mathrm{conc}}$, $\mathcal{C}_{\mathrm{aff}}$, and $\mathcal{C}_{\mathrm{cohcrit}}$ are removed, the inclusion leaves no recurrent critical-tail defect. □

**Lemma 9.92** (Bounded velocity gauges for realized blow-downs)**.** *Let $f : E \to \mathbb{R}^3$ be measurable on a finite positive-measure set $E$, and assume $|f| \le M$ a.e. Then the minimization problem*

$$\inf_{c\in\mathbb{R}^3} \int_E |f-c|^3$$

*has a minimizer in $\overline{B_M(0)}$. Consequently every affine-normalized blow-down of a sequence satisfying $\|U_n\|_{L^\infty} \le M$ may be represented, without increasing any local oscillation functional, by velocity gauges $c_n$ with $|c_n| \le M$. If the normalization is performed time-slice by time-slice, the same conclusion holds for a measurable choice $c_n(s)$, with $|c_n(s)| \le M$ for a.e. $s$.*

*Proof.* The functional is continuous and coercive in $c$, so a minimizer exists. Let $\pi : \mathbb{R}^3 \to \overline{B_M(0)}$ be the nearest-point projection. Since $\overline{B_M(0)}$ is closed and convex, for every $v \in \overline{B_M(0)}$ and every $c \in \mathbb{R}^3$,

$$|v - \pi(c)| \le |v - c|.$$

Taking $v = f(y)$ and integrating gives

$$\int_E |f - \pi(c)|^3 \le \int_E |f-c|^3.$$

Thus any minimizing sequence may be projected into $\overline{B_M(0)}$, and a minimizer can be chosen there. Applying this to the local velocity fields used to define the affine-normalized observer oscillations gives the asserted bounded representatives. The time-slice version is the same argument on each slice, with measurability obtained by the usual measurable-selection theorem for a continuous coercive finite-dimensional minimization problem. The projection step is pointwise in $s$, so the bound $|c_n(s)| \le M$ is preserved. □

**Lemma 9.93** (Domain realization dichotomy for critical-tail windows)**.** *Every critical-tail window used in the bounded-origin realization theorem is of exactly one of the following two admissible types.*

(i) *It is a window of a bounded ancient descendant $(U, \Pi) \in \mathfrak{X}_M$. Then every fixed macroscopic blow-down cylinder is contained in the full domain $\mathbb{R}^3 \times \mathbb{R}$.*

(ii) *It is a first-generation local window satisfying the local-window condition of Definition 9.7 on every fixed blow-down cylinder used in the extraction.*

*If neither condition holds, the window is not used in bounded-origin realization. It is assigned by Theorem 9.17 to the exterior/diffuse, pressure/noncompact, or critical-tail boundary branch before bounded-origin realization is invoked.*

*Proof.* Such windows are obtained after an ancient bounded profile has already been extracted, so their domains are global in centered variables. For a first-generation window, the local setup estimates are available precisely on the physical image of the local Type I cylinder. If every fixed blow-down cylinder remains in that image, the local-window condition gives the required compactness and pressure atlas. If some fixed blow-down cylinder leaves the local window, the sequence has escaped the first-generation setup domain; by Lemma 9.10, it must be assigned through the expanding-region protocol rather than treated as a bounded-origin compactness input. □

**Theorem 9.94** (Bounded-origin realization of coherent critical tails)**.** *Let a coherent critical tail be generated by a bounded normalized residual Navier–Stokes terminal sequence with:*

(i) *locally bounded velocity representatives;*
(ii) *pressure gauges bounded modulo time and affine functions on each fixed time window;*
(iii) *convergence on every fixed annulus/log-window;*
(iv) *the normalized support-profile property realized from the original blow-up sequence;*
(v) *the domain alternative in Lemma 9.93.*

*Then it is realized, after subsequence extraction and admissible gauges, as a weak-$*$ $L^\infty_{\mathrm{loc}}$ limit on full macroscopic compact cylinders, including cylinders meeting the macroscopic origin. In particular, every coherent critical tail has a representative which is locally bounded near $x = 0$ on compact macroscopic time intervals, and whose restrictions to all tail windows reproduce the coherent tail.*

*Remark* 9.95 (Use of the bounded-origin representative)*.* The bounded-origin representative is used only as a locally bounded extension of the annular coherent tail. The coherent transport equation and pressure normalization are required only on punctured annular cylinders $\{0 < r_0 < |x| < r_1 < \infty\} \times I$. None of the homogeneous, log-periodic, or aperiodic exclusions below uses a distributional equation through $x = 0$.

*Proof.* Let $(W, Q) \in \mathcal{C}_{\mathrm{cohcrit}}$ be a coherent critical tail generated by a bounded terminal sequence $(U_n, \Pi_n) \subset \mathfrak{X}_M$, with $\|U_n\|_{L^\infty} \le M$. By coherent extraction, after passing to a subsequence there are macroscopic scales $R_n \to \infty$, covariant observer centers $(y_n, \sigma_n)$, affine velocity gauges, and pressure gauges such that on every compact annular cylinder

$$A_{r_0,r_1,I} := \{r_0 < |x| < r_1\} \times I, \qquad 0 < r_0 < r_1 < \infty,$$

the fields

$$\widetilde{U}_n := \mathcal{R}_{(y_n,\sigma_n)} U_n, \qquad W_n(x, s) := \widetilde{U}_n(R_n x, s) - c_n(s)$$

converge to $W$ strongly in $L^3(A_{r_0,r_1,I})$ and weak-$*$ in $L^\infty(A_{r_0,r_1,I})$. By Lemma 9.93, the fixed macroscopic cylinders used below are contained in the domain either because the sequence is an already extracted ancient profile, or because the first-generation local-window condition holds. If this domain condition failed, the window would already belong to an exterior/diffuse or noncompact branch and would not be an input to bounded-origin realization.

We first check that the same gauges which realize the annular coherent tail are bounded on compact time intervals. Fix a compact interval $I$ and use the annulus $1 < |x| < 2$. The weak-$*$ $L^\infty$ convergence on $\{1 < |x| < 2\} \times I$ implies, after discarding finitely many $n$, that

$$\|W_n\|_{L^\infty(\{1<|x|<2\}\times I)} \le C_I.$$

For a.e. $s \in I$, if $|c_n(s)| > M + C_I$, then for every $x$ with $1 < |x| < 2$,

$$|\widetilde{U}_n(R_n x, s) - c_n(s)| \ge |c_n(s)| - M > C_I,$$

contradicting the preceding $L^\infty$ bound. Hence

$$|c_n(s)| \le M + C_I \qquad \text{for a.e. } s \in I.$$

This argument keeps the annular representative fixed. The bounded-gauge lemma above shows, in addition, that the affine-normalized construction may always be started with bounded canonical gauges.

Therefore, on every full macroscopic cylinder

$$K_{R,I} := B_R(0) \times I,$$

one has

$$|W_n(x,s)| \le |\widetilde{U}_n(R_n x, s)| + |c_n(s)| \le 2M + C_I$$

for a.e. $(x,s) \in K_{R,I}$. The domain realization dichotomy Lemma 9.93 is the hypothesis which ensures that $W_n$ is defined on $K_{R,I}$ for all sufficiently large $n$.

Choose an exhaustion of $\mathbb{R}^3 \times \mathbb{R}$ by full compact macroscopic cylinders $K_j = B_j(0) \times [-j,j]$. Banach–Alaoglu gives, by a diagonal extraction that does not change notation, a field

$$W^{\mathrm{full}} \in L^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R}),$$

such that

$$W_n \rightharpoonup^* W^{\mathrm{full}} \qquad \text{in } L^\infty(K_j)$$

for every $j$. The subsequence still has the annular strong $L^3_{\mathrm{loc}}$ convergence to the coherent tail $W$. Therefore, for every annular cylinder $A_{r_0,r_1,I}$,

$$W^{\mathrm{full}} = W \qquad \text{a.e. on } A_{r_0,r_1,I}.$$

Since the annuli exhaust $(\mathbb{R}^3 \setminus \{0\}) \times \mathbb{R}$, the annular coherent tail is the restriction of the full weak-$*$ limit $W^{\mathrm{full}}$ away from $x = 0$.

We take $W^{\mathrm{full}}$ as the full-cylinder representative of the coherent tail. It is locally bounded on every compact macroscopic cylinder, including cylinders meeting $x = 0$, and it realizes the same annular tail because it agrees with $W$ on all punctured compact cylinders. Pressure gauges are not needed for the boundedness conclusion; if one records the pressure as well, the local pressure-gauge compactness in the terminal profile class gives distributional pressure-gradient limits on the same full cylinders after the usual time-dependent gauges. The construction uses only the uniform local boundedness inherited from the Type I terminal profile class and finite-dimensional affine normalization; it uses no global critical norm or global tail budget. □

**Lemma 9.96** (Realized amplitude-normalized log-hull heredity)**.** *Let $(W,Q) \in \mathcal{C}_{\mathrm{cohcrit}}$ be generated by a bounded terminal sequence and be realized by Theorem 9.94. Write*

$$Z(\rho,\omega,s) = e^\rho W(e^\rho \omega, s).$$

*Let $Z_*$ be a nonzero element of the amplitude-normalized log/time hull, obtained as a local limit*

$$Z_*(\rho,\omega,s) = \lim_{j\to\infty} Z(\rho + a_j, \omega, s + b_j).$$

*Then one of the following alternatives occurs:*

(i) *the translated windows are realized by admissible critical-tail extractions from the original blow-up sequence, with bounded-origin representatives stable under the amplitude-normalized translation, and*

$$Z_*(\rho,\cdot,\cdot) \to 0 \qquad \text{as } \rho \to -\infty$$

*in the local logarithmic topology;*

(ii) *the failure of such realized bounded-origin heredity produces one of the already stratified lower alternatives:*

$$\mathcal{C}_{\mathrm{conc}}, \qquad \mathcal{C}_{\mathrm{aff}}, \qquad \mathcal{C}_{\log\mathrm{diff}}, \qquad \textit{critical-tail defect}, \qquad \textit{or hidden-scale variance.}$$

*Proof.* A translated log profile $Z(\rho + a_j, \omega, s + b_j)$ represents

$$e^{a_j} W(e^{a_j} x, s + b_j),$$

not merely $W(e^{a_j} x, s + b_j)$. Thus it is not admitted into the coherent critical-tail closure solely by changing the macroscopic scale in the no-amplitude blow-down. It is admitted only when the amplitude-normalized profile is itself realized by the normalized critical-tail extraction from the original terminal sequence.

Assume first that this realized amplitude-normalized extraction exists along the translated windows. The bounded-origin realization theorem applied to these realized representatives gives $L^\infty_{\mathrm{loc}}$ bounds on every full compact macroscopic cylinder, including cylinders meeting $x = 0$, with constants stable under the local limit. Hence $Z_*$ corresponds to a velocity

$$W_*(r\omega, s) = r^{-1} Z_*(\log r, \omega, s)$$

which is locally bounded near $r = 0$.

For $r = e^\rho \downarrow 0$, local boundedness gives

$$|Z_*(\rho, \omega, s)| = r\,|W_*(r\omega, s)| \le C_K r$$

on every compact angular-time set $K$. Thus $Z_*(\rho, \cdot, \cdot) \to 0$ locally as $\rho \to -\infty$, proving alternative (i).

It remains to identify the complementary case. If the amplitude-normalized translated windows cannot be realized with the same bounded-origin control, then one of the realized extraction inputs has failed. If positive unit concentration is regenerated, the branch is $\mathcal{C}_{\mathrm{conc}}$. If the failure is purely homogeneous or affine in the velocity/pressure quotient, it is $\mathcal{C}_{\mathrm{aff}}$. If mass escapes in logarithmic radius before a tight amplitude-normalized profile is obtained, it is a log-diffuse or critical-tail defect. If tightness holds but the Young measure is not Dirac on an annular window, it is hidden-scale variance. These are precisely the lower rows in the ordered critical-tail compactification, and all are sequence-realized support profiles by Lemmas 9.39, 9.51 and 9.76. This gives alternative (ii). □

**Lemma 9.97** (Homogeneous tails are incompatible with bounded-origin retention)**.** *Let $(W, Q) \in \mathcal{C}_{\mathrm{homcrit}}$ be a coherent critical tail obtained from the bounded-origin realization theorem. Assume it carries the retained non-affine annular activity required in the definition of the coherent tail stratum. Then no such $W$ exists.*

*Proof.* In the homogeneous critical-tail class,

$$W(r\omega, s) = r^{-1} A(\omega, s).$$

The bounded-origin realization gives a locally bounded representative on $\{0 < r < r_0\}$ for each compact angular-time set. Hence

$$|A(\omega, s)| = r|W(r\omega, s)| \le C_K r$$

on every compact angular-time set $K$. Letting $r \downarrow 0$ gives $A = 0$, and therefore $W \equiv 0$ on punctured compact cylinders. This is incompatible with the retained non-affine annular activity that defines a nonzero coherent critical tail. □

**Lemma 9.98** (Log-periodic tails are incompatible with bounded-origin retention)**.** *Let $(W, Q) \in \mathcal{C}_{\mathrm{logper}}$ be a coherent critical tail obtained from the bounded-origin realization theorem. Assume it carries retained non-affine annular activity. Then no such $W$ exists.*

*Proof.* Write
$$Z(\rho,\omega,s) = e^{\rho} W(e^{\rho}\omega, s).$$
Bounded-origin realization gives $Z(\rho,\cdot,\cdot) \to 0$ in the local logarithmic topology as $\rho \to -\infty$. If the log-period is $L \neq 0$, then
$$Z(\rho,\omega,s) = Z(\rho - kL,\omega,s) \qquad (k \in \mathbb{Z}).$$
For $L > 0$ take $k \to +\infty$, and for $L < 0$ take $k \to -\infty$. In both cases $\rho - kL \to -\infty$, so the right-hand side converges to zero. Thus $Z \equiv 0$, hence $W \equiv 0$ on punctured annular cylinders, contrary to retained non-affine annular activity. □

**Lemma 9.99** (Aperiodic coherent tails are incompatible with bounded-origin retention)**.** *Let $(W, Q) \in \mathcal{C}_{\mathrm{ap}}$ be a coherent critical tail obtained from the bounded-origin realization theorem. Assume it carries retained non-affine annular activity. Then no such $W$ exists.*

*Proof.* Let $Z(\rho,\omega,s) = e^{\rho}W(e^{\rho}\omega,s)$, and let $\mathcal{M}$ be the nonzero minimal aperiodic logarithmic subsystem in the definition of $\mathcal{C}_{\mathrm{ap}}$, taken inside the realized amplitude-normalized hull. Choose $Z_* \in \mathcal{M}$. By Lemma 9.96, either a lower stratum has already occurred, or $Z_*$ has a bounded-origin representative and hence
$$Z_*(\rho,\cdot,\cdot) \to 0 \qquad \text{as } \rho \to -\infty.$$
The lower-stratum alternatives are already removed before the coherent aperiodic row is closed. In the remaining case $0 \in \mathcal{H}(Z_*)$. Minimality gives $\mathcal{H}(Z_*) = \mathcal{M}$. Since the zero profile is a closed invariant subsystem, $\mathcal{M} = \{0\}$. This contradicts the retained non-affine annular activity that makes the aperiodic coherent tail a nonzero residual tail. □

**Corollary 9.100** (Coherent critical-tail alternative closure)**.** *One has*
$$\mathcal{C}_{\mathrm{homcrit}} = \varnothing, \qquad \mathcal{C}_{\mathrm{logper}} = \varnothing, \qquad \mathcal{C}_{\mathrm{ap}} = \varnothing.$$

*Proof.* Lemma 9.97 excludes $\mathcal{C}_{\mathrm{homcrit}}$, Lemma 9.98 excludes $\mathcal{C}_{\mathrm{logper}}$, and Lemma 9.99 excludes $\mathcal{C}_{\mathrm{ap}}$. Each contradiction uses the same retained non-affine annular activity that makes the coherent tail a nonzero realized residual object, so no coherent critical-tail lower stratum remains. □

**Corollary 9.101** (Critical-tail exclusion, complete form)**.** *No nonzero realized critical-tail profile remains after terminal concentration-set removal and affine/parasitic lower-stratum removal.*

*Proof.* Let $\mathcal{T} \in \mathfrak{T}_M$ be a remaining nonzero realized critical-tail profile. By Theorem 9.88, the ordered priority alternative is one of: regenerated concentration, affine/parasitic collapse, log-diffuse defect, Young critical tail, coherent homogeneous tail, coherent log-periodic tail, or coherent aperiodic tail.

Regenerated concentration contradicts terminal concentration-set removal. The affine/parasitic alternative is impossible by Theorem 9.89. The log-diffuse alternative is impossible by Corollary 9.54, and the Young alternative is impossible by Corollary 9.90. The three coherent alternatives are impossible by Corollary 9.100. All alternatives are therefore impossible, so no such $\mathcal{T}$ remains. □

**Theorem 9.102** (No mixed compact–diffuse residual profile)**.** *Let $(U,\Pi)$ be a retained terminal profile in the refined residual class, after the affine normalization of Definition 8.16. Then $U$ cannot have a diffuse exterior oscillation remainder, nor an escaping retained noncompact recentering remainder, coexisting with its compact retained core. Equivalently, the paired terminal configurations*
$$\textit{one compact concentration core} + \mathcal{R}_{\mathrm{diff}}, \qquad \textit{one compact concentration core} + \mathcal{R}_{\mathrm{noncomp}},$$
*and their finite, infinite, or continuum concentration-set analogues do not occur in the refined residual class.*

*Proof.* The noncompact escaping case is local. In this theorem the noncompact remainder is retained in the velocity or affine-oscillation sense. If an escaping recentering with retained unit velocity concentration loses the local dissipation or pressure-gauge bounds in Definition 9.3, then after its own local normalization it is a retained unit-velocity recentering sequence. This contradicts Corollary 9.11. Thus it remains to exclude a compact core plus diffuse exterior remainder.

Assume that such a diffuse exterior remainder exists. By the paired concentration-set decomposition Theorem 9.24, remove neighborhoods of the concentration set at every positive constant-in-time oscillation threshold. The mixed assumption says that a nonzero diffuse oscillation residual remains. Apply Theorem 9.36.

If regenerated concentration occurs, then a positive oscillatory observer survives after all positive-threshold concentration sets were removed, contradicting the definition of the diffuse residual. If affine/parasitic collapse occurs, then Definition 8.16 and Theorem 10.12 places the relevant descendant hull in an already closed lower stratum. This cannot occur inside $\mathcal{C}_{\mathrm{gen}}$.

The only remaining alternative is a critical diffuse tail. By Theorem 9.85, it generates a nonzero $\mathcal{T} \in \mathfrak{T}_M$. This contradicts Corollary 9.101. Hence no mixed compact–diffuse residual profile exists. □

**Definition 9.103** (Finite separated family of retained velocity concentration profiles)**.** A finite separated family of retained velocity concentration profiles is a finite family of recentering sequences. For the original terminal extraction these are the parabolic recenterings of Definition 9.1; for descendants of a centered profile they are the covariant tail recenterings of Definition 8.4. We write them uniformly as $g_{1,n}, \dots, g_{J,n}$, $J \geq 2$, such that:

(i) the recentering sequences are mutually separated;
(ii) each recentered sequence converges locally smoothly for velocity and locally in $L^{3/2}$ for pressure modulo gauges to a terminal profile $(U_j, \Pi_j)$;
(iii) each limiting profile has retained local velocity concentration:

$$\mathcal{V}_1(U_j; 0) \geq \eta_* \qquad (j = 1, \dots, J).$$

This is a finite local family, not a statement that all concentration profiles in the whole space have been counted. The separated-profile proof below treats the possible continuation of escaping recenterings by local concentration alternatives, not by a global counting estimate.

## 10. Separated concentration profiles, indecomposability, and sequence-$L^3$

We now close the terminal alternatives carrying retained local velocity concentration. The proof uses no global critical-mass budget. Instead, finite separated concentration families are treated through local recentering and compactness. Iterating retained tail limits from such a finite family has only local profile outcomes: an indecomposable retained velocity concentration profile, a mixed compact–diffuse or compact–noncompact successor, or a chain of retained concentrating descendants. The mixed successors are excluded by Theorem 9.102; infinite chains of retained concentrating descendants and finite loops are excluded by the compact recurrence theorem.

**Definition 10.1** (Tail limits and terminal indecomposability)**.** Let $(U, \Pi)$ be a terminal profile with retained local velocity concentration. A retained concentrating tail limit is a limit of covariant tail recenterings

$$\mathscr{T}_{(x_n,\tau_n)}(U, \Pi), \qquad |x_n| + |\tau_n|^{1/2} \to \infty,$$

in the local smooth velocity topology and the local $L^{3/2}$ pressure topology modulo gauges, with positive retained local velocity concentration at the origin, equivalently with retained local velocity concentration at some rational level $\eta \in \mathbb{Q}_{>0}$. An ordinary fixed spatial shift is not used in this definition; by Remark 8.3 and Definition 8.4, it would introduce an unbounded spurious drift for escaping centers. Diffuse exterior concentration means that there are $\tau_n \to -\infty$, $R_n \to \infty$, and $\eta > 0$, such that

$$\int_{|y|>R_n} |U(y,\tau_n)|^3 \, dy \geq \eta,$$

while every unit-scale exterior covariant recentering sequence based in the parabolic exterior windows

$$\{|y| > R_n\} \times (\tau_n - 1, \tau_n + 1)$$

has velocity concentration below the velocity threshold, unless that recentering sequence has loss of local compactness. If a pressure-inclusive CKN lower bound remains without velocity concentration, it is assigned by Lemma 9.5 to pressure loss, the affine/parasitic quotient, or neighboring velocity concentration. Thus the word "diffuse" is always parabolic: localization is tested by unit parabolic cylinders, not only by spatial balls on the distinguished time slices. An escaping non-compact recentering sequence is a covariant tail recentering sequence for which the local compactness, suitability, or pressure control needed to extract a terminal profile fails. A retained velocity concentration profile is terminally indecomposable if it has no covariant tail recentering limit carrying positive retained local velocity concentration at any rational level $\eta \in \mathbb{Q}_{>0}$, no diffuse exterior concentration, and no escaping non-compact recentering sequence.

**Lemma 10.2** (Compact state-space time-slice concentration)**.** *Let $\mathcal{K} \subset \mathfrak{X}_M^{\#}$ be compact in the normalized local profile topology, with pressure representatives taken from the fixed setup pressure atlas and with affine/parasitic modes already removed by the canonical representative or assigned to $\mathcal{C}_{\mathrm{aff}}$. For every $\varepsilon_0 > 0$ there are*

$$\delta_{\mathrm{sl}} = \delta_{\mathrm{sl}}(\mathcal{K}, \varepsilon_0) \in (0,1), \qquad c_{\mathrm{sl}} = c_{\mathrm{sl}}(\mathcal{K}, \varepsilon_0) > 0,$$

*such that every $(U,\Pi) \in \mathcal{K}$ satisfying*

$$\int_{B_1(0)} |U(y,0)|^3 \, dy \geq \varepsilon_0 > 0,$$

*also satisfies, after the standard local pressure gauge,*

$$\iint_{B_1(0)\times(-\delta_{\mathrm{sl}}^2,0)} |U|^3 \, dy \, d\tau \geq c_{\mathrm{sl}}, \qquad \text{and hence} \qquad \mathcal{V}_1(U;0) \geq c_{\mathrm{sl}}.$$

*The same constants apply to any covariant tail recentering family whose normalized pressure-atlas closure is contained in the same compact set $\mathcal{K}$.*

*Proof.* For $(U,\Pi) \in \mathcal{K}$ define

$$F(U,\tau) := \int_{B_1} |U(y,\tau)|^3 \, dy.$$

The topology of $\mathfrak{X}_M^{\#}$ gives local $C^0$ convergence of the velocity. Hence $(U,\tau) \mapsto F(U,\tau)$ is continuous on $\mathcal{K} \times [-1,1]$. Since this set is compact, the continuity is uniform. Choose $\delta_{\mathrm{sl}} \in (0,1)$ so small that, for every $(U,\Pi) \in \mathcal{K}$,

$$|F(U,\tau) - F(U,0)| \leq \frac{1}{2}\varepsilon_0 \qquad \text{whenever } -\delta_{\mathrm{sl}}^2 < \tau < 0.$$

If $F(U,0) \geq \varepsilon_0$, then $F(U,\tau) \geq \varepsilon_0/2$ throughout that time interval. Therefore

$$\iint_{B_1\times(-\delta_{\mathrm{sl}}^2,0)} |U|^3 \, dy \, d\tau \geq \frac{1}{2}\varepsilon_0 \delta_{\mathrm{sl}}^2.$$

Set $c_{\rm sl} := \varepsilon_0\delta_{\rm sl}^2/2$. The pressure contribution is not used to certify nontriviality; the pressure atlas is used only to define the compact normalized state space in which the uniform time modulus is valid.

The dependence on $\mathcal{K}$, rather than only on $M$, is essential. A spatially homogeneous parasitic family $U_N(y,\tau) = b_N(\tau)$, paired with the affine pressure $-(\partial_\tau b_N + \frac{1}{2}b_N)\cdot y$, can have a uniform velocity $L^\infty$ bound and fixed time-slice mass while losing every uniform backward time modulus. Such a family is not compact in the normalized pressure-atlas topology unless the homogeneous modes have a convergent modulus, and nonzero homogeneous modes are assigned to the affine/parasitic stratum $\mathcal{C}_{\rm aff}$. $\square$

**Lemma 10.3** (Quantitative exterior slice-to-defect alternative)**.** *Let $(U,\Pi) \in \mathfrak{X}_M^\#$ be represented after canonical affine/parasitic normalization. Let $\tau_n \to -\infty$, $R_n \to \infty$, and*

$$E_n := \{|y| > R_n\}$$

*satisfy*

$$\int_{E_n} |U(y,\tau_n)|^3\,dy \geq m_0 > 0.$$

*If the exterior covariant recenterings in the windows*

$$\{|y| > R_n - 2\} \times (\tau_n - 1, \tau_n + 1)$$

*do not have compact closure in the normalized local topology with the setup pressure atlas fixed, then alternative* (i) *below occurs. Otherwise, after passing to a subsequence, let $\mathcal{K}_{\rm ext} \subset \mathfrak{X}_M^\#$ be a compact set containing all these recentered profiles. Fix $\alpha > 0$, and let $c_{\rm sl}(\mathcal{K}_{\rm ext},\alpha) > 0$ be the parabolic lower bound given by Lemma 10.2 for the compact state class $\mathcal{K}_{\rm ext}$ and unit ball time-slice mass $\alpha$. Define*

$$m_n^{\rm vel} := \sup_{\substack{|x|>R_n-2\\ |s-\tau_n|\leq 1}} \mathcal{V}_1\big(\mathscr{T}_{(x,s)}U;0\big)\,.$$

*If $m_n^{\rm vel} < c_{\rm sl}(\mathcal{K}_{\rm ext},\alpha)$, then every unit ball with center in the exterior collar satisfies*

$$\int_{B_1(x)} |U(y,\tau_n)|^3\,dy < \alpha \qquad (|x| > R_n - 1).$$

*Consequently there are measurable exterior subsets $F_n \subset E_n$, escaping every compact set, such that*

$$m_0 \leq \int_{F_n} |U(y,\tau_n)|^3\,dy \leq 2m_0,$$

*and, after restricting $F_n$ to a bounded-overlap unit cover, the normalized exterior time-slice measures have no unit-observer atom larger than $C_{\rm ov}\alpha/m_0$, where $C_{\rm ov}$ is the overlap constant.*

*Moreover, at least one of the following alternatives occurs after passing to a subsequence:*

- (i) *an exterior covariant recentering has loss of local compactness or pressure-gauge compactness;*
- (ii) *a pressure-inclusive exterior unit cylinder has a lower bound not carried by velocity, and is assigned by Lemma 9.5;*
- (iii) *the exterior mass is critical only after the logarithmic or normalized-tail compactification and enters the corresponding critical-tail alternative;*
- (iv) *the remaining exterior mass is a diffuse exterior concentration in the sense of Definition 10.1.*

*Proof.* If the exterior recentering family has no compact closure in the normalized pressure-atlas topology, this is precisely the local compactness or pressure gauge failure alternative (i). We therefore work under the compact closure assumption and use the compact set $\mathcal{K}_{\mathrm{ext}}$ from the statement.

Since $|U(\cdot,\tau_n)|^3 dy$ is a nonatomic measure absolutely continuous with respect to Lebesgue measure, the lower bound on $E_n$ allows us to choose a measurable subset $F_n \subset E_n$ with

$$m_0 \le M_n := \int_{F_n} |U(y,\tau_n)|^3\, dy \le 2m_0.$$

For instance, take an exterior truncation with mass larger than $m_0$ and, if necessary, cut it by a level set of a smooth radial coordinate to reduce the mass below $2m_0$. Because $F_n \subset E_n$ and $R_n \to \infty$, these sets escape every compact set.

Let $\{B_1(x_{n,j})\}_j$ be a measurable unit-ball cover of $F_n$, chosen with centers $|x_{n,j}| > R_n - 1$ and with overlap at most $C_{\mathrm{ov}}$ after enlarging the balls by a fixed harmless factor. Suppose that for some $j$,

$$\int_{B_1(x_{n,j})} |U(y,\tau_n)|^3\, dy \ge \alpha.$$

Apply Lemma 10.2 to the covariantly recentered solution $\mathscr{T}_{(x_{n,j},\tau_n)}(U,\Pi)$, which belongs to $\mathcal{K}_{\mathrm{ext}}$. At local time 0 the ball $B_1(0)$ is the original ball $B_1(x_{n,j})$. Hence

$$\mathcal{V}_1\Big(\mathscr{T}_{(x_{n,j},\tau_n)}U;0\Big) \ge c_{\mathrm{sl}}(\mathcal{K}_{\mathrm{ext}},\alpha),$$

so the parabolic velocity part of the raw atlas defect measure on that unit cylinder has mass at least $c_{\mathrm{sl}}(\mathcal{K}_{\mathrm{ext}},\alpha)$. This is the only "thickening" used here: it applies to a unit ball with a quantitative time-slice lower bound, not to an arbitrarily diffuse exterior slice as a whole. The displayed lower bound contradicts $m_n^{\mathrm{vel}} < c_{\mathrm{sl}}(\mathcal{K}_{\mathrm{ext}},\alpha)$. This proves the stated unit-ball time-slice bound.

Define discrete exterior slice measures on the observer centers by

$$\lambda_n^{\mathrm{sl}} := \frac{1}{M_n}\sum_j \left(\int_{B_1(x_{n,j})\cap F_n} |U(y,\tau_n)|^3\, dy\right)\delta_{(x_{n,j},\tau_n)}.$$

Their total masses are bounded above and below by constants depending only on the overlap convention, and each unit observer ball has mass at most $C_{\mathrm{ov}}\alpha/m_0$. Since $R_n \to \infty$ and $\tau_n \to -\infty$, these normalized measures escape every compact subset of the covariant observer space. Passing to the observer compactification gives a nonzero boundary measure unless one of the local extraction inputs fails along some exterior observer sequence.

If local compactness or pressure-gauge compactness fails along such a sequence, we are in alternative (i). If compactness holds but a lower bound in a unit exterior cylinder is pressure-only, then Lemma 9.5 gives alternative (ii). If the mass is not seen at unit scale but becomes critical only after the logarithmic or normalized-tail compactification, it is the already listed critical-tail alternative (iii). In the complementary case the escaping exterior slice measure persists, all unit exterior velocity observers are below the velocity threshold fixed by $c_{\mathrm{sl}}(\mathcal{K}_{\mathrm{ext}},\alpha)$, pressure-only observers have been assigned to the pressure alternatives, and no noncompact recentering remains. The result is the parabolic diffuse exterior concentration alternative in Definition 10.1. The argument is local on unit covariant cylinders and uses no global $L^3$ or CKN budget. $\square$

**Lemma 10.4** (One retained tail limit gives two separated concentration profiles)**.** *Let $U$ be a retained velocity concentration terminal profile. If $U$ has a retained concentrating tail limit, then the corresponding terminal extraction contains a finite separated family of retained velocity concentration profiles.*

*Proof.* The origin recentering sequence carries retained velocity concentration:

$$\mathcal{V}_1(U;0) \ge \eta_*.$$

Let $W$ be a retained concentrating tail limit. By definition there are covariant relative tail recentering parameters $\zeta_n = (x_n, \tau_n)$, with $|x_n| + |\tau_n|^{1/2} \to \infty$, such that $\mathscr{T}_{\zeta_n}(U,\Pi) \to (W, \Pi_W)$ in the terminal topology and

$$\mathcal{V}_1(W;0) \ge \eta_*.$$

By strong local convergence of the velocity,

$$\mathcal{V}_1(\mathscr{T}_{\zeta_n}U;0) \ge \frac{1}{2}\eta_*$$

for all sufficiently large $n$. Thus, for large $n$, the origin recentering sequence and the covariant tail sequence $\zeta_n$ are separated and have retained local velocity concentration. They form a finite separated two-profile concentration family. □

**Proposition 10.5** (No finite separated family and local exclusions imply indecomposability)**.** *After finite separated-family, mixed compact–diffuse, and noncompact escaping alternatives have been excluded, every single retained velocity concentration terminal profile arising from a refined residual candidate is terminally indecomposable.*

*Proof deferred.* The proof is given immediately after Theorem 10.24. This proposition is not used in the proof of that theorem. □

**Lemma 10.6** (Indecomposable sequence-$L^3$ completeness)**.** *Let $(U,\Pi) \in \mathfrak{X}_M^\#$ be represented after canonical affine/parasitic normalization. Suppose $U$ has retained local velocity concentration and is terminally indecomposable in the sense of Definition 10.1. Then there exists a sequence $\tau_k \to -\infty$ such that*

$$\sup_k \|U(\cdot,\tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

*Proof.* We prove the contrapositive. Define

$$A(\tau) := \int_{\mathbb{R}^3} |U(y,\tau)|^3\,dy \in [0,\infty].$$

If the conclusion fails, then

$$\liminf_{\tau\to-\infty} A(\tau) = \infty.$$

Choose $\tau_n \to -\infty$ such that $A(\tau_n) \to \infty$, allowing the value $+\infty$. Since $U$ is bounded by some $M$, for each finite $R$,

$$\int_{|y|\le R} |U(y,\tau_n)|^3\,dy \le M^3|B_R|.$$

Choose $R_n \to \infty$ so slowly that, after discarding finitely many $n$,

$$\int_{|y|>R_n} |U(y,\tau_n)|^3\,dy \ge 1.$$

If $A(\tau_n) = +\infty$, take $R_n = n$; otherwise choose $R_n$ with $M^3|B_{R_n}| \le A(\tau_n)/2$.

Define the parabolic exterior velocity concentration level

$$m_n^{\mathrm{vel}} := \sup_{\substack{|x|>R_n-2 \\ |s-\tau_n|\le 1}} \mathcal{V}_1\left(\mathscr{T}_{(x,s)}U;0\right).$$

This is the quantity which can be passed to a limiting profile by strong local $L^3$ convergence. Pressure-inclusive CKN mass in the same exterior windows is handled separately by Lemma 9.5.

If some unit exterior covariant recentering sequence in the windows

$$\{|y| > R_n - 2\} \times (\tau_n - 1, \tau_n + 1)$$

has loss of local compactness or pressure-gauge compactness, terminal indecomposability is contradicted by the noncompact defect alternative. Hence we may assume that the normalized pressure-atlas closure of all such exterior recentering profiles is compact; denote one compact closure by $\mathcal{K}_{\text{ext}} \subset \mathfrak{X}_M^{\#}$.

If $\limsup_{n\to\infty} m_n^{\text{vel}} = \gamma > 0$, then, passing to a subsequence, choose $(x_n, s_n)$, with

$$|x_n| > R_n - 2, \qquad |s_n - \tau_n| \le 1,$$

such that

$$\mathcal{V}_1\big(\mathscr{T}_{(x_n,s_n)}U; 0\big) \ge \frac{\gamma}{2}.$$

Since $R_n \to \infty$ and $\tau_n \to -\infty$, these are escaping covariant tail recenterings. If they have loss of local compactness, terminal indecomposability is contradicted. Otherwise local compactness gives a subsequential terminal limit $(W, \Pi_W)$, and strong local $L^3$ convergence gives

$$\mathcal{V}_1(W; 0) \ge \frac{\gamma}{4} > 0.$$

Choose a rational $0 < \eta < \gamma/4$. Thus $W$ is a covariant tail limit with retained local velocity concentration at level $\eta$, contradicting the all-positive-threshold definition of terminal indecomposability. Therefore

$$m_n^{\text{vel}} \to 0.$$

Fix a small $\alpha_0 > 0$, and let

$$c_{\text{sl}}(\mathcal{K}_{\text{ext}}, \alpha_0) > 0$$

be the compact-state constant from Lemma 10.2, as used in Lemma 10.3. Since $m_n^{\text{vel}} \to 0$, for all large $n$,

$$m_n^{\text{vel}} < c_{\text{sl}}(\mathcal{K}_{\text{ext}}, \alpha_0).$$

If the velocity concentration is small but the pressure-inclusive CKN mass is not, Lemma 9.5 assigns the recentering either to pressure compactness failure, to the affine/parasitic pressure quotient, or to a neighboring unit cylinder with positive velocity concentration. Each case is already excluded by terminal indecomposability or by lower-stratum removal. If the exterior mass is critical only after log-window normalization, the resulting normalized support profile is a critical-tail defect and is one of the already excluded terminal alternatives. If no such recentering loses compactness and no critical-tail normalization is selected, then the exterior time-slice lower bound

$$\int_{|y|>R_n} |U(y, \tau_n)|^3 \, dy \ge 1$$

together with the parabolic velocity smallness $m_n^{\text{vel}} < c_{\text{sl}}(\mathcal{K}_{\text{ext}}, \alpha_0)$ is now handled quantitatively by Lemma 10.3. Apply that lemma with $m_0 = 1$, $\alpha = \alpha_0$, and the displayed inequality. If the lemma produces local compactness loss, pressure-gauge loss, pressure-only mass, or a critical-tail normalization, the branch is one of the already covered noncompact, pressure, affine/parasitic, or critical-tail alternatives. In the remaining case it produces diffuse exterior concentration in the sense of Definition 10.1. This again contradicts terminal indecomposability. Both parabolic alternatives are impossible, so the desired sequence exists. □

**Proposition 10.7** (Stability of the bounded mild formulation)**.** *Let $(U_m, \Pi_m)$ be normalized parasitic-free centered profiles whose physical pullbacks are bounded mild Navier–Stokes solutions on every finite terminal interval. Assume $U_m \to U$ locally smoothly in centered variables and that* *the $U_m$ are uniformly bounded on compact centered time intervals. Then the physical pullback of $U$ is also bounded mild on every finite terminal interval. Let*

$$u(x,t) := (-t)^{-1/2} U\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \qquad t<0,$$

*be the physical pullback. For each fixed $T<0$, define the finite terminal shift*

$$u^T(x,s) := u(x, T+s) = (-(T+s))^{-1/2} U\left(\frac{x}{\sqrt{-(T+s)}}, -\log(-(T+s))\right), \qquad s<0,$$

*Then, after translating the interval so that its terminal time is $0$, one has, for all $-\infty < s < t < 0$,*

$$u^T(t) = e^{(t-s)\Delta} u^T(s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P}\nabla \cdot (u^T \otimes u^T)(\sigma)\, d\sigma \tag{10.1}$$

*in the bounded mild duality sense used in the setup paper. The same conclusion is preserved by centered time translations, covariant translations, retained tail limits, concentrating descendants, recurrent-core limits, and affine normalization outside $\mathcal{C}_{\mathrm{aff}}$.*

*Proof.* For each $m$, the normalized setup representative satisfies the Duhamel formula on every finite physical interval. Fix $T<0$ and $-\infty < s < t < 0$. Pull the centered convergence back to the compact physical time interval $[s,t]$ after the finite terminal shift. The velocities converge locally smoothly and are uniformly bounded; hence

$$u_m^T \otimes u_m^T \to u^T \otimes u^T$$

locally in every finite $L^q$, and in particular in distributions. Testing the Duhamel identity for $u_m^T$ against a compactly supported smooth vector field and using the heat-kernel/Leray-projection duality gives convergence of the linear heat term and of the bilinear Duhamel term. The bounded mild duality formulation used in the setup paper then extends the identity from compactly supported tests to the standard bounded mild class, yielding (10.1) for $u^T$.

Centered time translations and covariant translations are exact symmetries of (1.2); in physical variables they are compositions of translations, scalings, and finite time shifts, which preserve the Duhamel identity. Retained tail limits, concentrating descendants, and recurrent-core limits are locally smooth limits of such normalized representatives, so the preceding limit argument applies. The affine normalization either removes a spatially homogeneous parasitic component or assigns the profile to $\mathcal{C}_{\mathrm{aff}}$; outside $\mathcal{C}_{\mathrm{aff}}$ no homogeneous parasitic forcing remains, and the same Duhamel identity holds for the canonical representative. ∎

**Theorem 10.8** (Parasitic-free mildness inheritance for normalized profiles)**.** *Let $(U,\Pi)$ be a retained terminal profile, retained covariant tail descendant, concentrating descendant limit, or recurrent-core limit represented in the normalized residual profile class generated by the Seregin extraction in the setup paper. Then exactly one of the following alternatives applies.*

(i) *$(U,\Pi)$ belongs to the affine/parasitic lower stratum of Definition 8.16;*

(ii) *after the canonical affine normalization already built into the residual representative, the physical pullback*

$$u(x,t) = (-t)^{-1/2} U\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \qquad t<0,$$

*is parasitic-free and has the following property: for every $T<0$, the finite-terminal shift*

$$u^T(x,s) := u(x,T+s), \qquad s<0,$$

*is a bounded mild ancient Navier–Stokes solution in the Albritton–Barker sense on $\mathbb{R}^3 \times (-\infty, 0)$.*

*Proof.* Affine-constant centered profiles are precisely the homogeneous parasitic profiles removed by Definition 8.16; they are assigned to alternative (i). We therefore consider a representative outside that lower stratum.

The setup paper constructs normalized Seregin limits as centered bounded mild ancient representatives, after removal of the spatially homogeneous parasitic mode. The stability statement Proposition 10.7 shows that the limiting operations used in this paper preserve the bounded mild formulation: exact centered symmetries preserve the Duhamel identity, local limits preserve it by bounded mild duality, and affine normalization either removes the homogeneous parasitic mode or places the profile in $\mathcal{C}_{\mathrm{aff}}$. Thus retained terminal profiles, covariant tail descendants, concentrating descendant limits, and recurrent-core limits remain parasitic-free mild unless they have fallen into the affine/parasitic lower stratum.

Finally fix $T < 0$. On $\mathbb{R}^3 \times (-\infty, T]$, the physical pullback

$$u(x,t) = (-t)^{-1/2} U\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right)$$

is smooth and bounded by $(-T)^{-1/2}\|U\|_{L^\infty}$, and it solves the physical Navier–Stokes equations by the direct self-similar change of variables. By the inherited parasitic-free mild formulation, for $-\infty < s < t \le T$,

$$u(t) = e^{(t-s)\Delta}u(s) - \int_s^t e^{(t-\sigma)\Delta}\mathbb{P}\nabla\cdot(u\otimes u)(\sigma)\,d\sigma.$$

After translating the finite terminal time $T$ to 0, this is the mild ancient formulation for $u^T$ on every finite forward interval $(-\infty, 0)$. The bound $\|u^T\|_{L^\infty(\mathbb{R}^3\times(-\infty,0))} \le (-T)^{-1/2}\|U\|_{L^\infty}$ is finite. Thus $u^T$ is a genuine bounded mild ancient solution in the sense required by the Albritton–Barker endpoint theorem. □

**Theorem 10.9** (Albritton–Barker endpoint Liouville theorem)**.** *Let $u \in L^\infty(\mathbb{R}^3 \times (-\infty, 0))$ be a bounded mild ancient solution of the three-dimensional Navier–Stokes equations. If there exists $t_k \to -\infty$ such that*

$$\sup_k \|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} < \infty,$$

*then $u \equiv 0$.*

*Proof.* This is the endpoint ancient Liouville theorem in the sequence-$L^3$ form proved by Albritton and Barker [1]. □

**Lemma 10.10** (No retained indecomposable concentration profile)**.** *No retained terminally indecomposable concentration profile exists.*

*Proof.* Let $U$ have retained local velocity concentration and be terminally indecomposable. By Lemma 10.6, choose $\tau_k \to -\infty$ such that

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3} < \infty.$$

Let $u$ be the physical pullback and set $t_k = -e^{-\tau_k}$. Since $\tau_k \to -\infty$, the sequence satisfies $t_k \to -\infty$, and critical scaling gives

$$\|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} = \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)}.$$

By Theorem 10.8, either $U$ is in the affine/parasitic lower stratum, which is not part of the normalized residual class, or else for each $T < 0$

$$u^T(x,s) := u(x, T+s), \qquad s < 0,$$

is a bounded mild ancient solution. Choose $T < 0$ and discard finitely many indices so that $t_k < T$. Then $s_k := t_k - T \to -\infty$, and

$$\|u^T(\cdot, s_k)\|_{L^3(\mathbb{R}^3)} = \|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} \leq \sup_j \|U(\cdot, \tau_j)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Theorem 10.9 gives $u^T \equiv 0$. Since $T < 0$ was arbitrary, the physical pullback vanishes on $\mathbb{R}^3 \times (-\infty, 0)$, and therefore $U \equiv 0$. This contradicts retained local velocity concentration. □

**Lemma 10.11** (Diagonal lifting of retained concentrating tail limits)**.** *Let $A = \{\mathfrak{s}_1, \ldots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained velocity concentration separated family for a refined residual candidate, represented by separated recentering sequences $g_{j,n} = (y_{j,n}, \tau_{j,n})$. Let $(U_j, \Pi_j)$ be the terminal profile obtained in the $j$-th recentering. If $(U_j, \Pi_j)$ has a retained concentrating tail limit $(W, \Pi_W)$, then there is a combined parabolic-covariant descendant recentering sequence $\mathfrak{d} = (d_n)$, in the sense of Definition 9.2, for the original terminal sequence such that:*

*(i) $\mathfrak{d}$ is asymptotically separated from $g_j$;*
*(ii) the recentered original sequence at $\mathfrak{d}$ converges locally to $(W, \Pi_W)$, after admissible pressure gauges;*
*(iii) $\mathfrak{d}$ has retained local velocity concentration.*

*Proof.* By definition of retained concentrating tail limit, there are relative covariant tail recenterings $\zeta_m = (\xi_m, \sigma_m)$, with

$$|\xi_m| + |\sigma_m|^{1/2} \to \infty,$$

such that

$$\mathscr{T}_{\zeta_m}(U_j, \Pi_j) \to (W, \Pi_W)$$

locally in the terminal topology, and

$$\mathcal{V}_1(W; 0) \geq \eta_*.$$

Fix $m$. Since $V_n^{g_j} \to U_j$ locally smoothly and $(P_n^{g_j}) \to \Pi_j$ locally in $L^{3/2}$ modulo gauges, the convergence also holds on the fixed covariant tube

$$\{(y + e^{s/2}\xi_m, s + \sigma_m) : (y, s) \in Q_m(0, 0)\}.$$

Therefore one can choose $n(m)$ so large that the covariantly recentered original fields

$$V_{n(m)}(y + y_{j,n(m)} + e^{s/2}\xi_m,\ s + \tau_{j,n(m)} + \sigma_m)$$

and the corresponding pressure gauges are within $m^{-1}$ of the $\mathscr{T}_{\zeta_m}$-recentered fields of $(U_j, \Pi_j)$ on $Q_m(0, 0)$.

Set

$$d_m := (g_{j,n(m)}; \xi_m, \sigma_m).$$

Relabeling the subsequence $n(m)$ as the terminal index gives a combined parabolic-covariant descendant recentering sequence $\mathfrak{d} = (d_m)$ for the original terminal sequence. Since the covariant parabolic distance from $d_m$ to $g_{j,n(m)}$ is $|\xi_m| + |\sigma_m|^{1/2}$, the sequence $\mathfrak{d}$ is separated from the $j$-th recentering sequence. The diagonal choice gives local convergence of the original sequence at $\mathfrak{d}$ to $(W, \Pi_W)$, after the admissible pressure gauges used above. The diagonal convergence gives strong local $L^3$-convergence of the velocity, so

$$\liminf_{m \to \infty} \mathcal{V}_1\big(V_{n(m)}^{\mathfrak{d}}; 0\big) \geq \mathcal{V}_1(W; 0) \geq \eta_*.$$

Thus $\mathfrak{d}$ has retained local velocity concentration. The pressure gauges are inherited through the same diagonal construction, but pressure-inclusive $\mathcal{E}_1$ is not used to certify retained nonzero concentration. □

**Theorem 10.12** (Heredity of retained descendants)**.** *Let $(U, \Pi)$ be a normalized retained residual profile in $\mathfrak{X}_M$, and let $(W, \Pi_W)$ be a retained covariant tail descendant of $(U, \Pi)$. Then $(W, \Pi_W) \in \mathfrak{X}_M$, it carries retained local velocity concentration at the descendant threshold fixed in Definition 3.4, and it has the same parasitic-free normalization as a terminal profile generated directly by the Seregin extraction in the setup paper. Consequently exactly one of the following ordered outcomes occurs:*

(i) $(W, \Pi_W)$ *belongs to one of the previously closed lower strata, in which case that descendant is already excluded;*

(ii) $(W, \Pi_W)$ *belongs to the normalized residual terminal profile class, so the concentrating descendant and terminal-indivisibility arguments may be iterated from $W$.*

*Proof.* The descendant is obtained as a limit of covariant tail recenterings $\mathscr{T}_{\zeta_n}(U, \Pi)$. By Lemma 8.9, the recentered sequence is locally precompact in $\mathfrak{X}_M$, and the limit remains a bounded smooth locally suitable centered solution with the same $L^\infty$ bound and compatible pressure gauges. Retained local velocity concentration is part of the definition of retained descendant; when the descendant is produced through a limit passage, the velocity semicontinuity in Lemma 9.4 fixes the threshold loss once and for all through Definition 3.4. The affine/parasitic normalization is canonical for the residual representative and is preserved by $T_a$, $\Theta_s$, and their composition $\mathscr{T}_{\zeta_n}$; if a limit is affine-parasitic, it is assigned to the affine-constant lower stratum of Definition 8.16. Otherwise it is represented in the same normalized residual profile class. The ordered residual decomposition then places the descendant either in a previously closed lower class or in the generic residual profile class. This is a local profile reduction and does not use a global critical norm or global tail budget. □

**Lemma 10.13** (Escaping recenterings in a separated family)**.** *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained velocity concentration separated family for a refined residual candidate. For each component $\mathfrak{s}_j$, the following ordered alternatives exhaust the possible successor profiles:*

*(i) the profile $(U_j, \Pi_j)$ is terminally indecomposable;*

*(ii) $(U_j, \Pi_j)$ has diffuse exterior concentration or an escaping recentering sequence with loss of local compactness;*

*(iii) $(U_j, \Pi_j)$ has a retained concentrating tail limit which lifts to a recentering sequence of the original terminal sequence with retained local velocity concentration.*

*Proof.* This is the definition of terminal indecomposability plus the diagonal lifting lemma, with no disjointness assertion. If $(U_j, \Pi_j)$ is terminally indecomposable, (i) holds. If it is not terminally indecomposable, at least one of the three excluded alternatives in Definition 10.1 occurs: a retained concentrating tail limit, diffuse exterior concentration, or an escaping recentering sequence with loss of local compactness. The diffuse exterior and local-compactness-failure cases give (ii). In the retained concentrating tail-limit case, Lemma 10.11 lifts the limit to a recentering sequence of the original terminal sequence with retained local velocity concentration, and Theorem 10.12 places the descendant either in a closed lower stratum or back in the normalized residual profile class, giving (iii) in the iterative residual case. If several successor profiles coexist, they are all recorded; the next lemma removes the nonconcentrating ones. □

**Lemma 10.14** (No diffuse or non-compact escaping recentering from a retained family)**.** *In the setting of Lemma 10.13, alternative* (ii) *cannot occur.*

*Proof.* Suppose that $(U_j, \Pi_j)$ has diffuse exterior concentration or an escaping recentering sequence with loss of local compactness. By the same diagonal procedure used in Lemma 10.11, this tail profile is realized by a terminal extraction of the original refined residual candidate, after passing to a subsequence and choosing the local pressure gauges supplied by the imported setup results. The fact that the diffuse or noncompact profile is not the sole source of the original compact concentration does not exclude coexistence with the compact core $(U_j, \Pi_j)$. The paired exclusion Theorem 9.102 rules out exactly this coexistence in the refined residual class. Therefore alternative (ii) cannot occur for a refined residual family with retained local velocity concentration. □

**Definition 10.15** (Closed retained velocity concentration successor relation)**.** For a fixed threshold $\eta > 0$, let

$$\mathfrak{A}_\eta := \mathfrak{A}_\eta^\# = \{U \in \mathfrak{X}_M^\# : \Phi(U) \geq \eta\}$$

be the compact non-affine oscillation set of Corollary 8.20. We write

$$U \, \mathcal{R}_\eta \, V$$

if $U, V \in \mathfrak{A}_\eta$ and $V$ is a retained concentrating covariant tail descendant of $U$.

**Lemma 10.16** (Closedness of the retained velocity concentration successor relation)**.** *The relation*

$$\mathcal{R}_\eta \subset \mathfrak{A}_\eta \times \mathfrak{A}_\eta$$

*is closed in the local $\mathfrak{X}_M$ topology.*

*Proof.* Let $U_m \to U$, $V_m \to V$ in $\mathfrak{X}_M^\#$, with $U_m \, \mathcal{R}_\eta \, V_m$. For each $m$, choose a retained covariant tail observer sequence $\zeta_{m,\ell}$ for $U_m$ whose recentered fields converge to $V_m$ on $Q_m(0,0)$ to accuracy $m^{-1}$, after admissible pressure gauges. A diagonal choice in $(m, \ell)$, using Lemmas 8.9 and 9.4, gives a single escaping covariant tail sequence for $U$ converging locally to $V$. Closedness of $\mathfrak{A}_\eta$ follows from Corollary 8.20. The retained descendant threshold is preserved by strong local $L^3$ convergence for the velocity part, or by closedness of the non-affine oscillation functional $\Phi$ when the successor relation is recorded in $\mathfrak{A}_\eta$. No pressure-only lower bound is used to keep the edge. Hence $U \, \mathcal{R}_\eta \, V$. □

**Theorem 10.17** (Recurrence for retained velocity concentration chains)**.** *Assume there exists an infinite chain of retained concentrating descendants*

$$U_0 \, \mathcal{R}_\eta \, U_1 \, \mathcal{R}_\eta \, U_2 \, \mathcal{R}_\eta \cdots .$$

*Then the path space*

$$\mathscr{P}_\eta := \Big\{(W_j)_{j\geq 0} \in \mathfrak{A}_\eta^{\mathbb{N}} : W_j \, \mathcal{R}_\eta \, W_{j+1} \text{ for all } j\Big\}$$

*is nonempty and compact, is invariant under the left shift*

$$S(W_0, W_1, W_2, \ldots) = (W_1, W_2, \ldots),$$

*and carries an $S$-invariant Borel probability measure $\mathbb{P}$. Its projected stationary profile measure $\nu = (\pi_0)_\# \mathbb{P}$ is supported entirely on $\mathfrak{A}_\eta$.*

*Proof.* The set $\mathfrak{A}_\eta = \mathfrak{A}_\eta^\#$ is compact by Corollary 8.20. Hence $\mathfrak{A}_\eta^{\mathbb{N}}$ is compact and metrizable. By Lemma 10.16,

$$\mathscr{P}_\eta = \bigcap_{j=0}^{\infty} \{(W_k)_{k\geq 0} : (W_j, W_{j+1}) \in \mathcal{R}_\eta\}$$

is closed, hence compact. The given infinite chain is an element of $\mathscr{P}_\eta$, so the path space is nonempty, and it is plainly invariant under the left shift.

For the path $p = (U_0, U_1, \ldots)$, define empirical measures

$$\mathbb{P}_N := \frac{1}{N} \sum_{k=0}^{N-1} \delta_{S^k p}.$$

Weak-* compactness of probability measures on compact $\mathscr{P}_\eta$ gives a convergent subsequence $\mathbb{P}_{N_\ell} \rightharpoonup \mathbb{P}$. For each continuous $F$ on $\mathscr{P}_\eta$,

$$\int F \circ S \, d\mathbb{P}_N - \int F \, d\mathbb{P}_N = \frac{F(S^N p) - F(p)}{N} \to 0.$$

Thus $\mathbb{P}$ is shift-invariant. Since every coordinate of every path in $\mathscr{P}_\eta$ belongs to $\mathfrak{A}_\eta$, the projected measure $\nu = (\pi_0)_\# \mathbb{P}$ is supported in $\mathfrak{A}_\eta$. □

**Lemma 10.18** (Concatenation of retained concentrating descendants)**.** *Let*

$$U_0 \, \mathcal{R}_\eta \, U_1 \, \mathcal{R}_\eta \cdots \mathcal{R}_\eta \, U_N$$

*be a finite chain of retained concentrating descendants. Then $U_N$ is obtained from $U_0$ by a single diagonal covariant recentering sequence, after admissible pressure gauges, and it remains in $\mathfrak{A}_\eta$. The pressure gauges in the intermediate limits can be chosen compatibly with the final diagonal limit.*

*Proof.* For $N = 1$ this is the definition of $\mathcal{R}_\eta$. Assume it holds up to $N - 1$. The relation $U_{N-1}\mathcal{R}_\eta U_N$ gives escaping covariant observers for $U_{N-1}$ which converge locally to $U_N$. Compose these observers with the diagonal observers which realize $U_{N-1}$ from $U_0$. The semidirect covariance identity in Lemma 8.5 shows that the composition is again a covariant observer sequence for $U_0$. A diagonal choice on $Q_m(0,0)$, together with the pressure-gauge compatibility in Lemma 9.4, gives local convergence to $U_N$. Since each link lies in $\mathfrak{A}_\eta$, the endpoint does too. □

**Lemma 10.19** (Recurrent path produces a compact retained velocity concentration core)**.** *Let $\mathbb{P}$ be a shift-invariant probability measure on the retained velocity concentration path space $\mathscr{P}_\eta$. For $\mathbb{P}$-a.e. recurrent path $(W_j)_{j \geq 0}$, the orbit closure of its coordinates, together with all covariant tail limits generated by the returning blocks, contains a nonempty compact invariant set $\mathcal{M}_\eta \subset \mathfrak{A}_\eta$. Every element of $\mathcal{M}_\eta$ is a retained descendant of the original profile realized from the original blow-up sequence.*

*Proof.* Poincaré recurrence for the left shift gives a full-measure set of paths which return arbitrarily close to themselves in the product topology. Fix such a path. For each returning block, use Lemma 10.18 to compose the finitely many concentrating successor observers into one covariant recentering of the initial profile. The endpoints of the returning blocks remain in the compact set $\mathfrak{A}_\eta$. Taking the closure of all such endpoints and their covariant tail limits inside the compact normalized profile class $\mathfrak{X}_M^{\#}$ gives a nonempty compact set. Closedness of $\mathcal{R}_\eta$ and compactness of $\mathfrak{A}_\eta$ keep the closure inside $\mathfrak{A}_\eta$. Intersecting over all closed invariant subsets of this compact set and applying Zorn's lemma gives a compact invariant minimal core $\mathcal{M}_\eta$. The diagonal construction in the concatenation lemma preserves realization from the original sequence and retained velocity concentration, so every element of $\mathcal{M}_\eta$ is realized from the original blow-up sequence. □

**Lemma 10.20** (Retention and gauges on the recurrent concentration core)**.** *The compact core $\mathcal{M}_\eta$ of Lemma 10.19 carries the same retained oscillation threshold:*

$$\Phi(U) \geq \eta \qquad (U, \Pi) \in \mathcal{M}_\eta,$$

*and its pressure gauges are compatible with the invariant averaging used in Theorem 8.29.*

*Proof.* The inequality $\Phi \geq \eta$ is closed under the local topology by Corollary 8.20, and all approximating profiles lie in $\mathfrak{A}_\eta$. Hence every element of the compact core lies in $\mathfrak{A}_\eta$. Gauge compatibility follows from Lemma 9.4: on each compact cylinder the pressure gauges are chosen by a fixed normalization, and diagonal limits preserve the normalized local $L^{3/2}$ pressure topology.

Therefore the invariant measures on $\mathcal{M}_\eta$ used in the recurrent-core rigidity theorem average genuine normalized profiles with retained oscillation, not untracked pressure representatives. $\square$

**Theorem 10.21** (No infinite retained velocity concentration chain)**.** *No infinite chain of retained concentrating descendants can occur in the refined residual class.*

*Proof.* Suppose an infinite chain of retained concentrating descendants exists. If $\limsup_j \Phi(U_j) = 0$, local compactness gives a subsequential limit $U_{j_k} \to U_\infty$ with $\Phi(U_\infty) = 0$. The diagonal lifting used in Lemma 10.11 shows that this limit is still a retained tail descendant of the original residual profile. By Theorem 8.21, $U_\infty$ belongs to the affine/parasitic lower stratum, contradicting descendant heredity in the refined residual class. Hence $\limsup_j \Phi(U_j) > 0$. Choose $\eta > 0$ and an infinite ordered subchain contained in $\mathfrak{A}_\eta$. By transitivity of retained tail descendants through Lemma 10.11, this ordered subchain is an $\mathcal{R}_\eta$-chain.

Theorem 10.17 gives a compact shift-invariant path space and a stationary path measure supported on $\mathfrak{A}_\eta$. Lemmas 10.18 to 10.20 turn a recurrent path into a compact tail-minimal recurrent core for (1.2) whose profiles remain in $\mathfrak{A}_\eta$ with compatible pressure gauges. The recurrent-core rigidity theorem Theorem 8.29 forces this core to be affine-constant after the canonical affine normalization. But affine/parasitic profiles have $\Phi = 0$, whereas $\mathfrak{A}_\eta = \{\Phi \geq \eta\}$. This contradiction rules out the infinite chain of retained concentrating descendants. $\square$

**Lemma 10.22** (Finite separated families produce indecomposable profiles or concentration chains)**.** *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained velocity concentration separated family for a refined residual candidate. Then either one of the terminal profiles $(U_j, \Pi_j)$ is terminally indecomposable, or the family generates an infinite chain of retained concentrating descendants. A finite recurrent loop is counted as an infinite chain by periodic repetition.*

*Proof.* Assume no $(U_j, \Pi_j)$ is terminally indecomposable. By Lemmas 10.13 and 10.14, each $(U_j, \Pi_j)$ has a retained concentrating tail limit which lifts to a recentering sequence of the original terminal sequence with retained local velocity concentration.

Starting from $A$, enlarge the separated family whenever a lifted retained tail descendant is separated from every profile already selected. If this enlargement can be carried out indefinitely, the successive lifted descendants already form an infinite chain of retained concentrating descendants. Otherwise the procedure stops after finitely many enlargements at a finite separated family $A'$ which is maximal for this descendant-lifting procedure.

Form a finite directed graph whose vertices are the retained profiles in $A'$. Draw an edge $i \to j$ if a retained tail descendant of $U_i$, lifted by Lemma 10.11, is equivalent to $U_j$. There is no escape edge by the construction of $A'$: an escape edge would be exactly another separated retained unit concentrating cylinder in the original sequence, so the enlargement procedure would not have stopped. Every decomposable vertex has an outgoing edge by the preceding paragraph.

A finite directed graph in which every vertex has an outgoing edge contains a directed cycle. Therefore, if no terminally indecomposable profile appears, the directed recentering relation contains a finite loop

$$\mathfrak{s}_{j_0} \to \mathfrak{s}_{j_1} \to \cdots \to \mathfrak{s}_{j_q} = \mathfrak{s}_{j_0}.$$

Repeating the loop periodically gives an infinite chain of retained concentrating descendants. The nonconcentrating successor alternatives at each vertex are excluded by Lemma 10.14, while coexistence of a compact core with a diffuse or noncompact remainder is handled separately by Theorem 9.102. This proof is graph/recurrence based and does not use a global counting or mass budget. $\square$

**Lemma 10.23** (Finite recurrent loops of retained tail limits are impossible)**.** *No refined residual terminal extraction contains a finite recurrent loop of retained concentrating tail limits.*

*Proof.* Let

$$\mathfrak{s}_{j_0} \to \mathfrak{s}_{j_1} \to \cdots \to \mathfrak{s}_{j_q} = \mathfrak{s}_{j_0}$$

be such a loop, and write $(U_\ell, \Pi_\ell)$ for the corresponding retained concentration terminal profiles. Each arrow means that $U_{\ell+1}$ is a retained concentrating tail limit of $U_\ell$. Thus, for each $\ell$, there are escaping tail recentering sequences whose recentered fields converge locally to $U_{\ell+1}$.

Let $\mathcal{K}$ be the closure, in the local smooth velocity topology and local $L^{3/2}$ pressure topology modulo gauges, of all profiles obtained from the finite set $\{U_0, \ldots, U_{q-1}\}$ by time translations and covariant terminal translations. The boundedness inherited from the Type I Seregin extraction theorem and the local pressure compactness from the imported setup results give compactness of $\mathcal{K}$ on every fixed terminal cylinder; a diagonal argument gives compactness of $\mathcal{K}$. The set is invariant under the same time and covariant translations by construction. The recurrent-loop relations imply that each $U_\ell$ returns to $\mathcal{K}$ through tail limits, so the closed invariant subset of $\mathcal{K}$ generated by the loop is nonempty, compact, invariant, and contains the retained velocity concentration profiles.

Repeating the loop periodically gives an infinite chain of retained concentrating descendants. This contradicts Theorem 10.21. Therefore no finite recurrent loop of retained concentrating tail limits exists. □

**Theorem 10.24** (Finite separated retained velocity concentration profile families are impossible)**.** *No refined residual candidate admits a finite separated family of retained velocity concentration profiles.*

*Proof.* Suppose, for contradiction, that a refined residual candidate admits a finite retained velocity concentration separated family $A = \{\mathfrak{s}_1, \ldots, \mathfrak{s}_J\}$, $J \geq 2$. By Lemma 10.22, either some component of the family is terminally indecomposable or the family generates an infinite chain of retained concentrating descendants. The first alternative contradicts Lemma 10.10. The second contradicts Theorem 10.21. Hence the finite retained velocity concentration separated family cannot exist. □

*Proof of Proposition 10.5.* Let $U$ be a single retained velocity concentration terminal profile after the finite separated-family, mixed compact–diffuse, and noncompact escaping alternatives have been excluded. If $U$ had a retained concentrating tail limit, Lemma 10.4 would produce a finite two-profile separated family with retained velocity concentration, contradicting Theorem 10.24. If $U$ had diffuse exterior concentration or an escaping non-compact recentering sequence, the corresponding terminal extraction would be a mixed profile configuration consisting of the retained compact core $U$ plus a diffuse or noncompact tail remainder. This is excluded by Theorem 9.102. Hence no retained concentrating tail limit, diffuse exterior concentration, or escaping non-compact recentering sequence exists, and $U$ is terminally indecomposable. □

The terminal endpoint argument uses the following acyclic order:

concentration-set extraction
$\Downarrow$
diffuse, pressure, noncompact, and critical-tail closures
$\Downarrow$
sequence-$L^3$ for terminally indecomposable atomic profiles
$\Downarrow$
no atomic active profile by bounded-mild endpoint Liouville
$\Downarrow$
no finite separated family and no infinite chain
$\Downarrow$
terminal stratification assembly.

In particular, Lemma 10.6 does not use Theorem 10.25; it uses only terminal indecomposability and the already covered exterior, diffuse, pressure/noncompact, and critical-tail alternatives.

Table 5: Acyclic dependency table for the terminal concentration argument.

| **Result** | **May use** | **Must not use** |
|---|---|---|
| Concentration-set extraction | Local Seregin compactness, pressure atlas, and the defect measures $\mu_n$. | Terminal assembly theorem. |
| Diffuse-defect compactness | Observer compactification, concentration-neighborhood removal, and amenable averaging. | Atomic Liouville contradiction. |
| Critical-tail closure | Mesoscopic reduction, affine quotient, bounded-origin realization, and coherent tail classification. | Terminal assembly theorem. |
| Mixed compact–diffuse exclusion | Concentration-set extraction, diffuse-defect compactness, and critical-tail closure. | Finite separated-family theorem. |
| Sequence-$L^3$ lemma | Terminal indecomposability and absence of retained velocity, diffuse, noncompact, pressure-only, and critical-tail descendants. | Albritton–Barker endpoint theorem. |
| Atomic contradiction | Sequence-$L^3$, finite-shift bounded mildness, and the Albritton–Barker endpoint theorem. | Terminal assembly theorem. |
| Finite separated-family exclusion | No atomic profile and no infinite retained velocity concentration chain. | Terminal assembly theorem. |
| Terminal stratification assembly theorem | All previous rows. | Itself. |

**Theorem 10.25** (Terminal stratification assembly theorem)**.** *Let $(V_n, P_n)$ be a terminal residual sequence realized from the original blow-up sequence and satisfying the imported setup admissibility hypotheses, the pressure atlas hypotheses, and a retained unit velocity concentration lower bound*

$$\liminf_{n\to\infty} \mathcal{V}_1(V_n; z_n) \geq \eta_* > 0.$$

*Assume all lower classes already excluded by the setup paper have been removed. The compact subthreshold regular residual alternative* (b) *of Theorem 9.17 is excluded before the terminal stratification by Lemma 9.18 (CKN $\varepsilon$-regularity); hence it does not appear as a terminal residual alternative below. Then the paired terminal extraction of Theorem 9.24 gives the following ordered alternatives:*

(1) *a compact concentrating descendant occurs;*
(2) *a finite separated family of retained velocity concentration profiles occurs;*
(3) *an infinite chain of retained concentrating descendants occurs;*
(4) *a diffuse exterior defect occurs after concentration-neighborhood removal;*
(5) *a critical-tail defect occurs;*
(6) *an affine/parasitic quotient profile occurs;*
(7) *an indecomposable atomic concentration profile occurs and satisfies a sequence-$L^3$ bound.*

*Alternative* (1) *is a reduction alternative: by descendant heredity it is iterated inside the same normalized profile class until one reaches* (2), (3), *a mixed diffuse/noncompact/critical-tail branch,* *or the atomic case* (7). *Alternatives* (2)–(6) *are excluded by the preceding branch closures and local terminal arguments. Alternative* (7) *is excluded by the endpoint bounded-mild ancient Liouville theorem after finite terminal shift. Hence no terminal residual profile satisfying the retained unit velocity concentration lower bound remains.*

*Proof.* This is the assembly of the local results and explicit profile-class arguments in Sections 9 and 10. The local concentration decomposition is the paired concentration-set/remainder decomposition Theorem 9.24. Closed or continuum active loci are reduced to the single-profile, finite-separated, or infinite-chain cases by Lemma 9.14, so no additional continuum active-locus branch is hidden in the compact concentration set. This theorem does not introduce an additional compactness principle; it only combines the preceding extraction, reduction, and closure results in the order recorded in Table 5. The pure radiative alternatives enter only as sole-source exclusions, and are used only in that form through Theorem 9.27. Coexistence of a compact concentration core with diffuse or noncompact tail remainder is separately ruled out by Theorem 9.102, using realized critical-tail rigidity Theorem 9.88 and Corollary 9.101. Infinite concentration chains and finite recurrent loops are ruled out by the retained-chain recurrence theorem Theorem 10.21, while finite separated concentration families are ruled out by Theorem 10.24. Therefore any retained concentration terminal profile that survives the ordered lower-class removals is single and has no retained concentrating descendant or diffuse/noncompact tail remainder. By Proposition 10.5 it is terminally indecomposable, and then Lemmas 10.6 and 10.10 and Theorem 10.8 give the sequence-$L^3$, parasitic-free finite-shift mildness, and endpoint contradiction. All estimates used in this assembly are localized to fixed terminal cylinders, except for the final sequence-$L^3$ conclusion, which is derived by contradiction from terminal indecomposability and is not assumed as a global bound. □

*Remark* 10.26 (No global budget in the terminal assembly theorem). The terminal stratification assembly theorem assumes no global $L^3$, global CKN budget, finite energy bound on expanding regions, or global counting measure for concentration profiles. Compactness and recurrence are local, normalized, or realized from the original blow-up sequence.

**Corollary 10.27** (Rotational residual closure)**.** *The ordered rotational relative-equilibrium residual class is empty:*

$$\mathcal{C}_{\mathrm{rot}}(\mathcal{S}; I, J) = \varnothing.$$

*Proof.* By Proposition 5.8, every R2 residual either belongs to a previously closed lower class or produces a terminal concentration profile realized from the original blow-up sequence with retained unit velocity concentration. The lower classes are empty by the imported setup results, and the terminal concentration case is empty by Theorem 10.25. Hence no R2 residual remains. □

**Corollary 10.28** (Stationary-hull residual closure)**.** *For every admissible stationary phase space* $\mathfrak{B}$*,*

$$\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}(\mathcal{S}; I, J; \mathfrak{B}) = \varnothing.$$

*Proof.* By Propositions 6.12 and 7.6, every R3 stationary-hull residual either lies in a lower class already excluded by the setup paper or produces a stationary terminal concentration profile realized from the original blow-up sequence. The lower classes are empty by the imported setup results. The stationary terminal profile satisfies the retained unit velocity concentration hypotheses of Theorem 10.25, so it is reduced to the atomic alternative only after all separated, diffuse, noncompact, critical-tail, and affine alternatives have been removed. In that atomic case Lemmas 10.6 and 10.10 and Theorem 10.8 give the sequence-$L^3$ bound, finite-shift bounded mildness, and the endpoint Albritton–Barker contradiction. Hence no stationary terminal concentration profile remains. This proves the deferred stationary concentration and degenerate stationary-family occurrence statements Remarks 7.4 and 7.5 and Proposition 6.10, and therefore the R3 class is empty. □

## 11. Closure of the generic terminal residual

**Definition 11.1** (Generic residual profile)**.** A generic residual profile is a smooth bounded normalized Seregin ancient solution of (1.2) which carries persistent compact velocity $L^3$ concentration and belongs to none of the previously closed alternatives, none of $\mathcal{C}_{\mathrm{ax}}, \mathcal{C}_{\mathrm{rot}}, \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$, and none of the structured or tight classes removed earlier, including the ordered critical-tail lower strata $\mathcal{C}_{\log\mathrm{diff}}, \mathcal{C}_{\mathrm{Young}}, \mathcal{C}_{\mathrm{homcrit}}, \mathcal{C}_{\mathrm{logper}}, \mathcal{C}_{\mathrm{ap}}$.

**Theorem 11.2** (Generic terminal exhaustion)**.** *Let $(U, \Pi) \in \mathcal{C}_{\mathrm{gen}}$ be represented after canonical affine/parasitic normalization and after removal of the lower alternatives already excluded by the setup paper. Then a terminal extraction of $(U, \Pi)$, with the priority order used throughout the terminal section, selects one of the following priority alternatives:*

(1) *a retained compact active descendant;*
(2) *a finite separated family of retained velocity profiles;*
(3) *an infinite chain of retained active descendants;*
(4) *a diffuse exterior defect after active-neighborhood removal;*
(5) *a noncompact or pressure-gauge defect;*
(6) *a critical-tail defect;*
(7) *a single terminally indecomposable retained velocity profile.*

*This is a concentration-compactness exhaustion of the realized terminal sequence, not a convention in the definition of $\mathcal{C}_{\mathrm{gen}}$.*

*Proof.* The generic representative is first put in the affine-normalized quotient. If all non-affine oscillation disappears, then Theorem 8.21 places the profile in the affine/parasitic lower stratum, contradicting the assumption that we are in $\mathcal{C}_{\mathrm{gen}}$. Thus a positive non-affine oscillation threshold survives, while the setup results supply retained compact velocity concentration.

Apply the paired terminal concentration-set/remainder decomposition Theorem 9.24 together with the compact-window to expanding-region protocol Theorem 9.17. The retained velocity mass forces at least one selected active or residual alternative by Corollary 9.25. If active neighborhoods persist, the priority order first records a compact retained descendant; iterating the same realized-descendant construction either produces a finite separated retained family, an infinite active chain, or terminates at a single retained profile with no further retained active descendant. If active neighborhoods do not exhaust the residual measure on the selected terminal regions, the protocol selects a diffuse exterior defect, a loss of local compactness or pressure-gauge compactness, or a scale-critical tail window. These alternatives are (4)–(6). If none of the separated, chained, diffuse, noncompact, or critical-tail remainders occurs, then the remaining single profile is terminally indecomposable by Proposition 10.5. The priority order makes the selected alternative unique. □

**Theorem 11.3** (Closure of the generic residual class by local concentration analysis)**.** *One has*

$$\mathcal{C}_{\mathrm{gen}} = \varnothing.$$

*Proof.* Suppose, for contradiction, that $V \in \mathcal{C}_{\mathrm{gen}}$. Apply Theorem 11.2. A retained compact active descendant is not a terminal stopping case: by descendant heredity it is reinserted into the same ordered terminal extraction until it gives a finite separated family, an infinite active chain, a diffuse/noncompact/critical-tail remainder, or a single indecomposable retained profile. Finite separated families are excluded by Theorem 10.24, and infinite active chains by Theorem 10.21. Diffuse and noncompact remainders are excluded by Theorems 9.27 and 9.102; critical tails are excluded by Theorem 9.88 and Corollary 9.101. The pressure-only case is assigned by Lemma 9.5 into the pressure/noncompact or affine/parasitic lower stratum, both already removed. In the remaining single-profile case, Proposition 10.5 gives terminal indecomposability and Lemma 10.10 gives the endpoint contradiction. Every selected alternative in the generic terminal exhaustion is impossible, so $\mathcal{C}_{\mathrm{gen}} = \varnothing$. ☐

## 12. Final assembly

The class closures above form the following complete dependency chain. The rotational relative-equilibrium arrow is the localized Coriolis-flux argument; the terminal concentration-set compactness, no finite separated-family, and indecomposability arrows are proved in this paper from local estimates:

$$\begin{aligned}
\mathcal{C}_{\mathrm{ax}} &\xrightarrow{\text{r-Liouville}} \varnothing,\\
\mathcal{C}_{\mathrm{rot}} &\xrightarrow{\text{local flux reduction + terminal assembly}} \varnothing,\\
\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}} &\xrightarrow{\text{stationary-hull reduction + terminal assembly}} \varnothing,\\
\mathcal{C}_{\mathrm{aff}} &\xrightarrow{\text{affine/parasitic normalization}} \text{no retained normalized representative},\\
\mathcal{C}_{\log\mathrm{diff}} &\xrightarrow{\text{log-window support profiles + hidden-scale closure}} \varnothing,\\
\mathcal{C}_{\mathrm{Young}} &\xrightarrow{\text{minimal mesoscopic-scale reduction}} \varnothing,\\
\mathcal{C}_{\mathrm{homcrit}} \cup \mathcal{C}_{\mathrm{logper}} \cup \mathcal{C}_{\mathrm{ap}} &\xrightarrow{\text{bounded-origin realization}} \varnothing,\\
\mathcal{C}_{\mathrm{gen}} &\xrightarrow{\text{critical-tail closure, finite separation, sequence-}L^3} \varnothing.
\end{aligned}$$

The positive local velocity concentration inherited by every normalized Seregin limit, after affine normalization through the non-affine oscillation functional $\Phi$, is the common contradiction target in the complete ordered decomposition.

Equivalently, the final exclusion is the composition of six steps:

(1) *Entry step:* The setup paper supplies a positive local velocity concentration sequence; the endpoint weak Serrin local Type I bounds Theorem 1.24 gives the terminal local energy/no-escape estimate in Theorem 1.26. Thus the sequence is admissible in the sense of Definition 1.17, and Theorem 1.5 turns it into a normalized Seregin profile with retained compact velocity concentration.

(2) *Stratification step:* the ordered residual decomposition places every remaining retained profile in the complete list of alternatives

$$\mathcal{C}_{\mathrm{ax}} \cup \mathcal{C}_{\mathrm{rot}} \cup \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\log\mathrm{diff}} \cup \mathcal{C}_{\mathrm{Young}} \cup \mathcal{C}_{\mathrm{homcrit}} \cup \mathcal{C}_{\mathrm{logper}} \cup \mathcal{C}_{\mathrm{ap}} \cup \mathcal{C}_{\mathrm{gen}}.$$

(3) *Closed-strata step:* the small, stationary $L^3$, tight, closed structured/decay, axisymmetric, rotational, and stationary-hull strata are excluded by the results cited above.

(4) *Lower-stratum step:* no retained normalized representative remains in $\mathcal{C}_{\mathrm{aff}}$, by oscillatory entry normalization. The alternative $\mathcal{C}_{\log\mathrm{diff}}$ is excluded by log-window support-profile selection plus hidden-scale compactness; see Corollary 9.54. The alternative $\mathcal{C}_{\mathrm{Young}}$ is excluded by the same hidden-scale compactness; see Corollary 9.90. The alternatives $\mathcal{C}_{\mathrm{homcrit}}, \mathcal{C}_{\mathrm{logper}}, \mathcal{C}_{\mathrm{ap}}$ are excluded by bounded-origin realization of coherent critical tails.

(5) *Residual terminal step:* the generic residual is reduced by the paired concentration-set decomposition, diffuse-defect compactness, diffuse-defect trichotomy, critical-tail compactification, realized critical-tail rigidity, descendant heredity, recurrence for concentration chains, finite separated-family exclusion, and terminal indecomposability.
(6) *Endpoint Liouville step:* a terminally indecomposable retained profile either belongs to the affine/parasitic lower stratum or has a parasitic-free bounded mild physical pullback with a backward $L^3$-bounded sequence. In the latter case it is zero by Albritton–Barker.

*Proof of Theorem 1.14, revisited.* Assume, for contradiction, that a retained admissible residual profile exists. By the refined ordered decomposition, after extraction it belongs to one of

$$\mathcal{C}_{\rm ax}, \mathcal{C}_{\rm rot}, \mathcal{C}^{\rm ret}_{\rm stat}, \mathcal{C}_{\rm aff}, \mathcal{C}_{\log\rm diff}, \mathcal{C}_{\rm Young}, \mathcal{C}_{\rm homcrit}, \mathcal{C}_{\rm logper}, \mathcal{C}_{\rm ap}, \mathcal{C}_{\rm gen}.$$

The axisymmetric class is empty by Theorem 4.4. The rotational class is reduced to the terminal stratification assembly theorem by Proposition 5.8 and is empty by Corollary 10.27. The stationary-hull class is reduced to the terminal stratification assembly theorem by Propositions 6.12 and 7.6 and is empty by Corollary 10.28. No retained normalized representative remains in $\mathcal{C}_{\rm aff}$ by Theorem 9.89. The log-diffuse and Young branches are empty by Corollaries 9.54 and 9.90. The coherent critical-tail classes are empty by Corollary 9.100. The generic terminal class is empty by Theorems 10.25 and 11.3.

Every refined residual alternative is impossible, contradicting the ordered decomposition. Hence $\mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing$, which is the residual hypothesis of the setup paper. □

**Theorem 12.1** (Paper IV residual closure)**.** *For every normalized Seregin collection $\mathcal{S}$ satisfying the imported setup admissibility hypotheses,*

$$\mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing.$$

*Equivalently, Paper IV proves exactly the missing setup hypothesis that this residual class is empty.*

*Proof.* This is the conclusion of Theorem 1.14, proved in the preceding final assembly. The statement is repeated here to separate the output of Paper IV from the final consequence imported from the setup paper. □

**Corollary 12.2** (Verification of the setup residual hypothesis)**.** *The residual-class hypothesis Hypothesis 16.1 of the setup paper holds.*

*Proof.* By Theorem 1.3, the setup residual hypothesis is the assertion that the retained residual class $\mathcal{R}^{\#}(\mathcal{S}; I, J)$ is empty after the lower setup classes are removed. This is Theorem 12.1. □

**Corollary 12.3** (Local pointwise Type-I exclusion from the two-paper chain)**.** *By the setup paper's conditional final assembly Theorem 33.1 of the setup paper and Corollary 12.2, no local pointwise Type I singularity exists under the stated admissibility hypotheses.*

*Proof.* This is the conditional final assembly imported in Theorem 1.3, with the residual hypothesis verified by Corollary 12.2. □

*Remark* 12.4 (Inherited inputs and new arguments)*.* The paper inherits the base setup, including local concentration, the local Type I Type II dichotomy, the Type I Seregin extraction theorem, and the initial structural decomposition, from the setup paper summarized in Theorem 1.3. All remaining case-closure steps are proved in this paper using local concentration and terminal dynamics:

- terminal concentration-set extraction;
- compactness of descendant/tail families and pressure controls;

- recurrence for chains of retained velocity concentration profiles;
- finite separated-family exclusion;
- oscillatory-entry normalization;
- hidden-scale compactness reduction for critical tails;
- bounded-origin realization and critical-tail alternative exclusion;
- and the compactness and alternative exclusions culminating in Theorem 11.3.

*Remark* 12.5 (No hidden global Liouville or critical-norm input). The residual closure does not invoke any global critical-norm bound on the original residual class, any global Coriolis–Leray Liouville theorem for bounded centered profiles, or any global Liouville theorem for bounded ancient solutions of the centered equation. The only external Liouville input used after residual closure is the Albritton–Barker endpoint Liouville theorem Theorem 10.9, applied in exactly two places. First, the local weak-Serrin Type I estimate Theorems 1.24 and 1.26 upgrades the local pointwise Type I envelope to terminal velocity no-escape; this is an entry implication, optional inside Paper IV in the sense of Section 1.4. Second, in Lemma 10.10, the endpoint ancient Liouville theorem is applied to a parasitic-free bounded mild physical pullback that carries a backward $L^3$-bounded sequence supplied by indecomposable sequence-$L^3$ completeness Lemma 10.6. No other global Liouville theorem and no global critical-norm estimate is used in any of the closure steps for $\mathcal{C}_{\mathrm{ax}}, \mathcal{C}_{\mathrm{rot}}, \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}, \mathcal{C}_{\mathrm{aff}}, \mathcal{C}_{\log\mathrm{diff}}, \mathcal{C}_{\mathrm{Young}}, \mathcal{C}_{\mathrm{homcrit}}, \mathcal{C}_{\mathrm{logper}}, \mathcal{C}_{\mathrm{ap}}, \mathcal{C}_{\mathrm{gen}}$. In particular, the rotational closure Corollary 10.27 proceeds through the localized Coriolis-flux identity and terminal concentration-set decomposition, and the variance/critical-tail closures proceed through the minimal mesoscopic-scale reduction theorem Theorem 9.81 together with bounded-origin realization Theorem 9.94.

## Appendix A. Setup Paper Label Dictionary

The imported setup results displayed in Theorem 1.3 are cited by their companion-paper theorem numbers. This table records the same dictionary in one place so that Paper IV can be read using the public theorem numbering and the imported statements displayed here.

| | |
|---|---|
| Theorem 5.7 of the setup paper | Positive local concentration at a Type I singular point. |
| Proposition 5.9 of the setup paper | Terminal concentration theorem. |
| Theorem 8.3 of the setup paper | Local weak-Serrin/Type I implication. |
| Corollary 8.4 of the setup paper | Admissibility of local Type I sequences. |
| Lemma 8.6 of the setup paper | Local Seregin extraction. |
| Theorem 9.1 of the setup paper | Normalized Seregin extraction theorem. |
| Proposition 11.12 of the setup paper | Ancient suitable limit. |
| Proposition 12.2 of the setup paper | Nonzero retained velocity limit. |
| Lemma 11.2 of the setup paper | Pressure compactness. |
| Lemma 11.3 of the setup paper | Pressure gauge compatibility. |
| Lemma 11.4 of the setup paper | Pressure atlas. |
| Lemma 15.3 of the setup paper | Mild ancient representation. |
| Lemma 20.4 of the setup paper | Mildness under limits. |
| Lemma 25.4 of the setup paper | Finite-shift physical mildness. |
| Theorem 15.8 of the setup paper | Small and stationary lower classes. |
| Theorem 15.10 of the setup paper | Known structured/decay classes. |
| Theorem 26.1 of the setup paper | No uniformly tight normalized collection. |
| Theorem 25.8 of the setup paper | Tight Liouville theorem. |
| Theorem 25.5 of the setup paper | Bounded mild ancient endpoint theorem. |
| Hypothesis 16.1 of the setup paper | Residual hypothesis proved here. |
| Theorem 33.1 of the setup paper | Conditional final assembly in the setup paper. |

Paper IV uses these results only through the imported theorem statements and the companion-paper numbering displayed in this appendix.

## Appendix B. Generator calculus on compact recurrent cores

This appendix collects the operator-theoretic facts used implicitly in the amenable-averaging and pressure-observable arguments of the body of the paper. The setting is a compact metrizable invariant minimal core $(M, \mu, \Phi_t)$ of the centered translation hull, where $\mu$ is an $\Phi_t$-invariant Borel probability measure (existence of such a measure is the standard Krylov–Bogolyubov consequence of Theorem 9.29; uniqueness is not used). The acting group is $G := \mathbb{R}^3 \rtimes \mathbb{R}$ (covariant translations and parabolic time shift).

B.1. **Hilbert space and unitary representation.** Let $H := L^2(M, \mu; \mathbb{C})$. The Koopman representation $U_g f := f \circ \Phi_g^{-1}$ $(g \in G)$ is a strongly continuous unitary representation of $G$ on $H$, by $\Phi_t$-invariance of $\mu$ and the dominated convergence theorem. Write $\Pi_0 : H \to \ker(\nabla_G)$ for the orthogonal projection onto the $G$-invariant subspace of $H$. If the chosen invariant measure is ergodic on the minimal core, this projection is the spatial mean $f \mapsto \int f \, d\mu$; the arguments below require only the projection $\Pi_0$, not ergodicity or uniqueness of $\mu$.

For each one-parameter subgroup $\{e^{tX_k}\}_{t\in\mathbb{R}}$ of $G$ ($k = 0, 1, 2, 3$: time translation $X_0$ and three covariant spatial translations $X_1, X_2, X_3$), Stone's theorem gives a self-adjoint generator $A_k$ on a dense domain $\mathcal{D}(A_k) \subset H$ with $U_{e^{tX_k}} = e^{itA_k}$. Define the non-negative self-adjoint operator $\mathcal{L}$ by the closed quadratic form

$$\mathfrak{q}(f) := \sum_{k=0}^{3} \|A_k f\|^2_{L^2(M,\mu)}, \qquad f \in \bigcap_{k=0}^{3} \mathcal{D}(A_k).$$

Thus $\mathcal{L}$ is the Friedrichs generator associated with the formal operator

$$\mathcal{L} = \sum_{k=0}^{3} A_k^2,$$

and $\mathcal{L}\Pi_0 = \Pi_0\mathcal{L} = 0$. All inverses below are understood as resolvent limits of $(\varepsilon + \mathcal{L})^{-1}(I - \Pi_0)$. Equivalently, with the skew-adjoint infinitesimal generators

$$D_k := iA_k,$$

the non-negative generator is

$$\mathcal{L} = -\sum_{k=0}^{3} D_k^2$$

on the common cylinder core. This is the operator denoted by $-\sum D_k^2$ in the recurrent-core pressure argument.

B.2. **Pressure observables and bounded continuous core.** The pressure observables used in Definition 9.31 (iv)–(v) (compatible-gauge averaged pressure differences $\Pi(\cdot) - a(\cdot)$ integrated against compactly supported test functions, plus the affine quotient observables $[\Pi] \in D_1$) define bounded measurable functions $b_{ij}^{(\varphi,a)} : M \to \mathbb{R}$ for each test function $\varphi \in C_c^\infty$ and each gauge selection $a$ compatible with Definition 8.14. These observables generate a unital $C^*$-subalgebra $\mathcal{B} \subset C(M)$ which is dense in $L^2(M, \mu)$. The restriction of the unitary representation $U_g$ to $\mathcal{B}$ is the Koopman action by composition. Affine pressure modes have already been separated into the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$ before this calculus is invoked; the observables below are therefore taken modulo the affine quotient fixed in Definition 8.14.

B.3. **Regularized inverse and distributional pressure identity.** For each $\varepsilon > 0$ and each generator $A_k$, define the regularized distributional pressure functional by

$$\left\langle q_{k,\varepsilon}^{(\varphi,a)}, F \right\rangle := \left\langle (I - \Pi_0) D_i D_j b_{ij}^{(\varphi,a)}, (\varepsilon + \mathcal{L})^{-1} D_k^* F \right\rangle$$

for cylinder test observables $F \in \mathcal{B}$. Here $D_k = iA_k$ is the skew-adjoint generator, $D_k^* = -D_k$, and $D_i D_j b_{ij}$ is interpreted in the weak generator sense on the cylinder core. The operator $(\varepsilon + \mathcal{L})^{-1}$ is bounded on $H$ with norm $\le \varepsilon^{-1}$ and commutes with $\Pi_0$; each $D_k$ is closed on $\mathcal{D}(A_k)$, and $(\varepsilon + \mathcal{L})^{-1} D_k^* F$ lies in the form domain. Therefore the displayed pairing is well-defined for every $\varepsilon > 0$, $\varphi$, $a$, and $F \in \mathcal{B}$. If the weak source $(I - \Pi_0) D_i D_j b_{ij}^{(\varphi,a)}$ is represented by an $L^2$ function, then this functional is represented by the corresponding $H$-element; the arguments in the paper need only the weak pairing. In the formal notation used in the body this is

$$q_{k,\varepsilon}^{(\varphi,a)} = D_k (\varepsilon + \mathcal{L})^{-1} (I - \Pi_0) D_i D_j b_{ij}^{(\varphi,a)}$$

understood through the preceding weak pairing.

**Lemma B.1** (Distributional pressure identity in the generator calculus)**.** *For every test function $\varphi \in C_c^\infty$ and every compatible gauge $a$, the family $\{q_{k,\varepsilon}^{(\varphi,a)}\}_{\varepsilon > 0}$ has a distributional limit along the cylinder core as $\varepsilon \downarrow 0$. The limit is independent of the chosen gauge $a$ in the affine quotient of Definition 8.14 and represents the centered distributional pressure identity* $\Pi(\cdot) - a(\cdot) = R_i R_j (\chi u_i u_j) + h$ *of Lemma 9.6 averaged against $\mu$.*

*Proof.* The spectral theorem applied to $\mathcal{L}$ gives

$$D_k (\varepsilon + \mathcal{L})^{-1} (I - \Pi_0) \xrightarrow{\varepsilon \downarrow 0} D_k \mathcal{L}^{-1} (I - \Pi_0)$$

in the weak resolvent sense on the form domain, after projecting away the invariant kernel. For $F \in \mathcal{B}$, the test $(\varepsilon + \mathcal{L})^{-1} D_k^* F$ converges in the form topology to $\mathcal{L}^{-1} D_k^* F$ modulo $\Pi_0 H$. Hence the displayed weak pairing with $(I - \Pi_0) D_i D_j b_{ij}^{(\varphi,a)}$ converges and defines the distributional limit $q_k^{(\varphi,a)}$. Gauge independence in the affine quotient is the statement that two compatible gauges $a, a'$ differ by an affine pressure mode whose distributional contribution to $D_i D_j b_{ij}$ vanishes (because affine functions are killed by two derivatives). The identification with the centered distributional pressure identity is the statement that, in a fixed local covariant chart, the operators $D_k$ on $M$ restrict to the distributional spatial derivatives along the chart and $A^{-1}(I - \Pi_0)$ restricts to the inverse Laplacian on the centered ancient solution modulo the kernel of $G$-invariant functions; this restriction is the content of the local pressure atlas (Theorem 1.3 (3)). ☐

*Remark* B.2 (Gauge independence in the affine quotient)*.* The limit $q_k^{(\varphi,a)}$ depends on the gauge $a$ only through its affine quotient $[a] \in D_1$ of Definition 8.14. If two compatible gauges $a, a'$ give the same affine quotient, then $a - a'$ is the time-only contribution allowed by the gauge atlas, and the pressure identity in Lemma B.1 differs by a quantity in the kernel of $D_i D_j$, which is killed in the regularized formula and therefore does not contribute to the limit. This is the precise statement underlying the use of pressure observables in the amenable-averaging arguments of Theorem 9.32 and Theorem 9.29: averaging is performed on the gauge-quotient pressure observables, and the regularized inverse $q_{k,\varepsilon}$ provides a regularized weak representative for each fixed $\varepsilon$ before the limit $\varepsilon \downarrow 0$ gives the gauge-independent pressure identity.

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